YILDIZ TEKNİK ÜNİVERSİTESİ

İNŞAAT FAKÜLTESİ - İNŞAAT MÜHENDİSLİĞİ BÖLÜMÜ
YAPI ANABİLİM DALI

YAPI STATİĞİ I DERS NOTLARI

Prof. Dr. Ali KOÇAK

MART, 2021

YAPI STATİĞİ I DERS NOTLARI

Kaynaklar

1. Yapı Statiği I.....(İ. EKİZ)
2. Yapı Statiği.....(İ. GÖĞÜŞ)
3. Yapı Statiği (Cilt I-II).....(A. ÇAKIROĞLU- E. ÇETMELİ)
4. Yapı Mühendisliğine Giriş....(F. KARADOĞAN, S.PALA, E.YÜKSEL, Y. DURGUN) (Yapısal Çözümleme)

I. GENEL BİLGİLER

I.1) Yapı Mühendisliğinin Amacı

Yapı Mühendisliğinin amacı, yapıları belirli bir emniyet ve yeterli rijitlikte, ekonomik ve amaca en uygun şekilde tasarlamak ve boyutlandırmaktır.

I.2) Yapı Mühendisliğinde İzlenen Yol

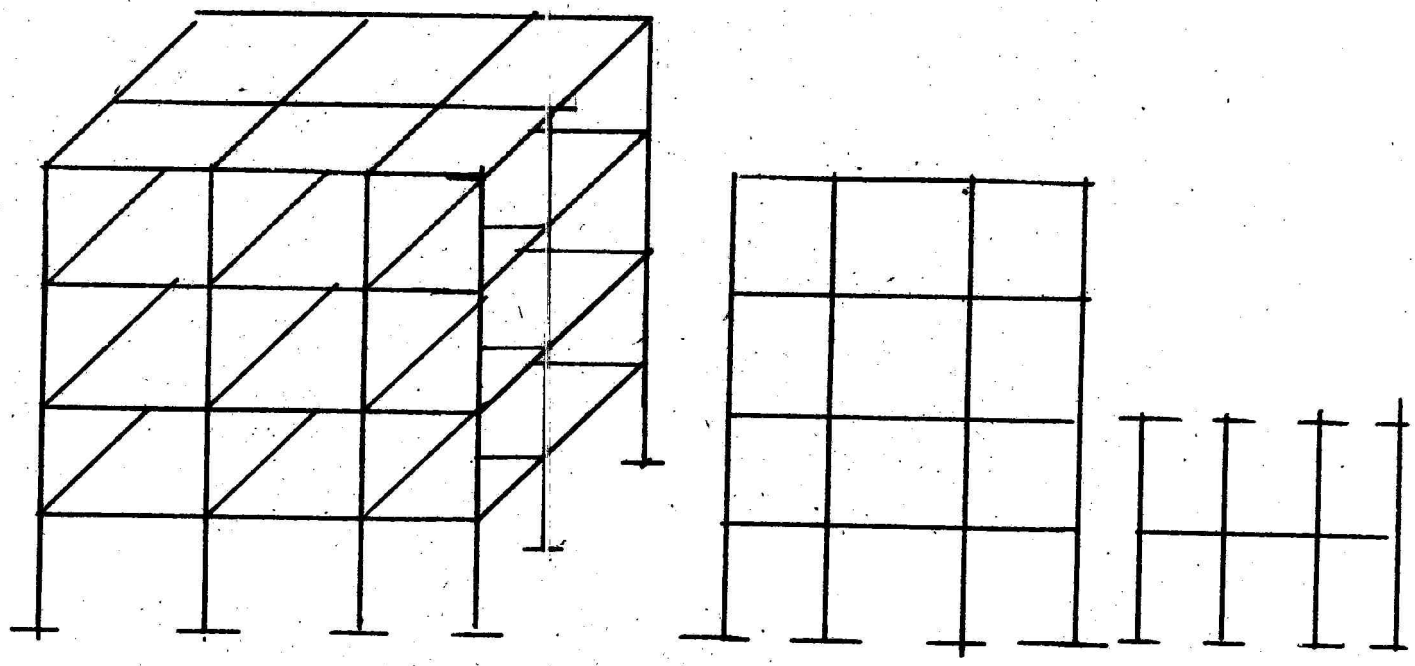
Bina tasarımı ve imalatı süreci;

- Planlama (Ülke, Kent, İlçe, Ada)
- Yer seçimi (Amaca uygun bina yapılabilecek arsa)
- Mimari Tasarım
- Taşıyıcı Sistem Tasarımı
    - Sistem seçimi
    - Yükler
    - Ön boyutlar
    - Sistem analizi
    - Kontrol
- Bina İmalatı şeklindedir.

Yapı mühendisinin görevi, mimari tasarım evresinden başlamakta, bina imalatının bitimine kadar sürmektedir. Yapı mühendisliğinde izlenen yolu ve evreleri şu şekilde verebiliriz:

a) Amaca en uygun olan yapı biçimi (çubuk sistem, plâk, kabuk, dolu sistem, kafes sistem vb). ve malzeme cinsi (betonarme, ön gerilmeli beton, çelik v.b) seçilir.

b) Yapının biçimi, nesnelleri ve malzemenin özellikleri için kabuller yaparak yapı için ideal bir sistem seçilir. Bu sistem üç boyutlu veya düzlem sistem şeklinde olur.



Üç Boyutlu Sistem (Gerçek Model)

Düzlem Bina Çerçevesi

Düzlem Kat Çerçevesi

Yalnız sonuçlarda yapının bu idealleştirilmesi göz önünde bulundurulmalı ve buna göre bir emniyet payı bırakılmalıdır.

c) Yapıyı etkileyecek işletme, deprem vs. yüklerin şiddetleri ve cinsleri saptanır.

d) Yapıda kullanılan malzemelerin mekanik özellikleri belirlenir.

$$f_{cd} = \frac{f_{ck}}{\gamma_{mc}} \quad , \quad f_{yd} = \frac{f_{yk}}{\gamma_{ms}} \quad , \quad \epsilon = \frac{\sigma}{E} \quad , \quad \gamma = \frac{Z}{G}$$

e) İdealleştirilmiş sistemin boyutları bazı ampirik formüllere göre boyutlandırılır.

f) İdealleştirilmiş sisteme, dış yükler altında çözülecek elemanların kesit tesirleri, yer değiştirmeleri ve gerilmeleri hesaplanır. Bulunan kesit tesirlerine göre boyut analizi ve sistemin yapısına göre diğer elemanları (donatı, bağlantı elemanları, birleştirme elemanları v.b) tahkik edilir.

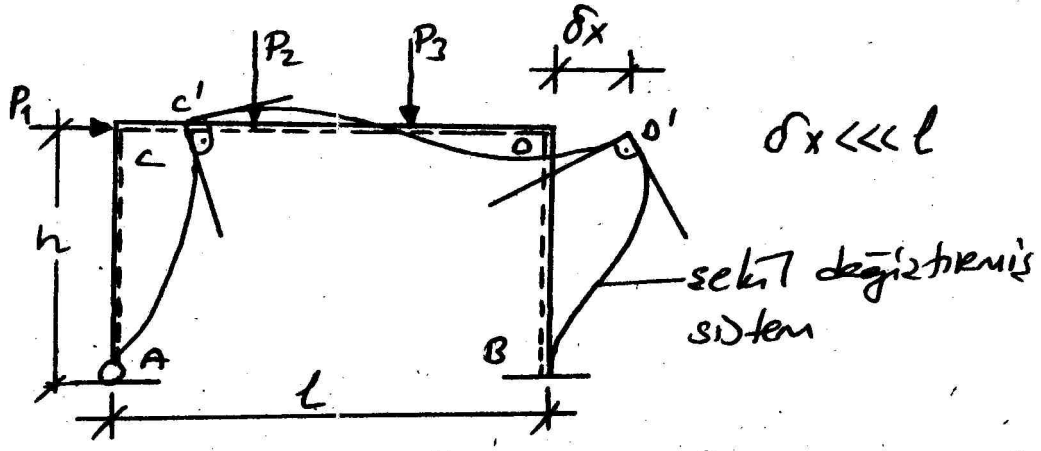
g) Yukarıdaki şartları sağlayan çeşitli çözümler vardır. Ancak bunların arasından en ekonomik olanı seçilir. Bu seçim sırasında amaca uygunluk, estetik olanı tercih edilir.

I.3) Yapı Statiğinde Yapılan Varsayımlar

a) Yapı statiğinde incelenecek sistemler yükün şekline ve şiddetine bağlı değildir.

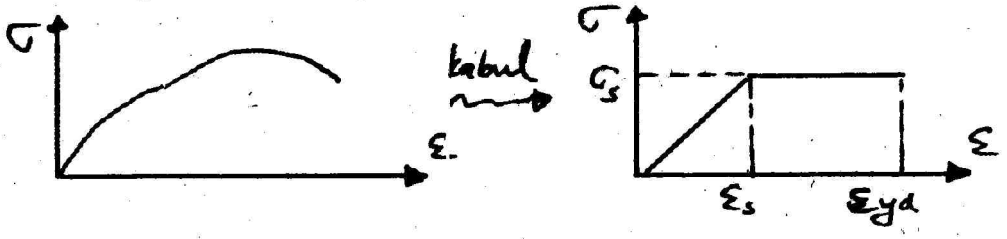
b) Sisteme etkiyen bütün yükler ayrı ayrı sisteme etki ettirilip, sonuç değerleri toplanabilir. (Süperpozisyon kuralı geçerlidir.)

c) Dış yüklerden meydana gelen yer değiştirmeler taşıyıcı elemanın boyutları yanında ihmal edilebilecek kadar küçüktür.

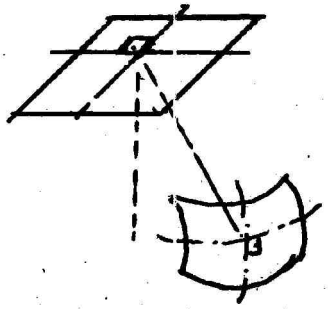


Yer değiştirmelerin çok küçük olması sebebiyle hesaplar deformasyona uğramamış sistem üzerinde yapılır. Bu teoriye I. Mertebe Teorisi denir. Süperpozisyon kuralı geçerlidir. Deformasyona uğramış sistem üzerinde yapılan hesaplar II. Mertebe Teorisidir, ancak bu teoride süperpozisyon kuralı geçerli değildir.

d) Malzeme lineer elastiktir (Malzeme homojen, izotrop ve Hooke Kanunu'na uygundur.)

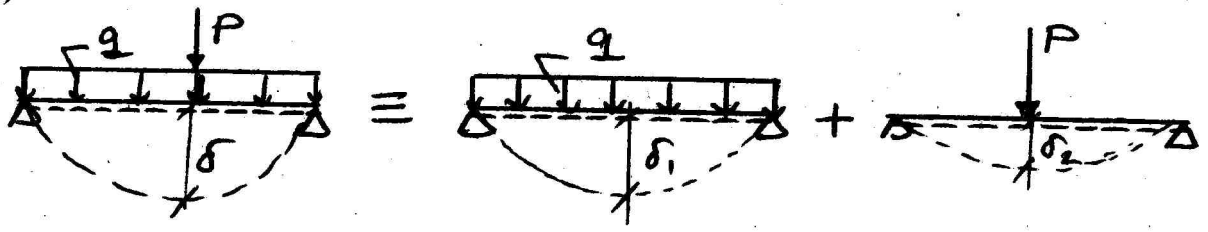


e) Bernoulli Hipotezi geçerlidir. Kesitler şekil değiştirdikten sonra kendi düzlemlerine veya ekseriyetle dik veya düzlem kalırlar.



f) Sistem statik denge durumundadır

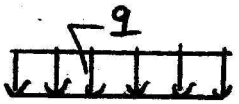
g) Superpozisyon kuralı geçerlidir.



$$\delta \equiv \delta_1 + \delta_2$$

I.4) YÜKLER

Sistemde iç kuvvet ve deformasyon meydana getiren etkilere tümüne (dış yük, ısı değişimi, rötre, şişme, mesnet çökmesi) yük denir. Yüklerin sınıflandırılması aşağıdaki şekildedir:

1) a) Sabit yük: (öz ağırlık) (tekil yük),  Yayılı yük

b) İlave yükler: zamanla yapıya etkiyen yükler (rüzgar, kar, mesnet çökmesi)

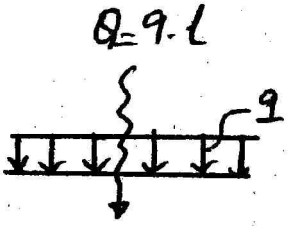
c) Toplam yük ($P=1,4G+1,6Q$; $P=G+Q$; $0,9G$)

2) a) Sabit yükler

b) Hareketli yükler

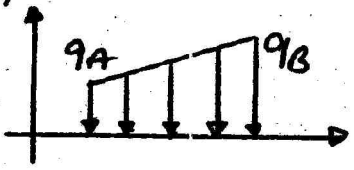
3) a) Tekil yükler: Sisteme tek bir noktada etkiyen yükler ($\downarrow P$)

b) Yayılı yükler

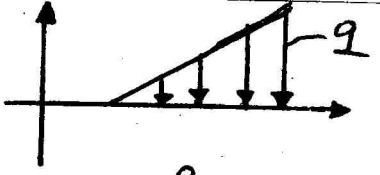


— Düzgün yayılı yükler

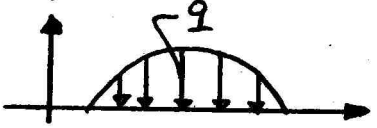
— Trapez " "



— Üçgen yayılı yük



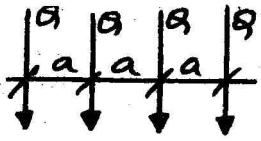
— Parabol " "

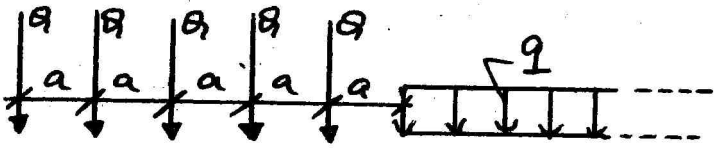


Yayılı yük çizgi boyunca etki ediyorsa: kN/m

" " düzlem " " " : kN/m^2

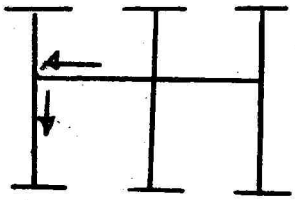
c) Yük Kateği



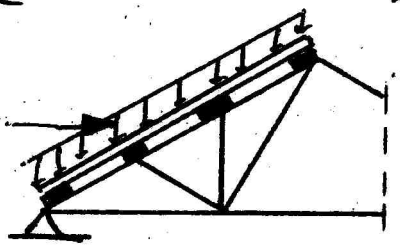


4) Sisteme Aktarılmasına Göre

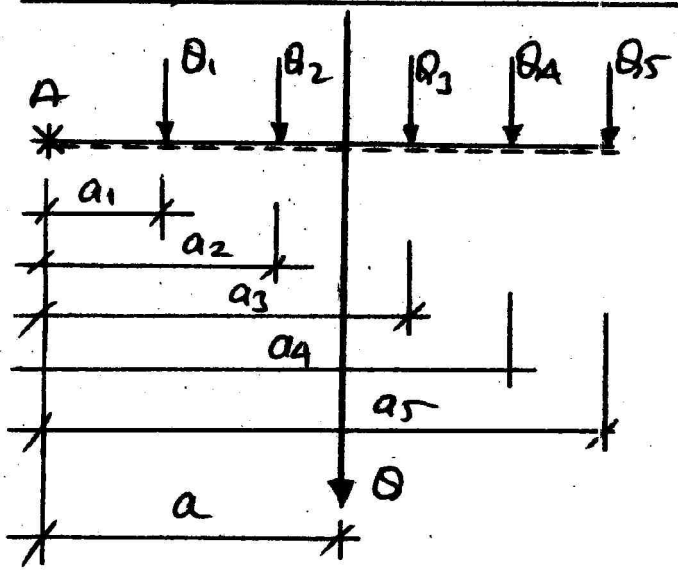
a) Doğrudan yükler



b) Dolaylı "



I.4.1) Bileşke Kuvvetin Yeri ve Değeri



$\Sigma M_A = 0$

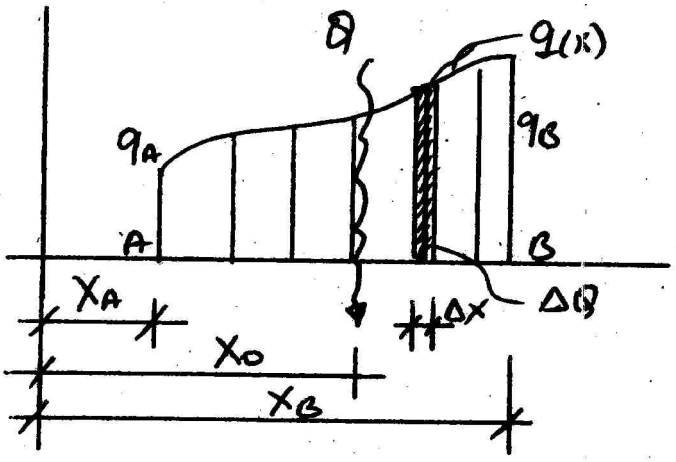
$Q_1 \cdot a_1 + Q_2 \cdot a_2 + Q_3 \cdot a_3 + Q_4 \cdot a_4 + Q_5 \cdot a_5 + \dots =$

$= Q \cdot x$

$\Sigma Q_i \cdot a_i = \Sigma Q_i \cdot x$

$$x = \frac{\Sigma Q_i \cdot a_i}{\Sigma Q_i}$$

Yayılı yükün bileşkesi ve şiddeti



1) $q_{(x)} = \lim_{\Delta x \to 0} \frac{\Delta Q}{\Delta x}$

2) $Q = \sum q_{(x)} . \Delta x$

3) $Q = \sum_{A}^{B} \lim_{\Delta x \to 0} \frac{\Delta Q}{\Delta x} . \Delta x$

$$Q = \int_{x_A}^{x_B} q_{(x)} dx$$

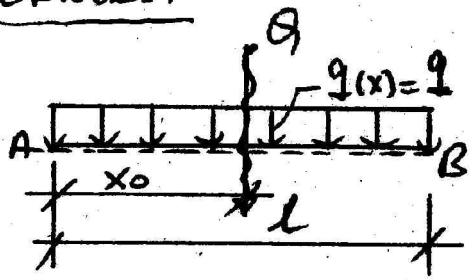
Yayılı yükün bileşkesi

4) $x_o = \frac{\sum (\Delta Q . x)}{\sum \Delta Q}$

$$x_o = \frac{\int_{x_A}^{x_B} q_{(x)} . x \, dx}{\int_{x_A}^{x_B} q_{(x)} dx}$$

Bileşke kuvvetin yeri

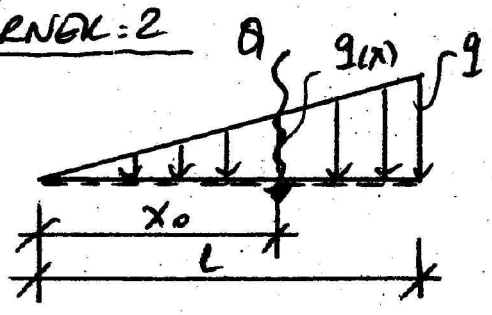
ÖRNEK: 1



$$Q = \int_0^l q \, dx = q.x \Big|_0^l = q.l$$

$$x_o = \frac{\int_0^l q . x \, dx}{\int_0^l q \, dx} = \frac{q \frac{x^2}{2} \Big|_0^l}{q.l} = \frac{l}{2}$$

ÖRNEK: 2



$q_{(x)} = Ax + B$

$$q_{(x)} = \frac{q}{L} x$$

$x = 0 \rightarrow q_{(x)} = 0 \rightarrow 0 = A.0 + B$

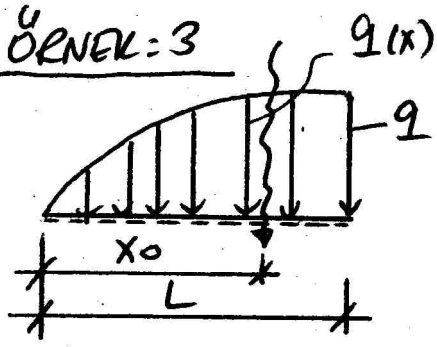
$B = 0$

$x = L \rightarrow q_{(x)} = q \rightarrow q = AL \rightarrow A = \frac{q}{L}$

$$Q = \int_0^L \frac{q}{L} x \, dx = \frac{q}{L} \frac{x^2}{2} \Big|_0^L = \frac{qL}{2}$$

$$x_o = \frac{\int_0^l \frac{q}{L} x . x \, dx}{qL/2} = \frac{\frac{q}{L} \frac{x^3}{3} \Big|_0^L}{\frac{qL}{2}} = \frac{2}{3} l$$

ÖRNEK: 3



$q(x) = Ax^2 + Bx + C$

Sınır Koşulları

$x=0 \rightarrow q(x)=0 \rightarrow A0+B0+C=0 \rightarrow C=0$

$x=L \rightarrow q(x)=q \rightarrow AL^2+BL=q$

$x=L \rightarrow q(x')=0 \rightarrow 2Ax+B=q(x')$

$2AL+B=0$

$A=-\frac{q}{L^2}, \quad B=\frac{2q}{L}$

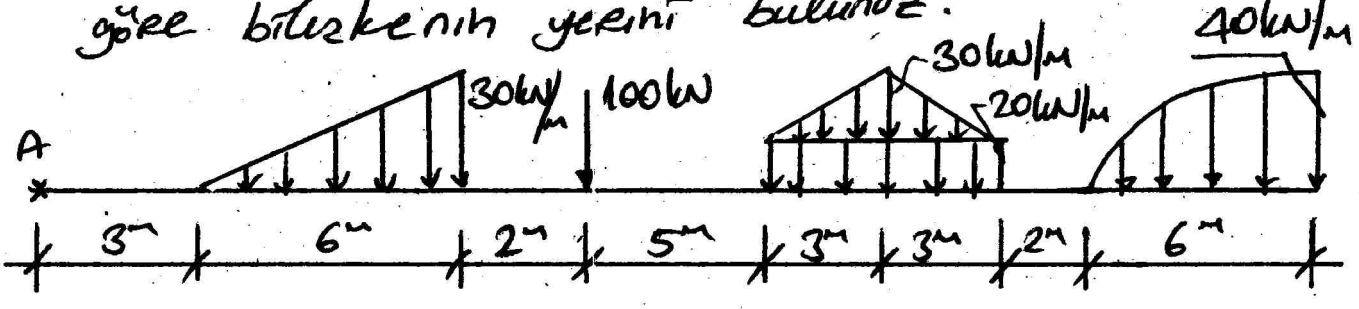
$$q(x) = -\frac{q}{L^2}x^2 + \frac{2q}{L}x$$

$$\boxed{q(x) = \frac{q}{L^2}x(2L-x)}$$

$$Q = \int_0^L q(x)\,dx = \int_0^L \left(-\frac{q}{L^2}x^2 + \frac{2q}{L}x\right)dx = \left|-\frac{q}{L^2}\frac{x^3}{3} + \frac{2q}{L}\frac{x^2}{2}\right|_0^L = \frac{2}{3}qL$$

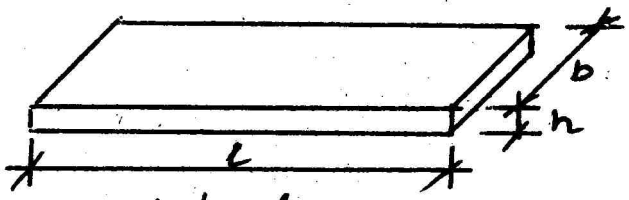
$$x_0 = \frac{\int_0^L \left(-\frac{q}{L^2}x^3 + \frac{2q}{L}x^2\right)dx}{\frac{2}{3}qL} = \frac{\left|-\frac{q}{L^2}\frac{x^4}{4} + \frac{2q}{L}\frac{x^3}{3}\right|_0^L}{\frac{2}{3}qL} = \frac{5}{8}L$$

ÖDEV: Şekilde verilen yüklerin bileşkesini ve A noktasına göre bileşkenin yerini bulunuz.

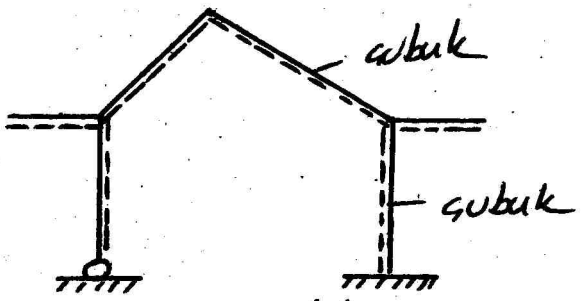


I.5) YAPI SİSTEMLERİNİN SINIFLANDIRILMASI

a) Bir Boyutlu Sistemler: Bu sistemler, iki boyutu üçüncü boyutuna göre çok küçük olan sistemlerdir. Bunlara çubuk sistemler denir.

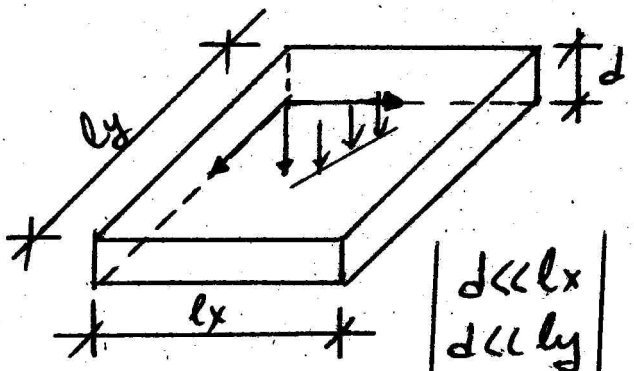


çubuk eleman



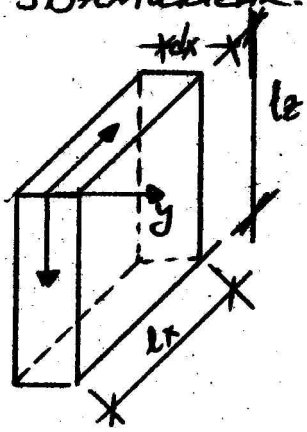
çerçeve sistem

b) İki Boyutlu Sistemler: Kalınlıkları diğer 2 boyutuna nazaran küçük olan yapı sistemleridir.

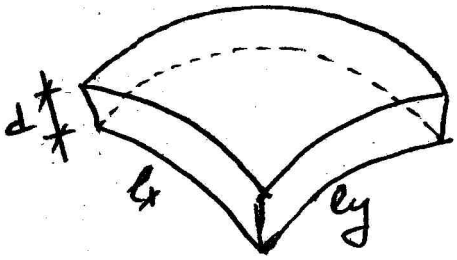


plâk

$d \ll l_x$
$d \ll l_y$
$l_x \simeq l_y$

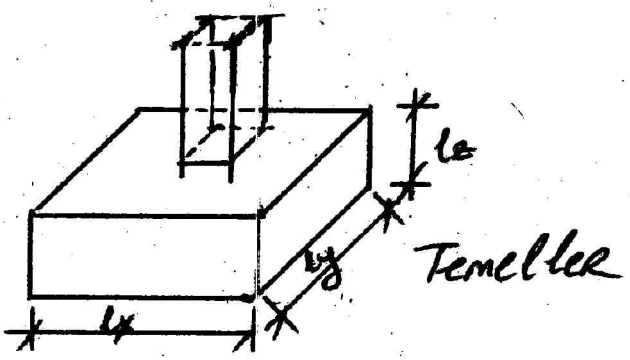


levha



kabuk

c) Üç Boyutlu Sistemler: Üç boyutta birbirine yakın olan sistemlerdir.



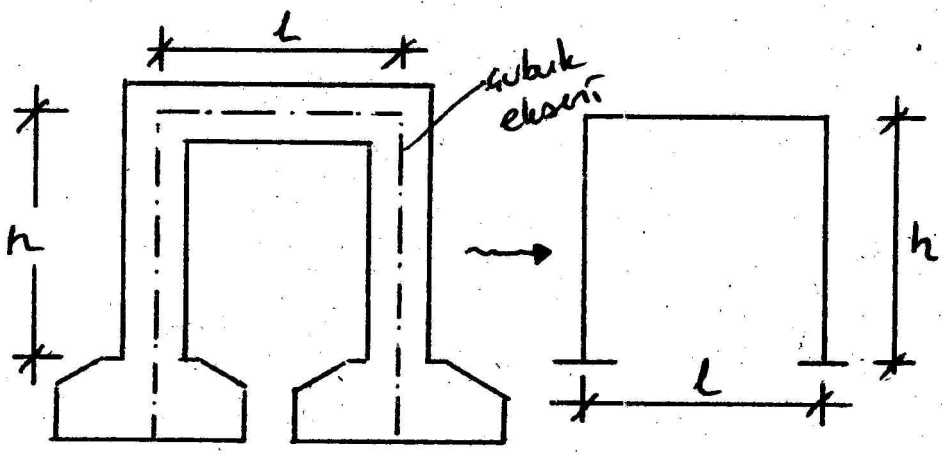
Temeller

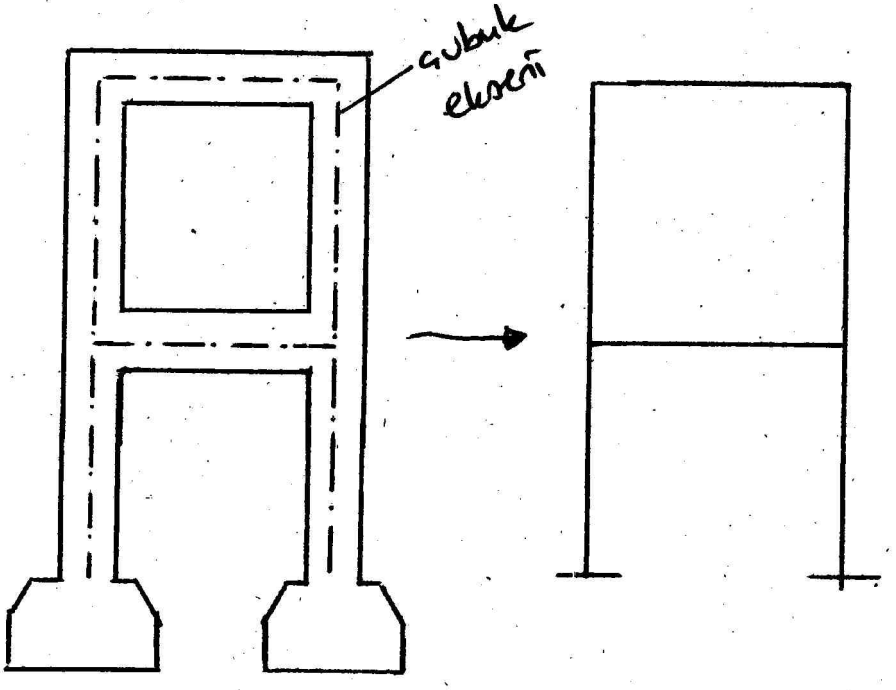
* Yapı statiğinde yalnız bir boyutlu taşıyıcı sistemler incelenecektir.

I.6) BİR BOYUTLU TAŞIYICI ELEMANLAR (ÇUBUKLAR)

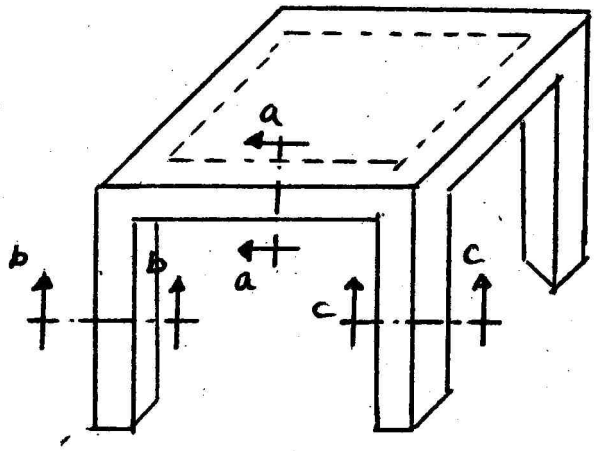
Bir boyutlu taşıyıcı sistemler; düğüm noktası, mesnet ve çubuklardan oluşmaktadır. Sistem hesapları çubuk eksenlerine göre ve çubuk kesitleri dikkate alınarak yapılır.

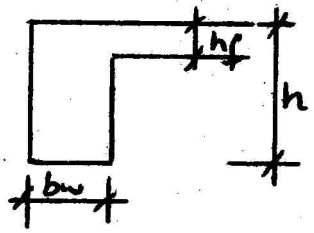
I.6.1) Çubuk Ekseni: Çubuğa dayatılan dik kesitle çubuk ara kesitinin ağırlık merkezini birleştiren noktaların geometrik yeridir.



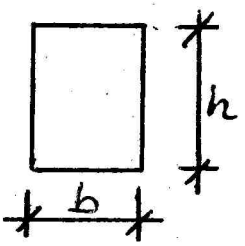


I.6.2) Çubuk Kesiti: Çubuğa dik olarak alınan kesittir.

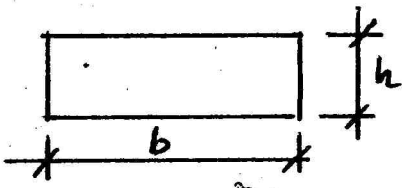




a-a kesiti



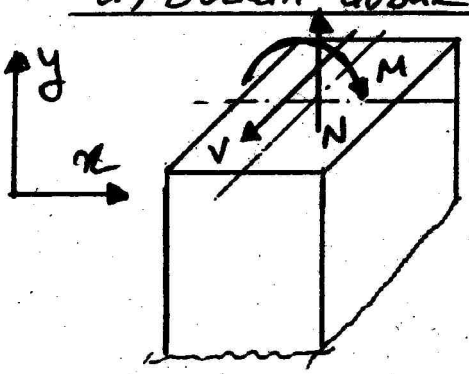
b-b kesiti

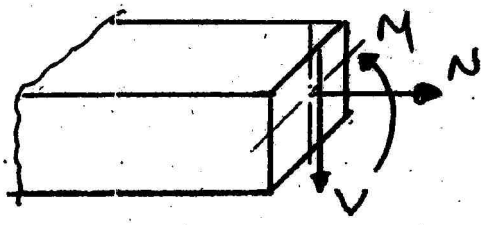


c-c kesiti

Eksenleri bir düzlem içinde bulunan çubuklara düzlem çubuklar, eksenleri ve dış kuvvetleri aynı düzlem içinde olan sistemlere düzlem sistemler denir. Sistemlerin 3 boyutlu olarak ele alınması sonucu oluşan sistemlere uzay sistemler, bu sistemdeki çubuklara da uzay çubuk adı verilir.

a) Düzlem Çubuk

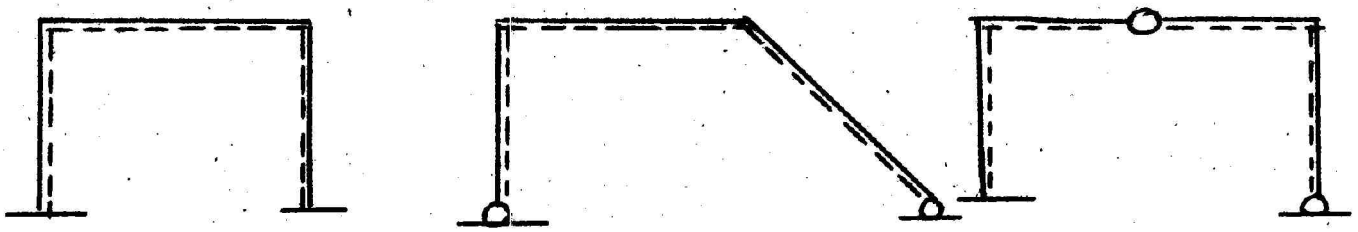




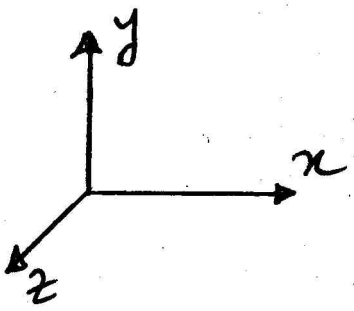
Bilinmeyen Sayısı:

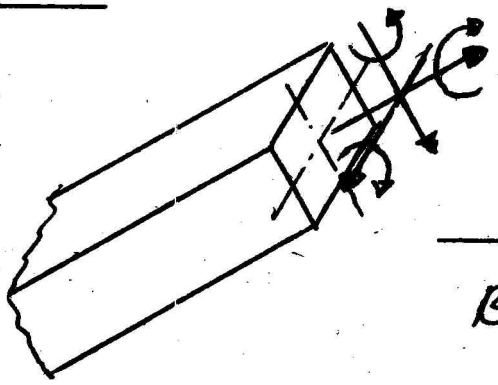
Bir uçta: $3\begin{pmatrix} M_i \\ N_i \\ V_i \end{pmatrix}$

b) Düzlem Sistem



b) Uzay Çubuk

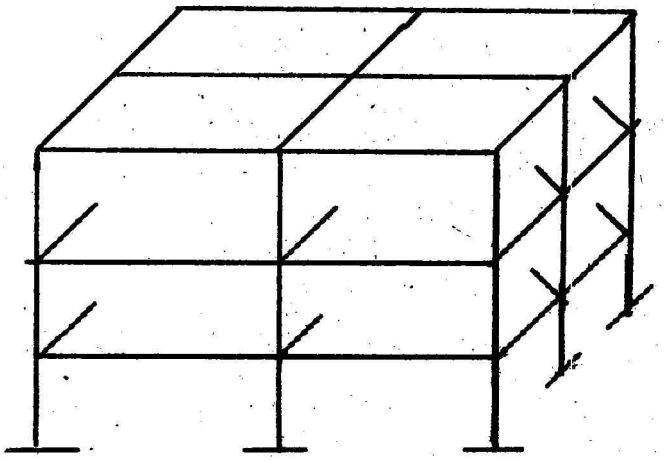




Bilinmeyen Sayısı:

Bir uçta: 6

d) Uzay Sistem

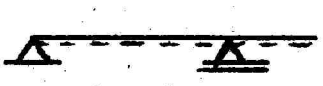


I.6.3) Düzlem Sistemler

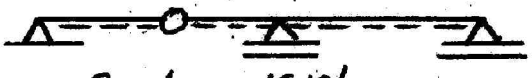
a) Dolu Gövdeli Sistemler: Eğilme ve uzama+kısalma etkisinde olan sistemlerdir. Sistemde N,N,V kesit tesiri oluşmaktadır.

Basit Kiriş

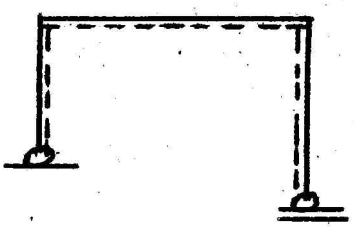
Konsol Kiriş



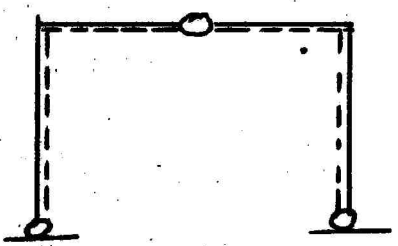
Çıkmalı Kiriş



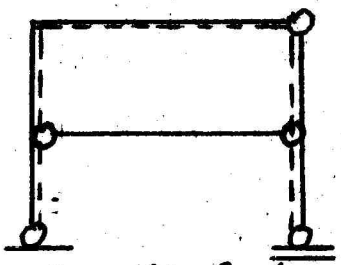
Gerber Kiriş



Basit Çerçeve Sistem

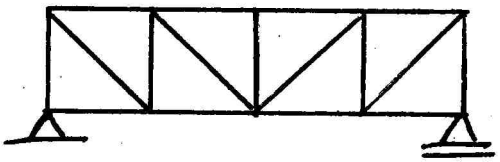


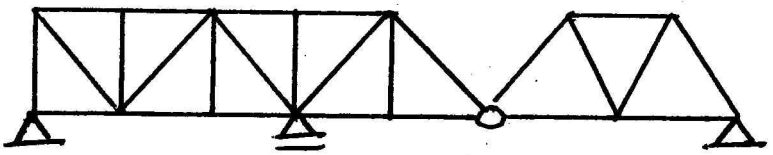
Çok Katlı Sistem



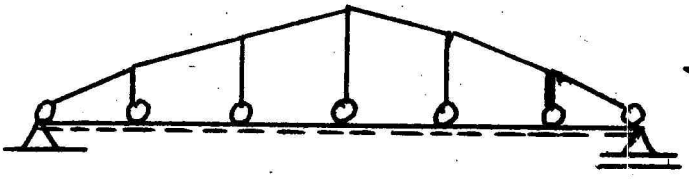
Gergili Sistem

b) Kafes Sistemler : Yalnız uzama+kısalma etkisinde olan sistemlerdir. Sistemde normal kuvvet (N) oluşur.





c) Karma Sistemler



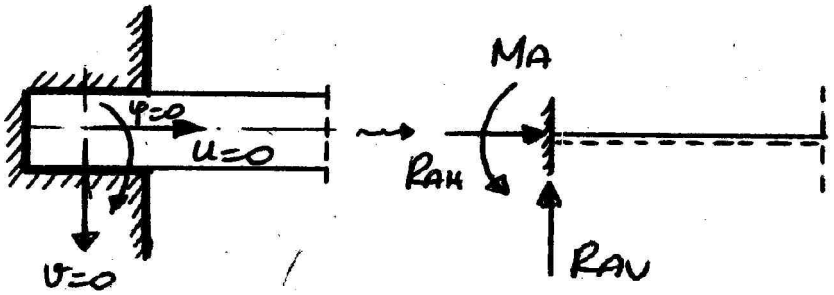
Langer Kirişi

I.6.4) Mesnetler ve Düğüm Noktaları

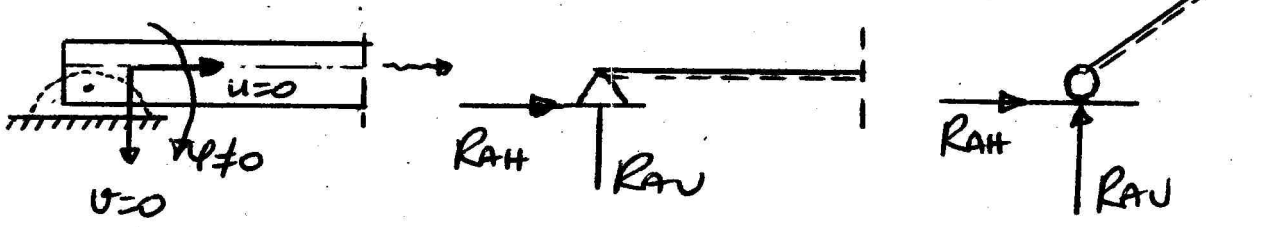
Mesnetler, yapıların dış ortamla birleştiği, düğüm noktaları ise çubukların birbirleri ile birleştiği yerlerdir. Başka bir deyişle;

a) Mesnet : Bir taşıyıcı elemandan diğer bir taşıyıcı elemana veya temele yük aktaran yapı elemanıdır.

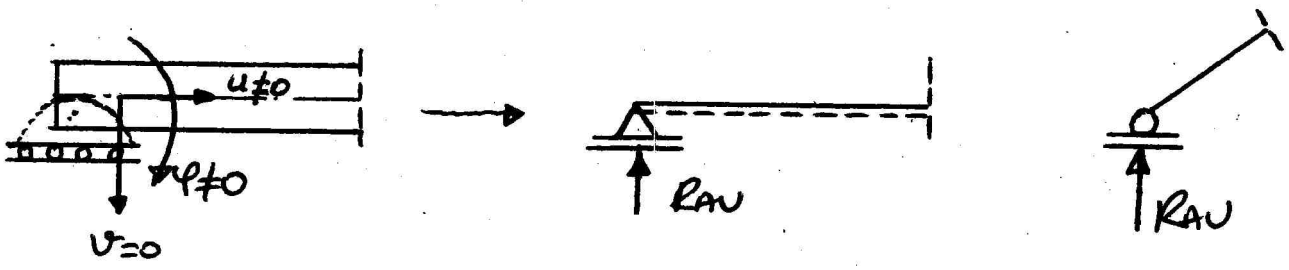
a.1) Ankastre Mesnet



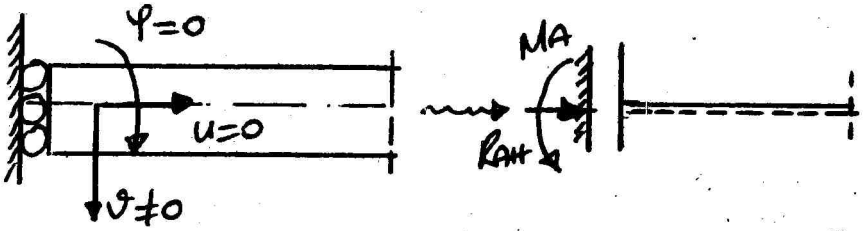
a.2) Sabit Mesnet



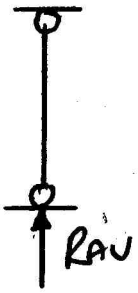
a.3) Hareketli Mesnet



a.4) Kılavuz Mesnet

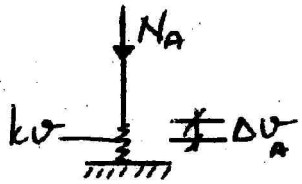


a.5) Pandül Mesnet



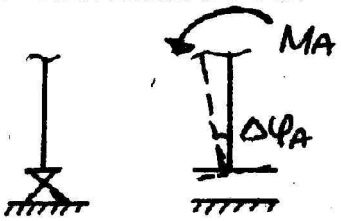
a.6) Elastik Ankastre Mesnet

a.6.1) Elastik Göken Mesnet



$$\Delta v_A = \frac{N}{k_v}$$

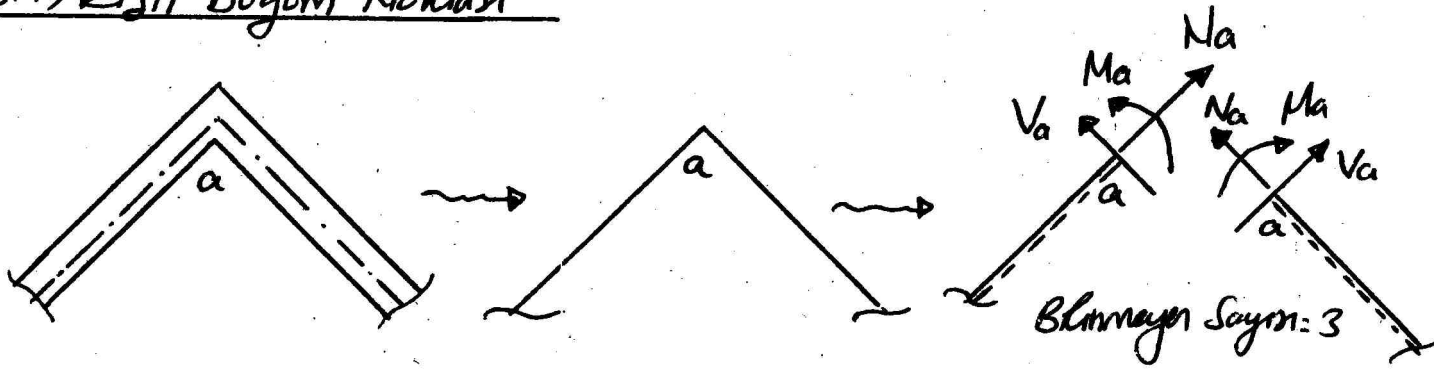
a.6.2) Elastik Dönen Mesnet



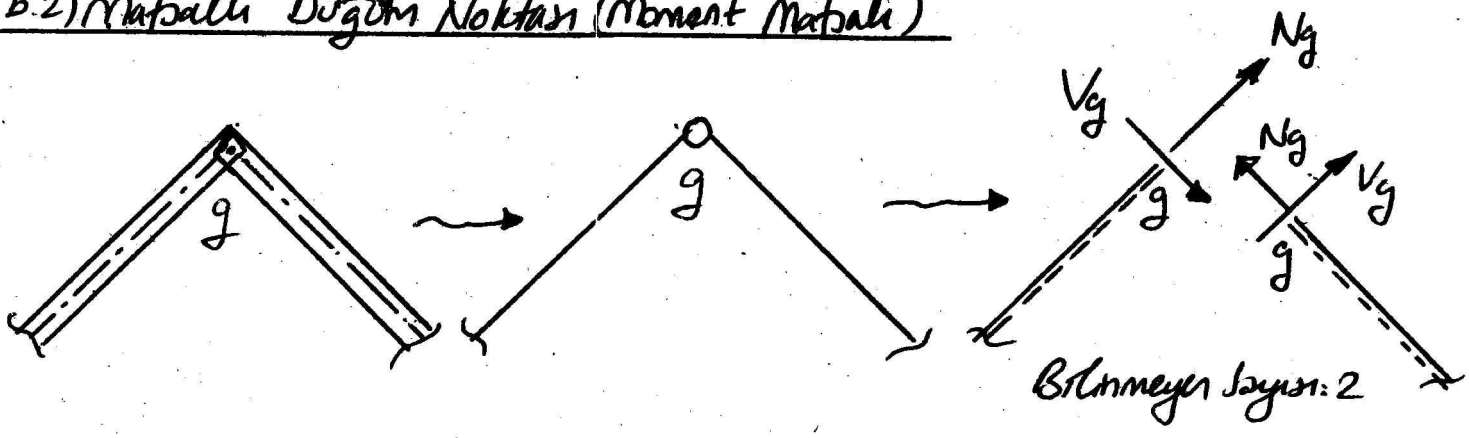
$$\Delta \varphi_A = \frac{M_A}{k_\varphi}$$

b) Düğüm Noktası: Sistemi meydana getiren çubukların birleştiği noktalardır.

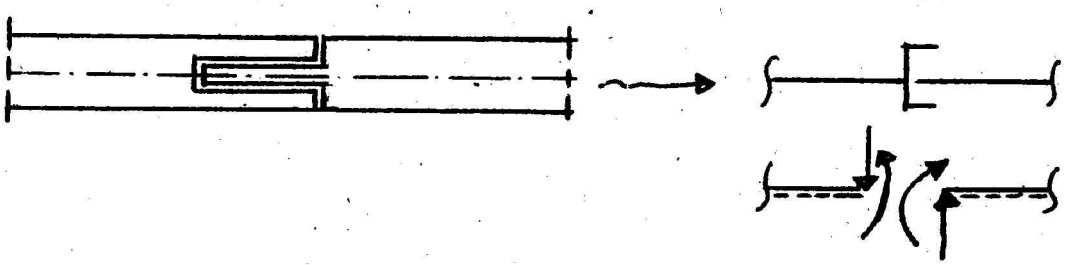
b.1) Rijit Düğüm Noktası



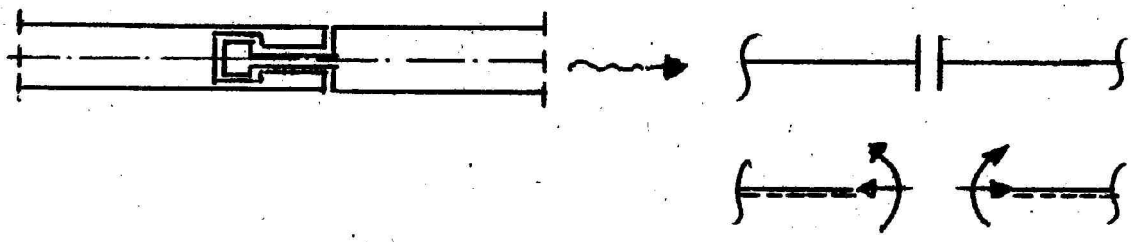
b.2) Mafsallı Düğüm Noktası (Moment Mafsalı)



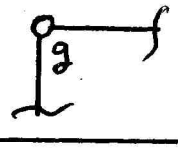
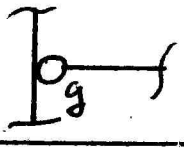
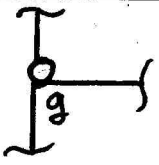
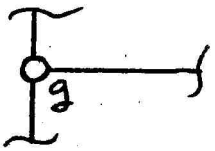
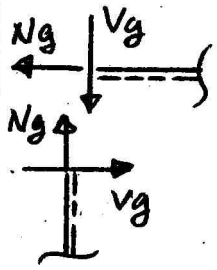
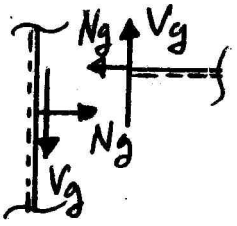
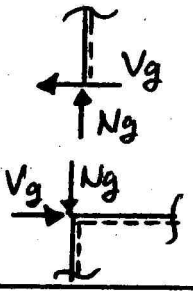
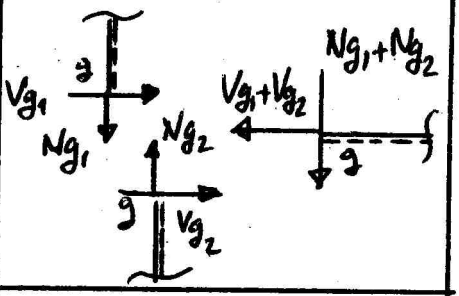
b.3) Normal Kuvvet Mafsalı



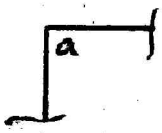
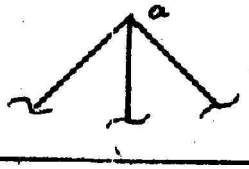
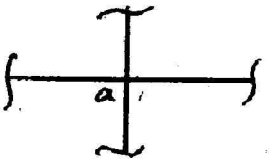
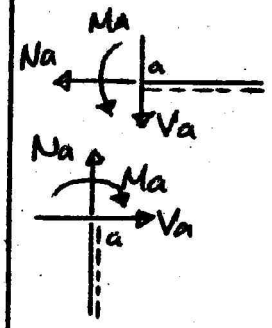
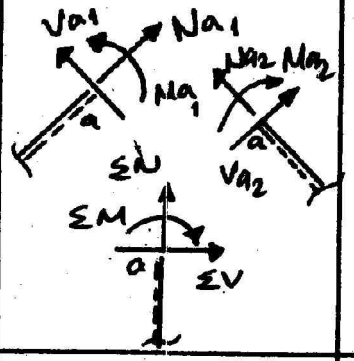
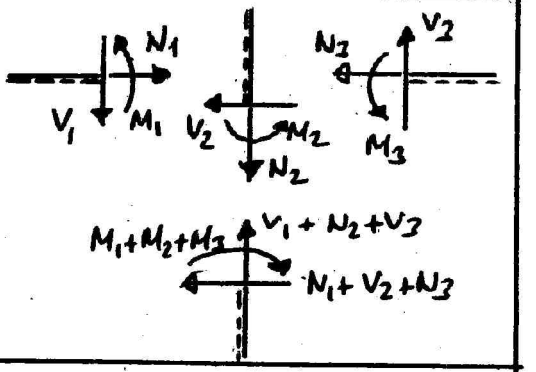
b.4) Kesme Kuvveti Mafsalı



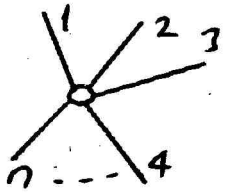
Mafsallı Sistemler

Şekil				
Bilinmeyenler				
Bilinmeyen Sayısı	2	2	2	4
Serbestlik Derecesi	1	1	1	2

Rijit Sistemler

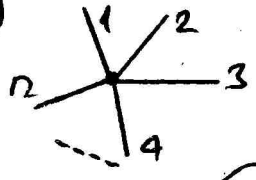
Şekil			
Bilinmeyenler			
Bilinmeyen Sayısı	3	6	9

- Mafsallı birleşimlerde bilinmeyen sayısı : $2(n-1)$

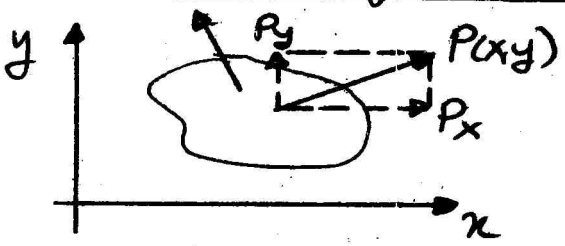


- Rijit düğüm noktasında bilinmeyen sayısı : $3(n-1)$

(n: çubuk sayısı)

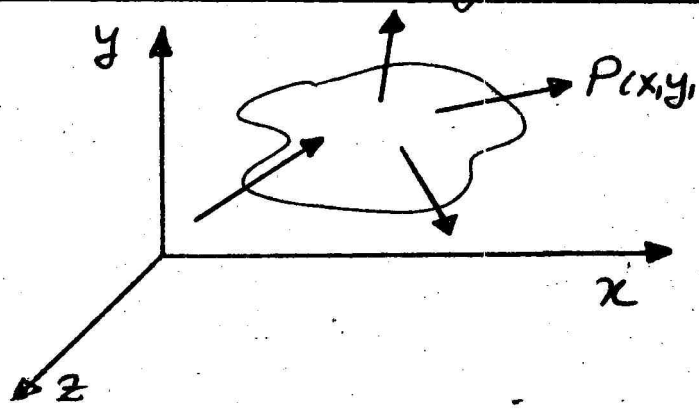


I.6.5) Düzlem Sistemde Denge Şartı



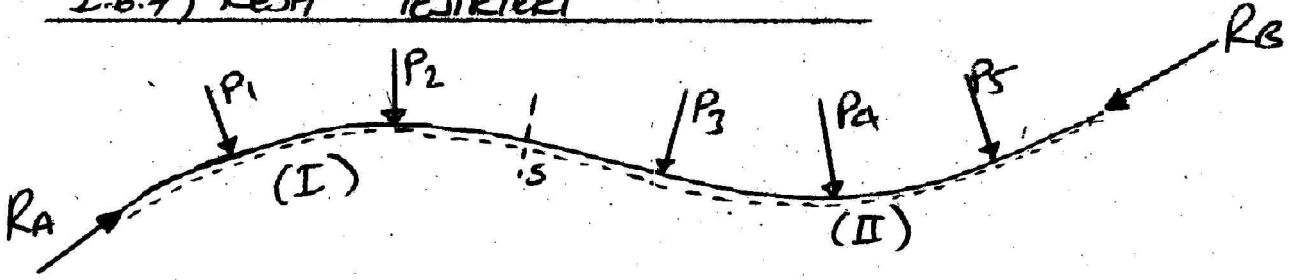
1) $\Sigma X = 0$
2) $\Sigma Y = 0$
3) $\Sigma M = 0$

I.6.6) Uzay Sistemde Denge Şartı

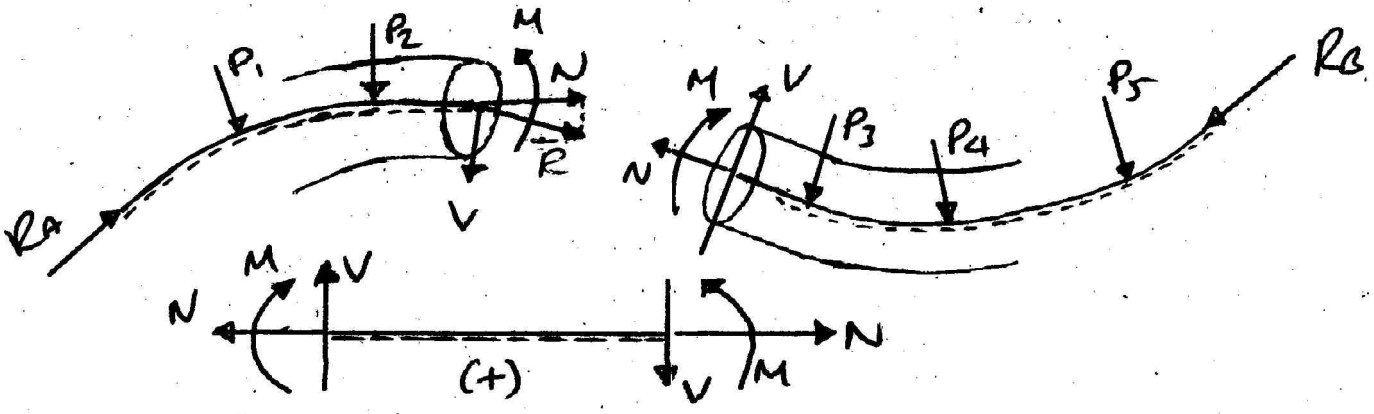


1) $\Sigma X = 0$
2) $\Sigma Y = 0$
3) $\Sigma Z = 0$
4) $\Sigma M_x = 0$
5) $\Sigma M_y = 0$
6) $\Sigma M_z = 0$

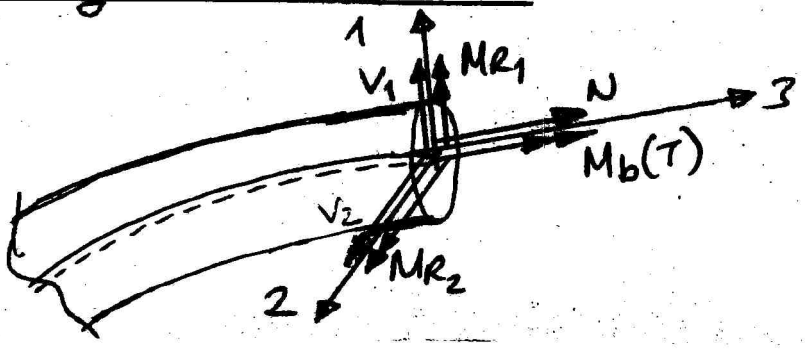
I.6.7) Kesit Tesirleri



a) Düzlem Sistemler



b) Uzay Sistemler



II) SİSTEMLERİN HİPERSTATİKLİK DERECESİNİN BELİRLENMESİ

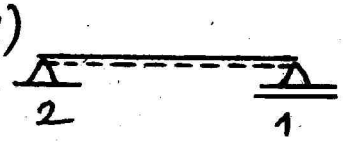
Mühendislikte ele alınan sistemler kararlı (stabil) olarak teşkil edilirler. Mesnetlere göre rölatif durumlarını değiştirmeyen sistemlere kararlı sistem denir. Ayrıca sistemler labil olarak (hareketli) da teşkil edilemezler. Oluşturulan sistemler izostatik veya hiperstatik olmalıdır.

Mesnet tepkilerini ve iç kuvvetleri denge denklemleri ile çözebildiğimiz sistemlere izostatik sistemler denir. Denge denklemleri ile çözülemeyen sistemlere ise hiperstatik sistemler denir. Mesnet tepkilerinin denge denklemleri ile bulunabildiği, ancak kesit tesirlerinin denge denklemleri ile bulunamadığı sistemlere dıştan izostatik, içten hiperstatik sistemler denir.

II.1) Dolu Gövdeli Sistemlerde Hiperstatiklik Derecesinin Belirlenmesi

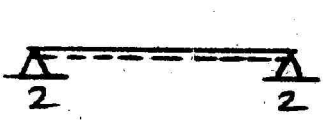
n<0 → labil
n=0 → izostatik
n>0 → hiperstatik

1)



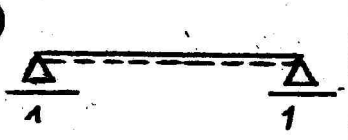
$n=3-3=0$

2)



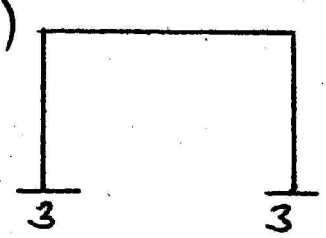
$n=4-3=1^{\circ}$

3)



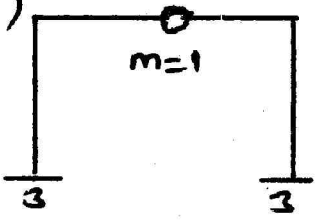
$n=2-3=-1^{\circ}$

4)



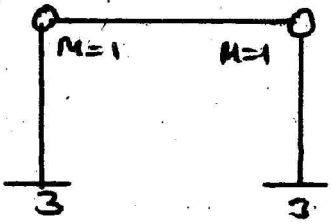
$n=6-3=3^{\circ}$

5)



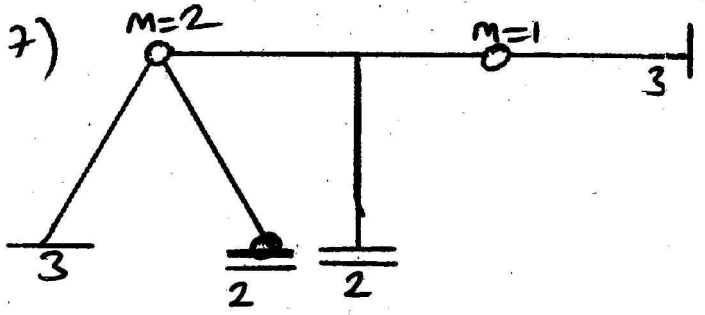
$n=6-3-1=2$

6)



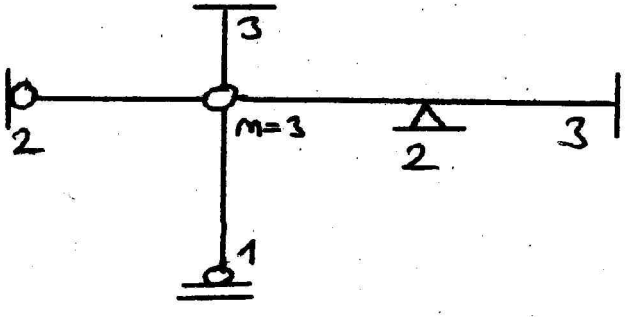
$n=6-3-2=1^{\circ}$

7)



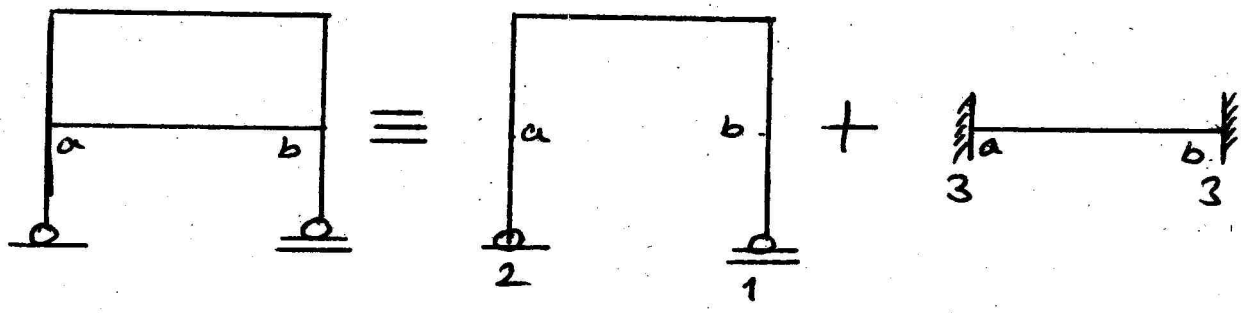
$n=10-3-(2+1)=4^{\circ}$

8)



$n=11-3-3=5^{\circ}$

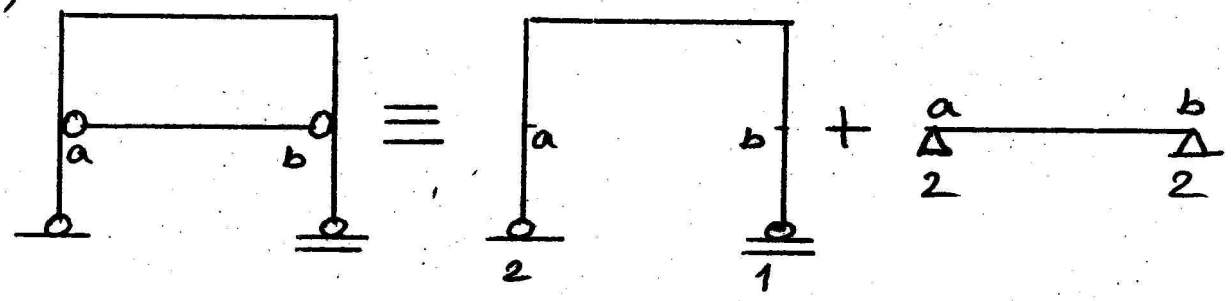
9)



$n_1 = 3-3=0$ $\quad n_2 = 6-3=3^\circ$

$\Sigma n = n_1 + n_2 = 3^\circ$

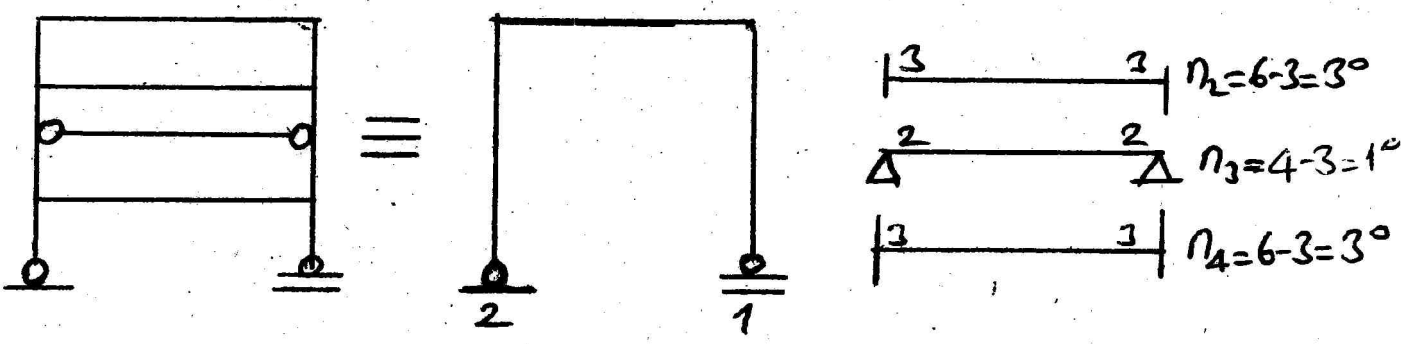
10)



$n_1 = 3-3=0$ $\quad n_2 = 4-3=1^\circ$

$\Sigma n = n_1 + n_2 = 1^\circ$

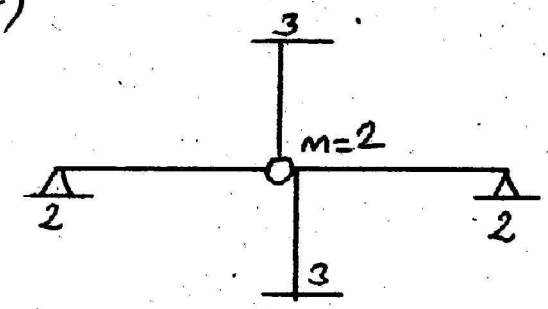
11)



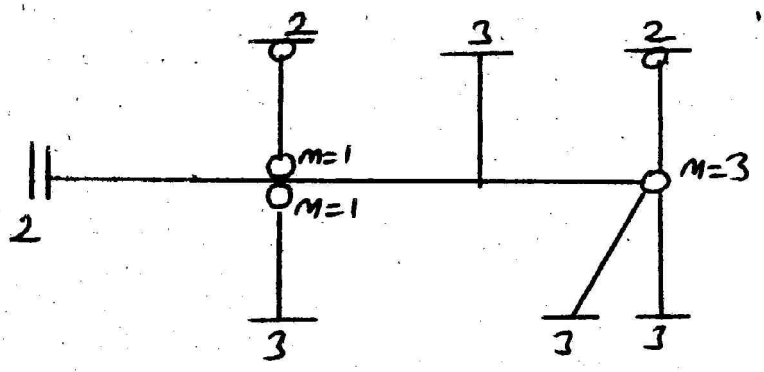
$n_1 = 3-3=0$

$\Sigma n = n_1 + n_2 + n_3 + n_4 = 7^\circ$

12)

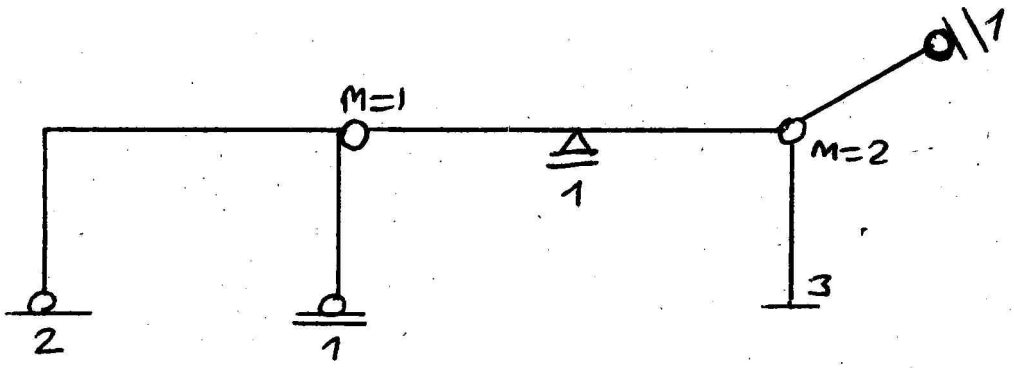


$n = 10-3-2=5^\circ$

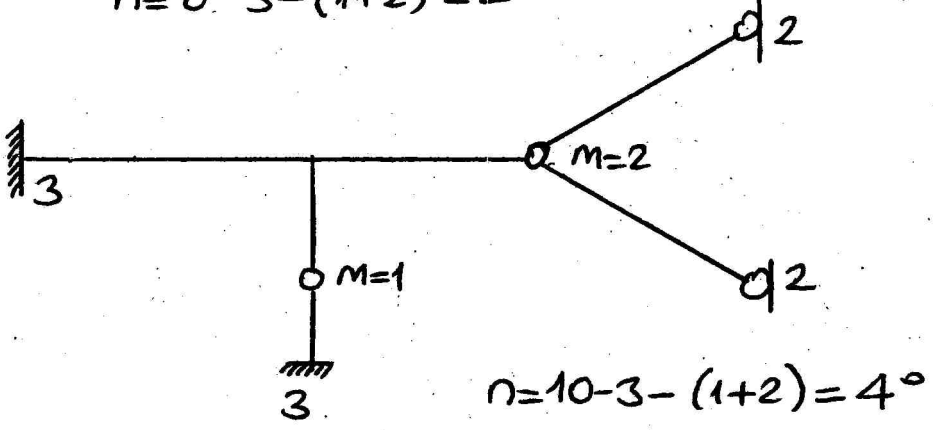


$n = 18-3-5=10^\circ$

a) Yalnız Tek Bir Kesimle Her Noktadan Tüye Ayrılabilen Sistemler



$n = 8 - 3 - (1+2) = 2^\circ$



$n = 10 - 3 - (1+2) = 4^\circ$

b) Genel Durum

Birbirleriyle sürekli veya noktasal olarak birleştirilmiş çubuklardan oluşan sistemlerin hiperstatiklik derecesi şu şekilde hesaplanır:

1) Sistem noktasal veya uygun yerlerden geçirilen kesitlerle basit elemanlara ayrılır. Ayrılan her eleman kendisine etkiyen dış kuvvetler ve ayırma kesitindeki kesit tesirleri ile dengededir. Ayırma kesitleri noktasallardan geçirildiğinde, her kesim yerinde noktasal şarttına bağlı olarak moment sıfır olacağından 2 kesit tesiri, rijit bir noktadan veya çubuktan geçirildiğinde her kesim yerinde 3 kesit tesiri vardır. Her eleman dış kuvvetler ve ayırma kesitindeki kesit tesirleri altında dengede olacağından her eleman için 3 ayrı denge denklemi yazılabilir. Sistem "S" sayıda parçaya ayrılmasa tüm sistem için "3S" kadar denge şartı elde edilebilir. (20)

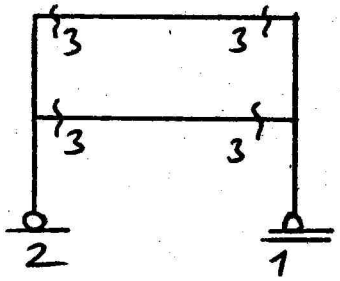
2) Sistemdeki (r) mesnet tepkisi sayısı belirlenir.

3) Kesilmiş yapılan kısımlardaki bilinmeyen sayısı (z) belirlenir.

4) $n = z + r - 3s$ ile sistemin hiperstatiklik derecesi belirlenir.

ÖRNEKLER:

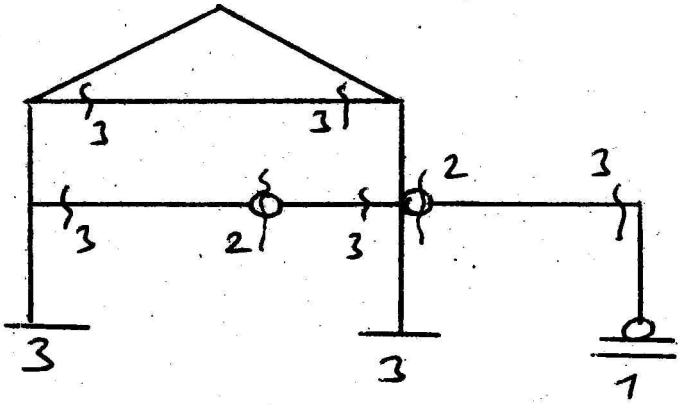
1)



$z=12,\ r=3,\ s=4$

$n = 12+3-3\times4 = 3^\circ$

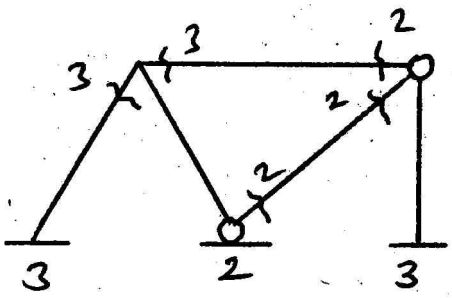
2)



$z=19,\ r=7,\ s=6$

$n = 19+7-3\times6 = 8^\circ$

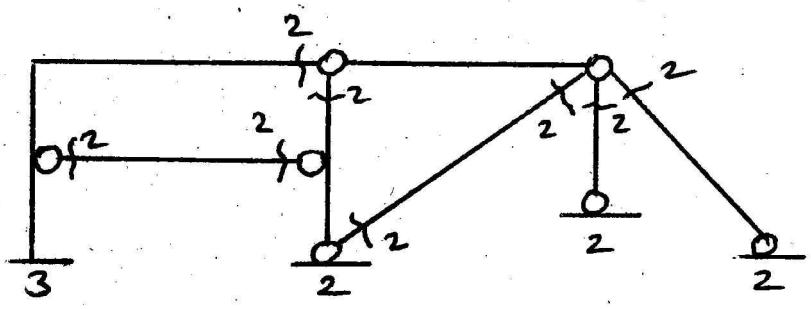
3)



$z=12,\ r=8,\ s=5$

$n = 12+8-3\times5 = 5^\circ$

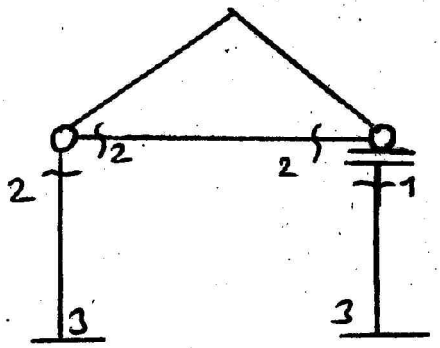
4)



$z=16,\ r=9,\ s=7$

$n = 16+9-3\times7 = 4^\circ$

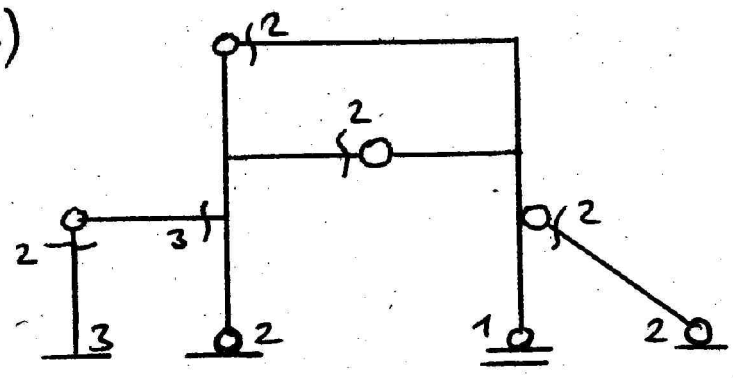
5)



$z=7,\ r=6,\ s=4$

$n = 7+6-3\times4 = 1^\circ$

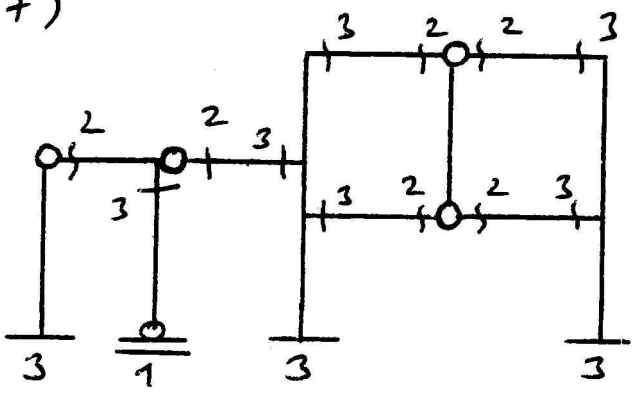
6)



$z=11,\ r=8,\ s=5$

$n = 11+8-3\times5 = 4^\circ$

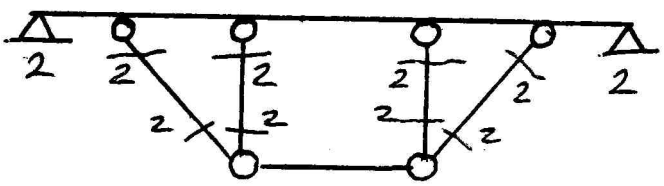
7)



$z=30$, $r=10$, $s=11$

$n=30+10-3\times11=7^{\circ}$

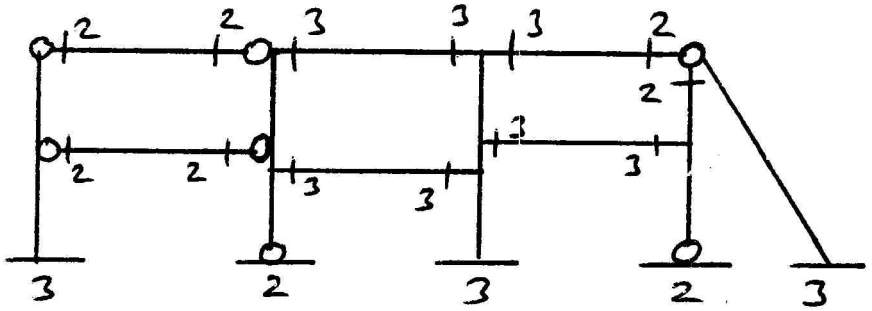
8)



$z=16$, $r=4$, $s=6$

$n=16+4-3\times6=2^{\circ}$

9)

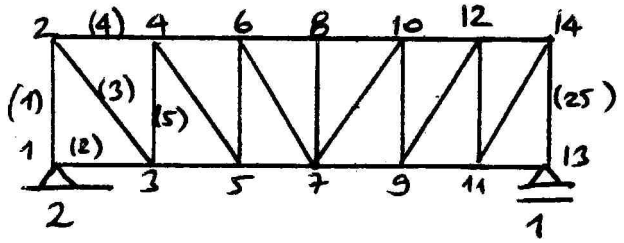


$z=33$, $r=13$, $s=11$

$n=33+13-3\times11=13^{\circ}$

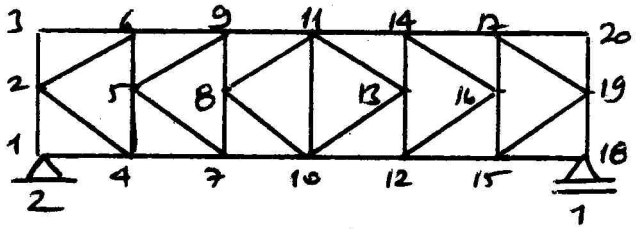
II.2) Kafes Sistemlerin Hiperstatiklik Derecesinin Belirlenmesi

Kafes kirişlerde yükler açıklık vasıtasıyla düğüm noktalarına etkir. Her çubukta yalnızca normal kuvvet (N) bulunmaktadır. Bilinmeyen sayısı çubuk sayısı kadardır. Bilinmeyen sayısı $n=r+q-2d$ ile hesaplanır. Burada; r: mesnet tepkisi sayısı, q: çubuk sayısı, d= düğüm noktası sayısıdır.



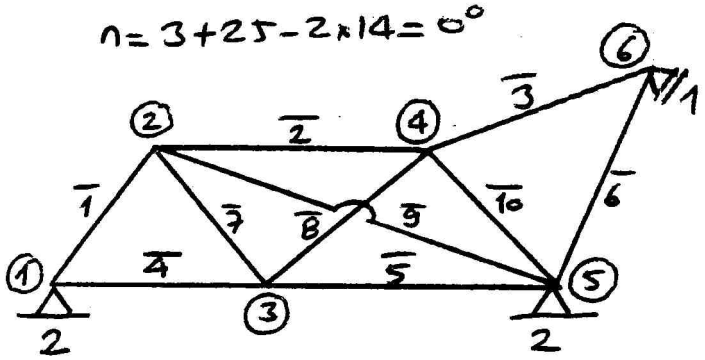
$r=3$, $q=25$, $d=14$

$n=3+25-2\times14=0^{\circ}$



$r=3$, $q=37$, $d=20$

$n=3+37-2\times20=0^{\circ}$



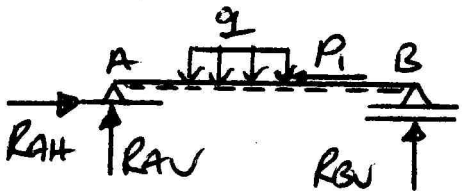
$r=5$, $q=10$, $d=6$

$n=5+10-2\times6=3^{\circ}$

III- SİSTEMLERİN SABİT YÜKLERE GÖRE HESABI

III.1) Mesnet Tepkileri

a) Basit Sistemler

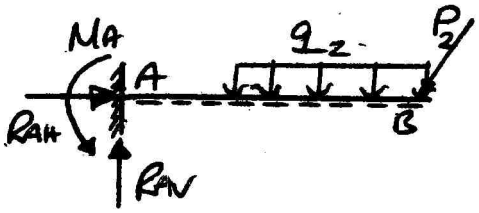


$\Sigma X=0 \rightarrow R_{AH}=!$

$\Sigma M_A=0 \rightarrow R_{BV}=!$

$\Sigma M_B=0 \rightarrow R_{AV}=!$

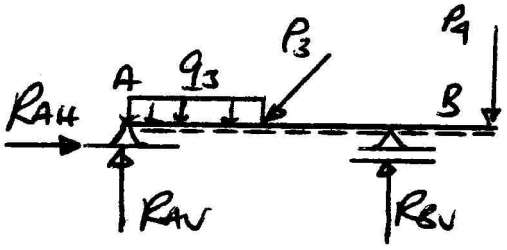
$\Sigma Y=0 \rightarrow$ Kontrol



$\Sigma X=0 \rightarrow R_{AH}=!$

$\Sigma Y=0 \rightarrow R_{AV}=!$

$\Sigma M_A=0 \rightarrow M_A=!$

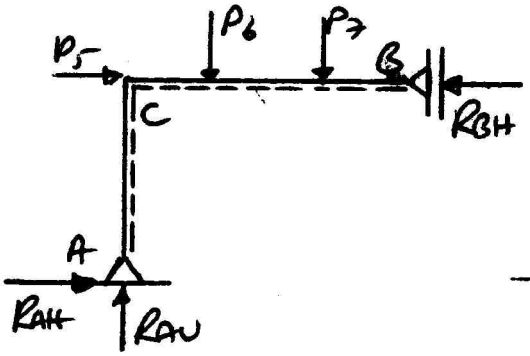


$\Sigma X=0 \rightarrow R_{AH}=!$

$\Sigma M_A=0 \rightarrow R_{BV}=!$

$\Sigma M_B=0 \rightarrow R_{AV}=!$

$\Sigma Y=0 \rightarrow$ Kontrol

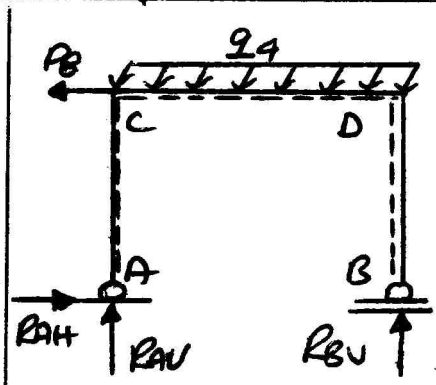


$\Sigma Y=0 \rightarrow R_{AV}=!$

$\Sigma M_B=0 \rightarrow R_{AH}=!$

$\Sigma M_A=0 \rightarrow R_{BH}=!$

$\Sigma X=0 \rightarrow$ Kontrol



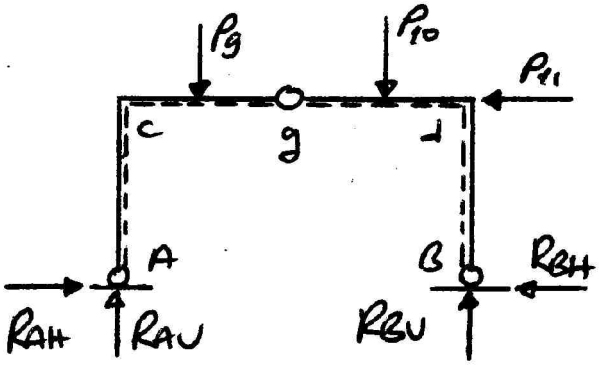
$\Sigma X=0 \rightarrow R_{AH}=!$

$\Sigma M_A=0 \rightarrow R_{BV}=!$

$\Sigma M_B=0 \rightarrow R_{AV}=!$

$\Sigma Y=0 \rightarrow$ Kontrol

b) Ara Mafsallı Sistemler

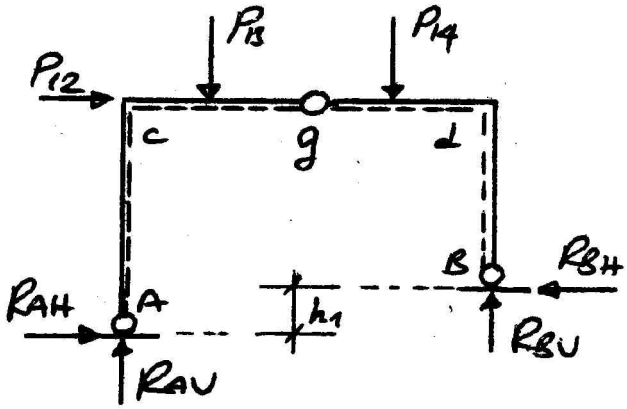


$\Sigma M_B=0 \rightarrow R_{AV}=!$ } $\Sigma Y=0$ (Kontrol)

$\Sigma M_A=0 \rightarrow R_{BV}=!$

$\Sigma M_{g_{sol}}=0 \rightarrow R_{AH}=!$ } $\Sigma X=0$ (Kontrol)

$\Sigma M_{g_{sag}}=0 \rightarrow R_{BH}=!$



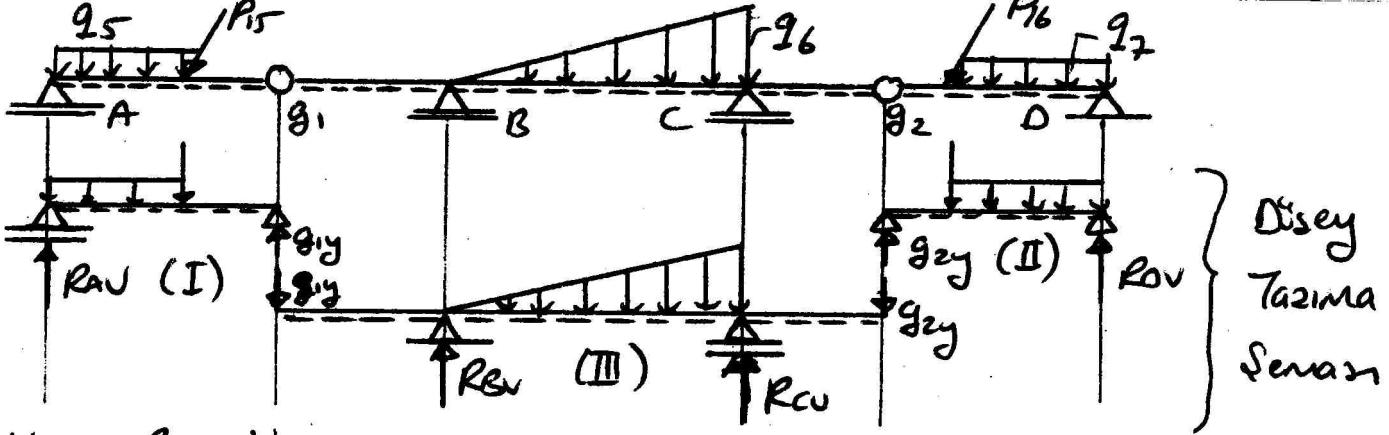
$\Sigma M_B=0 \rightarrow f(R_{AV}, R_{AH})$ } Tüm denklemin çözümünden; $R_{AV}=!$, $R_{AH}=!$

$\Sigma M_{g_{sol}}=0 \rightarrow f(R_{AV}, R_{AH})$

$\Sigma M_A=0 \rightarrow f(R_{BV}, R_{BH})$ } $R_{BV}=!$

$\Sigma M_{g_{sag}}=0 \rightarrow f(R_{BV}, R_{BH})$ $R_{BH}=!$

$\Sigma X=0, \Sigma Y=0 \rightarrow$ Kontrol

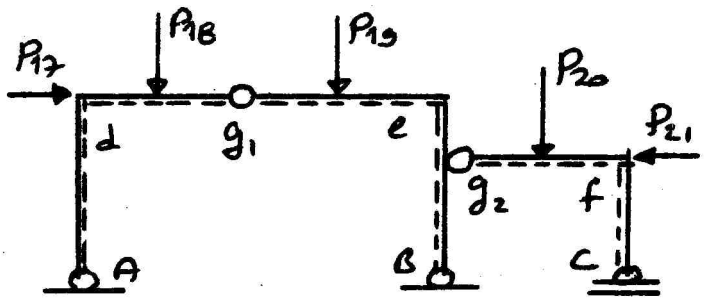


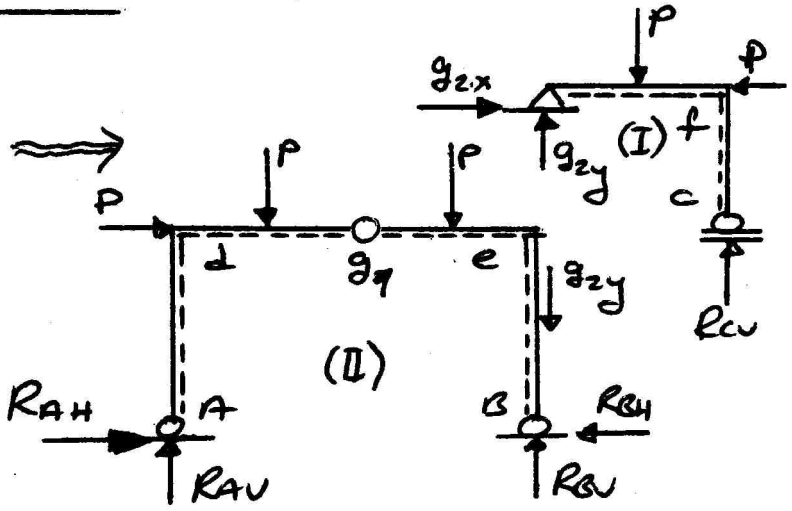
$\Sigma M_{g_1}=0 \rightarrow R_{AV}=!$
$\Sigma M_A=0 \rightarrow g_{1y}=!$
$\Sigma Y=0 \rightarrow$ Kontrol

$\Sigma M_C=0 \rightarrow R_{BV}=!$
$\Sigma M_B=0 \rightarrow R_{CV}=!$
$\Sigma Y=0 \rightarrow$ Kontrol

$\Sigma M_{g_2}=0 \rightarrow R_{DV}=!$
$\Sigma M_D=0 \rightarrow g_{2y}=!$
$\Sigma Y=0 \rightarrow$ Kontrol

c) Birleşik Sistemler



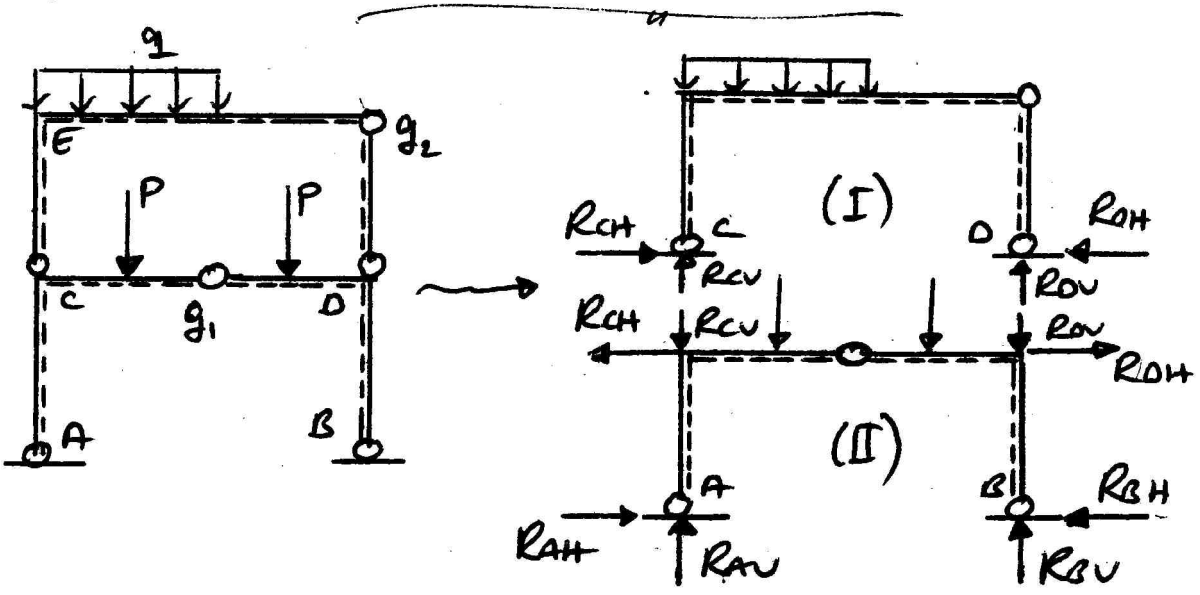


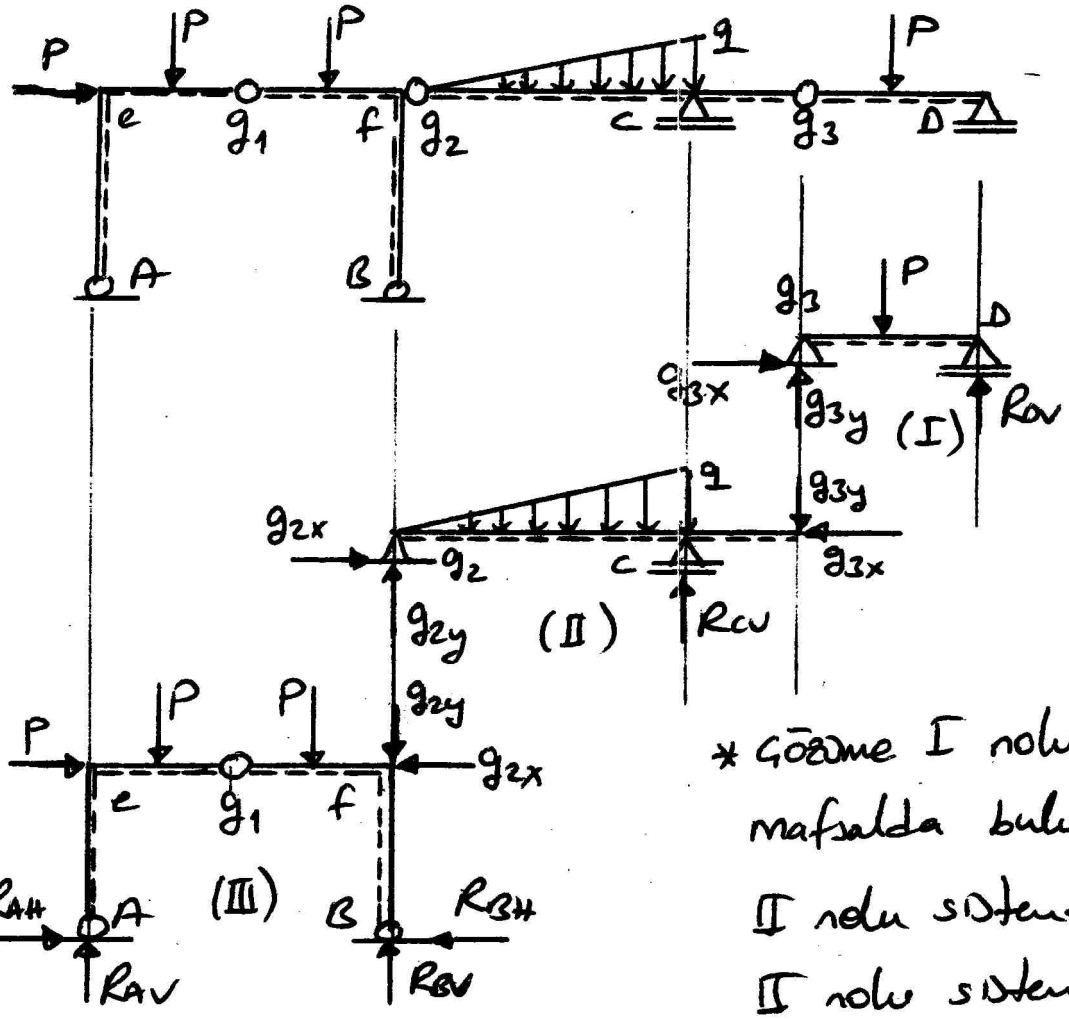
(I) Sistemi

$\Sigma X=0 \rightarrow g_{2x}=!$
$\Sigma M_C=0 \rightarrow g_{2y}=!$ } $\Sigma Y=0$ Kontrol
$\Sigma M_{g_2}=0 \rightarrow R_{CV}=!$

(II) Sistemi

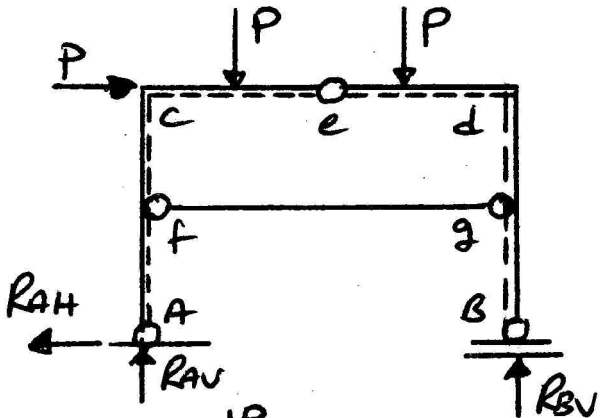
$\Sigma M_B=0 \rightarrow R_{AV}=!$; $\Sigma M_A=0 \rightarrow R_{BV}=!$ } $\Sigma X=0$, $\Sigma Y=0$ Kontrol
$\Sigma M_{g_1}=0 \rightarrow R_{AH}=!$; $\Sigma M_{g_1}=0 \rightarrow R_{BH}=!$





* Çözüme I nolu sistemden başlanır. Mafsalda bulunan reaksiyonlar II nolu sisteme aktarılır. Çözülen II nolu sistemdeki mafsal kuvvetleri III nolu sisteme aktarılır ve III nolu sistem çözülür.

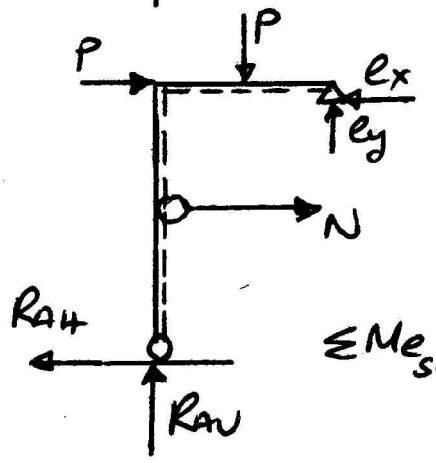
d) Gergili Sistemler



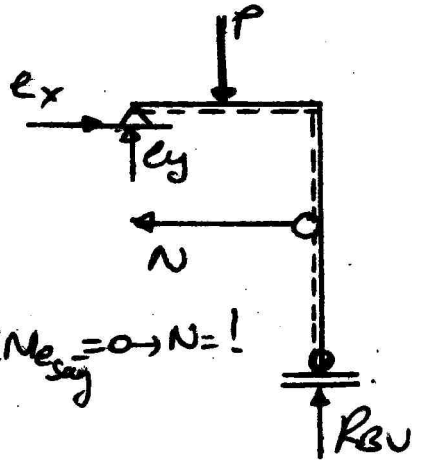
$\Sigma X=0 \rightarrow R_{AH}=!$

$\Sigma M_B=0 \rightarrow R_{AV}=!$ } $\Sigma Y=0$ Kontrol

$\Sigma M_A=0 \rightarrow R_{BV}=!$ }



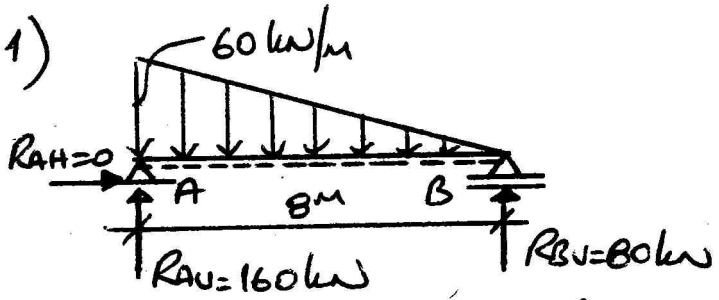
$\Sigma M_{e_{sol}}=0 \rightarrow N=!$



$\Sigma M_{e_{sag}}=0 \rightarrow N=!$

ÖRNEKLER :

1)



$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma M_B=0 \rightarrow R_{AV}=\dfrac{\frac{60\times8}{2}\times\left(\frac{2}{3}\times8\right)}{8}=160\,kN$

$\Sigma M_A=0 \rightarrow R_{BV}=\dfrac{\frac{60\times8}{2}\times(8/3)}{8}=80\,kN$

(Kontrol) $\Sigma Y=0 \rightarrow \dfrac{60\times8}{2}=160+80 \rightarrow 240=240$

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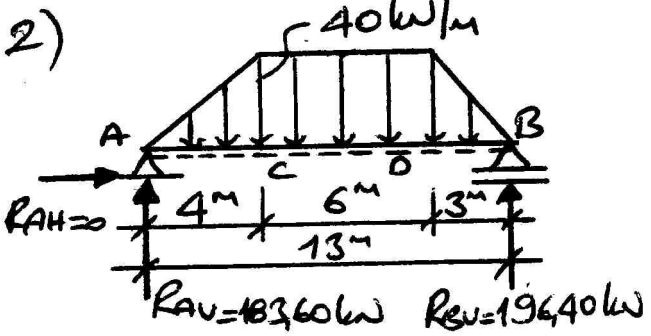
2)



$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma M_B=0 \rightarrow R_{AV}=\dfrac{\frac{40\times4}{2}\left(\frac{4}{3}+9\right)+40\times6\times6+\frac{40\times3}{2}\left(\frac{2}{3}\times3\right)}{13}$

$=183,60\,kN$

$\Sigma M_A=0 \rightarrow R_{BV}=\dfrac{\frac{40\times4}{2}\left(\frac{2}{3}\times4\right)+40\times6\times7+\frac{40\times3}{2}\left(\frac{3}{3}+10\right)}{13}$

$=196,40\,kN$

(Kontrol) $\Sigma Y=0 \rightarrow \dfrac{40\times4}{2}+40\times6+\dfrac{40\times3}{2}=183,60+196,40$

$380=380$

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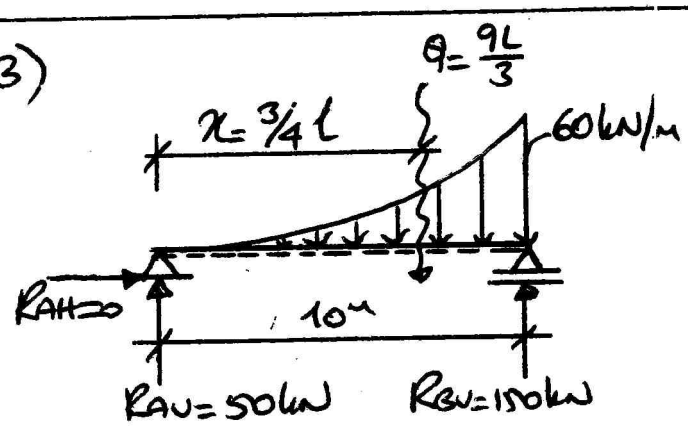
3)



$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma M_B=0 \rightarrow R_{AV}=\dfrac{\frac{9L}{3}\times L/4}{L}=\dfrac{9L}{12}=\dfrac{60\times10}{12}$

$=50\,kN$

$\Sigma M_A=0 \rightarrow R_{BV}=\dfrac{\frac{9L}{3}\times\frac{3}{4}L}{L}=\dfrac{9L}{4}=\dfrac{60\times10}{4}$

$=150\,kN$

(Kontrol) $\Sigma Y=0 \rightarrow \dfrac{60\times10}{3}=50+150$

$200=200$

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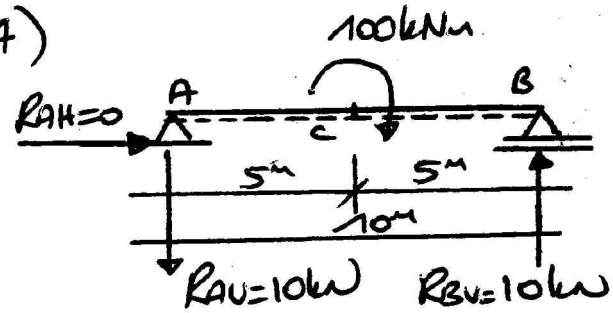
4)



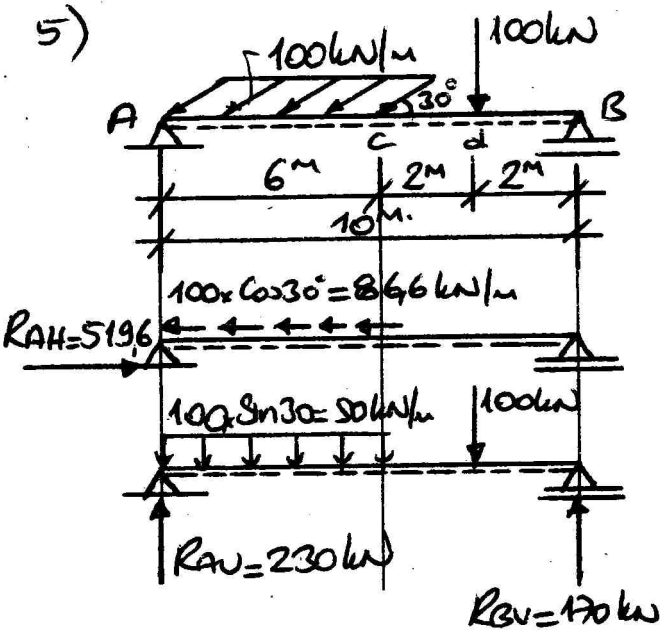
$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma M_B=0 \rightarrow R_{AV}=\dfrac{100}{10}=10\,kN$

$\Sigma M_A=0 \rightarrow R_{BV}=\dfrac{100}{10}=10\,kN$

$\Sigma Y=0 \rightarrow 10=10$

5)



$\Sigma X=0 \rightarrow R_{AH} = 86{,}6 \times 6 = 519{,}6\ kN$

$\Sigma M_B=0 \rightarrow R_{AV} = \dfrac{50\times6\times7+100\times2}{10} = 230\ kN$

$\Sigma M_A=0 \rightarrow R_{BV} = \dfrac{50\times6\times3+100\times8}{10} = 170\ kN$

$\Sigma Y=0 \rightarrow 50\times6+100 = 230+170$

$400 = 400$

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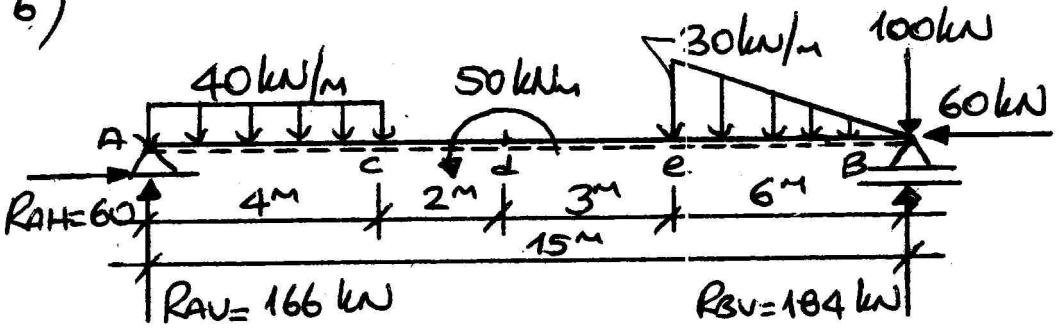
6)



$\Sigma X=0 \rightarrow R_{AH} = 60\ kN$

$\Sigma M_B=0 \rightarrow R_{AV} = \dfrac{40\times4\times13+50+\frac{30\times6}{2}\left(\frac{2}{3}\times6\right)}{15} = 166\ kN$

$\Sigma M_A=0 \rightarrow R_{BV} = \dfrac{40\times4\times2-50+\frac{30\times6}{2}\left(\frac{6}{3}+9\right)}{15} + 100 = 184\ kN$

$\Sigma Y=0 \rightarrow 166+184 = 40\times4+\dfrac{30\times6}{2}+100 = 166+184$

(Kontrol)

$350 = 350$

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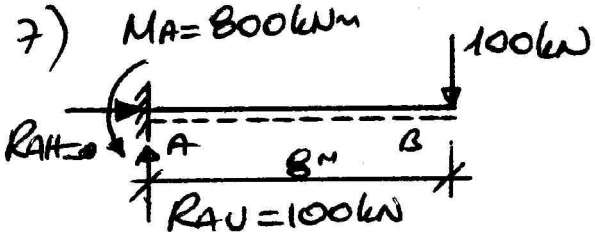
7)



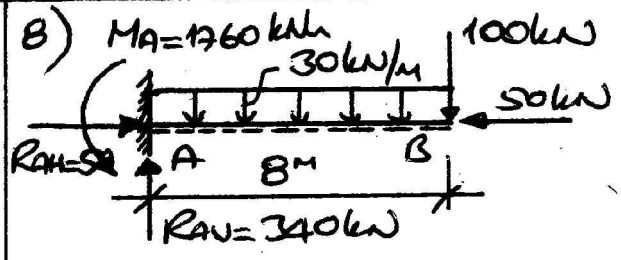
$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma Y=0 \rightarrow R_{AV}=100\ kN$

$\Sigma M_A=0 \rightarrow M_A = 100\times8$

$= 800\ kNm$

8)



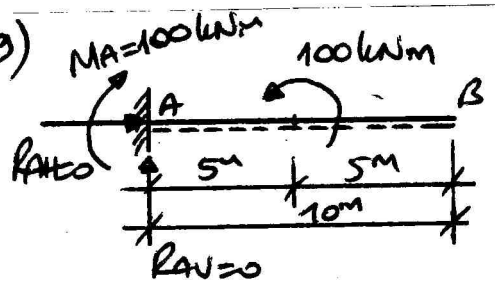
$\Sigma X=0 \rightarrow R_{AH}=50\ kN$

$\Sigma Y=0 \rightarrow R_{AV}=30\times8+100=340\ kN$

$\Sigma M_A=0 \rightarrow M_A = 30\times8\times4+100\times8$

$= 1760\ kNm$

9)

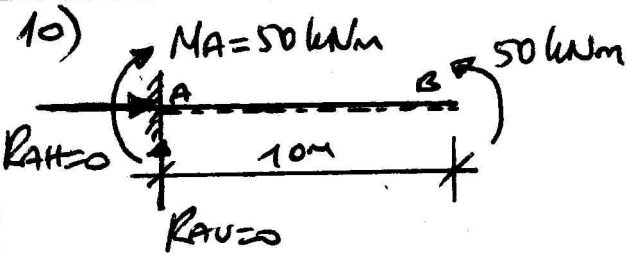


$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma Y=0 \rightarrow R_{AV}=0$

$\Sigma M_A=0 \rightarrow M_A=100\ kNm$

10)

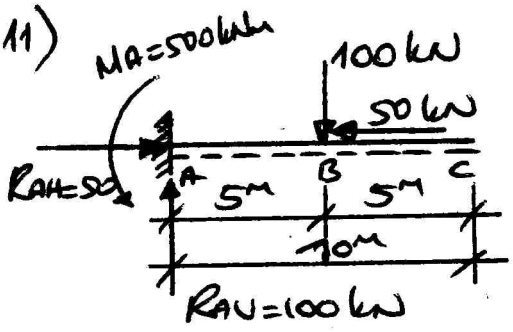


$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma Y=0 \rightarrow R_{AV}=0$

$\Sigma M_A=0 \rightarrow M_A=50\ kNm$

11)

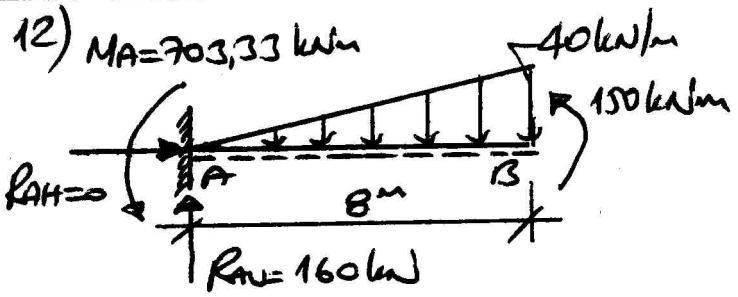


$\Sigma X=0 \rightarrow R_{AH}=50\ kN$

$\Sigma Y=0 \rightarrow R_{AV}=100\ kN$

$\Sigma M_A=0 \rightarrow M_A=100\times5=500\ kNm$

12) $M_A=703{,}33\ kNm$

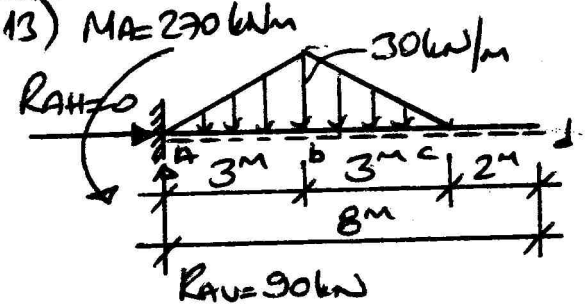


$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma Y=0 \rightarrow R_{AV}=\frac{40\times8}{2}=160\ kN$

$\Sigma M_A=0 \rightarrow M_A=\frac{40\times8}{2}\left(\frac{2}{3}8\right)-150=703{,}33\ kNm$

13) $M_A=270\ kNm$

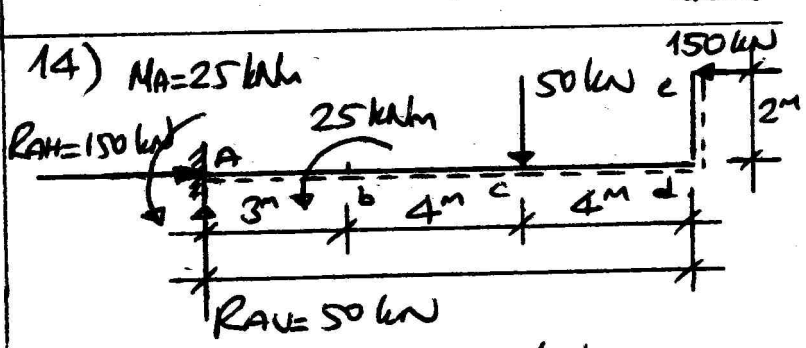


$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma Y=0 \rightarrow R_{AV}=\frac{30\times6}{2}=90\ kN$

$\Sigma M_A=0 \rightarrow M_A=\frac{30\times6}{2}\times3=270\ kNm$

14) $M_A=25\ kNm$



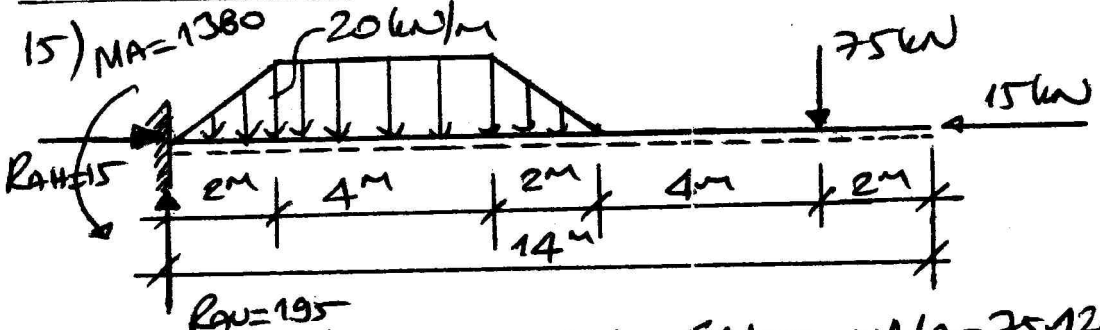
$\Sigma X=0 \rightarrow R_{AH}=150\ kNm$

$\Sigma Y=0 \rightarrow R_{AV}=50\ kN$

$\Sigma M_A=0 \rightarrow M_A=-25+50\times7-150\times3$

$=25\ kNm$

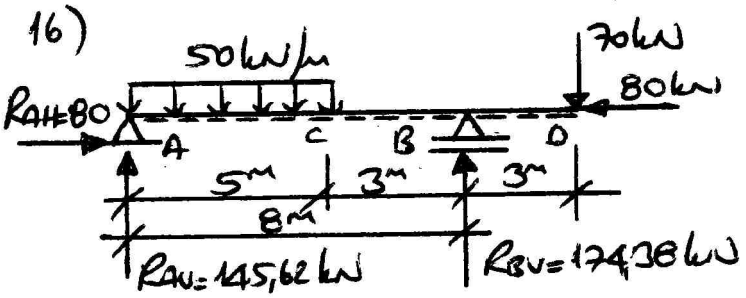
15) $M_A=1380$



$\Sigma X=0 \rightarrow R_{AH}=15\ kN$; $\Sigma M_A=0 \rightarrow M_A=75\times12+\frac{20\times2}{2}\times\left(\frac{2}{3}\times6\right)+\dots=1380\ kNm$

$\Sigma Y=0 \rightarrow R_{AV}=\frac{20\times2}{2}+20*4+\frac{20\times2}{2}+75=195\ kN$

16)



$\Sigma X=0 \rightarrow R_{AH}=80\,kN$

$\Sigma M_B=0 \rightarrow R_{AV}=\frac{50\times5\times5,5-70\times3}{8}$

$=145,62\ kN$

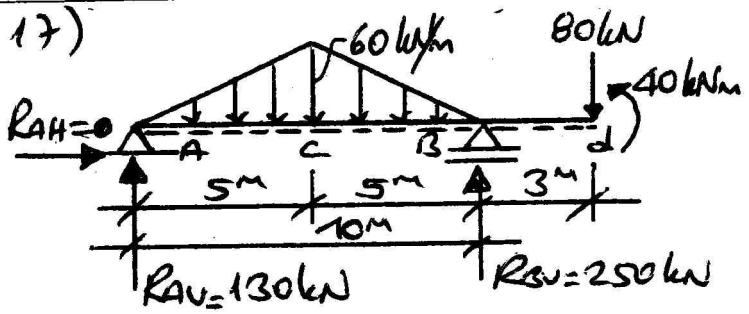
$\Sigma M_A=0 \rightarrow R_{BV}=\frac{50\times5\times2,5+70\times11}{8}$

$=174,38\ kN$

$\Sigma Y=0 \rightarrow 50\times5+70=145,62+174,38$

$320=320$

17)



$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma M_B=0 \rightarrow R_{AV}=\frac{\frac{60\times10}{2}\times5-80\times3+40}{10}$

$=130\,kN$

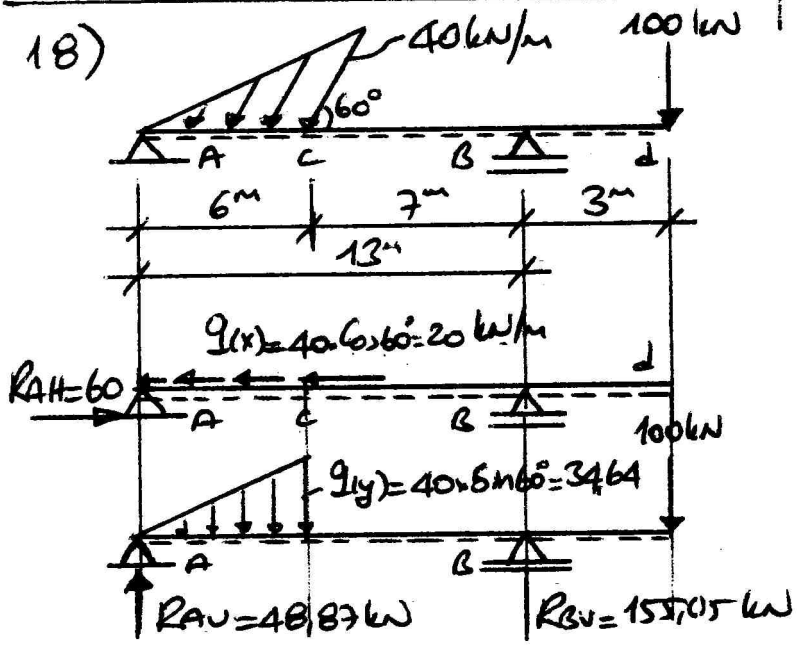
$\Sigma M_A=0 \rightarrow R_{BV}=\frac{\frac{60\times10}{2}\times5+80\times13-40}{10}$

$=250\,kN$

$\Sigma Y=0 \rightarrow \frac{60\times10}{2}+80=130+250$

$380=380$

18)



$\Sigma X=0 \rightarrow R_{AH}=\frac{20\times6}{2}=60\,kN$

$\Sigma M_B=0 \rightarrow R_{AV}=\frac{\frac{34,64\times6}{2}\left(\frac{6}{3}+7\right)-100\times3}{13}$

$=48,87\ kN$

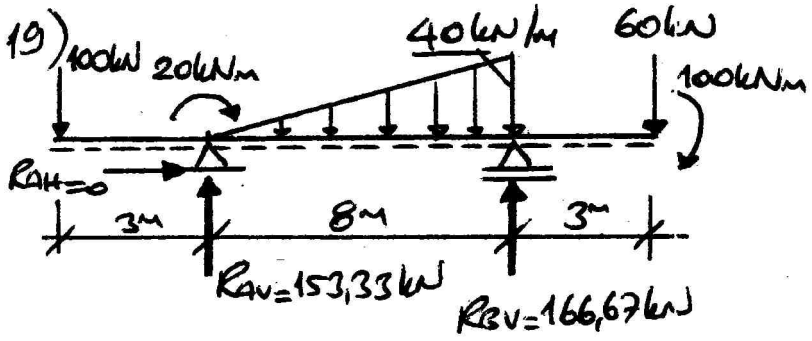
$\Sigma M_A=0 \rightarrow R_{BV}=\frac{\frac{34,64\times6}{2}\left(\frac{2}{3}\times6\right)+100\times16}{13}$

$=155,05\ kN$

$\Sigma Y=0 \rightarrow \frac{34,64\times6}{2}+100=48,87+155,05$

$203,92=203,92$

19)



$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma M_B=0 \rightarrow R_{AV}=\frac{100\cdot11-20+\frac{40\cdot8}{2}\cdot\frac{8}{3}-60\cdot3-100}{8}$

$=153,33\ kN$

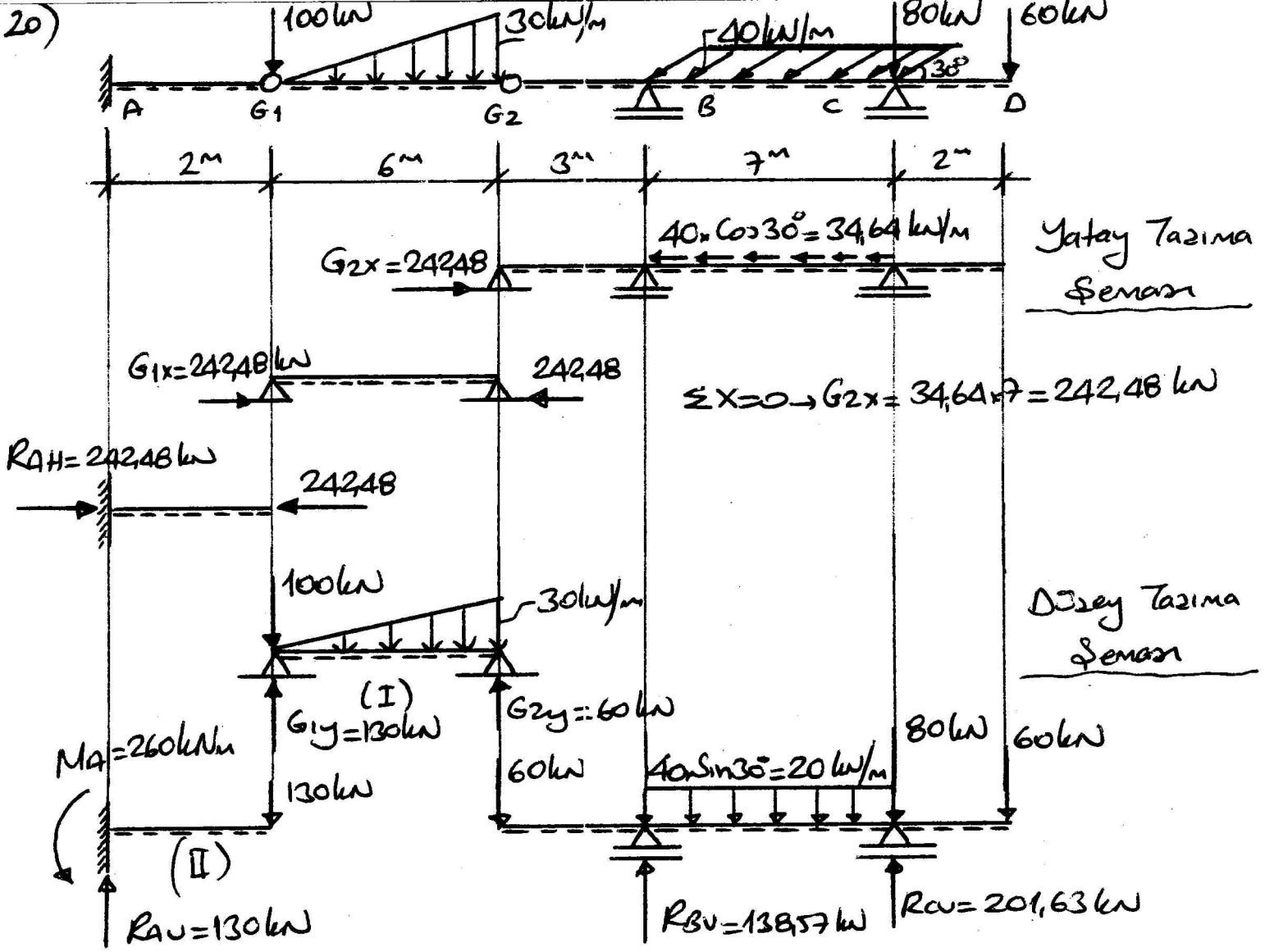
$\Sigma M_A=0 \rightarrow R_{BV}=\frac{-100\times3+20+\frac{40\cdot8}{2}\left(\frac{2}{3}\cdot8\right)+60\cdot11+10}{8}$

$=166,67\ kN$

$\Sigma Y=0 \rightarrow 153,33+166,67=100+\frac{40\cdot8}{2}+60$

$320=320$

20)



(I)nci sistem

$$\Sigma M_{G_2}=0 \rightarrow G_{1y}=\frac{\frac{30\cdot 6}{2}\cdot\frac{6}{3}}{6}+100=130\,kN$$

$$\Sigma M_{G_1}=0 \rightarrow G_{2y}=\frac{\frac{30\cdot 6}{2}\left(\frac{2}{3}\cdot 6\right)}{6}=60\,kN$$

$$\Sigma Y=0 \rightarrow 100+\frac{30\cdot 6}{2}=130+60$$

$$190=190$$

(II) nci sistem

$$\Sigma Y=0 \rightarrow R_{AV}=130\,kN$$

$$\Sigma M_A=0 \rightarrow M_A=130\cdot 2=260\ kNm$$

(III) nci sistem

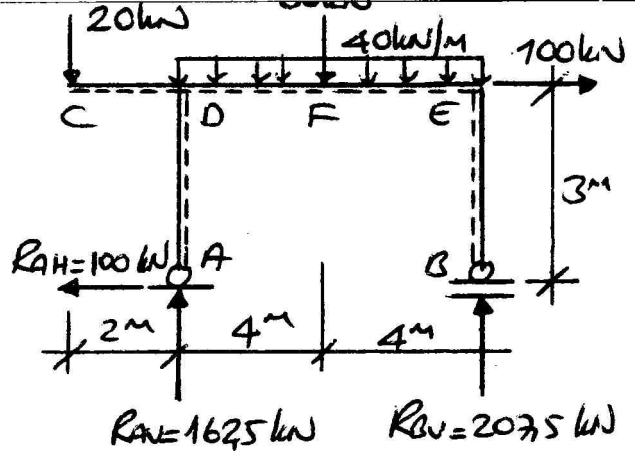
$$\Sigma M_C=0 \rightarrow R_{BV}=\frac{20\cdot 7\cdot 3{,}5+60\cdot 10-60\cdot 2}{7}=138{,}57\,kN$$

$$\Sigma M_B=0 \rightarrow R_{CV}=\frac{20\cdot 7\cdot 3{,}5-60\cdot 3+60\cdot 9}{7}+80=201{,}43\,kN$$

$$\Sigma Y=0 \rightarrow 138{,}57+201{,}43=60+20\cdot 7+80+60$$

$$340=340$$

21)



$\Sigma X=0 \rightarrow R_{AH}=100\ kN$

$\Sigma M_B=0 \rightarrow R_{AV}=\frac{20\times10+40\times8\times4+30\times4-100\times3}{8}$

$=162{,}5\ kN$

$\Sigma M_A=0 \rightarrow R_{BV}=\frac{-20\times2+40\times8\times4+30\times4+100\times3}{8}$

$=207{,}5\ kN$

$\Sigma Y=0 \rightarrow 20+40\times8+30=162{,}5+207{,}5$

$370=370$

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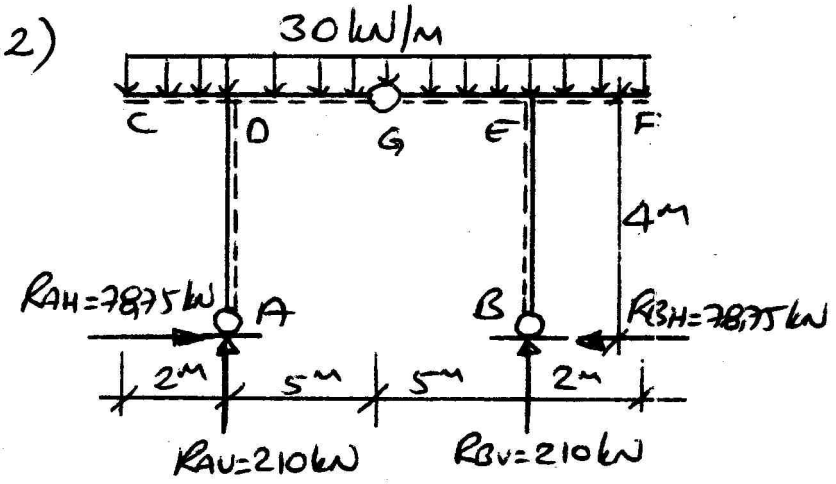
22)



$\Sigma M_B=0 \rightarrow R_{AV}=\frac{30\times14\times5}{10}=210\ kN$

$\Sigma M_A=0 \rightarrow R_{BV}=\frac{30\times14\times5}{10}=210\ kN$

$\Sigma M_{G_{sol}}=0 \rightarrow R_{AH}=\frac{210\times5-30\times7\times3{,}5}{4}$

$=78{,}75\ kN.$

$\Sigma M_{G_{sağ}}=0 \rightarrow R_{BH}=\frac{210\times5-30\times7\times3{,}5}{4}$

$=78{,}75\ kN$

$\Sigma X=0 \rightarrow 78{,}75=78{,}75$

$\Sigma Y=0 \rightarrow 30\times14=210+210$

$420=420$

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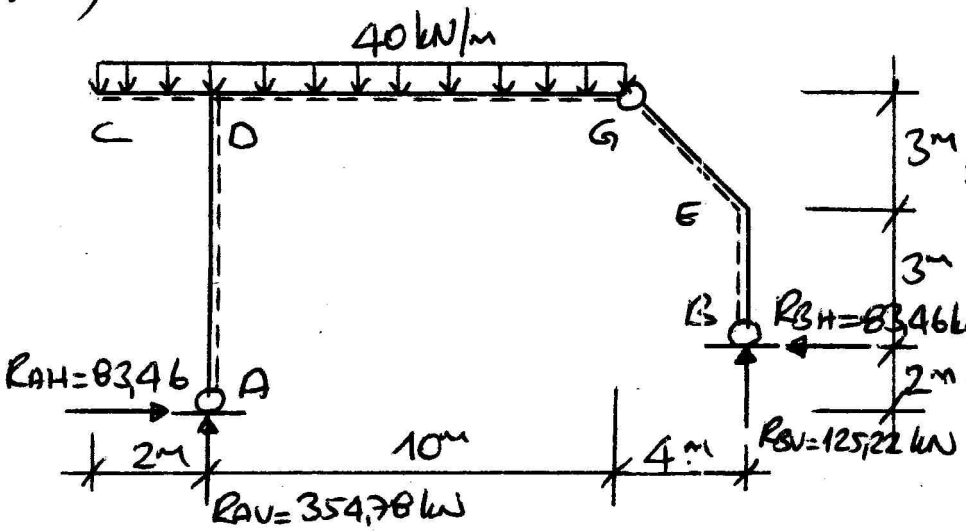
23)



$\Sigma M_B=0 \rightarrow R_{AV}.14-R_{AH}.2-40\times12\times10=$

$7R_{AV}-R_{AH}=2400$

$\Sigma M_{G_{sol}}=0 \rightarrow R_{AV}.10-R_{AH}.8-40\times12\times6=0$

$5R_{AV}-4R_{AH}=1440$

$R_{AV}=354{,}78\ kN;\ R_{AH}=83{,}46\ kN$

$\Sigma M_A=0 \rightarrow R_{BV}.14+R_{BH}.2=40\times10.5-$

$-40.2.1$

$7R_{BV}+R_{BH}=960$

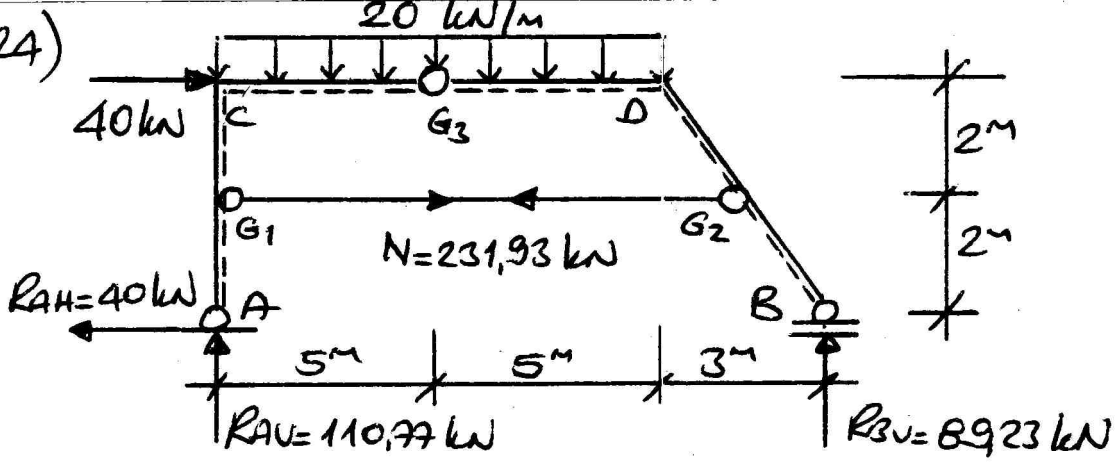
$\Sigma M_{G_{sağ}}=0 \rightarrow R_{BV}.4-R_{BH}.6=0$

$R_{BV}=125{,}22\ kN;\ R_{BH}=83{,}46\ kN$

$\Sigma X=0 \rightarrow R_{AH}=R_{BH}$

$\Sigma Y=0 \rightarrow 40\times12=354{,}78+125{,}22 \rightarrow 480=480$

24)

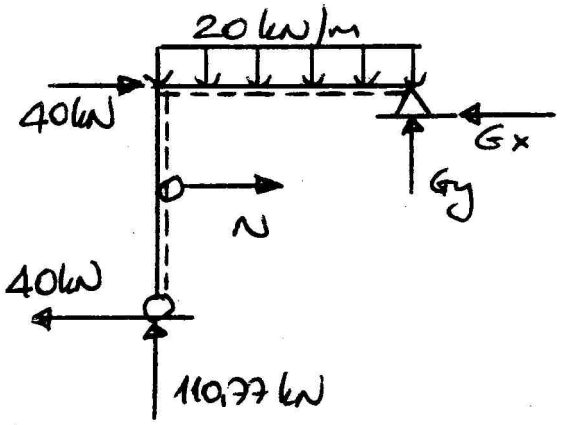


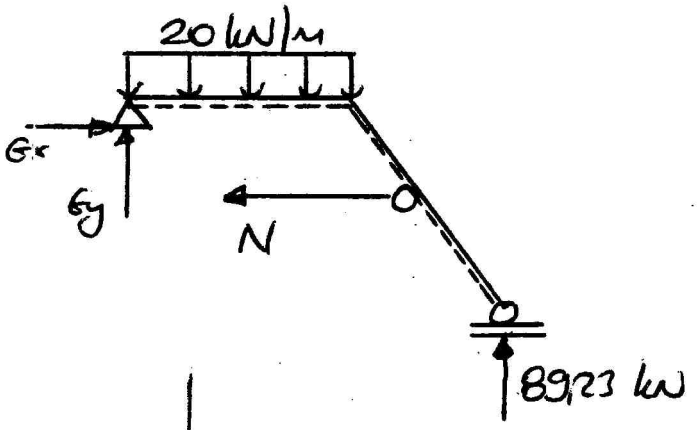
$\Sigma X=0 \rightarrow R_{AH}=40\ kN$

$\Sigma M_B=0 \rightarrow R_{AV}=\frac{20\times10\times8-40\times4}{13}=110,77\ kN$

$\Sigma M_A=0 \rightarrow R_{BV}=\frac{20\times10\times5+40\times4}{13}=89,23\ kN$

$\Sigma Y=0 \rightarrow 20\times10=110,77+89,23 \Longrightarrow 200=200$





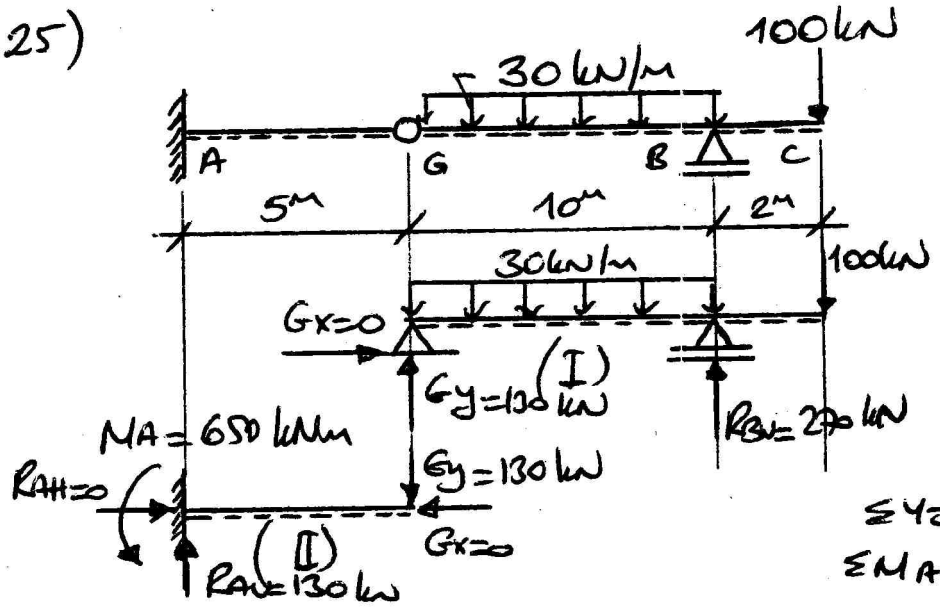
$\Sigma M_{G_3}=0$ sol $\rightarrow N=\frac{110,77\times5+40\times4-20\times5\times2,5}{2}$

$N=231,93\ kN$

$\Sigma M_{G_3}=0$ sağ $\rightarrow N=\frac{89,23\times8-20\times5\times2,5}{2}$

$N=231,93\ kN$

25)



(I) Sistemi

$\Sigma X=0 \rightarrow G_x=0$

$\Sigma M_B=0 \rightarrow G_y=\frac{30\times10\times5-100\times2}{10}=130\ kN$

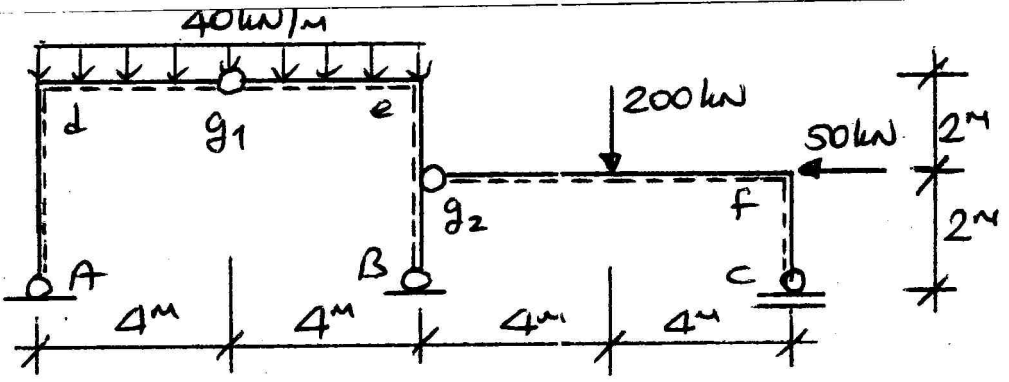
$\Sigma M_C=0 \rightarrow R_{BV}=\frac{30\times10\times5+100\times12}{10}=270\ kN$

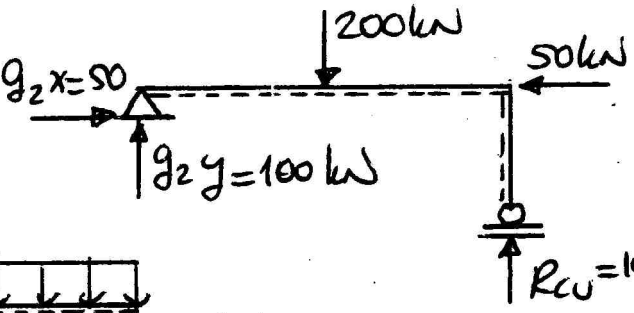
(II) Sistemi

$\Sigma Y=0 \rightarrow R_{AV}=130\ kN$

$\Sigma M_A=0 \rightarrow M_A=130\times5=650\ kNm$

26)





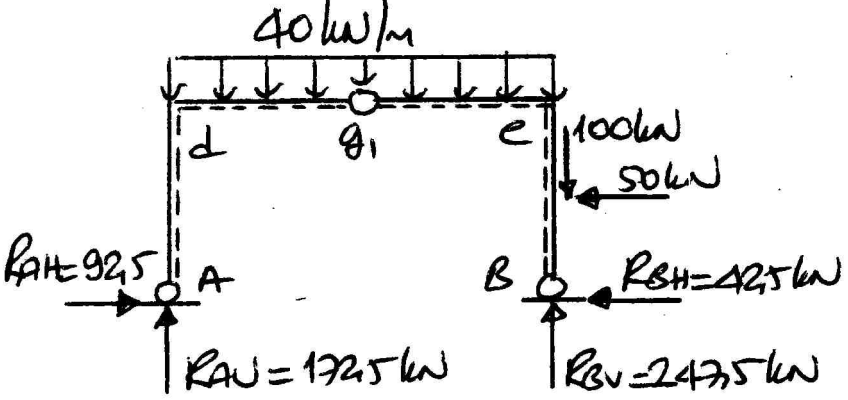
$\Sigma X=0 \rightarrow G_{2x}=50kN$

$\Sigma M_1=0 \rightarrow g_{2y}=\frac{200\times4-50\times2+50\times2}{8}$

$=100kN$

$\Sigma M_{g_2}=0 \rightarrow R_{CV}=\frac{200\times4}{8}$

$=100kN$



$\Sigma M_B=0 \rightarrow R_{AV}=\frac{40\times8\times4+50\times2}{8}=172,5kN$

$\Sigma M_A=0 \rightarrow R_{BV}=\frac{40\times8\times4-50\times2}{8}+100=247,5kN$

$\Sigma Y=0 \rightarrow 172,5+247,5=40\times8+100$

$420=420$

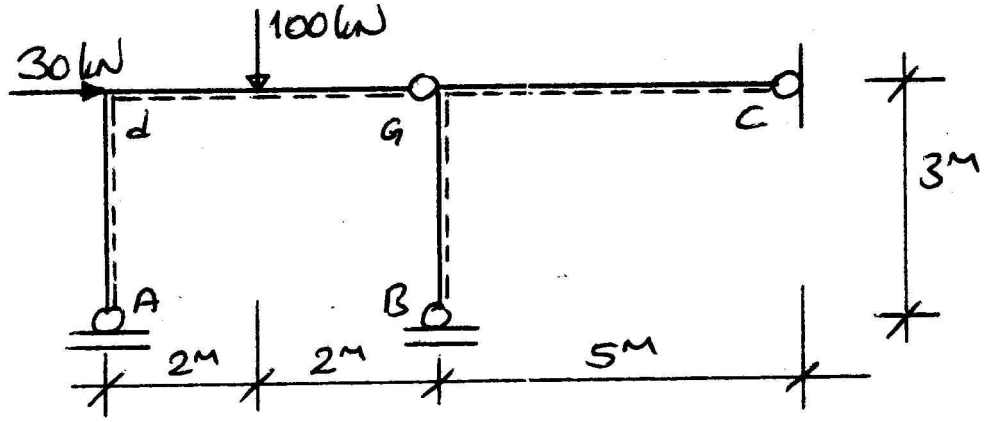
$\Sigma M_{g_1}=0$ sol $\rightarrow R_{AH}=\frac{172,5\times4-40\times4\times2}{4}=92,5kN$

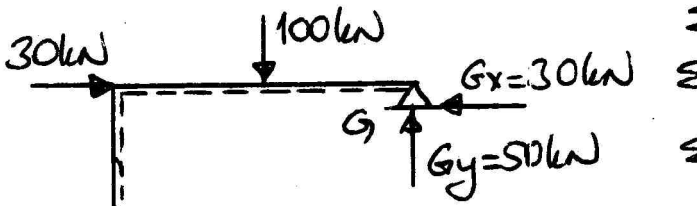
$\Sigma M_{g_1}=0$ sağ $\rightarrow R_{BH}=\frac{247,5\times4-40\times4\times2-100\times4-50\times2}{4}=42,5kN$

$\Sigma X=0 \rightarrow 92,5=50+42,5$

$92,5=92,5$

27)

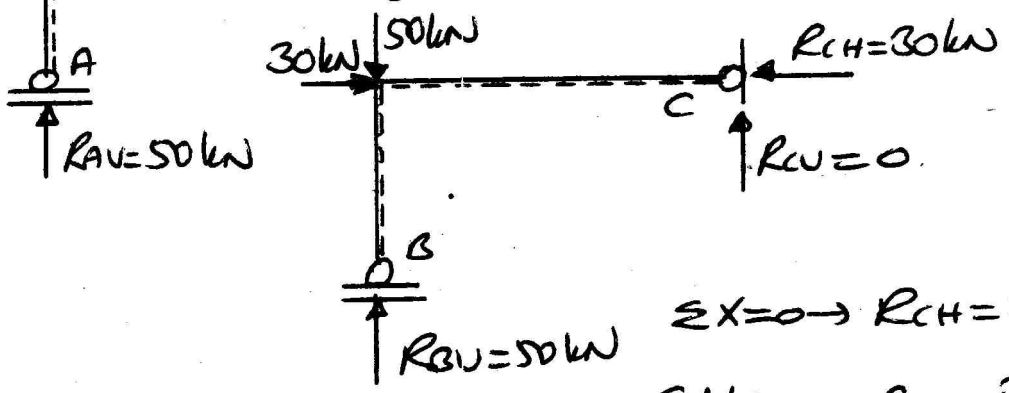




$\Sigma X=0 \rightarrow G_x=30kN$

$\Sigma M_A=0 \rightarrow G_y=\frac{100\times2+30\times3-30\times3}{4}=50kN$

$\Sigma M_G=0 \rightarrow R_{AV}=\frac{100\times2-30\times3+30\times3}{4}=50kN$



$\Sigma X=0 \rightarrow R_{CH}=30kN$

$\Sigma M_B=0 \rightarrow R_{CV}=\frac{30\times3-30\times3}{5}=0$

$\Sigma M_C=0 \rightarrow R_{BV}=50kN$

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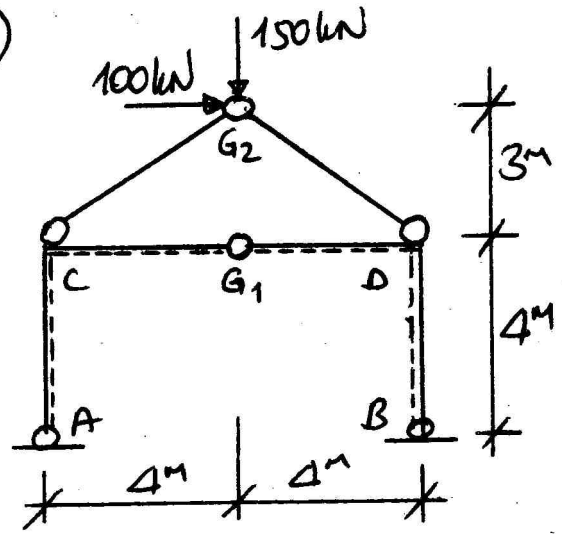
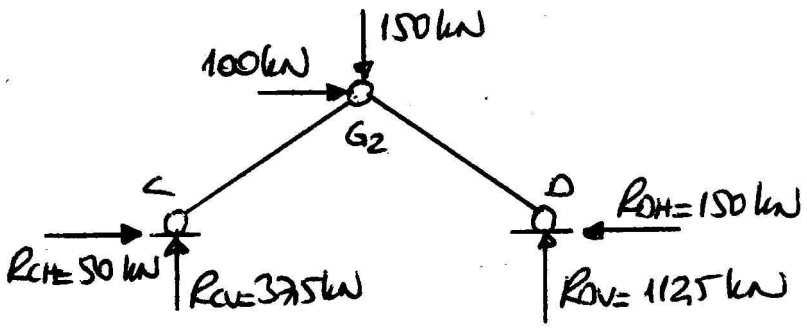
28)





$\Sigma M_D=0 \rightarrow R_{CV}=\frac{150\times4-100\times3}{8}=37,5kN$

$\Sigma M_C=0 \rightarrow R_{OV}=\frac{150\times4+100\times3}{8}=112,5kN$

$\Sigma Y=0$

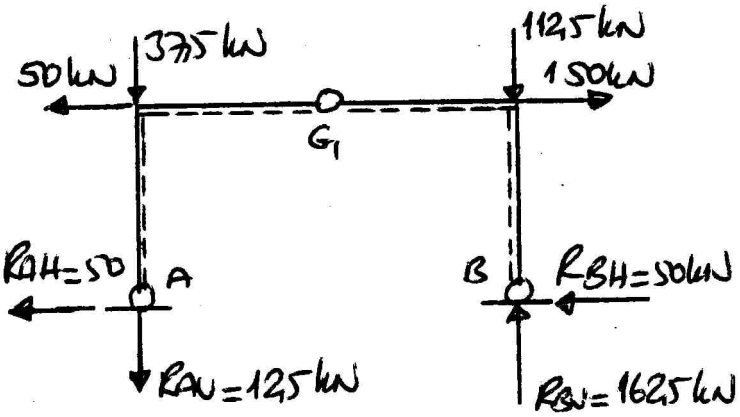
$R_{AV}+R_{BV}=150$

$\Sigma M_{G_2\,sol}=0 \rightarrow R_{CH}=\frac{37,5\times4}{3}=50kN$

$\Sigma M_{G_2\,sag}=0 \rightarrow R_{OH}=\frac{112,5\times4}{3}=150kN$

$\Sigma X=0$

$R_{CH}+R_{OH}=100$



$\Sigma M_B=0 \rightarrow R_{AV}=\frac{150\times4-50\times4}{8}-37,5=125\,kN$

$\Sigma M_A=0 \rightarrow R_{BV}=\frac{150\times4-50\times4}{8}+112,5=162,5\,kN$

$\Sigma Y=0 \rightarrow 37,5+112,5=-12,5+162,5$

$150=150$

$\Sigma M_{G_1\,jobb}=0 \rightarrow R_{AH}=\frac{37,5\times4+12,5\times4}{4}=50\,kN$

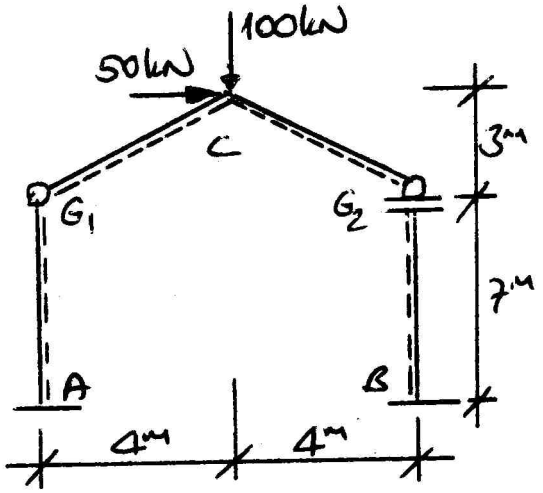
$\Sigma M_{G_1\,bal}=0 \rightarrow R_{BH}=\frac{162,5\times4-112,5\times4}{4}=50\,kN$

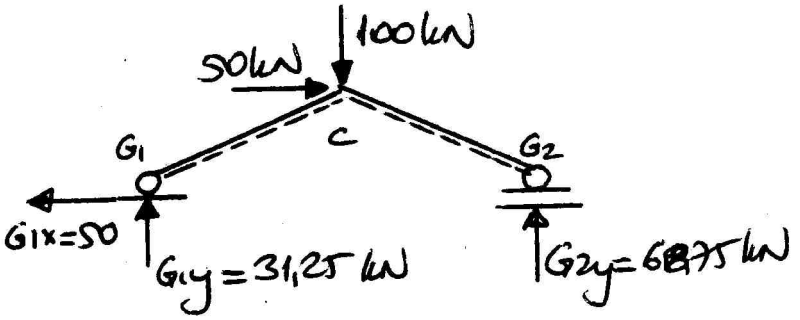
$\Sigma X=0 \rightarrow 50+50=150-50$

$100=100$

---

29)



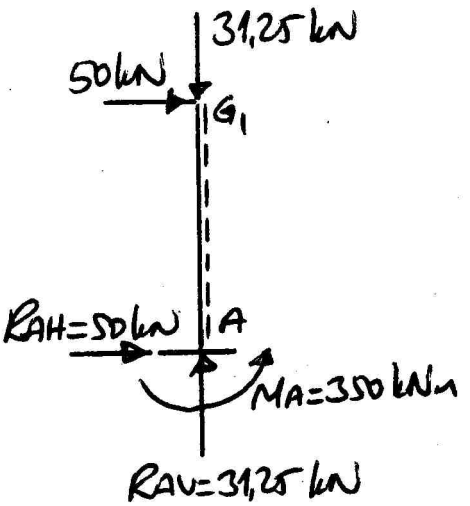


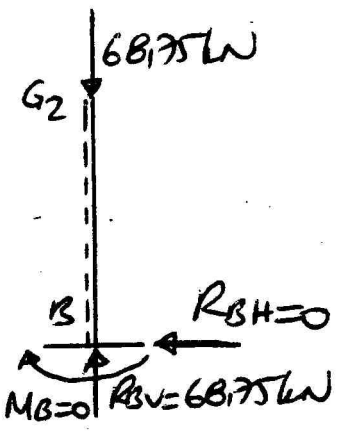
$\Sigma X=0 \rightarrow G_{1x}=50\,kN$

$\Sigma M_{G_2}=0 \rightarrow G_{1y}=\frac{100\times4-50\times3}{8}=31,25\,kN$

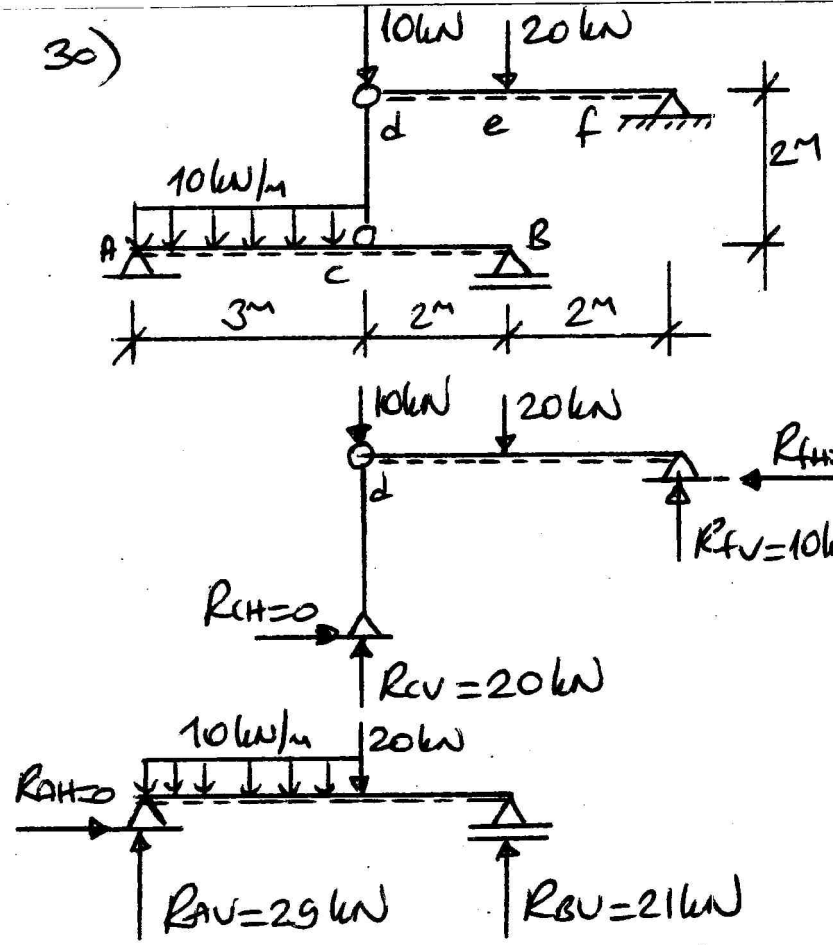
$\Sigma M_{G_1}=0 \rightarrow G_{2y}=\frac{100\times4+50\times3}{8}=68,75\,kN$

$\Sigma Y=0 \rightarrow 100=31,25+68,75$





30)



$\Sigma M_{d\,ljol} = 0 \rightarrow R_{CH}=0$ (Pandul Ajuk)

$\Sigma X=0 \rightarrow R_{fH}=0$

$\Sigma M_C=0 \rightarrow R_{fV}=\frac{20\times 2}{4}=10kN$

$\Sigma M_f=0 \rightarrow R_{CV}=\frac{20\times 2}{4}+10=20kN$

$\Sigma Y=0 \rightarrow 10+20=10+20$

$30=30$

$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma M_B=0 \rightarrow R_{AV}=\frac{10\times 3\times 3{,}5+20\times 2}{5}=29kN$

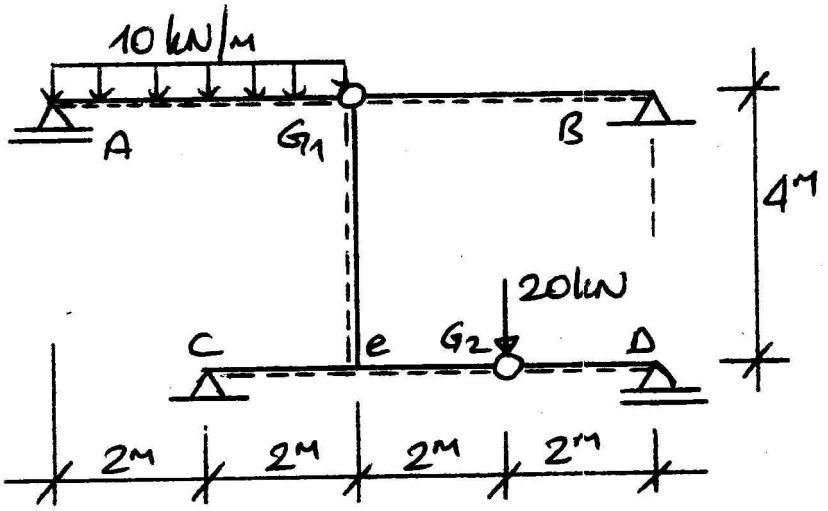
$\Sigma M_A=0 \rightarrow R_{BV}=\frac{10\times 3\times 1{,}5+20\times 3}{5}=21kN$

$\Sigma Y=0 \rightarrow 10\times 3+20=29+21$

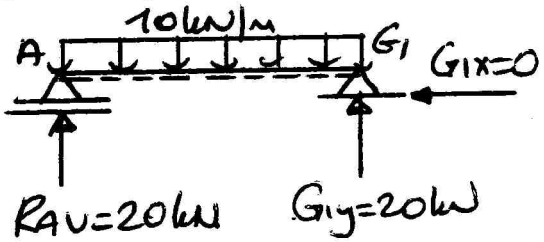
$50=50$

---

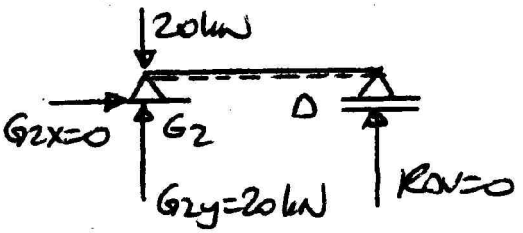
31)



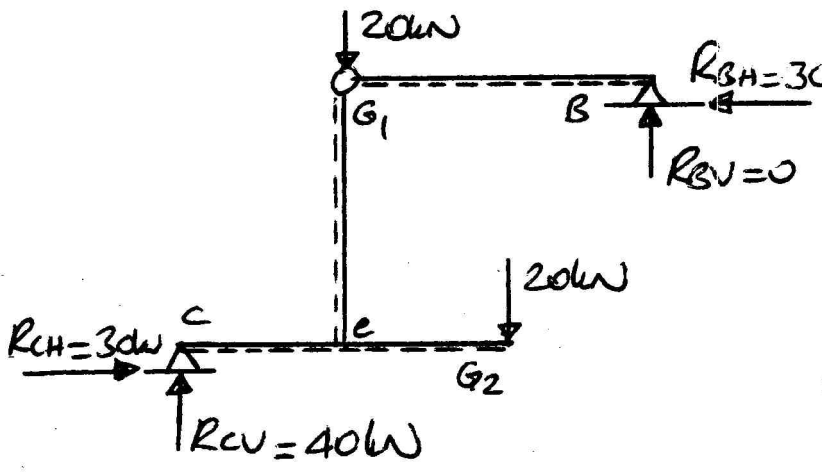
(I) rolu sistem



(II) rolu sistem



(III) nolu sistem



$\Sigma M_{G_{1sag}}=0 \rightarrow R_{BV}=0$

$\Sigma M_C=0 \rightarrow R_{BH}=\frac{20\times2+20\times4}{4}=30kN$

$\Sigma M_B=0 \rightarrow R_{CV}\cdot 6-R_{CH}\cdot 4=20\times4+20\times2$

$\Sigma M_{G_{sol}}=0 \rightarrow R_{CV}\cdot 2-R_{CH}\cdot 4=-20\times2$

$R_{CV}=40kN,\ R_{CH}=30kN$

$\Sigma X=0 \rightarrow 30=30$

$\Sigma Y=0 \rightarrow 40+0=20+20$

} Kontrol

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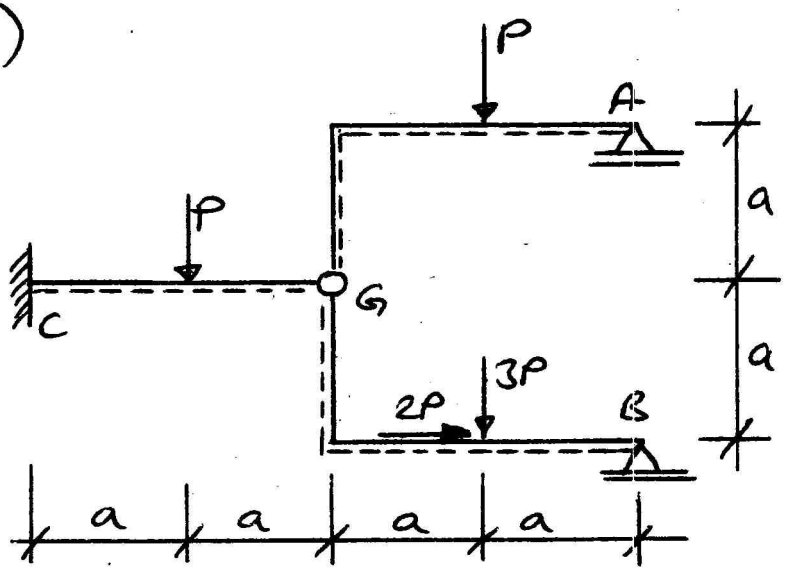
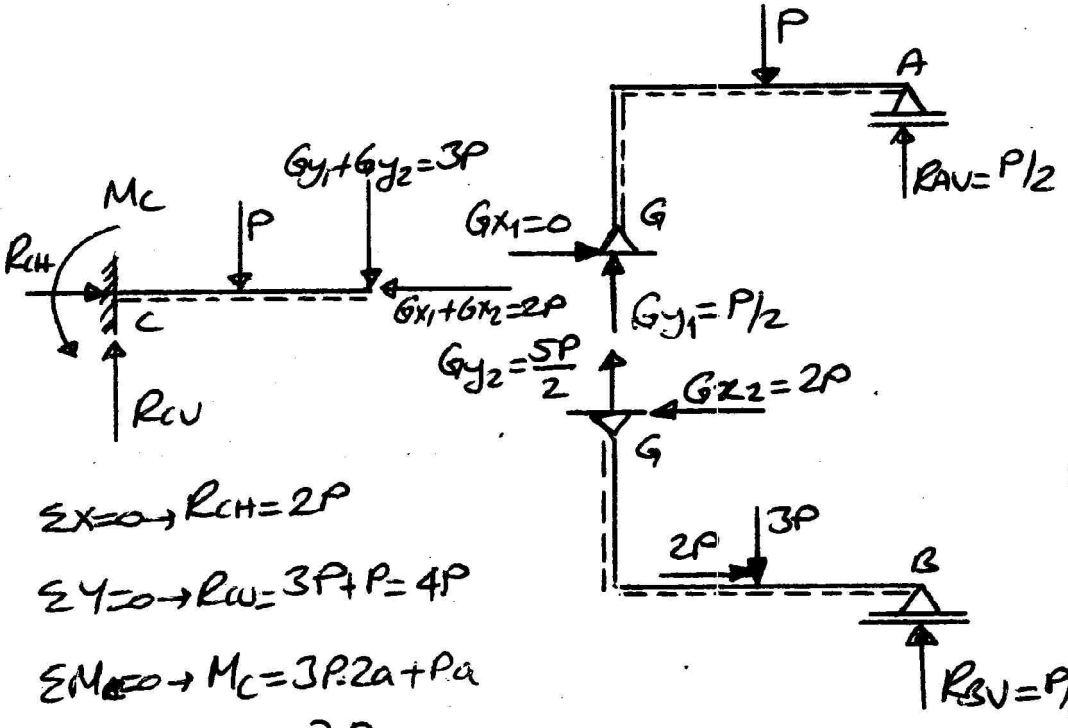
32)





$\Sigma X=0 \rightarrow G_{x_1}=0$

$\Sigma M_G=0 \rightarrow R_{AV}=P/2$

$\Sigma M_A=0 \rightarrow G_{y_1}=P/2$

$\Sigma X=0 \rightarrow R_{CH}=2P$

$\Sigma Y=0 \rightarrow R_{CV}=3P+P=4P$

$\Sigma M_C=0 \rightarrow M_C=3P\cdot 2a+Pa$

$=7Pa$

$\Sigma X=0 \rightarrow G_{x_2}=2P$

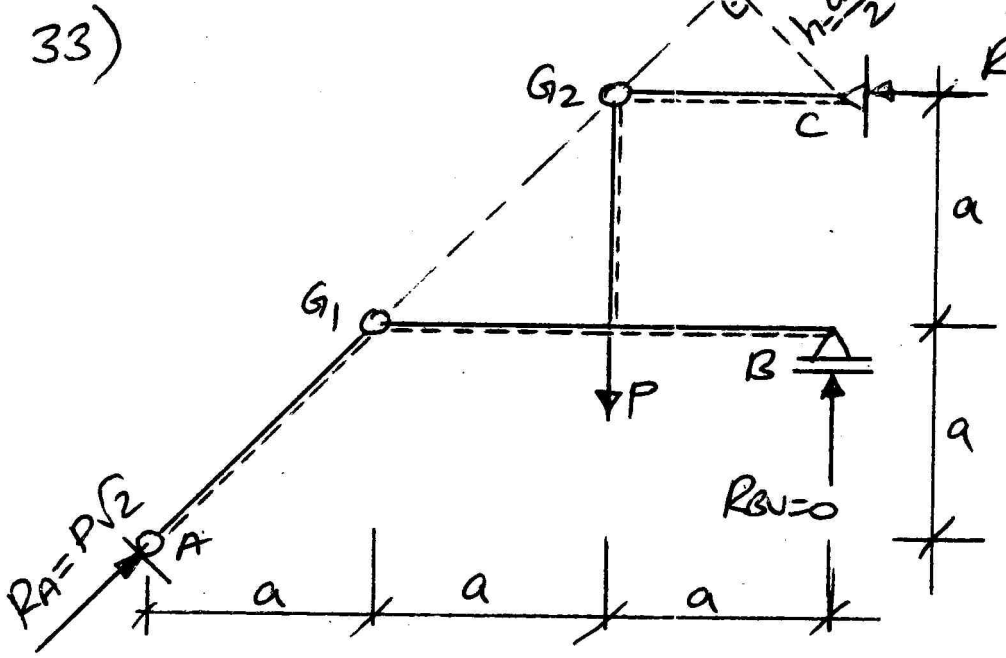
$\Sigma M_G=0 \rightarrow R_{BV}=\frac{3Pa-2P\cdot a}{2a}$

$=P/2$

$\Sigma M_B=0 \rightarrow G_{y_2}=\frac{2P\cdot a+3P\cdot a}{2a}$

$=\frac{5}{2}P$

33)



$\Sigma M_{G_2}=0 \rightarrow R_{BV}=0$

$\Sigma M_C=0 \rightarrow R_A \cdot h = P \cdot a$

$R_A \cdot \frac{a\sqrt{2}}{2} = P \cdot a$

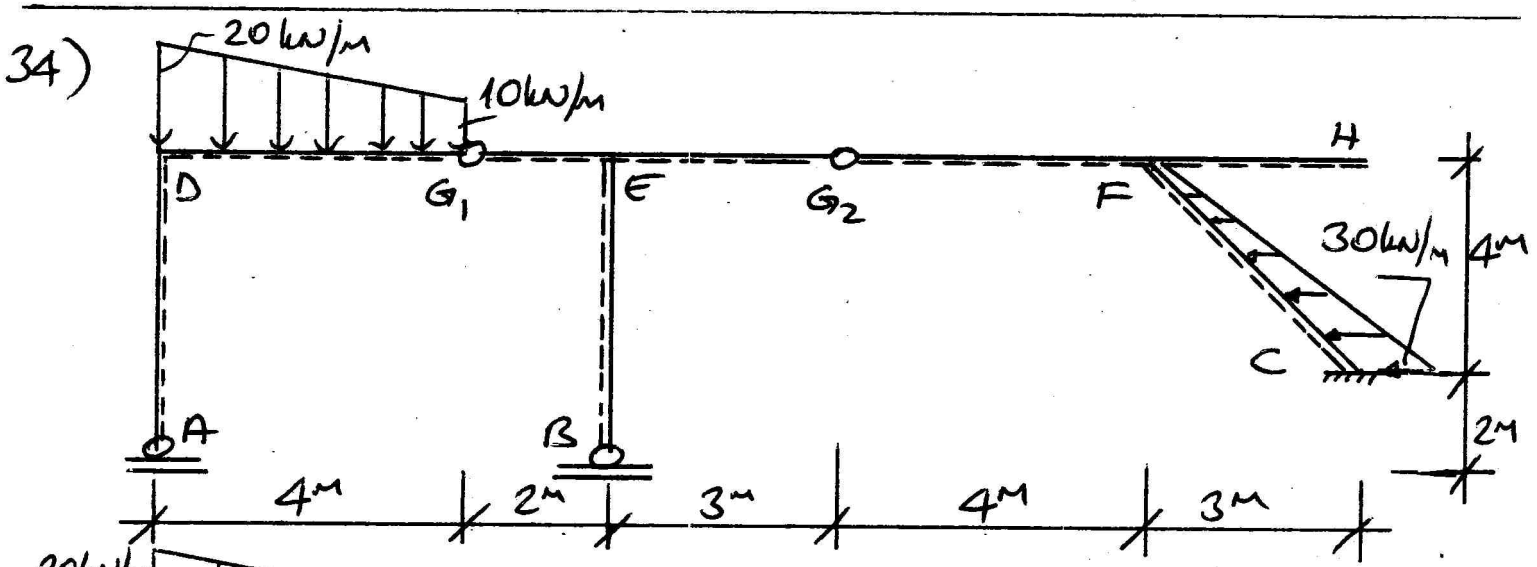
$R_A = P\sqrt{2}$

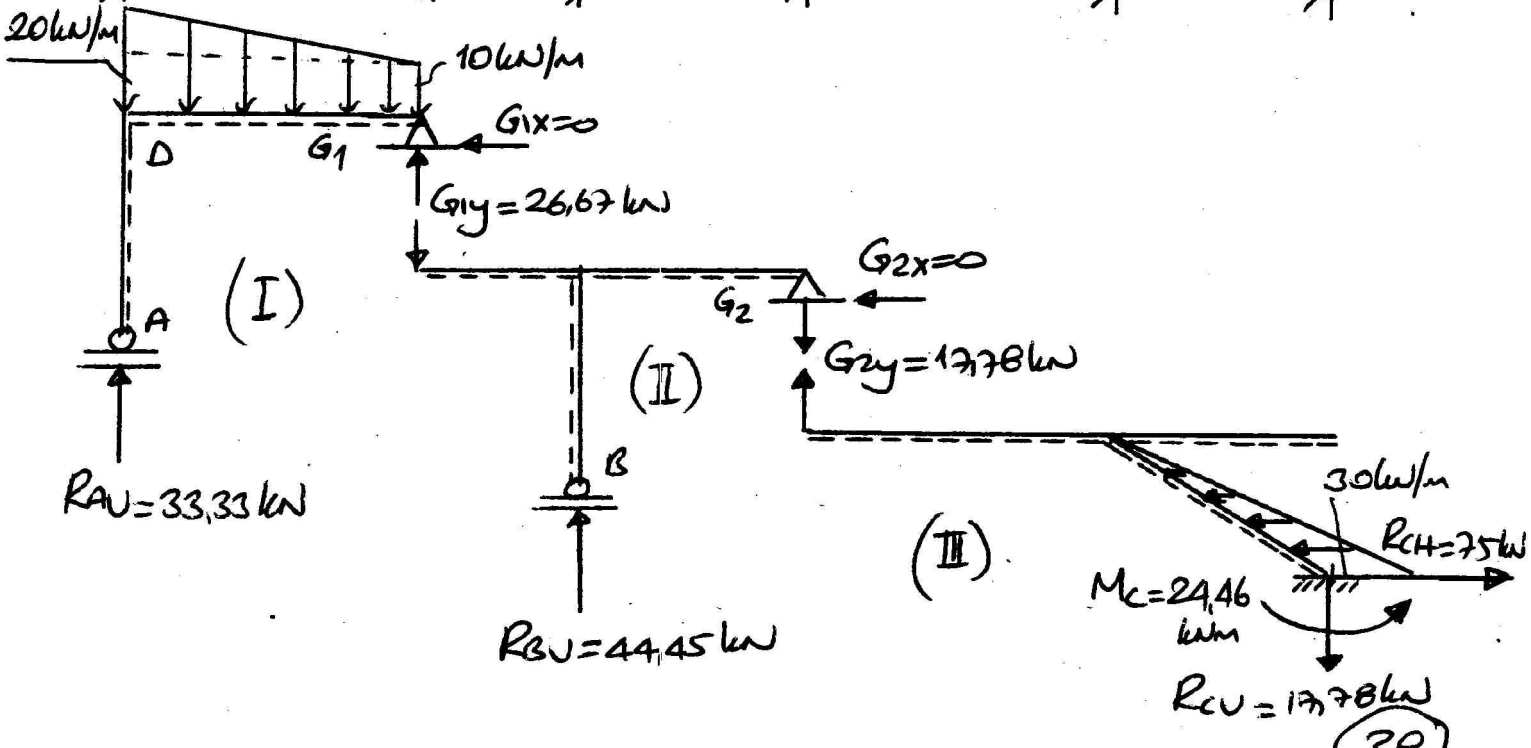
$\Sigma M_{G_1}=0 \rightarrow R_{CH} \cdot a - P \cdot a = 0$

$R_{CH} = P$

---

34)





(I) NOLU SİSTEM

$\Sigma X=0 \rightarrow G_{1x}=0$

$\Sigma M_{G_1}=0 \rightarrow R_{AV}=\dfrac{10\times4\times2+\frac{10\times4}{2}\cdot\left(\frac{2}{3}\times4\right)}{4}=33{,}33\ kN$

$\Sigma M_A=0 \rightarrow G_{1y}=\dfrac{10\times4\times2+\frac{10\times4}{2}(4/3)}{4}=26{,}67\ kN$

$\Sigma Y=0 \rightarrow 10\times4+\dfrac{10\times4}{2}=33{,}33+26{,}67$

$60=60$

(II) NOLU SİSTEM

$\Sigma X=0$

$\Sigma M_{G_2}=0 \rightarrow R_{BV}=\dfrac{26{,}67\times5}{3}=44{,}45\ kN$

$\Sigma M_B=0 \rightarrow G_{2y}=\dfrac{26{,}67\times2}{3}=17{,}78\ kN$

$\Sigma Y=0 \rightarrow 26{,}67+17{,}78=44{,}45$

$44{,}45=44{,}45$

(III) NOLU SİSTEM

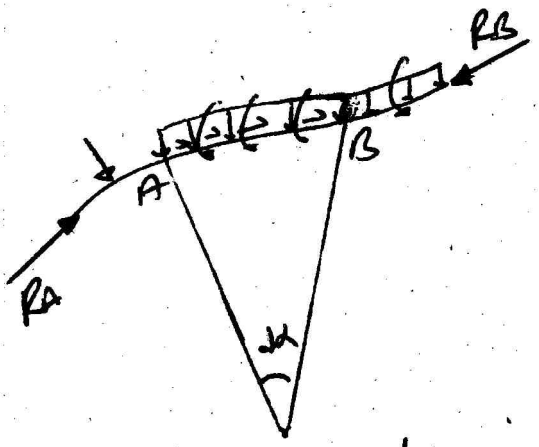
$\Sigma Y=0 \rightarrow R_{CV}=17{,}78\ kN$

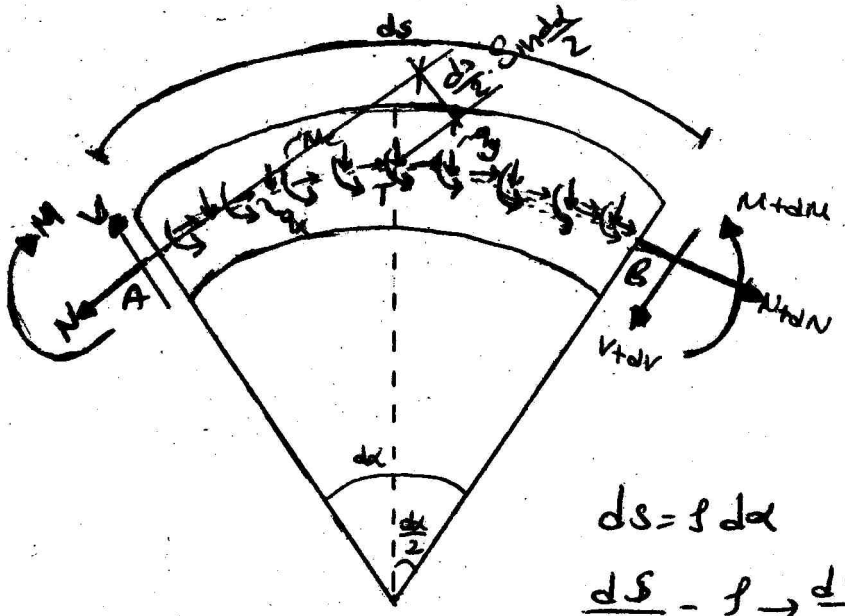
$\Sigma X=0 \rightarrow R_{CH}=30\times5\times\frac{1}{2}=75\ kN$

$\Sigma M_C=0 \rightarrow M_C=17{,}78\times7-\dfrac{30\times5}{2}\cdot\dfrac{4}{3}=24{,}46\ kNm$

III.2) KESİT TESİRLERİNİN HESABI (M,N,V DİYAGRAMLARI)

a) Yükler ve Kesit Tesirleri Arasındaki Bağıntılar





$d\alpha \neq 0 \rightarrow \cos\frac{d\alpha}{2} = 1$

$\sin\frac{d\alpha}{2} = \frac{d\alpha}{2}$

$ds = \rho\, d\alpha$

$\frac{ds}{d\alpha} = \rho \rightarrow \frac{d\alpha}{ds} = \frac{1}{\rho}$

$$\xrightarrow{+} \quad \Sigma P_x = 0 \longrightarrow -N\cdot\cos\frac{d\alpha}{2} + (N+dN)\cos\frac{d\alpha}{2} - V\sin\frac{d\alpha}{2} - (V+dV)\sin\frac{d\alpha}{2} + q(x)\,ds = 0$$

$$\overset{+}{\circlearrowleft}\Sigma M_T=0 \rightarrow M-(M+dM)+V.\rho.\sin\frac{d\alpha}{2}+(V+dV)\rho\sin\frac{d\alpha}{2}-N\frac{ds}{2}\sin\frac{d\alpha}{2}+$$

$$+(N+dN)\frac{ds}{2}.\sin\frac{d\alpha}{2}-Me.ds=0$$

$$-dM+2V\rho.\sin\frac{d\alpha}{2}+dV.\rho.\sin\frac{d\alpha}{2}+dN.\frac{ds}{2}.\sin\frac{d\alpha}{2}-Me\,ds=0$$

$$-dM+2V\rho\frac{d\alpha}{2}+dV.\rho.\frac{d\alpha}{2}+dN.\frac{ds}{2}.\frac{d\alpha}{2}-Me.ds=0$$

$$-dM+V.ds+\underbrace{dV\frac{ds}{2}}_{\approx 0}+\underbrace{dN.ds.\frac{d\alpha}{4}}_{\approx 0}-Me.ds=0$$

$$-\frac{dM}{ds}+V-Me=0$$

$$\boxed{\frac{dM}{ds}+Me-V=0} \quad \text{.....(3)}$$

(1), (2) ve (3) bağıntıları eğri eksenli çubuklarda yükler ve kesit tesirleri arasındaki bağıntılardır.

b) Düz Yüzeyli Çubuklarda Yükler ve Kesit Tesirleri Arasındaki Bağıntılar

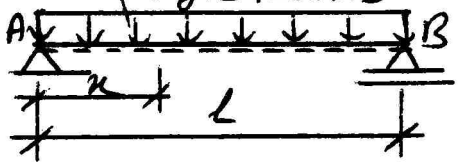
(4), (5) ve (6) bağıntıları çubuk eksenli sistemlerde yükler ve kesit tesirleri arasındaki bağıntılardır.

$$\frac{dV}{dx} = -q(y) \longrightarrow V = \int -q(y)\,dx$$

$$\frac{dM}{dx} = V \longrightarrow M = \int V_x\,dx = \iint -q(y)\,dx$$

$$\frac{d^2M}{dx^2} = \frac{dV}{dx} = -q(y)$$

ÖRNEK:1) Şekilde verilen basit kirişin kesit tesirlerinin fonksiyonunu yazınız.



1) $\frac{dN}{dx} \neq 0 \rightarrow N = C_1 \rightarrow C_1 = N = 0$

2) $\frac{dV}{dx} = -q(y) \rightarrow V = \int -q(y)\,dx \rightarrow V_x = -q.x + C_1$

3) $\frac{dM}{dx} = -q.x + C_1 \longrightarrow M = \int (-qx + C_1)\,dx$

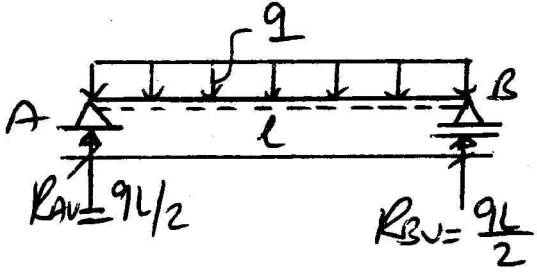
$$M_x = -\frac{qx^2}{2} + C_1 x + C_2$$

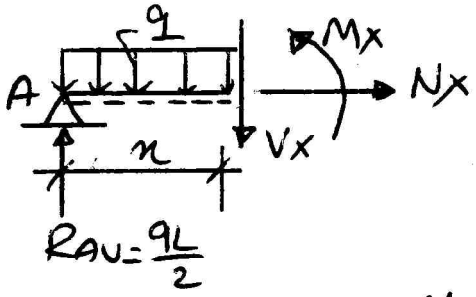
Sınır Koşulları

$x = 0 \longrightarrow M_x = 0 \longrightarrow 0 = -q.0 + C_1.0 + C_2 \rightarrow C_2 = 0$

$x = L \longrightarrow M_x = 0 \longrightarrow 0 = -\frac{qL^2}{2} + C_1 L \longrightarrow C_1 = \frac{qL}{2}$

$$V_x = \frac{qL}{2} - q.x \qquad M_x = \frac{qL}{2}x - \frac{qx^2}{2}$$



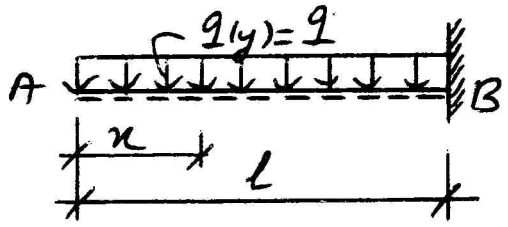


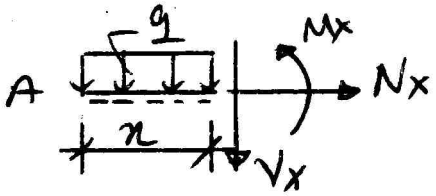
$$V_x = \frac{qL}{2} - q.x$$

$$M_x = \frac{qL}{2}.x - q.x.\frac{x}{2} = \frac{qL}{2}x - \frac{qx^2}{2}$$

$$\max M \rightarrow V_x = 0 \rightarrow x = \frac{L}{2} \rightarrow \max M = \frac{qL^2}{8}$$

ÖRNEK: 2)





$$V_x = -q.x$$

$$m_x = -qx.\frac{x}{2} = -\frac{qx^2}{2}$$

1) $\frac{dV}{dx} = -q \rightarrow V_x = -qx + C_1$

2) $\frac{dM}{dx} = V \rightarrow M_x = -\frac{qx^2}{2} + C_1x + C_2$

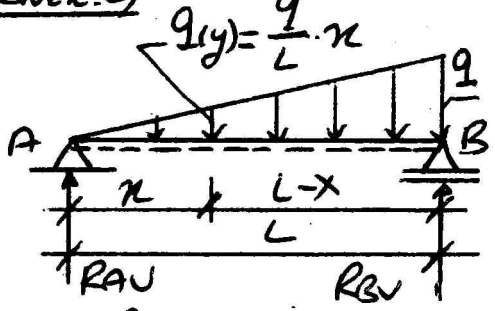
Sınır Koşulları

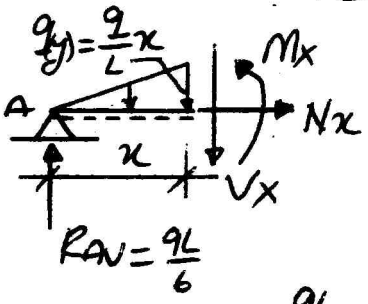
$x=0 \rightarrow V_x = 0 \rightarrow 0 = -q.0 + C_1 \rightarrow C_1 = 0$

$x=0 \rightarrow M_x = 0 \rightarrow 0 = -\frac{q.0}{2} + C_1.0 + C_2 \rightarrow C_2 = 0$

$$\boxed{V_x = -qx} \qquad \boxed{M_x = -\frac{qx^2}{2}}$$

ÖRNEK:3)





$R_{AV}=\frac{9L}{6}$

$\Sigma M_B=0 \rightarrow R_{AV}=\dfrac{\frac{9L}{2}\cdot\left(\frac{L}{3}\right)}{L}$

$R_{AV}=\frac{9L}{6}$

$\Sigma M_A=0 \rightarrow R_{BV}=\dfrac{\frac{9L}{2}\left(\frac{2}{3}L\right)}{L}$

$R_{BV}=\frac{9L}{3}$

$V_x=\frac{9L}{6}-\frac{\left(\frac{9}{L}x\right)\cdot x}{2}$

$=\frac{9L}{6}-\frac{9x^2}{2L}$

$M_x=\frac{9L}{6}x-\frac{\left(\frac{9}{L}x\right)x}{2}\cdot\frac{x}{3}$

$=\frac{9L}{6}x-\frac{9x^3}{6L}$

1) $\frac{dV}{dx}=-q_{(y)} \rightarrow V_x=\int -\frac{9}{L}x\,dx$

$$V_x=-\frac{9x^2}{2L}+C_1$$

2) $\frac{dM}{dx}=V_x \rightarrow M_x=\int\left(-\frac{9x^2}{2L}+C_1\right)dx$

$$M_x=-\frac{9x^3}{6L}+C_1x+C_2$$

Sınır Koşulları

$x=0 \rightarrow M_x=0 \rightarrow 0=-0+C_1\cdot 0+C_2 \rightarrow C_2=0$

$x=L \rightarrow M_x=0 \rightarrow -\frac{9L^3}{6L}+C_1L=0 \rightarrow C_1=\frac{9L}{6}$

$$\boxed{V_x=\frac{9L}{6}-\frac{9x^2}{2L}} \qquad \boxed{M_x=\frac{9L}{6}x-\frac{9x^3}{6L}}$$

$\max M \rightarrow V_x=0 \rightarrow \frac{9L}{6}-\frac{9x^2}{2L}=0$

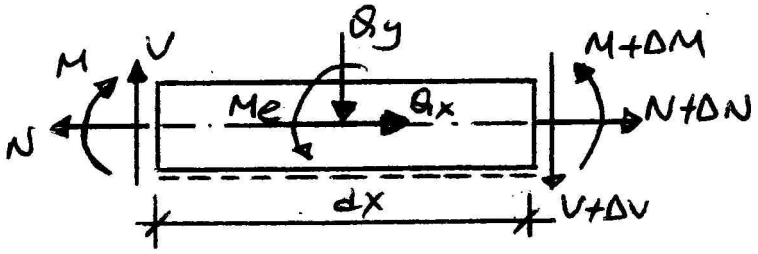
$x=\frac{\sqrt{3}}{3}L$

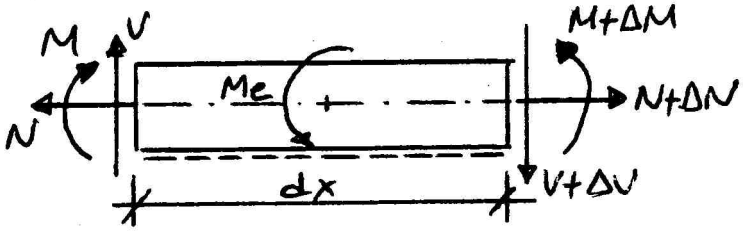
$\max M=\frac{9L}{6}\left(\frac{\sqrt{3}}{3}L\right)-\frac{9}{6L}\left(\frac{\sqrt{3}}{3}L\right)^3$

$=\frac{\sqrt{3}}{27}9L^2$

$=0{,}06415\,9L^2$

b2) Tekil Yükler için M,N,V Bağıntıları





$\Sigma X=0 \rightarrow -N+N+\Delta N+q_x=0$

$$\boxed{\Delta N=-q_x}$$

$\Delta V=0$, $\Delta N=0$

$$\boxed{\Delta M=-M_e}$$

$\Sigma Y=0 \rightarrow -V+V+\Delta V+q_y=0$

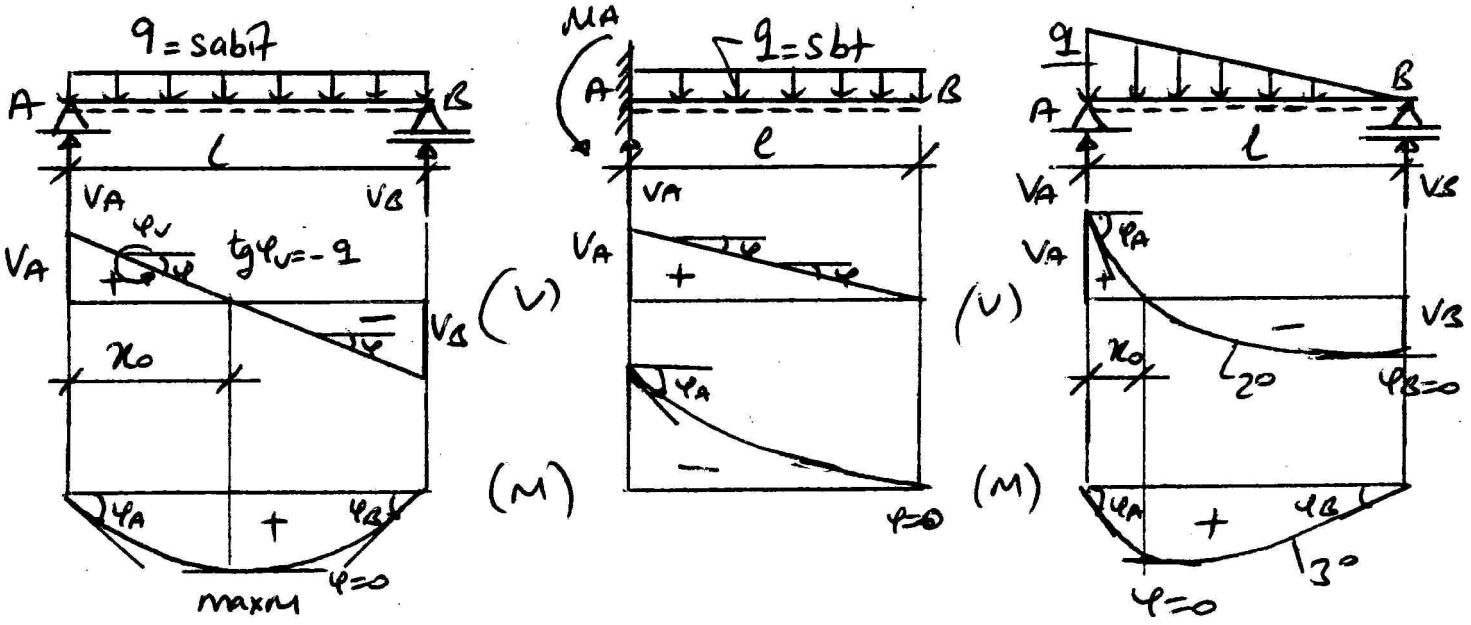
$$\boxed{\Delta V=-q_y}$$

III.3) Kesit Tesirleri Diyagramları

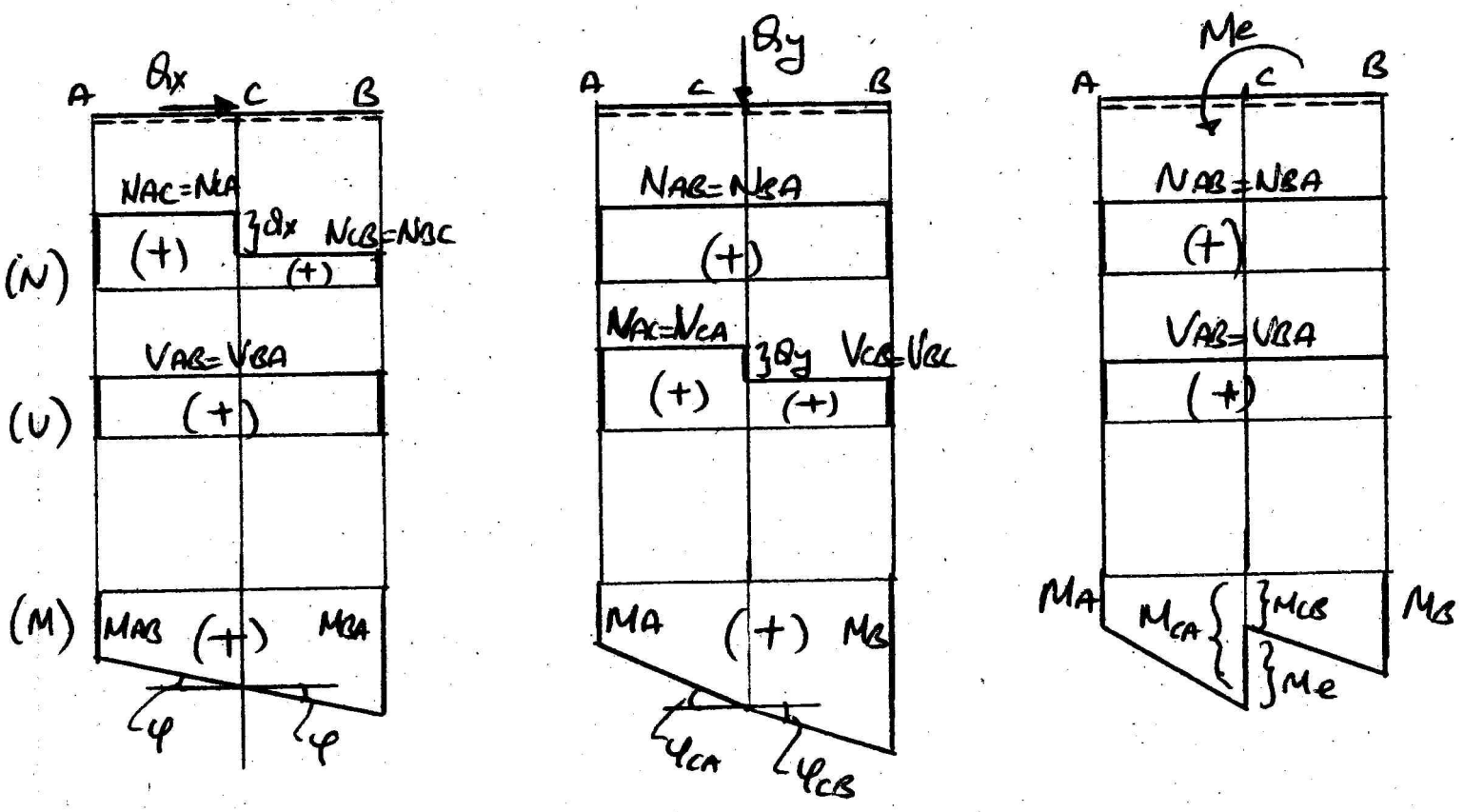
Dış yüklerden meydana gelen kesit tesiri fonksiyonlarının grafiklerine kesit tesiri diyagramları denir. Kesit tesirleri diyagramları belirli bir yükleme altında belirli bir kesit tesiri büyüklüğünün çubuk ekseri boyunca değişimini gösterir.

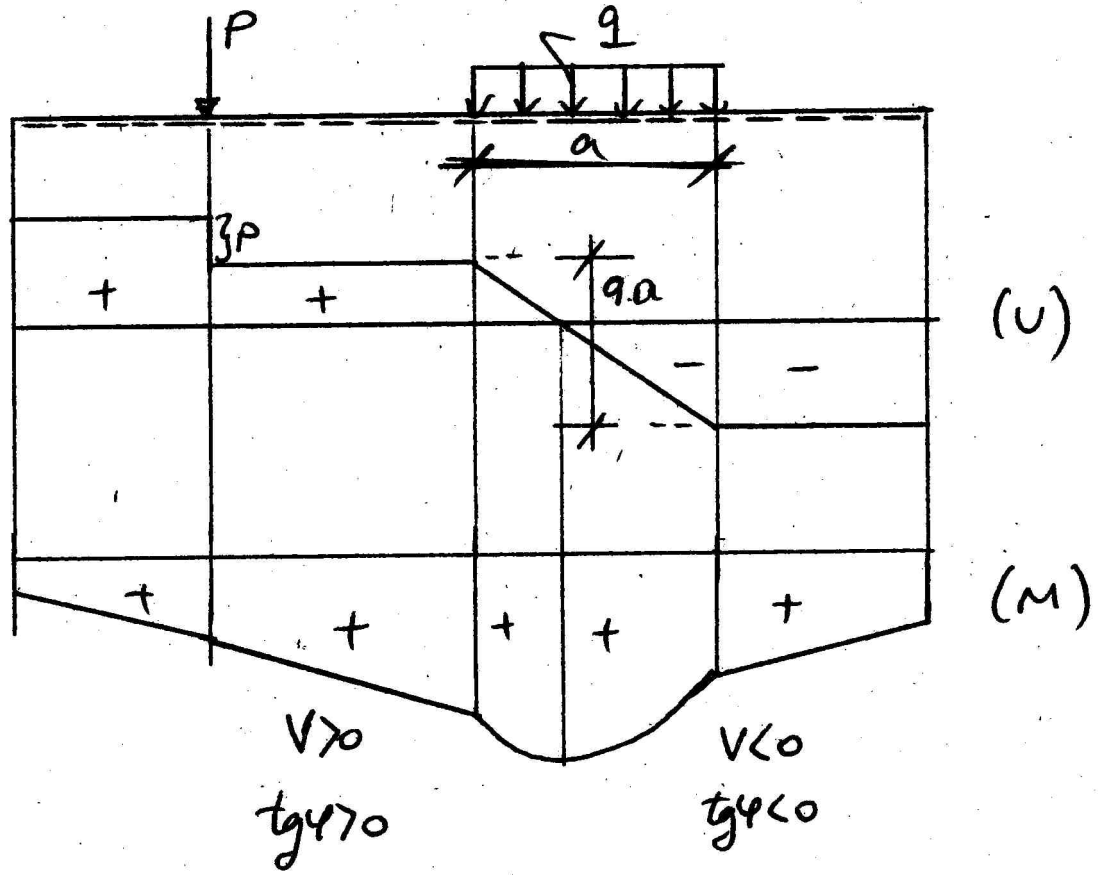
Kesit tesirleri diyagramları 2 yolla çizilebilir

III.3.1) Yayılı Yük Durumunda Kesit Tesiri Diyagramları



III.3.2) Tekil Yük Durumunda Kesit Tesiri Diyagramları





III.3.3) Yükler ve Kesit Tesirleri Diyagramları Arasındaki Bağıntılar

1) Düzgün Yayılı Yük ($q_{(x)}, q_{(y)}$) Durumu

a) Çubuk boyunca herhangi bir kesitteki normal kuvvetin diyagramının eğimi bu noktadaki $q_{(x)}$'in ters işaretlisidir. $\left(\frac{dN}{dx} = -q_{(x)}\right)$

b) Çubuk boyunca herhangi bir kesitteki kesme kuvveti diyagramının eğimi bu noktadaki $q_{(y)}$'nin ters işaretlisidir. $\left(\frac{dV}{dx} = -q_{(y)}\right)$

c) Çubuk boyunca herhangi bir kesitteki eğilme momenti diyagramının eğimi bu noktadaki kesme kuvvetinin değerine eşittir. $\left(\frac{dM}{dx} = V_x\right)$

2) Tekil Yük (Q_x, Q_y) Durumu

a) Çubuk eksenine paralel tekil yük (Q_x) etkisindeki bir çubukta tekil yükün etkidiği kesitte normal kuvvette ani bir değişim meydana gelir. Bu değişimin değeri Q_x'in ters işaretlisidir. $(\Delta = -Q_x)$

b) Çubuk eksenine dik tekil yük (Q_y) etkisindeki bir çubukta tekil yükün etkidiği kesitte kesme kuvvetinde ani bir değişim meydana gelir. Bu değişimin değeri Q_y'nin ters işaretlisidir. $(\Delta V = -Q_y)$

c) Q_x ve Q_y tekil yüklerinden dolayı eğilme momenti diyagramında bir değişim olmaz. (Q_y'de eğim değişir.)

3) Tekil Moment Etkisi Durumu

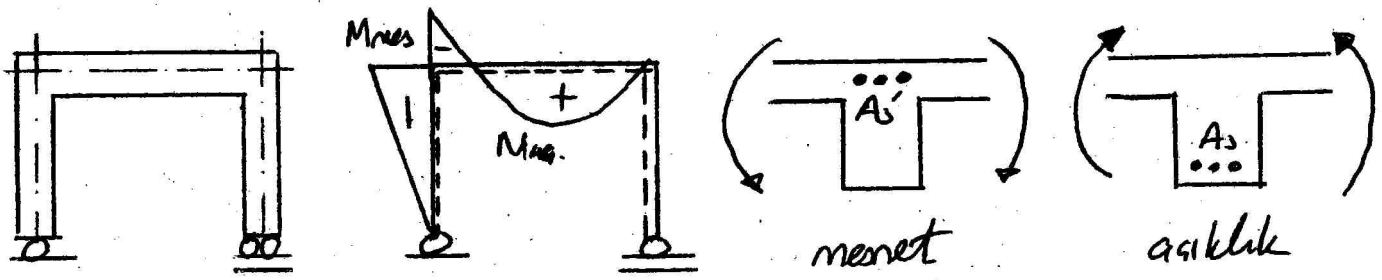
a) Herhangi bir kesite etkiyen tekil momentten dolayı normal kuvvet ve kesme kuvvetinde bir değişim meydana gelmez.

b) Tekil moment etkisindeki çubukta, bu momentin etkidiği kesitte eğilme momentinde ani bir değişim meydana gelir. Bu değişimin değeri dış tekil momentin ters işaretlisidir. ($\Delta M = -Me$)

III. 4) Kesit Tesirleri Hesabında İzlenecek Yol

1- Mesnet tepkileri bulunur.

2- Sistem üzerinde bakış yönü belirlenir.



3- Sistem üzerinde karakteristik (kritik) noktalar belirlenir. Bu karakteristik noktalar;

a) Tekil yüklerin etkidiği kesitler,

b) Yayılı yüklerin süreksizlik noktaları,

c) Çubukların süreksizlik noktaları (mafsallar, düğüm noktaları, mesnetler) dir.

4- Kritik noktalardaki kesit tesirleri hesaplanır. Kesit tesirlerinin hesaplanması analitik yolla ve denge denklemleri ile yapılmaktadır.

5- En büyük kesit tesirleri ve yerleri belirlenir.

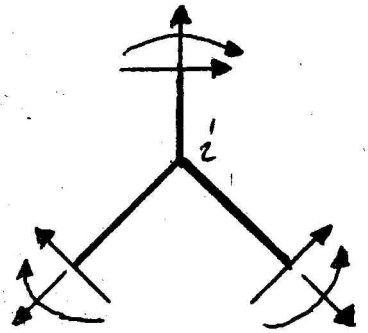
6- Gerektiği durumlarda yayılı yük bölgelerinde ara bölgeler hesaplanır.

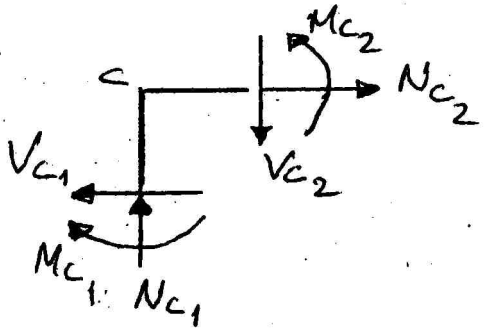
7- Bulunan değerlere göre diyagramlar çizilir.

8- Bulunan değerler şu şekilde kontrol edilir;

a) Göz önüne alınan kesitlerde sağ veya soldan indirgenerek bulunan kesit tesirleri birbirlerine eşit olmalıdır.

b) Düğüm noktalarında bulunan değerler denge şartlarını sağlamalıdır.





$\Sigma X_i = 0$

$\Sigma Y_i = 0$

$\Sigma M_i = 0$

$\Sigma X = 0 \rightarrow V_{C_1} = N_{C_2}$

$\Sigma Y = 0 \rightarrow N_{C_1} = V_{C_2}$

$\Sigma M = 0 \rightarrow M_{C_1} = M_{C_2}$

IV. DOLU GÖVDELİ SİSTEMLER

1_ BASİT KİRİŞLER

2_ KONSOL "

3_ ÇIKMALI "

4_ EĞİK "

5_ GERBER "

6_ GERGEVELER

6.1) BASİT GERGEVELER

6.2) ÜÇ MAFSALLI "

6.3) GERGİLİ "

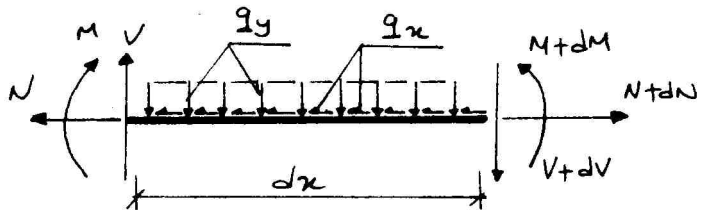
6.4) BİRLEŞİK "

7_ KEMERLER

7.1) BASİT KEMERLER

7.2) ÜÇ MAFSALLI "

III_1) BASİT KİRİŞLER



$$\frac{dN}{dx} + q_x = 0 \longrightarrow \boxed{\frac{dN}{dx} = -q_x}$$

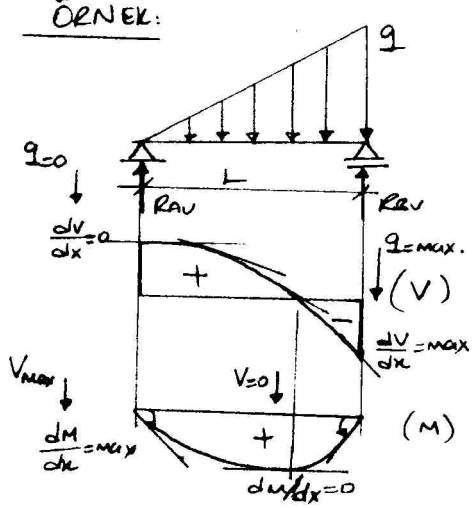
$$\frac{dV}{dx} + q_y = 0 \longrightarrow \boxed{\frac{dV}{dx} = -q_y}$$

$$\frac{dM}{dx} - V = 0 \longrightarrow \boxed{\frac{dM}{dx} = V}$$

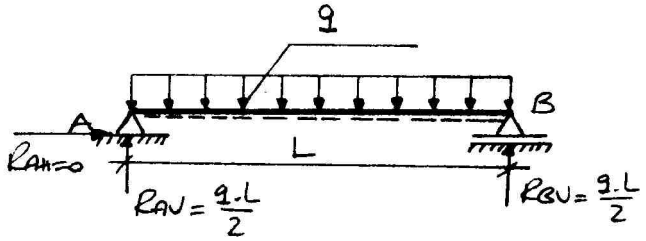
* Yükün eğimi sıfır ise kesme kuvvetinin x'e göre teğetin eğimi sıfırdır.

* Kesme kuvveti diyagramının eğimi artan ise, moment diyagramının eğimi de artacak; azalan ise diyagramın eğimi de azalacaktır.

ÖRNEK:



- Düzgün Yayılı Yükle Yüklü Basit Kirişin Kesit Tesiri



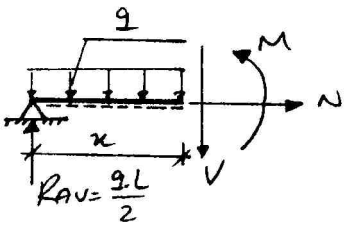
$\Sigma M_B = 0 \rightarrow R_{AV}.L - q.L.\frac{L}{2} = 0$

$R_{AV} = \frac{q.L}{2}$

$\Sigma M_A = 0 \rightarrow R_{BV}.L - q.L.\frac{L}{2} = 0$

$R_{BV} = \frac{q.L}{2}$

$\Sigma X = 0 \rightarrow R_{AH} = 0$



$\Sigma x = 0 \rightarrow N = 0$ (Yatay yük olmadığından sıfır)

$\Sigma Y = 0 \rightarrow V_x - \frac{q.L}{2} + q.x = 0$

$$\boxed{V_x = \frac{qL}{2} - q.x}$$

$\Sigma M = 0 \rightarrow M_x - \frac{qL}{2}.x - q.x.\frac{x}{2} = 0$

$$\boxed{M_x = \frac{qL}{2}x - q.\frac{x^2}{2}}$$

$\max M \Rightarrow V_x = 0$ olmalı. $\Rightarrow \frac{qL}{2} - q.x = 0 \Rightarrow x_0 = \frac{q.L/2}{q}$

$\frac{q.L}{2} = V_A$ ise ; $\boxed{x_0 = \frac{V_A}{q}}$ ------ (1)

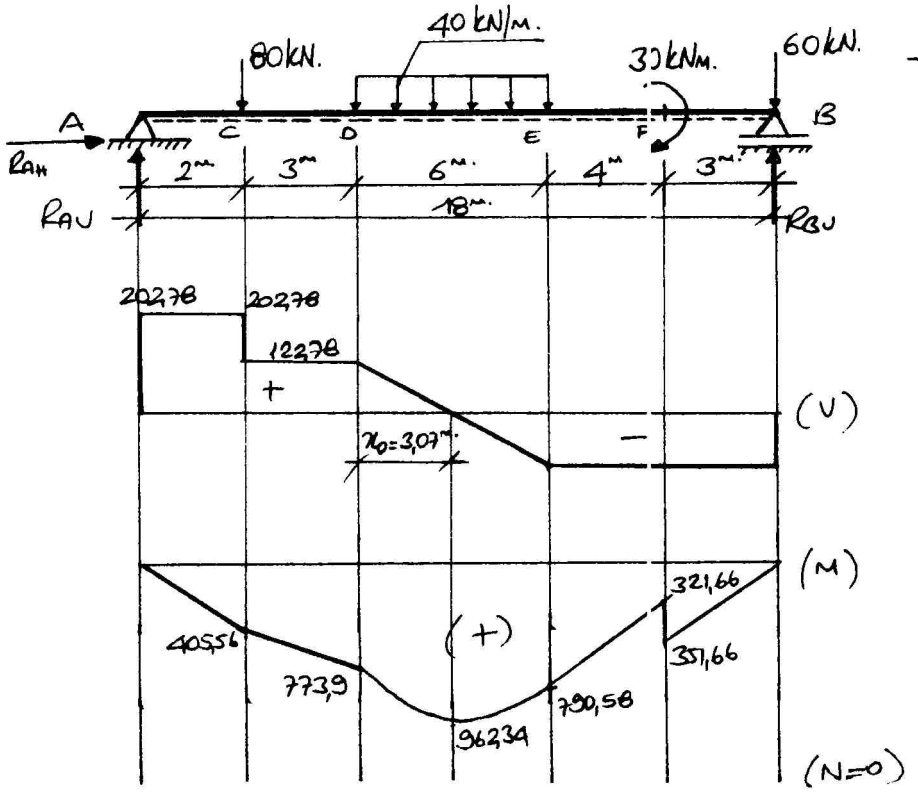
$$\max M = V_A.\frac{V_A}{q} - q.\frac{\left(\frac{V_A}{q}\right)^2}{2} = \frac{V_A^2}{q} - \frac{V_A^2}{2q} = \underline{\frac{V_A^2}{2q}}$$

A mesnedinde moment olması durumunda;

$$\boxed{\max M = \frac{V_A^2}{2q} \mp M}$$ ---------- (2)

* Bu iki formül yalnız düzgün yayılı yük durumu için geçerlidir.

Örnek-1) Şekildeki sistemin M,N,V diyagramlarını çizelim.



Mesnet Tepkileri:

$\Sigma M_B=0 \rightarrow R_{AV}=\frac{80.16+40.6.10-30}{18}$

$R_{AV}=202,78\ kN.$

$\Sigma M_A=0 \rightarrow R_{BV}=\frac{80.2+40.6.8+30+60.18}{18}$

$R_{BV}=177,22\ kN.$

$\Sigma Y=0$ (Kontrol):

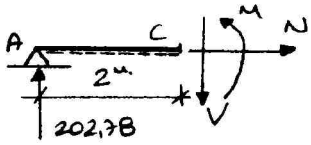
$177,22+202,78=80+40.6+60$

$380=380$ ✓

$\Sigma X=0 \rightarrow R_{AH}=0$

Kesit Tesirleri:

A-C arası

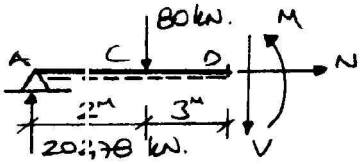


$N_{AC}=N_{CA}=0$

$V_{AC}=V_{CA}=202,78\ kN.$

$M_C=202,78.2=405,56\ kNm.$

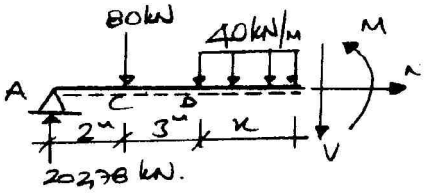
C-D arası



$V_{CD}=V_{DC}=202,78-80=122,78\ kN.$

$M_D=202,78.5-80.3=773,9\ kNm$

D-E arası



$V_{DE}=122,78\ kN.$

$V_{ED}=122,78-40.6=-117,22\ kN$

max M var.

$x_0=\frac{122,78}{40}=3,07\ m.$

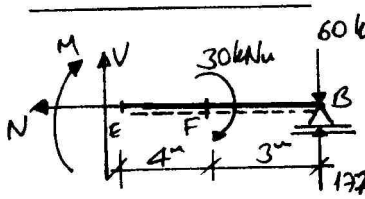
$M_{max}=\frac{122,78^2}{2.40}+773,9$

$=962,34\ kNm$

$M_E=202,78.11-80.9-40.6.3$

$=790,58\ kNm.$

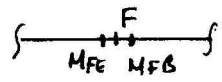
E-F arası



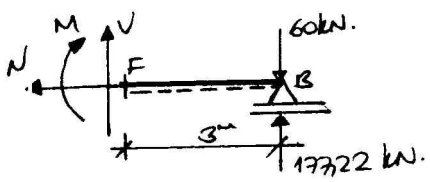
$V_{EF}=V_{FE}=-177,22-60=-117,22\ kN.$

$M_{FE}=177,22.3-60.3-30=321,66\ kNm.$

$M_{FB}=177,22.3-60.3=351,66\ kNm.$



F-B arası

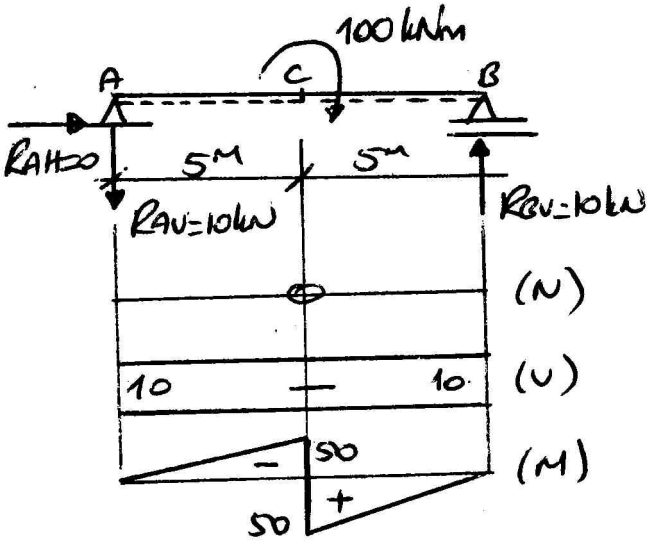


$V_{FB}=V_{BF}=-177,22-60$

$=117,22\ kN.$

$M_{FB}=(177,22-60).3=351,66\ kNm.$

ÖRNEK-2)



Mesnet Tepkileri

$R_{AH}=0$, $R_{AV}=R_{BV}=10kN$

Kesit Tesirleri

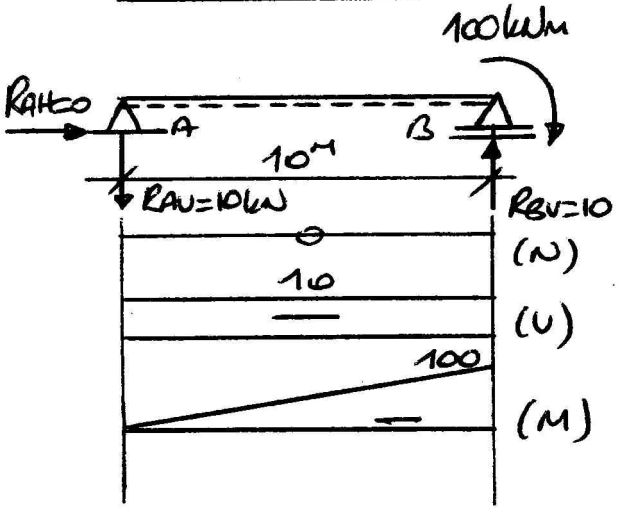
$N_{AB}=N_{BA}=0$

$V_{AB}=V_{BA}=-10\,kN$

$M_{CA}=-5\times10=-50\,kNm$ (sol parçanın dengesi)

$M_{CB}=5\times10=50\,kNm$ (sağ " ")

ÖRNEK-3



Mesnet Tepkileri

$R_{AH}=0$, $R_{AV}=R_{BV}=10kN$

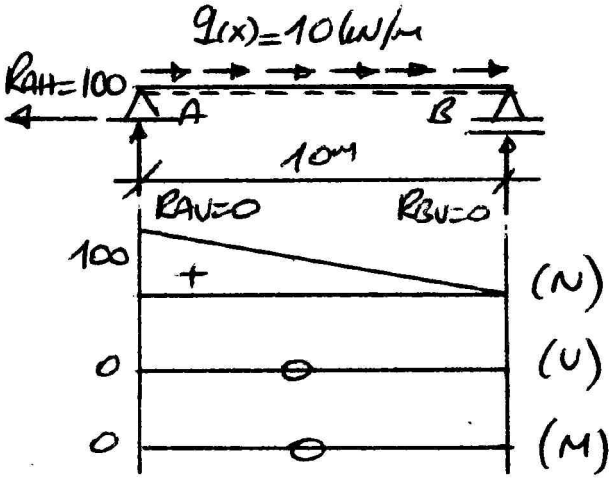
Kesit Tesirleri

$N_{AB}=N_{BA}=0$

$V_{AB}=V_{BA}=-10kN$

$M_{AB}=0$, $M_{BA}=-100\,kNm$

ÖRNEK-4



Mesnet Tepkileri

$\Sigma X=0 \rightarrow R_{AH}=10\times10=100\,kN$

$\Sigma M_B=\Sigma M_A=0 \rightarrow R_{AV}=R_{BV}=0$

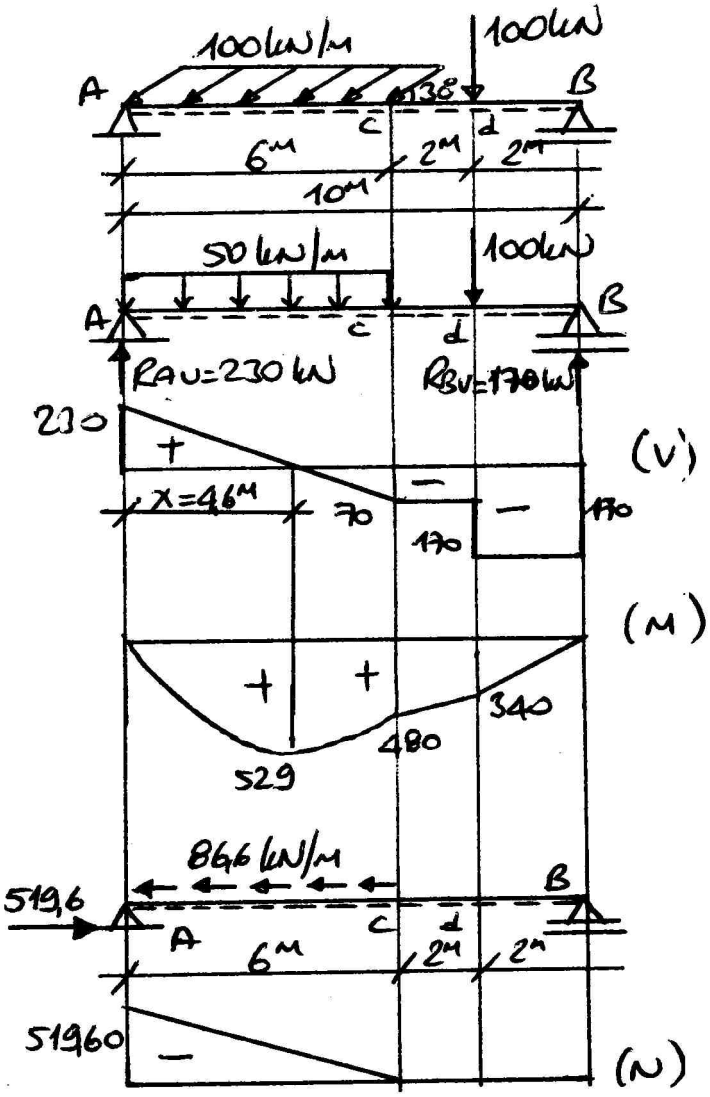
Kesit Tesirleri

$V_{AB}=V_{BA}=0$

$M_{AB}=M_{BA}=0$

$N_{AB}=100\,kN$, $N_{BA}=0$

ÖRNEK-5



Mesnet Tepkileri

$R_{AV}=230kN$, $R_{BV}=170kN$

$R_{AH}=519,6\ kN$

Kesit Tesirleri

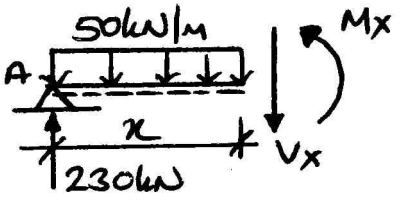
$V_{Ac}=230\ kN$

$V_{CA}=V_{Cd}=V_{dC}=230-50\times6=-70\ kN$

$V_{dB}=V_{Bd}=-70-100=-170\ kN$

$M_C=230\times6-50\cdot6\times3=480\ kNm$

$M_d=170\times2=340\ kNm$



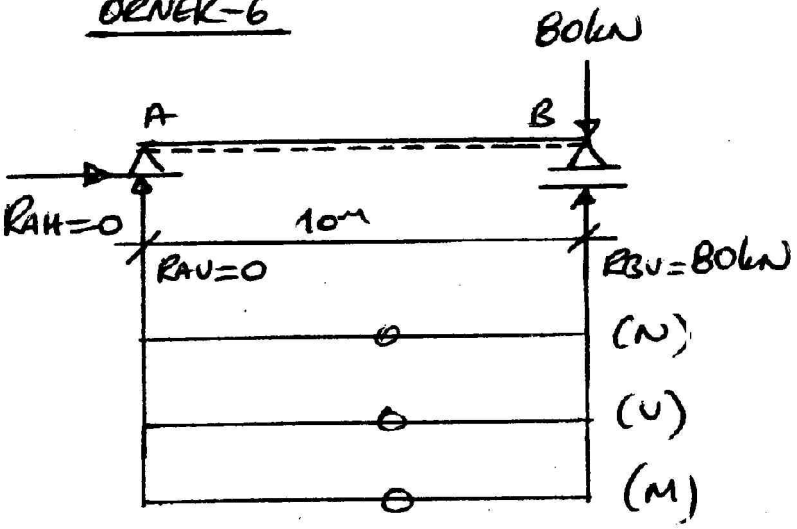
$V_x=230-50x$

$M_x=230x-\frac{50x^2}{2}$

$maxM \rightarrow V_x=0 \rightarrow x=4,60m$

$maxM=529\ kNm$

ÖRNEK-6



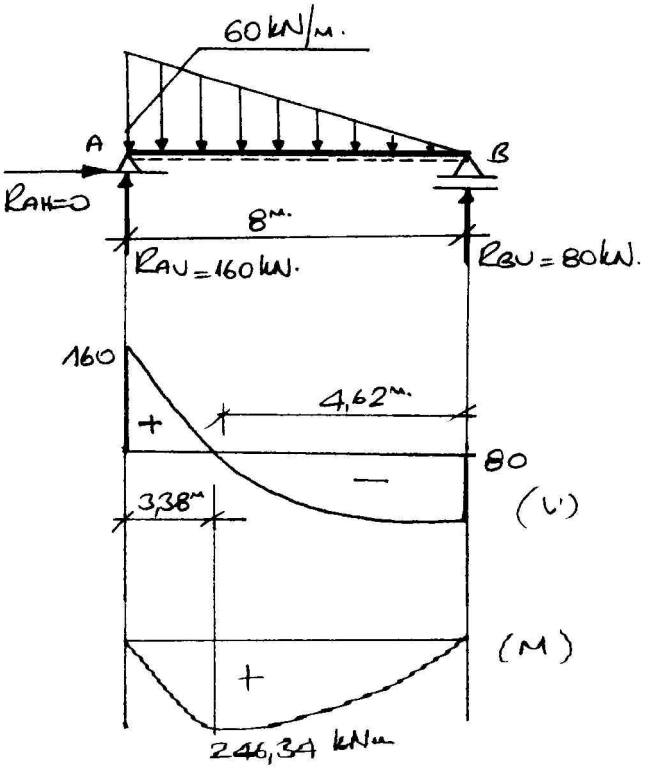
Mesnet Tepkileri

$\Sigma M_B=0 \rightarrow R_{AV}=0$

$\Sigma M_A=0 \rightarrow R_{BV}=80\ kN$

$\Sigma X=0 \rightarrow R_{AH}=0$

ÖRNEK-7



$\Sigma X=0 \rightarrow R_{AH}=0$

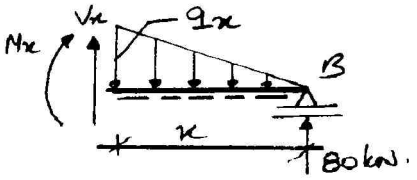
$\Sigma M_B=0 \rightarrow R_{AV}=\dfrac{\frac{60.8}{2}\cdot\frac{2}{3}\cdot 8}{8}=160\ kN.$

$\Sigma M_A=0 \rightarrow R_{BV}=\dfrac{\frac{60.8}{2}\cdot\frac{1}{3}\cdot 8}{8}=80\ kN.$

$\Sigma Y=0$ (Kontrol) $\rightarrow \dfrac{60.8}{2}=160+80$

$240=240$ ✓

Kesit Tesirleri



$\dfrac{60}{8}=\dfrac{9x}{x}$

$9x=7,5x$

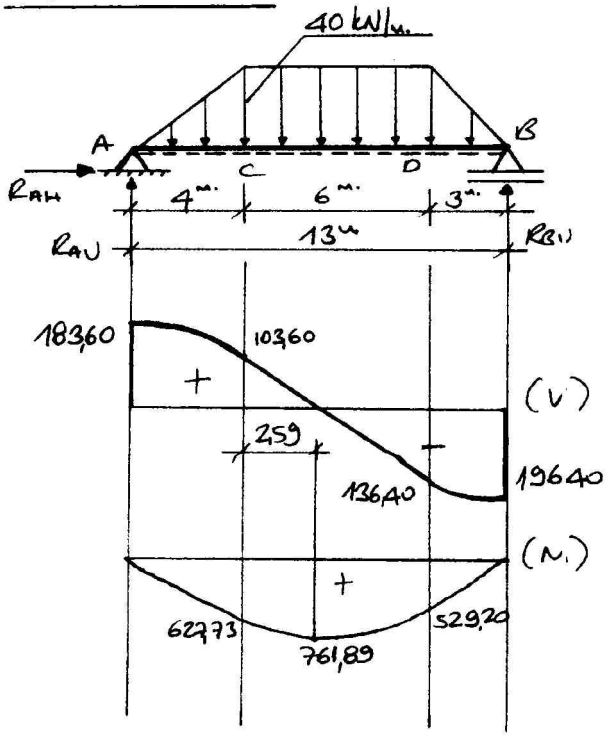
$V_x=-80+9x\cdot\dfrac{x}{2}=-80+3,75x^2$

$M_x=80.x-9x\cdot\dfrac{x}{2}\cdot\dfrac{x}{3}=80x-1,25x^3$

$max M \rightarrow V_x=0 \Rightarrow -80+3,75x^2 \Rightarrow x_0=4,62m$

$max M=80.4,62-1,25.(4,62)^3=246,34\ kNm.$

ÖRNEK-8



$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma M_B=0 \rightarrow R_{AV}=\dfrac{\frac{40.4}{2}\left(\frac{4}{3}+9\right)+40.6.6+\frac{40.3}{2}\cdot\frac{2}{3}\cdot 3}{13}=$

$R_{AV}=183,60\ kN.$

$\Sigma M_A=0 \rightarrow R_{BV}=\dfrac{\frac{40.4}{2}\cdot\frac{2}{3}\cdot 4+40.6.7+\frac{40.3}{2}\left(\frac{1}{3}\cdot 3+10\right)}{13}=$

$R_{BV}=196,40\ kN.$

$\Sigma Y=0 \rightarrow \dfrac{40.4}{2}+40.6+\dfrac{40.3}{2}=183,60+196,40$

$380=380$ ✓

Kesit Tesirleri

$V_{AC}=183,60\ kN,\ V_{CA}=183,60-\dfrac{40.4}{2}=103,60\ kN.$

$V_{OC}=V_{OB}=103,60-40.6=-136,40\ kN.$

$V_{BO}=-196,40\ kN.$

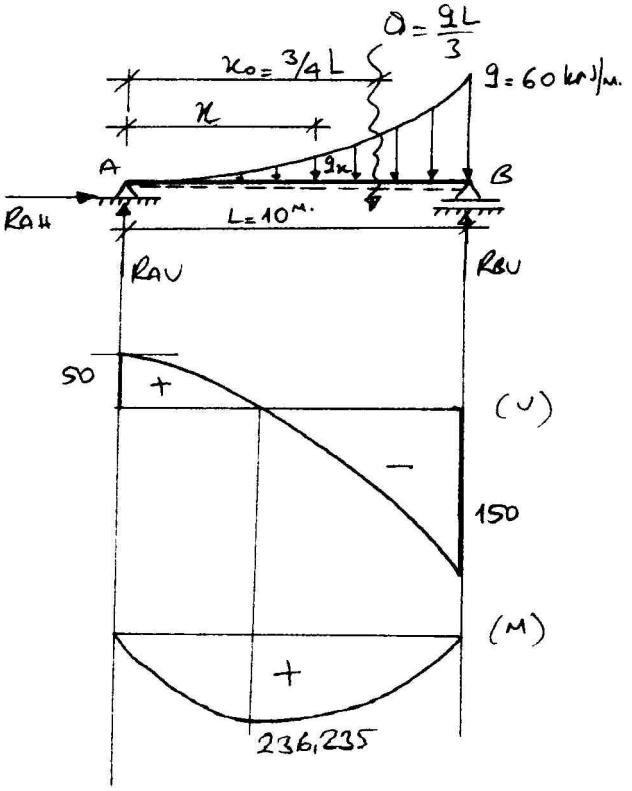
$M_C=183,60.4-\dfrac{40.4}{2}\cdot\dfrac{4}{3}=627,73\ kNm$

$M_D=196,40.3-\dfrac{40.3}{2}\cdot\dfrac{3}{3}=529,20\ kNm.$

$x_0=\dfrac{V_{CO}}{g}=\dfrac{103,60}{40}=2,59m.$

$max M=\dfrac{(103,6)^2}{2.40}+627,73=761,89\ kNm.$

ÖRNEK-9)



Mesnet Reaksiyonları

$\Sigma X=0 \rightarrow R_{AH}=0$

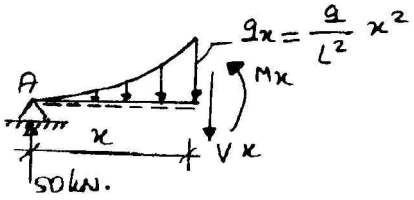
$\Sigma M_B=0 \rightarrow R_{AV}=\dfrac{9.L/3.\ 1/4L}{L}=\dfrac{9L}{12}=\dfrac{60.10}{12}=50\,kN$

$\Sigma M_A=0 \rightarrow R_{BV}=\dfrac{9.L/3.\ 3/4L}{L}=\dfrac{9L}{4}=\dfrac{60.10}{4}=150\,kN$

$\Sigma Y=0 \rightarrow \dfrac{9L}{3}=\dfrac{9L}{12}+\dfrac{9L}{4} \Rightarrow \dfrac{9L}{3}=\dfrac{9L}{3}$ ✓

Kesit Tesirleri

$V_{AB}=50\,kN,\ V_{BA}=-150\,kN.$



$$V_x=50-\int_0^x q_x\,dx=50-\int_0^x \frac{q}{L^2}x^2\,dx=50-\left|\frac{q}{L^2}\frac{x^3}{3}\right|_0^x=50-\frac{q}{L^2}\frac{x^3}{3}$$

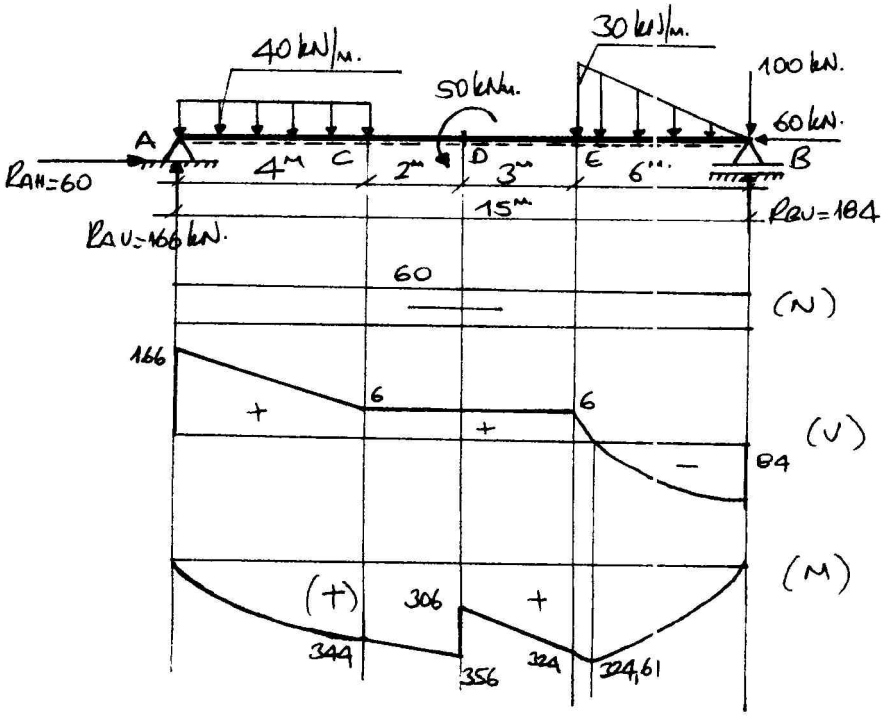
$$\max M \rightarrow V_x=0 \Rightarrow 50-\frac{60}{100}\cdot\frac{x^3}{3}=0 \Rightarrow x_0=\sqrt[3]{\frac{50.300}{60}}$$

$$\int\frac{dM}{dx}=\int V \rightarrow \int dM=\int V\,dx$$

$x_0 \cong 6{,}30$ m.

$$M=\int 50\,dx-\int_0^x \frac{q}{L^2}\frac{x^3}{3} \Rightarrow \max M=50.x-\int_0^{6{,}30}\frac{q}{L^2}\frac{x^3}{3}\,dx=50x-\left|\frac{q}{L^2 3.4}x^4\right|_0^{6{,}30}=236{,}235\ kNm$$

ÖRNEK-10)



Mesnet Tepkileri

$\Sigma X=0 \rightarrow R_{AH}=60\,kN.$

$\Sigma M_B=0 \rightarrow R_{AV}=\dfrac{40.4.13+50+\frac{30.6}{2}.\frac{2}{3}.6}{15}$

$R_{AV}=166\,kN.$

$\Sigma M_A=0 \rightarrow R_{BV}=\dfrac{40.4.2-50+\frac{30.6}{2}\left(\frac{6}{3}+9\right)}{15}+$

$+100=$

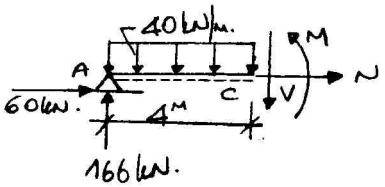
$R_{BV}=184\,kN.$

$\Sigma Y=0 \rightarrow 166+184=40.4+30.\frac{6}{2}+100$

$350=350$ ✓

Kesit Tesirleri

A-C arası

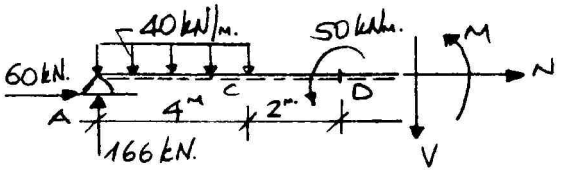


$N_{AC} = N_{CA} = -60\ kN.$

$V_{AC} = 166\ kN,\ V_{CA} = 166 - 40.4 = 6\ kN.$

$M_C = 166.4 - 40.4.2 = 344\ kNm.$

C-D-E arası



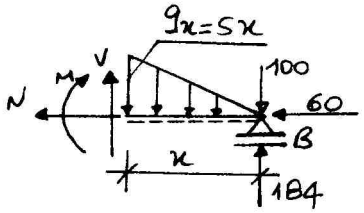
$N_{CD} = N_{DC} = N_{DE} = N_{ED} = -60\ kN.$

$V_{CD} = V_{DC} = V_{DE} = V_{ED} = 166 - 40.4 = 6\ kN.$

$M_{DC} = 166.6 - 40.4.4 = 356\ kNm,\ M_{DE} = 356 - 50 = 306\ kNm$

$M_E = 166 * 9 - 40.4.7 - 50 = 324\ kNm.$

E-B arası



$V_{BE} = -18c. + 100 = -84\ kN.$

$V_{EB} = -8c. + \frac{30.6}{2} = 6\ kN.$

$V_x = -84 + 5x.\frac{x}{2} = -84 + 2{,}5x^2$

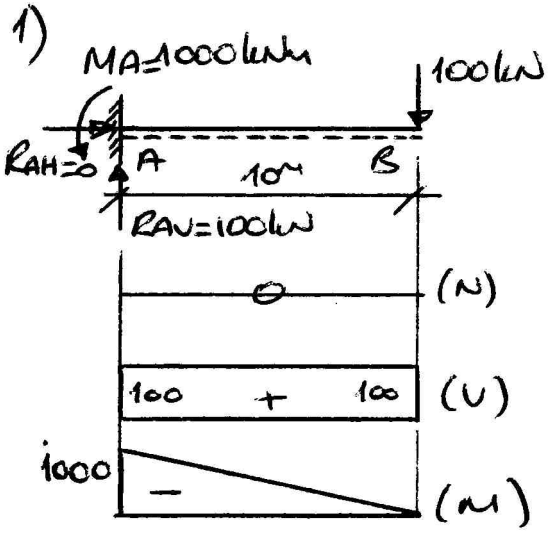
$M_x = (184 - 100).x - 5x.\frac{x}{2}.\frac{x}{3} = 84x - \frac{2{,}5}{3}x^3$

$\max M \longrightarrow V_x = 0 \longrightarrow x_0 = \sqrt{\frac{84}{2{,}5}} = 5{,}80\ m.$

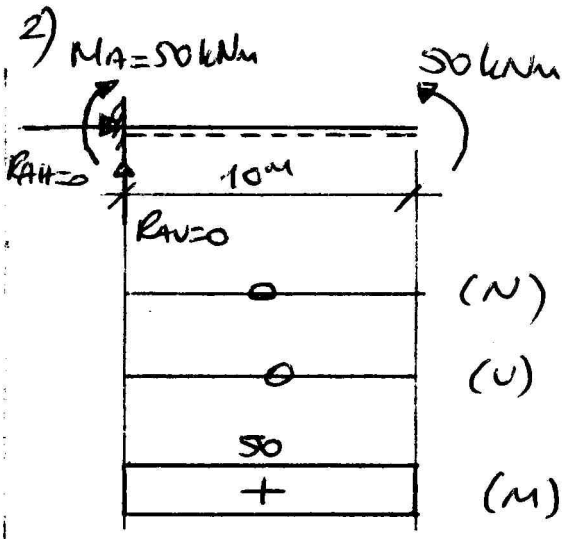
$\max M = 84 * 5{,}80 - \frac{2{,}5}{3} * 5{,}80^3 = 324{,}61\ kNm.$

IV - 2) KONSOL KİRİŞLER

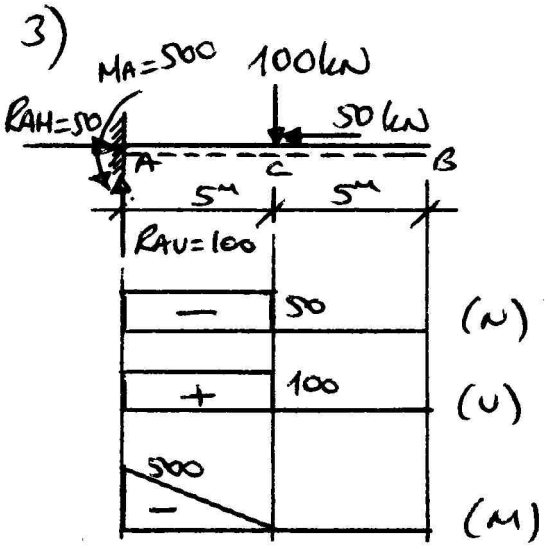
1)



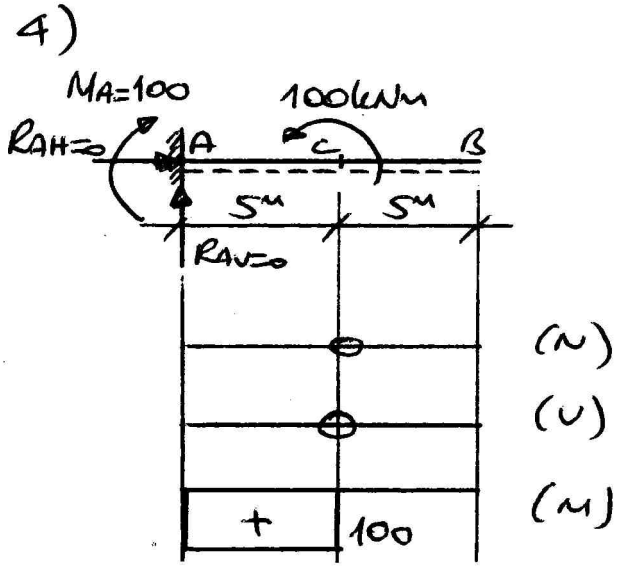
2)



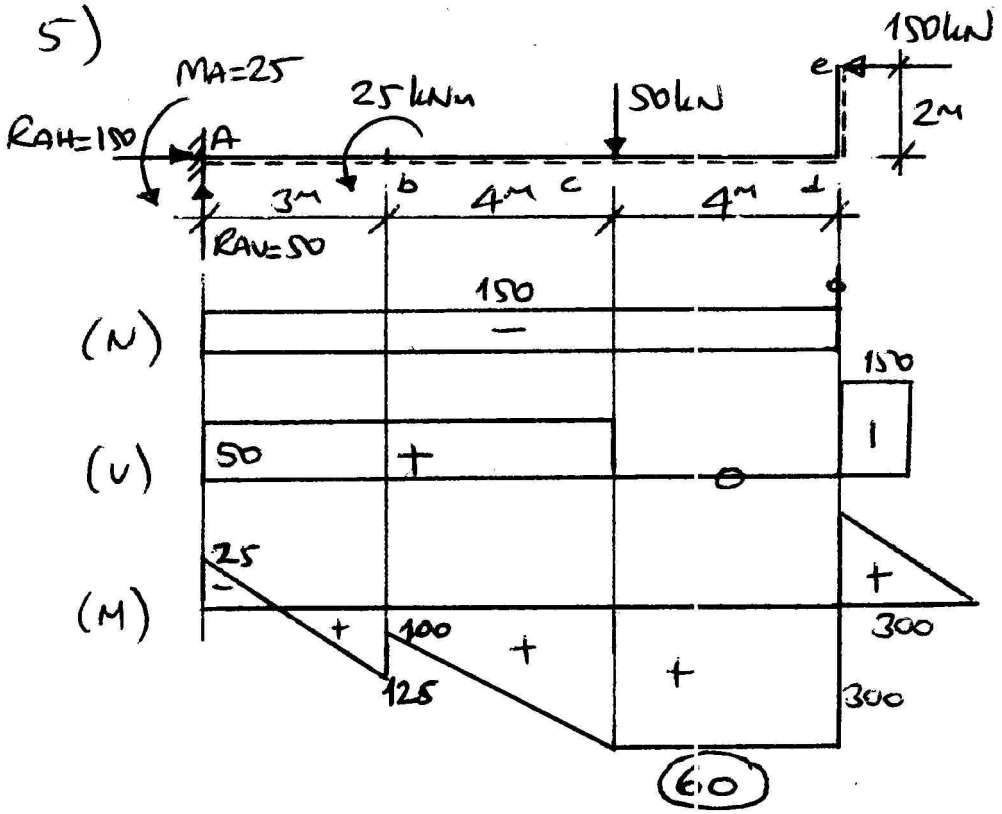
3)



4)



5)



Kesit Tesirleri

$N_{Ad} = N_{dA} = -150\ kN$

$V_{AC} = V_{CA} = 50\ kN$

$V_{cd} = V_{dc} = 0$

$N_{de} = N_{ed} = 0$

$V_{de} = V_{ed} = -150\ kN$

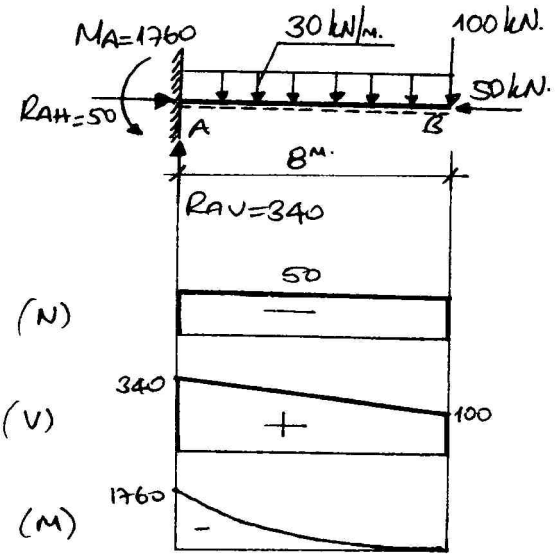
$M_{de} = +150 \times 2 = 300\ kNm$

$M_{Ab} = -25\ kNm$

$M_{bA} = 50 \times 3 - 25 = 125\ kNm$

$M_{bc} = 125 - 25 = 100\ kNm$

6)



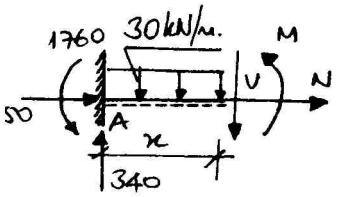
Mesnet Tepkileri

$\Sigma X=0 \rightarrow R_{AH}=50\ kN$

$\Sigma Y=0 \rightarrow R_{AV}=30.8+100=340\ kN.$

$\Sigma M_A=0 \rightarrow M_A=+30.8.4+100.8=1760\ kNm$

Kesit Tesirleri



$N_{AB}=N_{BA}=-50\ kN.$

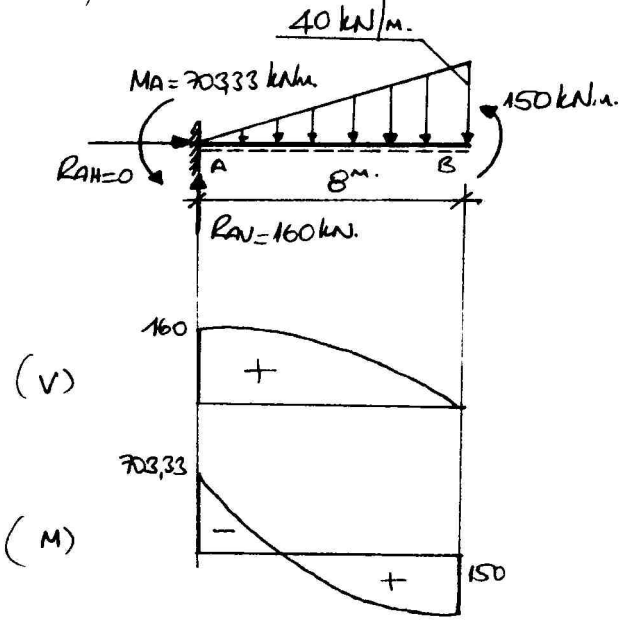
$V_x=340-30.x$

$M_x=340.x-30.x.\frac{x}{2}-1760$

$x=0 \Rightarrow \begin{cases} V_{AB}=340\ kN. \\ M_A=-1760\ kNm. \end{cases}$

$x=8^m \Rightarrow \begin{cases} V_{BA}=100\ kN. \\ M_B=0 \end{cases}$

7)



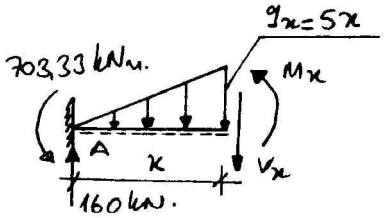
Mesnet Tepkileri

$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma Y=0 \rightarrow R_{AV}=\frac{40.8}{2}=160\ kN.$

$\Sigma M_A=0 \rightarrow M_A=\frac{40.8}{2}.\frac{2}{3}.8-150=703,33\ kNm.$

Kesit Tesirleri



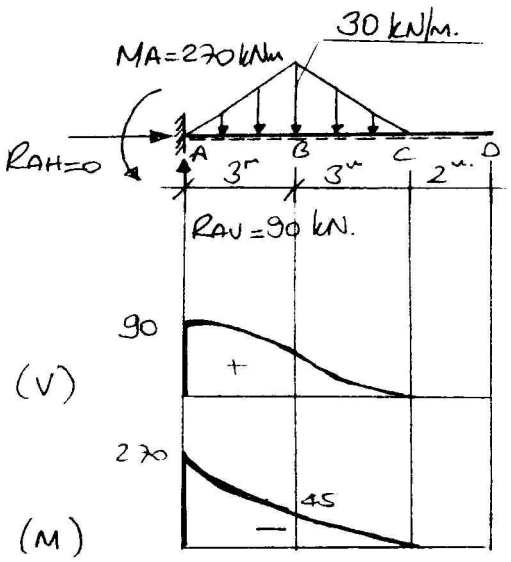
$V_x=160-5x.\frac{x}{2}=160-2,5x^2$

$M_x=-703,33+160.x-5x.\frac{x}{2}.\frac{x}{3}=-703,33+160x-\frac{5}{6}x^3$

$x=0 \rightarrow \begin{cases} V_{AB}=160\ kN. \\ M_A=-703,33\ kNm. \end{cases}$

$x=8^m \rightarrow \begin{cases} V_{BA}=160-2,5.(8)^2=0 \\ M_B=-703,33+160.8-\frac{5}{6}.(8)^3=150\ kNm \end{cases}$

8)



Mesnet Tepkileri

$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma Y=0 \rightarrow R_{AV}=\frac{30.6}{2}=90\ kN.$

$\Sigma M_A=0 \rightarrow M_A=\frac{30.6}{2}.3=270\ kNm.$

Kesit Tesirleri

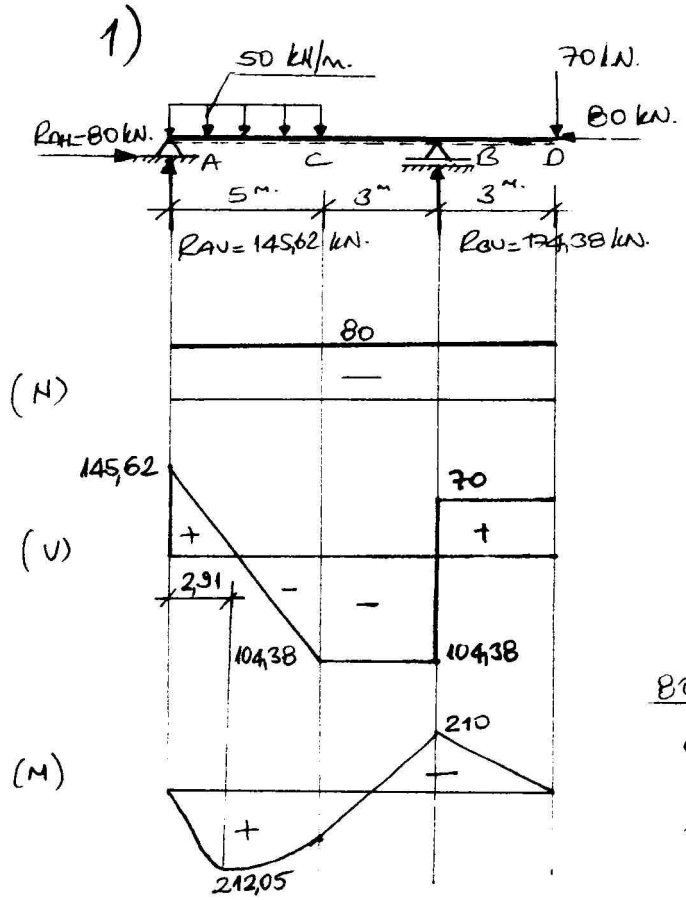
$V_{AB}=90\ kN,\ V_{BA}-V_{BC}=90-\frac{30.3}{2}=45\ kN.$

$M_B=-270+90.3-\frac{30.3}{2}.\frac{1}{3}.3=-45\ kNm.$

IV_3) GIKMALI KIRIŞLER

* Gikmali kirişlerin çözümünde sistem bütün olarak da, parçalara ayrılarak da çözülebilir. Sistem parçalara ayrılarak çözülürse, konsollar ayrı olarak ele alınır ve burada bulunan reaksiyonlar yük olarak kirişe yüklenir. Aşağıdaki örneklerde sistem bütün olarak ele alınmıştır.

1)



Mesnet Reaksiyonları

$\Sigma X=0 \rightarrow R_{AH}=80\ kN.$

$\Sigma M_A=0 \rightarrow R_{BV}=\frac{50.5.2,5+70.11}{8}=174,38$

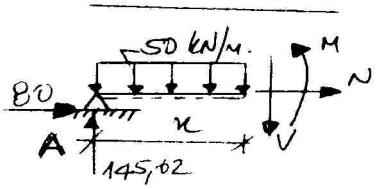
$\Sigma M_B=0 \rightarrow R_{AV}=\frac{50.5.5,5-70.3}{8}=145,62$

$\Sigma Y=0 \rightarrow 145,62+174,38=50.5+70$

$320=320$ ✓

Kesit Tesirleri

A-C arası



$N_{AC}=N_{CA}=-80\ kN.$

$V_{AC}=145,62\ kN.$

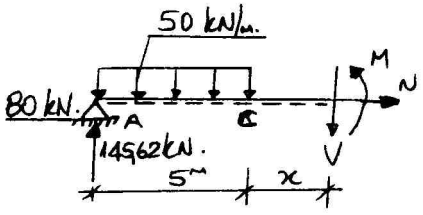
$V_{CA}=145,62-50.5=$

$=-104,38\ kN.$

$x_0=\frac{145,62}{50}=2,91\ m.$

$max M=\frac{145,62^2}{2.50}=212,05\ kNm.$

C-B arası

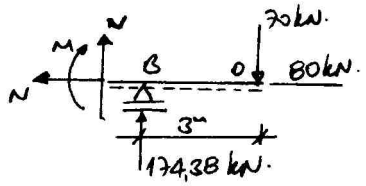


$N_{BC} = N_{CB} = -80\ kN.$

$V_{BC} = V_{CB} = 145,62 - 50.5 = -104,38\ kN.$

$M_C = 145,62.5 - 50.5.2,5 = 103,10\ kNm.$

B-D arası

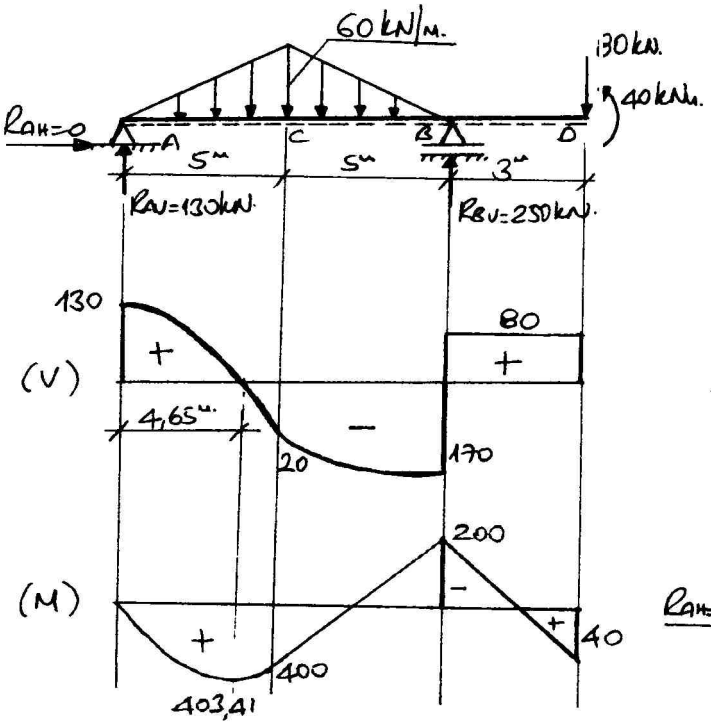


$N_{BD} = N_{DB} = -80\ kN.$

$V_{DB} = 70\ kN.$, $V_{BD} = 70\ kN.$

$V_{BC} = 70 - 174,38 = -104,38\ kN.$

2)



Mesnet Tepkileri

$\Sigma X = 0 \rightarrow R_{AH} = 0$

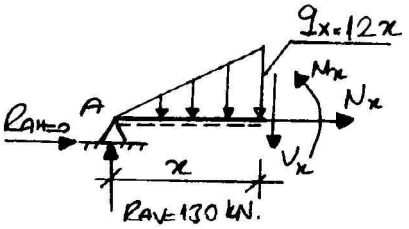
$\Sigma M_B = 0 \rightarrow R_{AV} = \dfrac{\frac{60.10}{2}.5 - 80.3 + 40}{10} = 130\ kN.$

$\Sigma M_A = 0 \rightarrow R_{BV} = \dfrac{\frac{60.10}{2}.5 + 80.13 - 40}{10} = 250\ kN.$

$\Sigma Y = 0 \rightarrow 130 + 250 = \dfrac{60.10}{2} + 80 \Rightarrow 380 = 380$ ✓

Kesit Tesirleri

A-C arası



$N_{AD} = N_{DA} = 0$

$V_x = 130 - 12x.\frac{x}{2} = 130 - 6x^2$

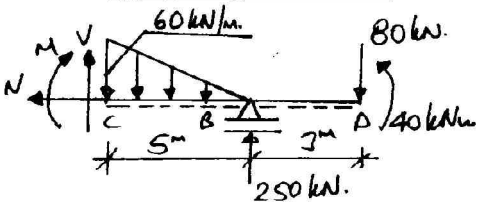
$M_x = 130x - 12x.\frac{x}{2}.\frac{x}{3} = 130x - 2x^3$

$x=0 \begin{cases} V_{AC} = 130\ kN. \\ M_A = 0 \end{cases}$ $x=5^m \begin{cases} V_{CA} = V_{CB} = -20\ kN. \\ M_C = 400\ kNm. \end{cases}$

$max M \rightarrow V_x = 0 \Rightarrow x_0 = \sqrt{\frac{130}{6}} = 4,65^m$

$max M = 130.4,65 - 2.(4,65)^3 = 403,41\ kNm.$

C-B arası



$V_{CB} = V_{BC} = 80 - 250 + \dfrac{60.5}{2} = -20\ kN.$

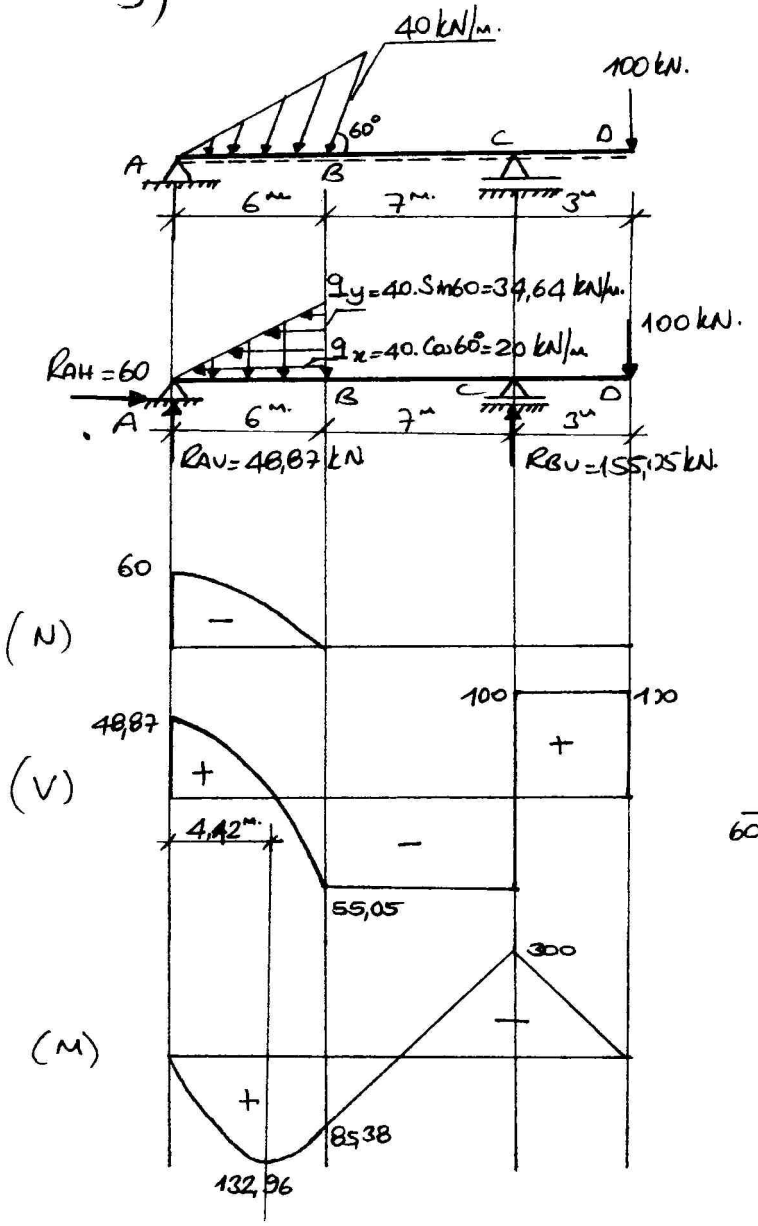
$M_C = 40 - 80.8 + 250.5 - \dfrac{60.5}{2}.\dfrac{5}{3} = 400\ kNm.$

B-D arası

$V_{BD} = V_{DB} = 80\ kN.$

$M_B = -80.3 + 40 = -200\ kNm.$

3)



Mesnet Tepkileri

$\Sigma X=0 \rightarrow R_{AH}=\frac{20.6}{2}=60\ kN.$

$\Sigma M_B=0 \rightarrow R_{AV}=\frac{34,64.6/2(6/3+7)-100.3}{13}=48,87$

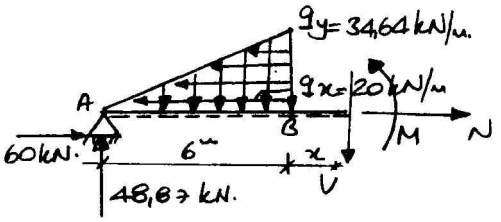
$\Sigma M_A=0 \rightarrow R_{BV}=\frac{34,64.6/2.(2/3.6)+100.16}{13}=155,05$

$\Sigma Y=0 \rightarrow 34,64.6/2+100=48,87+155,05$

$203,92=203,92$ ✓

Kesit Tesirleri

A-B arası



$q_{y_x}=\frac{34,64}{6}x = 5,77x$

$N_{AB}=N_{BA}=$, $N_{BA}=N_{BC}=0$

$V_{AB}=48,87\ kN$, $V_{BA}=48,87-\frac{34,64.6}{2}=-55,05\ kN$

$V_x=48,87-5,77x.\frac{x}{2}$

$M_x=48,87x-5,77x.\frac{x}{2}.\frac{x}{3}$

$max M \rightarrow V_x=0 \Rightarrow x_0=\sqrt{\frac{48,87.2}{5,77}}=4,42\ m.$

$max M=48,87.4,42-5,77.(4,42)^3.1/6$

$=132,96\ kNm.$

$M_B=48,87.6-\frac{34,64.6}{2}.\frac{6}{3}=85,38\ kNm.$

B-C arası

$N_{BC}=N_{CB}=0$

$V_{BC}=V_{CB}=48,87-\frac{34,64.6}{2}=-55,05\ kN.$

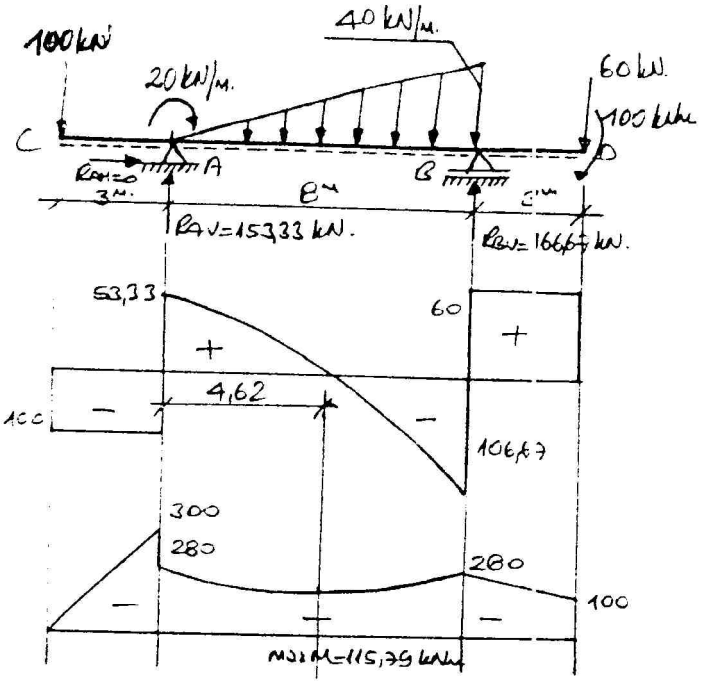
$M_C=-100.3=-300\ kNm.$

C-D arası

$N_{CD}=N_{DC}=0$

$V_{CD}=V_{DC}=100\ kN.$

4)



Mesnet Tepkileri

$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma M_B=0 \rightarrow R_{AV}=\dfrac{100.11-20+\frac{40.8}{2}.\frac{1}{3}.8-60.3-100}{8}$

$R_{AV}=153,33\ kN.$

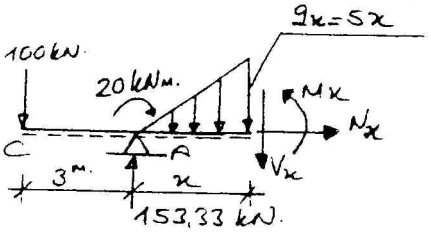
$\Sigma M_A=0 \rightarrow R_{BV}=\dfrac{-100.3+20+\frac{40.8}{2}.\frac{2}{3}.8+60.11+100}{8}$

$R_{BV}=166,67\ kN.$

$\Sigma Y=0 \rightarrow 153,33+166,67=100+\frac{40.8}{2}+60$

$320=320$ ✓

Kesit Tesirleri



C-A arası / A-B arası

$V_{CA}=V_{AC}=-100\ kN.$

$V_x=-100+153,33-\dfrac{5x.x}{2}$

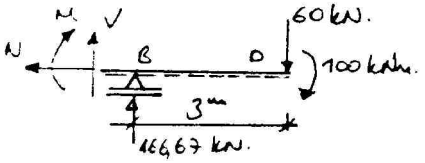
$M_x=-100(3+x)+20+153,33.x-\dfrac{5x.x}{2}.\dfrac{x}{3}$

$x=0 \Rightarrow \begin{cases} V_{AB}=53,33\ kN. \\ M_A=-300+20=-280\ kNm. \end{cases}$

$x=8^m \Rightarrow \begin{cases} V_{BA}=-100+153,33-25.(8)^2=-106,67\ kN. \\ M_B=-100.11+20+153,33.8-25.\frac{(8)^3}{3}=-280\ kNm. \end{cases}$

$max M \rightarrow V_x=0 \Rightarrow -100+153,33-25x^2=0 \rightarrow x_0=4,62^m \Rightarrow max M=-115,79\ kNm$

B-O arası

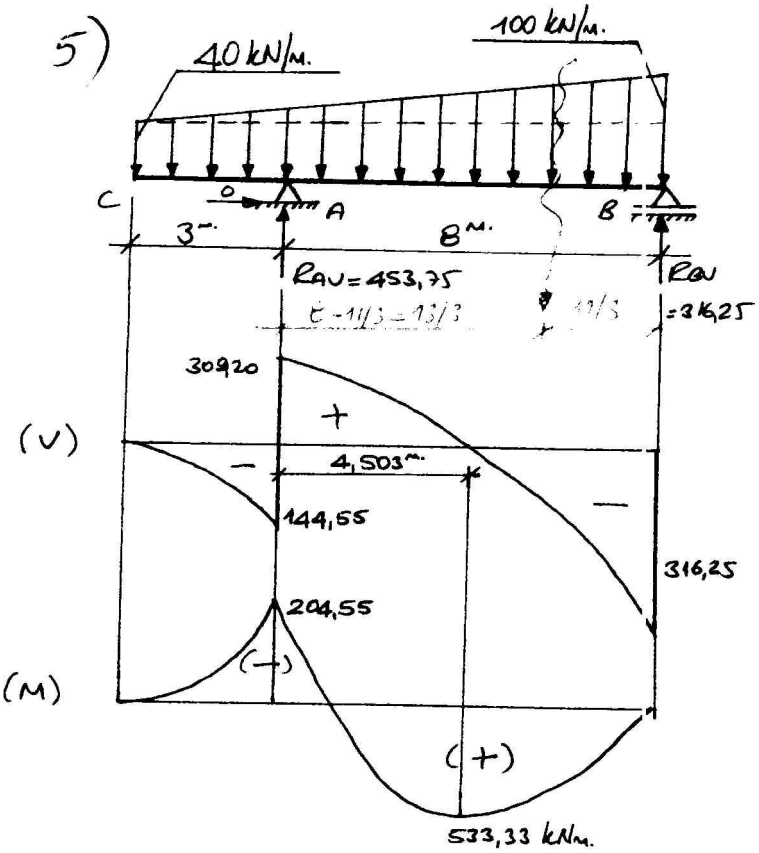


$V_{BO}=V_{OB}=60\ kN.$

$M_O=-100\ kNm,\ M_B=-60.3-100=-180\ kNm$

$V_{BA}=60-166,67=-106,67\ kN.$

5)



Mesnet Tepkileri

$$R_{AV} = \frac{40\cdot 11\cdot 5{,}5 + \frac{60\cdot 11}{2}\cdot\frac{11}{3}}{8} = 453{,}75 \text{ kN.}$$

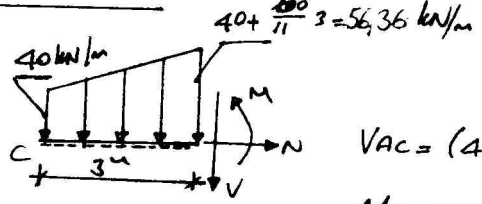
$$R_{BV} = \frac{40\cdot 11\cdot 2{,}5 + 60\cdot 11/2\cdot 13/3}{8} = 316{,}25 \text{ kN}$$

$$\Sigma Y = 0 \rightarrow 453{,}75 + 316{,}25 = \frac{40+100}{2}\cdot 11$$

$$770 = 770 \checkmark$$

Kesit Tesirleri

C-A arası



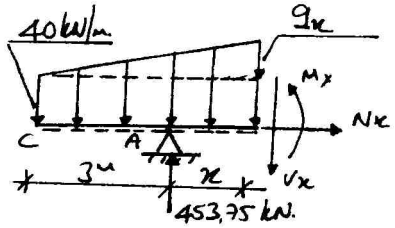
$N_{CA} = N_{AC} = 0$

$V_{CA} = 0,$

$$V_{AC} = (40+56{,}36)\cdot\frac{1}{2}\cdot 3 = 144{,}55 \text{ kN.}$$

$$M_A = -40\cdot 3\cdot 1{,}5 - \frac{16{,}36\cdot 3}{2}\cdot\frac{1}{3}\cdot 3 = -204{,}55 \text{ kNm.}$$

A-B arası



$$\frac{60}{11} = \frac{9_x}{3+x} \Rightarrow 9_x = 16{,}364 + 5{,}455x$$

$$V_x = -40(3+x) - 9_{(x)}\cdot(3+x)\cdot\frac{1}{2} + 453{,}75$$

$$V_x = -120 - 40x - (16{,}364 + 5{,}455x)(3+x)\frac{1}{2} + 453{,}75$$

$$V_x = 453{,}75 - 144{,}55 - 56{,}73x - 2{,}73x^2$$

$$M_x = 453{,}75x - \frac{40(3+x)^2}{2} - \frac{9_x\cdot(3+x)}{2}\cdot\frac{(3+x)}{3}$$

$$M_x = -204{,}55 + 309{,}20x - 28{,}182x^2 - 0{,}909x^3$$

$x=0$ { $V_{AB} = 309{,}20$ kN. ; $M_A = -204{,}55$ kNm.

$x=8^m$ { $V_{BA} = -316{,}25$ kN. ; $M_B = 0$

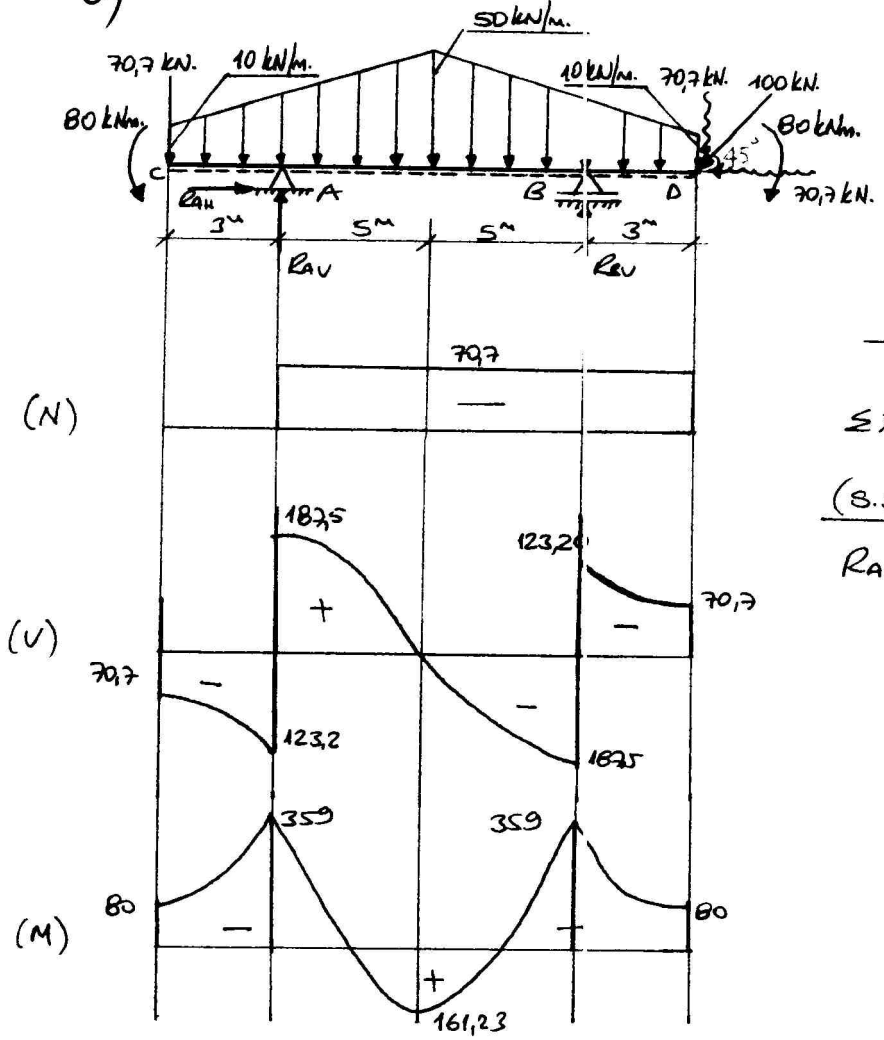
$\max M \rightarrow V_x = 0 \Rightarrow 309{,}20 - 56{,}37x - 2{,}73x^2 =$

$$x_{1,2} = \frac{56{,}37 \mp \sqrt{56{,}37^2 + 4\cdot 2{,}73\cdot 309{,}2}}{-2\cdot 2{,}73}$$

$x_1 = 4{,}503^m$, $x_2 = -25{,}15$

$\max M = 533{,}33$ kNm

6)



Mesnet Tepkileri

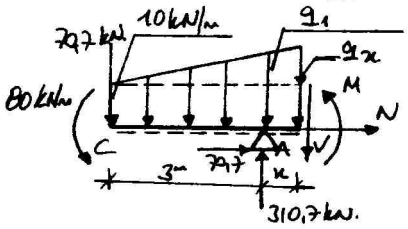
$\Sigma X=0 \rightarrow R_{AH}=70,7$ kN.

(S.S+Y.S)

$$R_{AV}=R_{BV}=70,7+\frac{10.16+\frac{40.16}{2}}{2}=310,7 \text{ kN.}$$

Kesit Tesirleri

A-C arası



$q_1=10+\frac{40}{8}\cdot 3=25$ kN/m.

$N_{CA}=N_{AC}=0$, $V_{CA}=-70,7$ kN., $V_{AC}=-70,7+\frac{(10+25)}{2}\times 3=-123,20$ kN

$V_{AB}=-123,20+310,7=187,5$ kN.

$M_A=-80-70,7.3-10.3.1,5-\frac{15.3}{2}.1=-359$ kNm.

A-B arası

$q_x=\frac{40}{8}(3+x)=15+5x$, $V_x=-70,7+310,7-10(3+x)-\underbrace{q_x}_{(15+5x)}\frac{(3+x)}{2}$

$x=0\begin{cases}V_{AB}=187,5\text{ kN.}\\ M_A=-359\text{ kNm.}\end{cases}$

$=-70,7+310,7-30-10x-22,5-7,5x-7,5x-2,5x^2$

$=187,5-25x-2,5x^2$

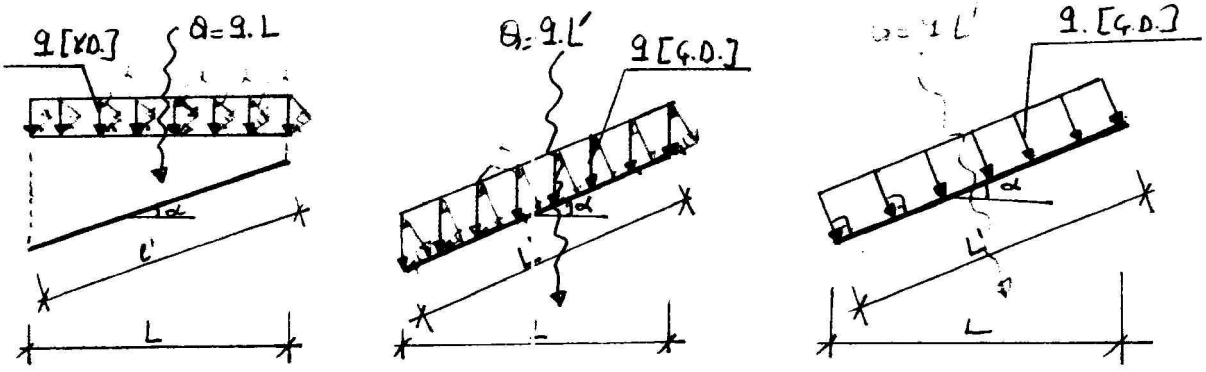
$x=10^-\begin{cases}V_{BA}=-187,5\text{ kN.}\\ M_B=-359\text{ kNm.}\end{cases}$

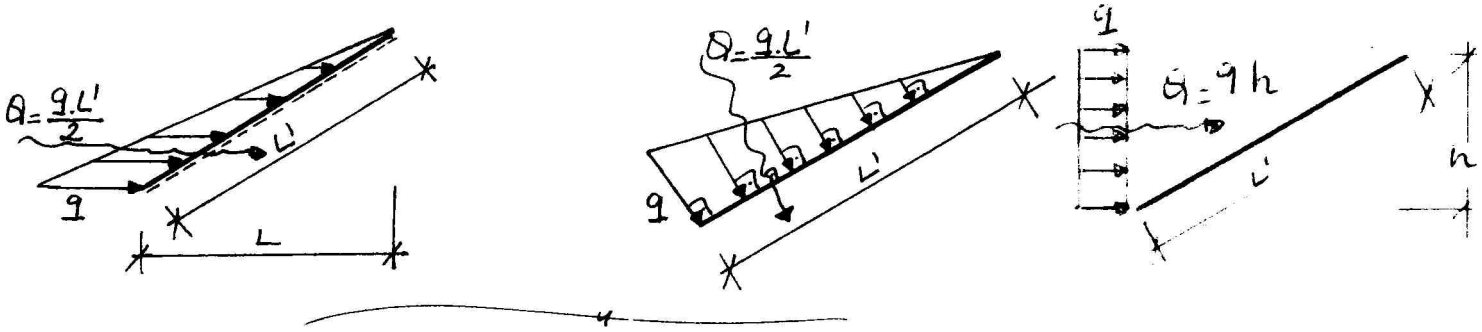
$M_x=-80-70,7(3+x)-10\frac{(3+x)^2}{2}-q_x\frac{(3+x)}{2}.\frac{(3+x)}{3}+310,7.x$

$\max M \rightarrow V_x=0 \Rightarrow x_0=5^m \Rightarrow \max M=161,23$ kNm.

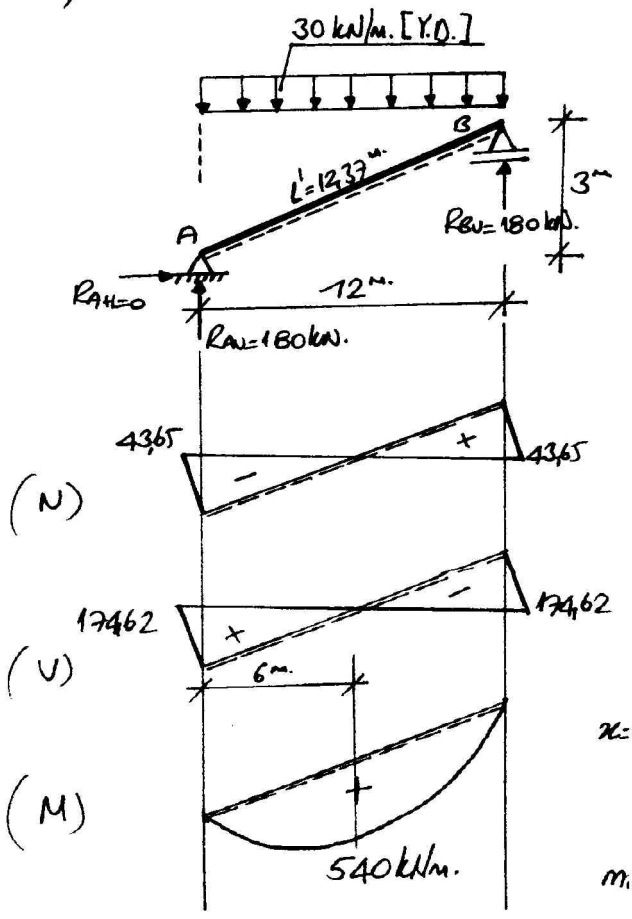
(M) K

IV - 4) EĞİK KİRİŞLER





1)



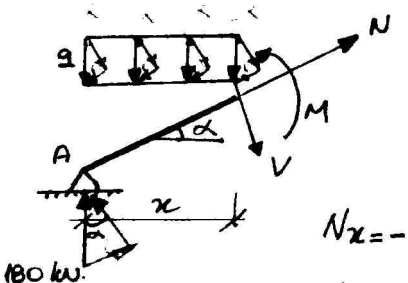
Mesnet Tepkileri

$\Sigma X = 0 \Rightarrow R_{AH} = 0$

$\Sigma M_B = 0 \rightarrow R_{AV} = \frac{30.12.6}{12} = 180\ kN$, $R_{BV} = 180\ kN.$

$\Sigma Y = 0 \rightarrow$ (Kontrol) : $180 + 180 = 30.12 \Rightarrow 360 = 360$ ✓

Kesit Tesirleri



$\sin\alpha = 3/12{,}37$

$\cos\alpha = 12/12{,}37$

$N_x = -180.\sin\alpha + 30.x.\sin\alpha$

$V_x = 180.\cos\alpha - 30.x.\cos\alpha$

$M_x = 180x - 30.x.\frac{x}{2} = 180x - 15x^2$

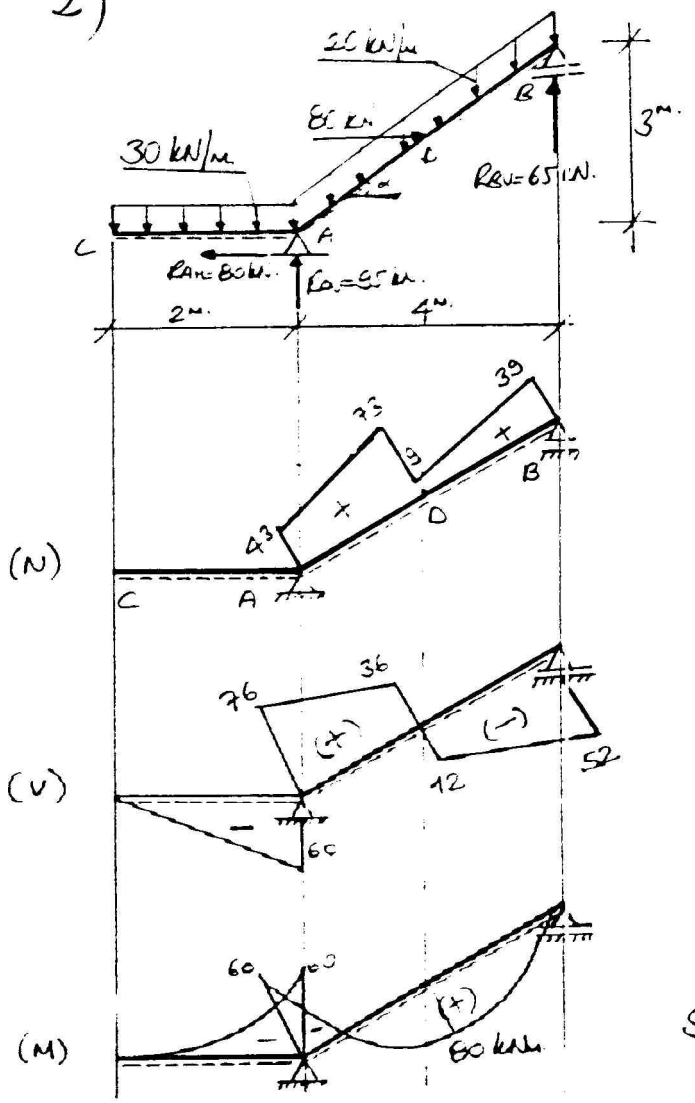
$x = 0 \Rightarrow \begin{cases} N_{AB} = -43{,}65\ kN. \\ V_{AB} = 17{,}462\ kN. \end{cases}$ $\quad x = 12^m \begin{cases} N_{BA} = 43{,}65\ kN. \\ V_{BA} = -17{,}462\ kN. \end{cases}$

$max M \rightarrow V_x = 0 \Rightarrow x_0 = \frac{180.\cos\alpha}{30.\cos\alpha} = 6^m.$

$max M = 540\ kNm.$

veya $max M = \frac{30.12^2}{8} = 540\ kNm$

2)



Mesnet Tepkileri

$\Sigma X=0 \rightarrow R_{AH}=80\,kN.$

$\Sigma M_B=0 \rightarrow R_{AV}=\dfrac{30.2.5-80.3+80.1.5+20.5.2}{4}$

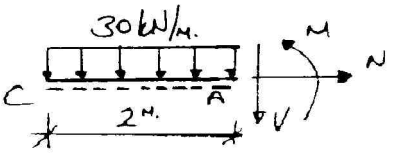
$R_{AV}=95\,kN.$

$\Sigma M_A=0 \rightarrow R_{BV}=\dfrac{-30.2.1+80.1.5+20.5.2}{4}$

$R_{BV}=65\,kN.$

$\Sigma Y=0 \rightarrow 95+65=30.2+20.5 \Rightarrow 160=160$ ✓

Kesit Tesirleri

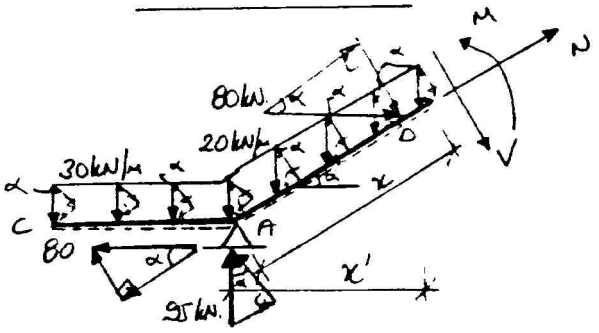


$V_{AC}=-30.2=-60\,kN$, $V_{CA}=0$

$N_{CA}=N_{AC}=0$, $M_A=-30.2.1=-60\,kNm.$

$\sin\alpha=3/5$, $\cos\alpha=4/5$

A-B aralığı



* A-D aralığı

$N_x=30.\sin\alpha.2+80.\cos\alpha-95.\sin\alpha+20.x.\sin\alpha=43+12x$

$V_x=-30.\cos\alpha.2+80.\sin\alpha+95.\cos\alpha-20.x.\cos\alpha=76-16x$

$x=0\begin{cases}N_{AD}=43{,}00\,kN.\\V_{AD}=76\,kN.\end{cases}$ $x=2{,}5^m\begin{cases}N_{DA}=73\,kN.\\V_{DA}=36\,kN.\end{cases}$

* D-B aralığı

$N_x=30.\sin\alpha.2+80.\cos\alpha-95.\sin\alpha+20.x.\sin\alpha-80.\cos\alpha=-21+12x$

$V_x=-30.\cos\alpha.2+80\sin\alpha+95.\cos\alpha-20.x.\cos\alpha-80.\sin\alpha=28-16x$

$x=2{,}5^m\begin{cases}N_{DB}=9\,kN.\\V_{DB}=-12\,kN.\end{cases}$ $x=5^m\begin{cases}N_{BD}=39\,kN.\\V_{BD}=-52\,kN.\end{cases}$

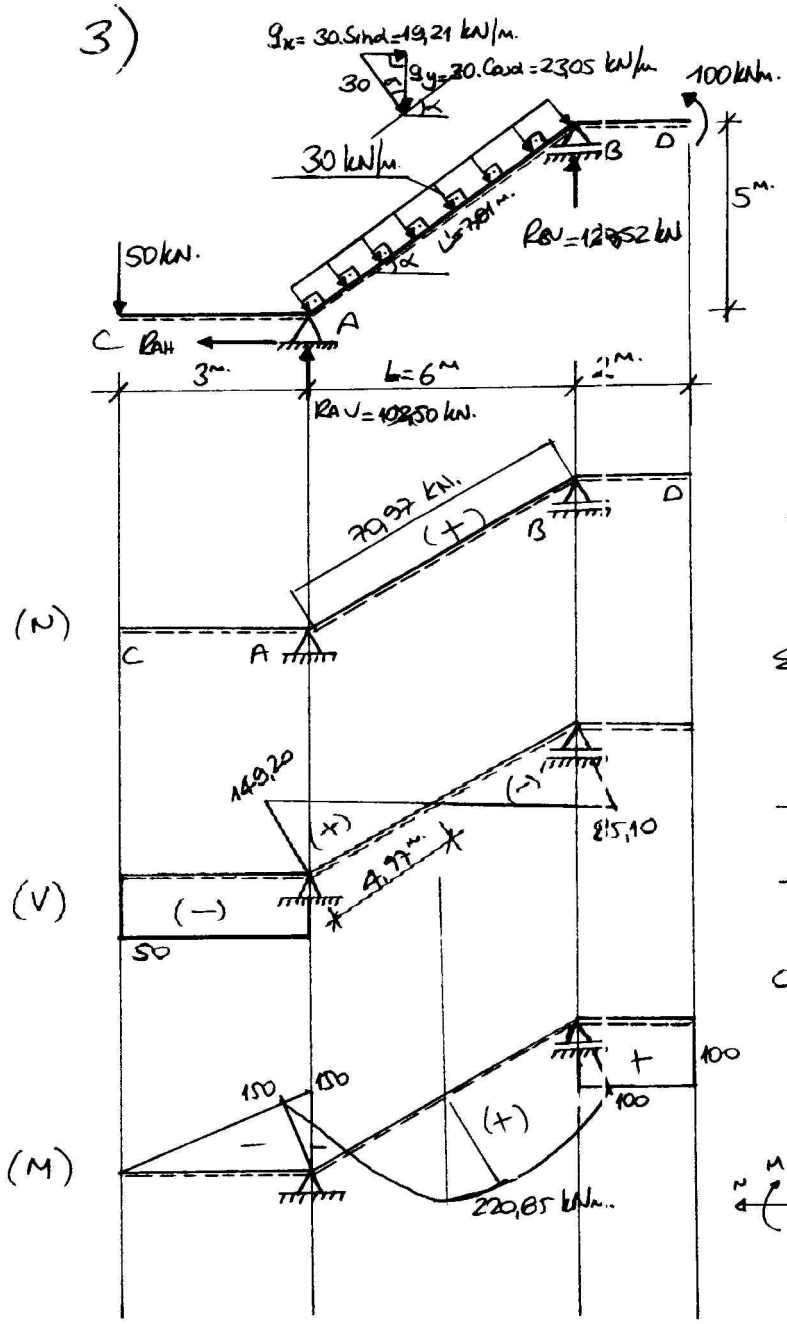
maxM D noktasında;

$\max M=-30.2.3+80.1.5+95.2-20.2{,}5.1$

$=80\,kNm.$

$M_A=-30.2.1=-60\,kNm.$

3)



$\sin\alpha = 5/7{,}81$, $\cos\alpha = 6/7{,}81$

Mesnet Tepkileri

$\Sigma X = 0 \rightarrow R_{AH} = 19{,}21 \cdot 7{,}81 = 150{,}03$ kN.

$\Sigma M_B = 0 \rightarrow R_{AV} = \dfrac{50.9 - 150{,}03.5 + 30.7{,}81.7{,}81/2 + 100}{6}$

$R_{AV} = 119{,}18$ kN.

$\Sigma M_A = 0 \rightarrow R_{BV} = \dfrac{-50.3 + 30.7{,}81.7{,}81/2 - 100}{6}$

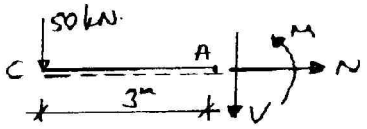
$R_{BV} = 110{,}84$ kN.

$\Sigma Y = 0 \rightarrow 50 + 23{,}05.7{,}81 = 119{,}18 + 110{,}84$

$230{,}02 = 230{,}02$ ✓

Kesit Tesirleri

C-A arası

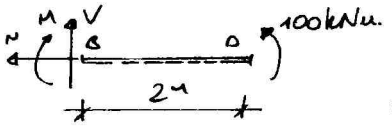


$N_{CA} = N_{AC} = 0$

$V_{CA} = V_{AC} = -50$ kN.

$M_A = -50.3 = -150$ kN

B-D arası

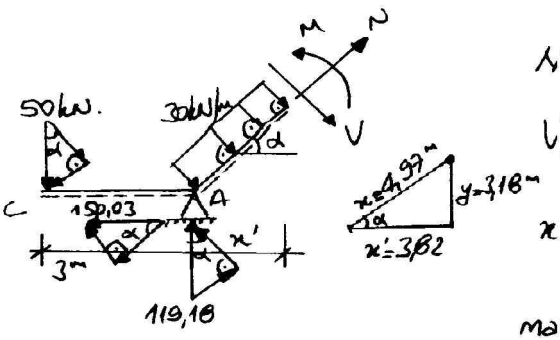


$N_{BD} = N_{DB} = 0$

$V_{BD} = V_{DB} = 0$

$M_{BD} = M_{DB} = 100$ kNm

A-B arası



$N_x = 50.\sin\alpha + 150{,}03.\cos\alpha - 119{,}18.\sin\alpha = 150{,}03\cos\alpha - 69{,}18\sin\alpha$

$N'_{AB} = N_{BA} = 70{,}97$ kN.

$V'_x = -50.\cos\alpha + 150{,}03.\sin\alpha + 119{,}18.\cos\alpha - 30.x = 149{,}20 - 30.x$

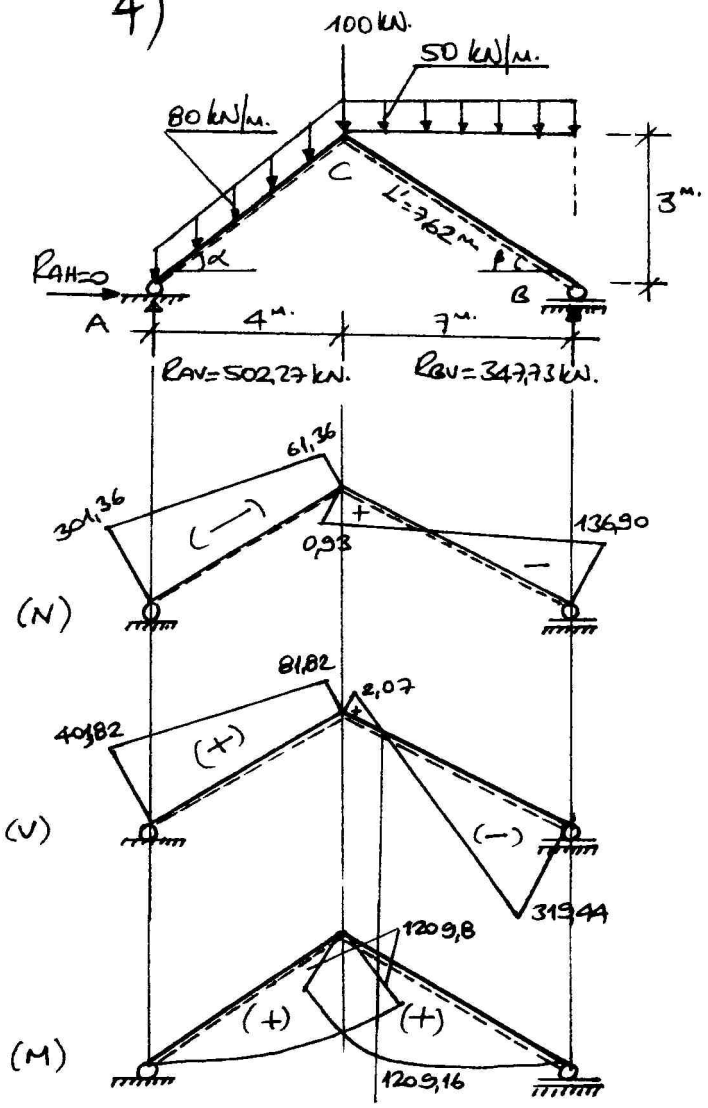
$x = 0 \begin{cases} V_{AB} = 149{,}20 \text{ kN} \\ \end{cases}$, $x = 7{,}81$ m $\begin{cases} V_{BA} = -85{,}10 \text{ kN.} \end{cases}$

$\max M \rightarrow V_x = 0 \Rightarrow x_0 = \dfrac{149{,}20}{30} = 4{,}97$ m.

$\max M = -50.6{,}82 + 150{,}03.3{,}18 + 119{,}18.3{,}82 - 30.\overline{4{,}97}^2.1/2 =$

$\max M = 220{,}85$ kNm.

4)



MESNET TEPKİLERİ

$\Sigma X=0 \rightarrow R_{AH}=0$

$\Sigma M_B=0 \rightarrow R_{AV}=\frac{80.5.9+100.7+50.7.3,5}{11}=502,27\ kN$

$\Sigma M_A=0 \rightarrow R_{BV}=\frac{80.5.2+100.4+50.7.7,5}{11}=347,73\ kN$

$\Sigma Y=0 \rightarrow 80.5+100+50.7=502,27+347,73$

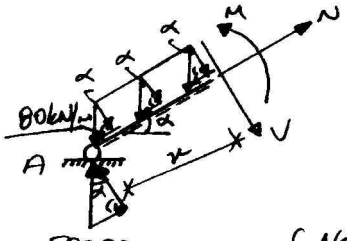
$850=850$ ✓

$\sin\alpha=3/5$, $\sin\beta=3/7,62$

$\cos\alpha=4/5$, $\cos\beta=7/7,62$

KESİT TESİRLERİ

A-C arası



$N_x=-502,27.\sin\alpha+80.x.\sin\alpha$

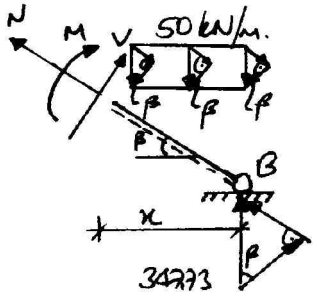
$V_x=502,27.\cos\alpha-80.x.\cos\alpha$

$x=0 \begin{cases} N_{AC}=-301,36\ kN. \\ V_{AC}=401,82\ kN. \end{cases}$

$x=5\ m. \begin{cases} N_{CA}=-61,36\ kN. \\ V_{CA}=81,82\ kN. \end{cases}$

$M_C=502,27.4-80.5.2=1209,08\ kNm.$

C-B arası



$N_x=-347,73.\sin\beta+50.\sin\beta.x=-136,90+19,69x$

$V_x=-347,73.\cos\beta+50.\cos\beta.x=-319,44+45,93x$

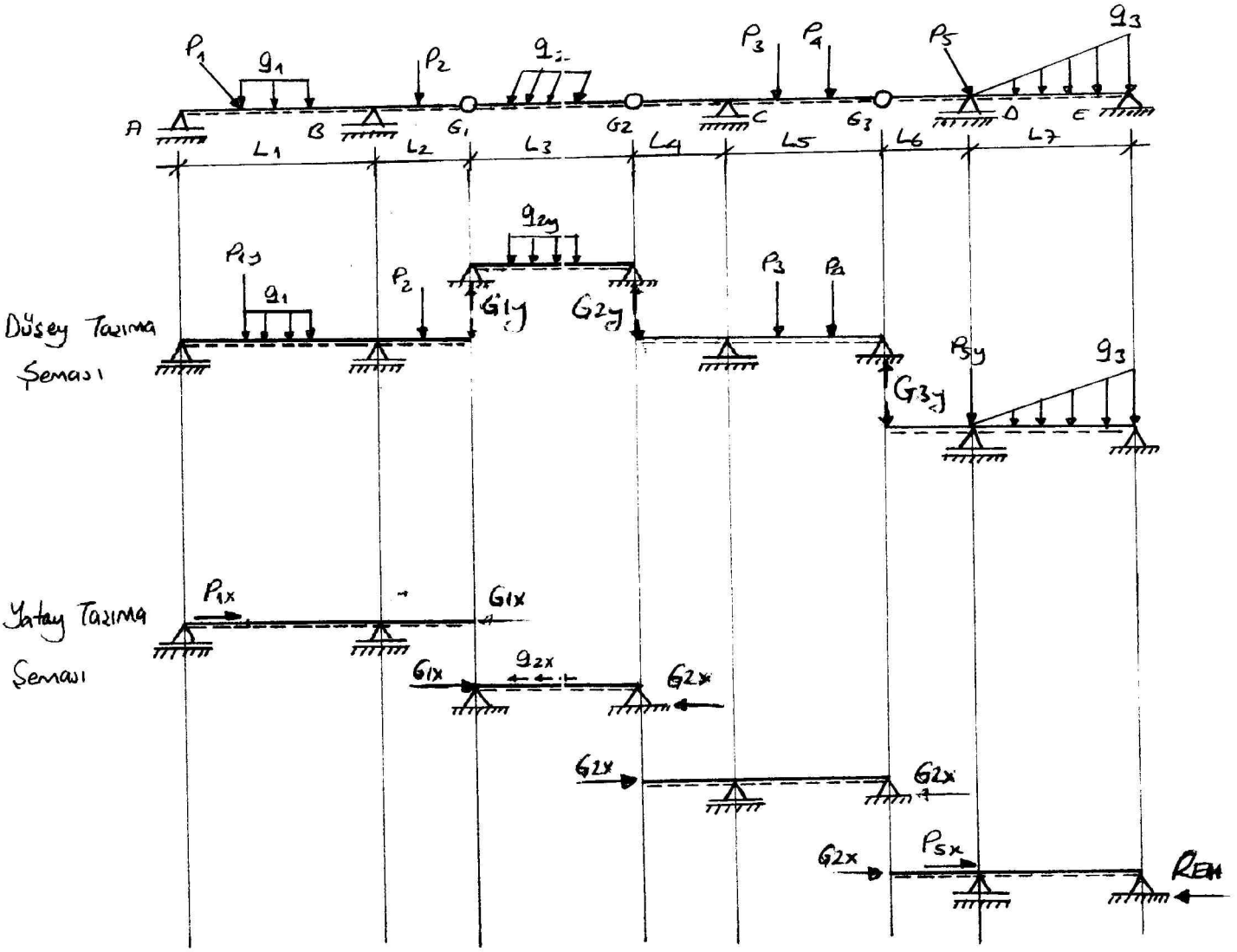
$x=0 \begin{cases} N_{BC}=-136,90\ kN. \\ V_{BC}=-319,44\ kN. \end{cases}$ $\quad x=7\ m. \begin{cases} N_{CB}=0,93\ kN. \\ V_{CB}=2,07\ kN. \end{cases}$

$\max M \rightarrow V_x=0 \Rightarrow x_0=\frac{319,44}{45,93}=6,95\ m.$

$\max M=347,73.6,95-50.6,95.\frac{6,95}{2}=1209,16\ kNm.$

IV-5) GERBER KİRİŞLER

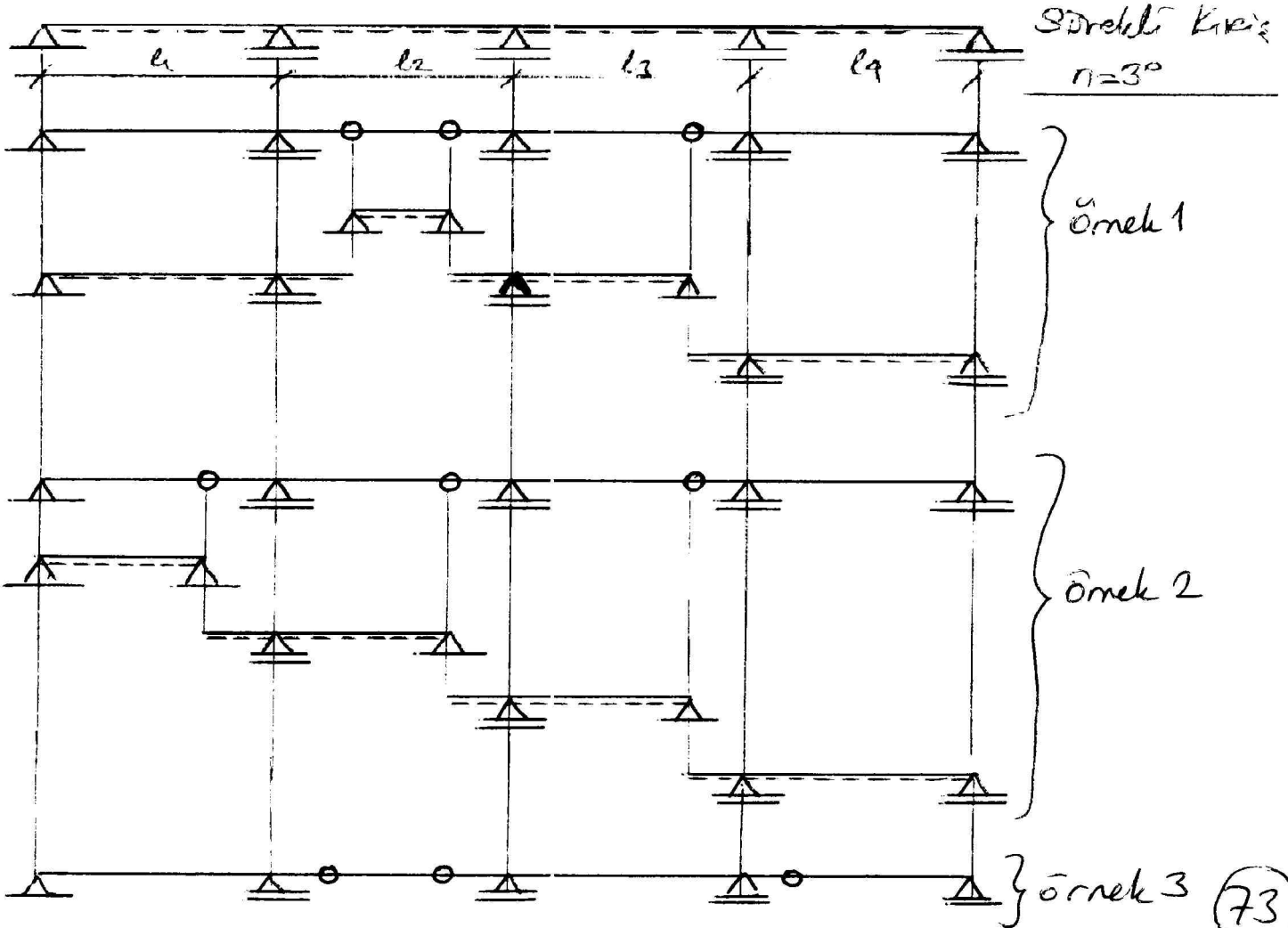
- Gerber kirişlerin çözümünde sistem, mafsallardan ayrılarak basit ve çıkmalı kirişler halinde çözüm yapılır. Ancak çözüm için sistemin düşey ve yatay taşıma şemaları çıkartılır. Yatay kuvvetin sistem içinde moment etkisi yoksa sistem, düşey kuvvetler ve yatay kuvvetler olarak ayrı ayrı çözülebilir.



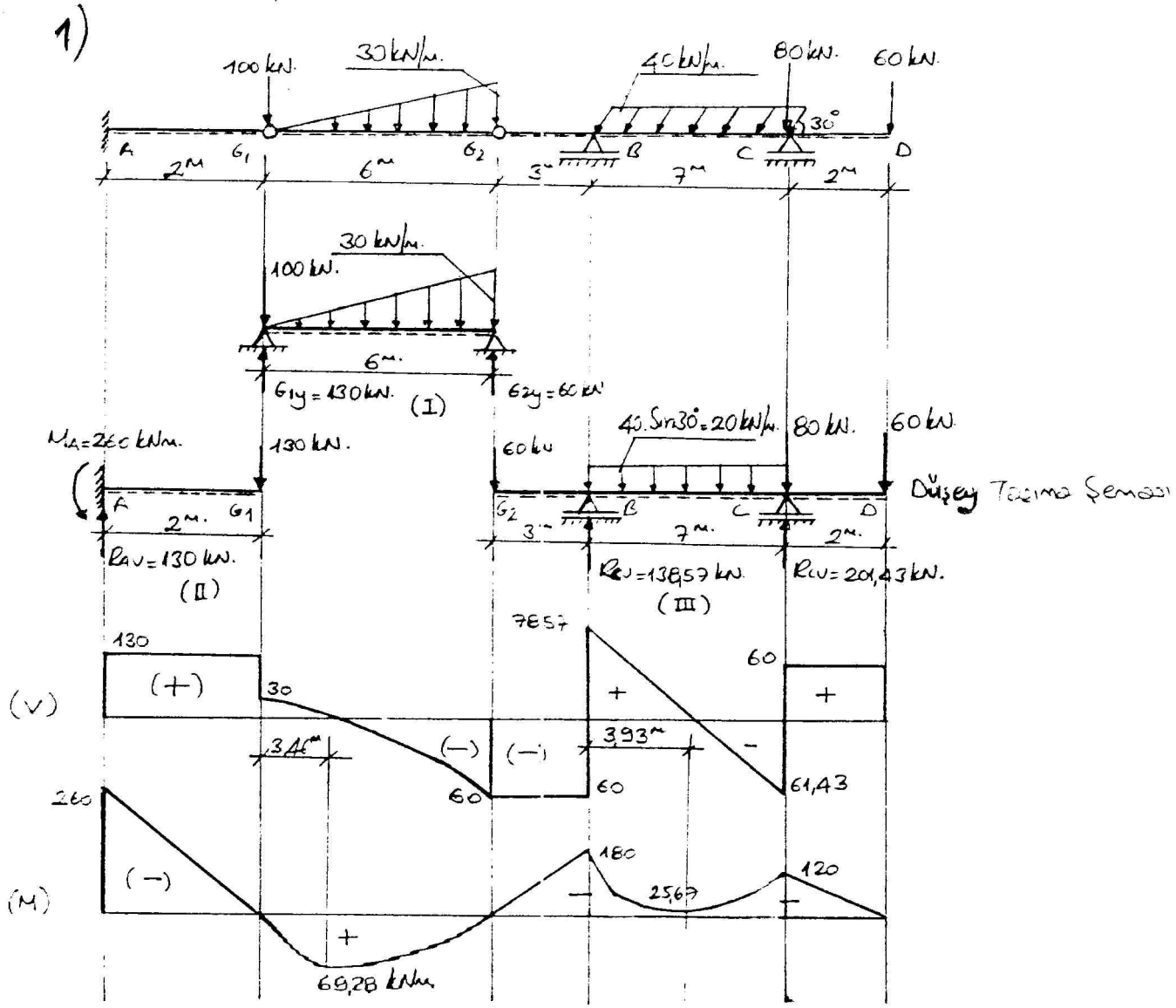
Hiperstatik sürekli kirişlerin izostatik hale getirilmesinde genellikle mafsallardan yararlanırız. Mafsal yerleştirilerek izostatik hale getirilen sürekli kirişlere gerber kirişler denir. Ancak sürekli bir kirişe gelişigüzel mafsal yerleştirilemez. Mafsal yerleşiminden sonra sistemin mafsallardan ayrılan her parçası sistemin kendisi gibi izostatik olmalıdır.

Mafsal Yerleştirme Teknikleri

Sürekli kirişin herhangi bir orta açıklığına en çok 2 mafsal yerleştirilebilir. Kenar açıklığa en fazla 1 mafsal konulabilir. Ancak kenar açıklıktaki kenar mesnet ankastre ise bu açıklığa en çok 2 mafsal konulabilir. Bir orta açıklıkta 2 mafsal varsa bunun yanındaki açıklıklara en çok 1 mafsal konulabilir. Her iki açıklığa ise birer tane mafsal konulabilir.



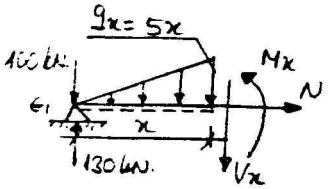
1)



(I) PARÇASI

$\Sigma M_{G_2} = 0 \rightarrow G_{1y} = \frac{30.6.\frac{1}{3}.6}{6} + 100 = 130\ kN.$

$\Sigma M_{G_1} = 0 \rightarrow G_{2y} = \frac{30.6/2.\frac{2}{3}.6}{6} = 60\ kN.$



$V_x = 130 - 100 - 5x.\frac{x}{2} = 30 - 2{,}5x^2$

$M_x = (130-100)x - 5x.\frac{x}{2}.\frac{x}{3}$

$= 30x - \frac{5}{6}x^3$

$x=0 \begin{cases} V_{G_1} = 30\ kN. \\ M_{G_1} = 0 \end{cases} \quad x = 6^m. \begin{cases} V_{G_2 G_1} = -60\ kN. \\ M_{G_2} = 0 \end{cases}$

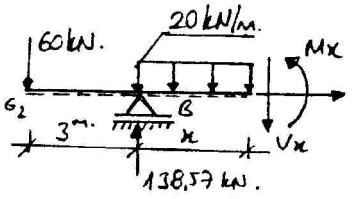
$max M \rightarrow V_x = 0 \Rightarrow x_0 = \sqrt{30/2{,}5} = 3{,}46^m.$

$max M = 30.3{,}46 - \frac{5}{6}.(3{,}46)^3 = 69{,}28\ kNm.$

(II) PARÇASI

$V_{AG_1} = V_{G_1 A} = 130\ kN,\ M_A = -260\ kNm.,\ M_{G_1} = 0$

(III) PARÇASI



$V_{G_2 B} = V_{B G_2} = -60\ kN.$

$M_B = -60.3 = -180\ kNm.$

$V_x = -60 + 138{,}57 - 20.x$

$M_x = -60(3+x) + 138{,}57.x - 20.x.\frac{x}{2}$

$x=0 \begin{cases} V_{BC} = 78{,}57\ kN. \\ M_B = -180\ kNm. \end{cases}$

$x = 7^m. \begin{cases} V_{CB} = -61{,}43\ kN. \\ M_C = -120\ kNm. \end{cases}$

$\Sigma M_C = 0 \rightarrow R_{BV} = \frac{60.10 - 60.2 + 20.7.3{,}5}{7}$

$R_{BV} = 138{,}57\ kN.$

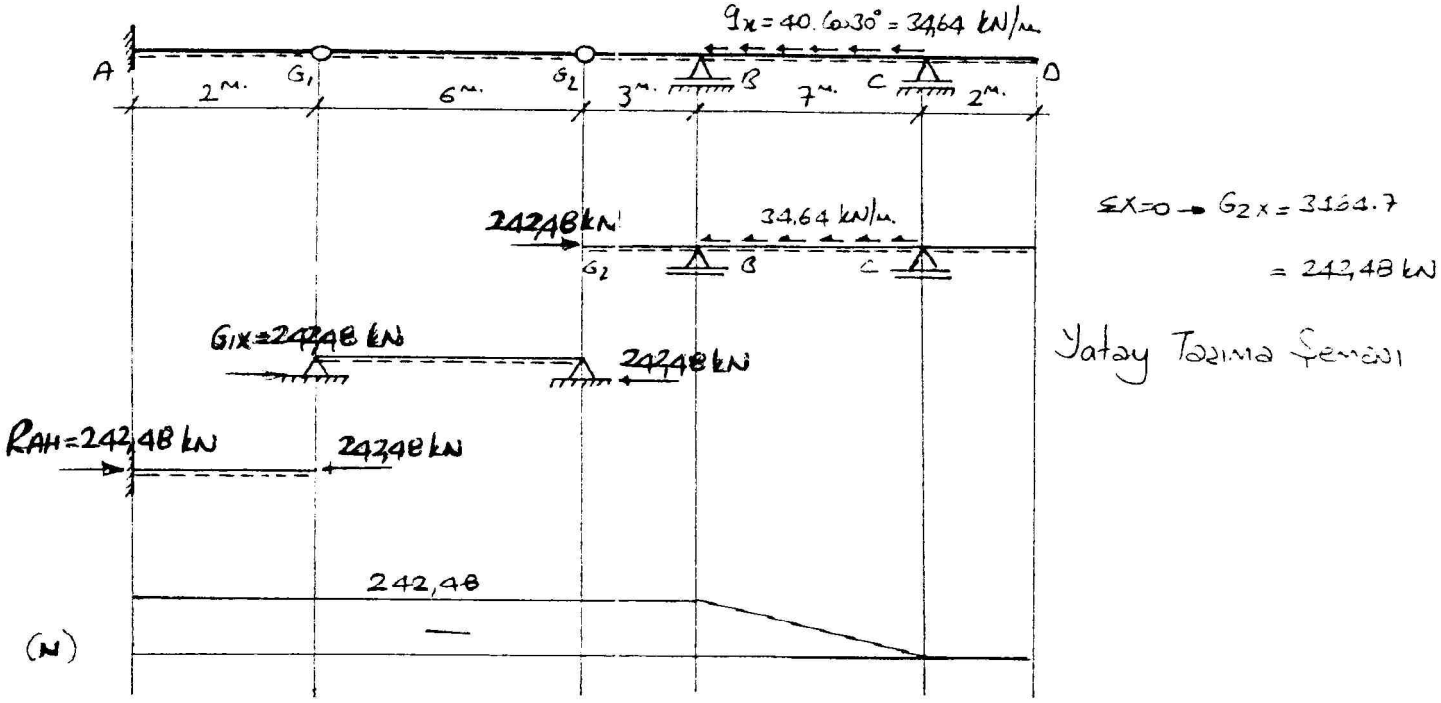
$\Sigma M_B = 0 \rightarrow R_{CV} = \frac{60.3 + 20.7.3{,}5 + 60.9}{7}$

$R_{CV} = 201{,}43\ kN.$

$\left.\begin{array}{l} max M \rightarrow V_x = 0 \Rightarrow x_0 = 3{,}93^m \\ max M = -25{,}67\ kNm. \end{array}\right\}$

$V_{DC} = V_{CD} = 60\ kN.$

$M_C = -60.2 = -120\ kNm.$

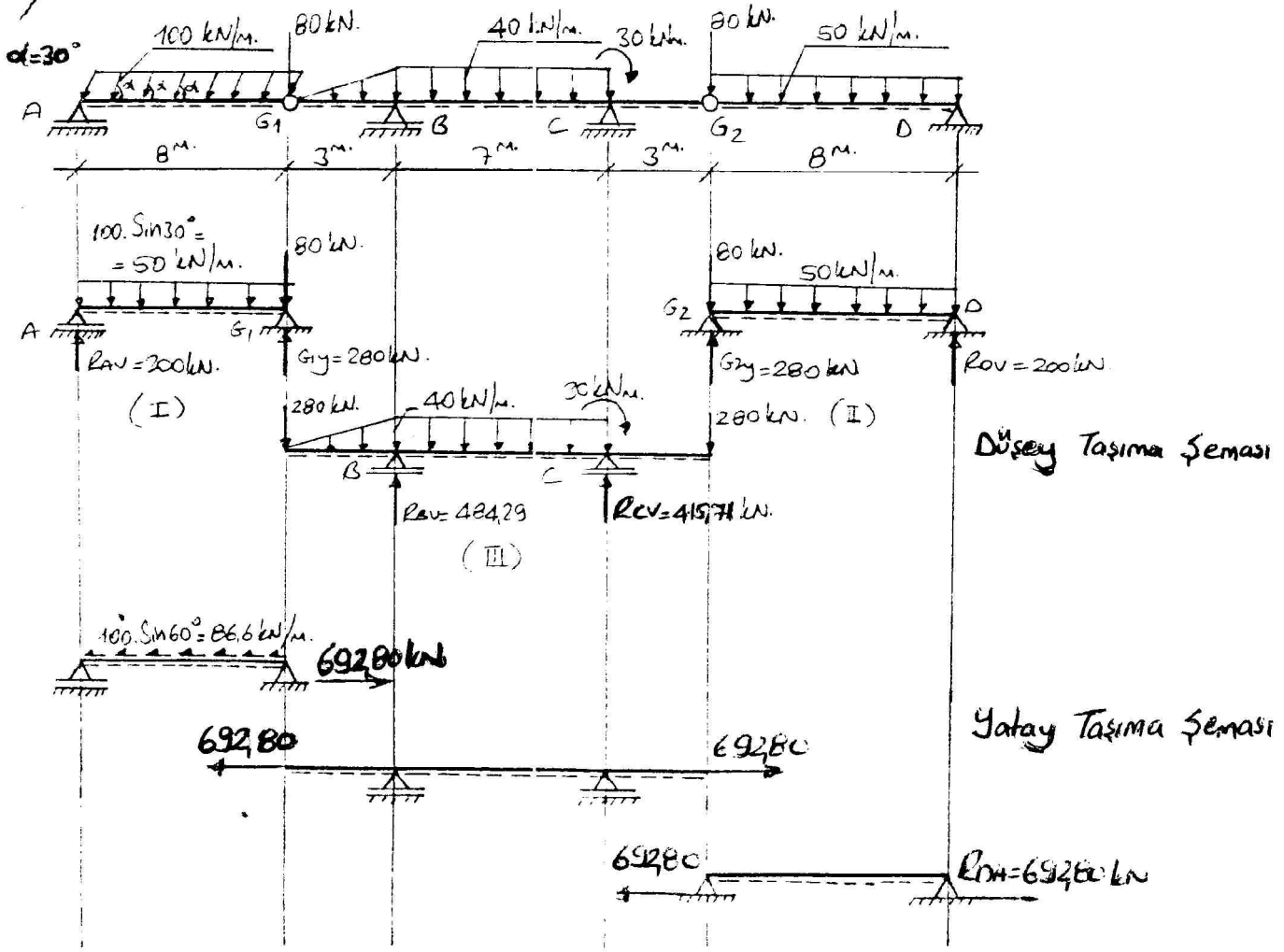


$\Sigma X = 0 \rightarrow G_2x = 3164.7$

$= 242{,}48$ kN

Yatay Taşıma Şeması

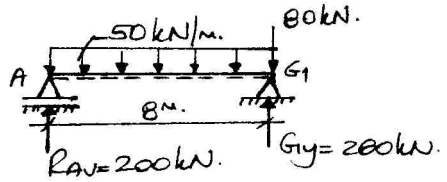
2)



Düşey Taşıma Şeması

Yatay Taşıma Şeması

(I)-(II) Parçası Çözümü



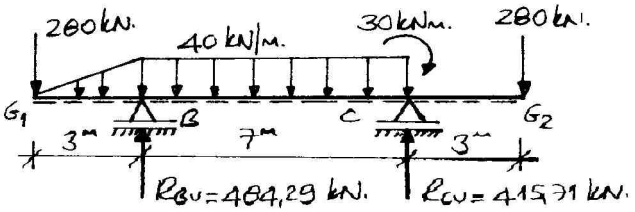
$\Sigma M_{G_1}=0 \rightarrow R_{AV}=\frac{50.8.4}{8}$ = 200 kN.

$\Sigma M_A=0 \rightarrow G_{1V}=\frac{50.8.4}{8}+80=280\,kN.$

$V_{AG_1}=200\,kN,\ V_{G_1A}=200\,kN,\ x_0=\frac{200}{50}=4{,}00^{m}.$

$max M=\frac{200^2}{2.50}=400\,kNm.$

(III) Parçası Çözümü



Mesnet Tepkileri

$\Sigma M_C=0 \rightarrow R_{BV}=\frac{280.10+\frac{40.3}{2}.8+40.7.3{,}5-30-280.3}{7}=$

$R_{BV}=484{,}29\,kN.$

$\Sigma M_B=0 \rightarrow R_{CV}=\frac{-280.3-\frac{40.3}{2}.1+40.7.3{,}5+30+280.10}{7}=$

$R_{CV}=415{,}71\,kN.$

$\Sigma Y=0 \rightarrow 280+\frac{40.3}{2}+40.7+280=484{,}29+415{,}71$

$900=900$ ✓

Kesit Tesirleri

G_1-B arası

$V_{G_1B}=-280\,kN,\ V_{BG_1}=-280-\frac{40.3}{2}=-340\,kN.$

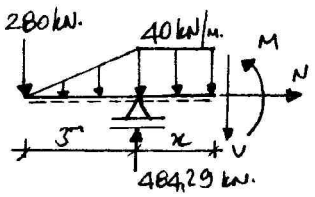
$M_B=-280.3-\frac{40.3}{2}.1=-900\,kNm.$

C-G_2 arası

$V_{G_2C}=V_{CG_2}=280\,kN,\ M_C=-280.3=-840\,kNm.$

$M_{CB}=-840-30=-870\,kNm.$

B-C arası



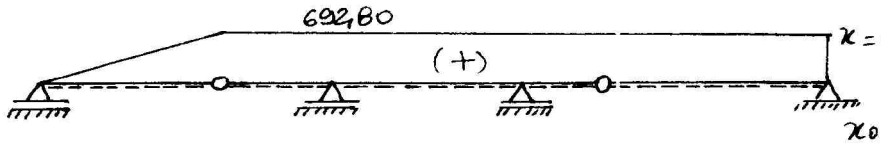
$V_x=-280+484{,}29-\frac{40.3}{2}-40x=144{,}29-40x$

$M_x=-280(3+x)-\frac{40.3}{2}.(1+x)+484{,}29.x-40.x.\frac{x}{2}$

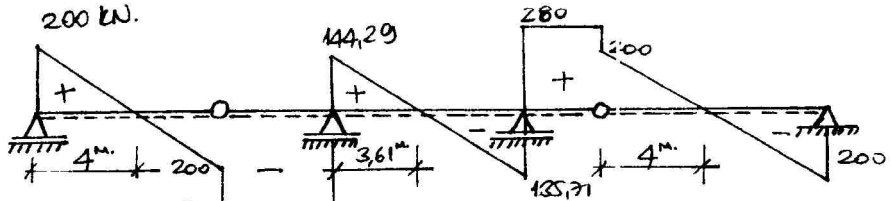
$x=0\begin{cases}V_{BC}=144{,}29\,kN.\\ M_B=-900\,kNm.\end{cases}\quad x=7^{m}\begin{cases}V_{CB}=-135{,}71\,kN\\ M_C=-870\,kNm.\end{cases}$

$x_0=\frac{144{,}29}{40}=3{,}61\,m. \rightarrow max M=-639{,}76\,kNm$

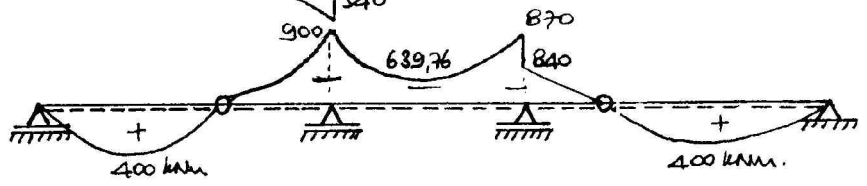
(N)



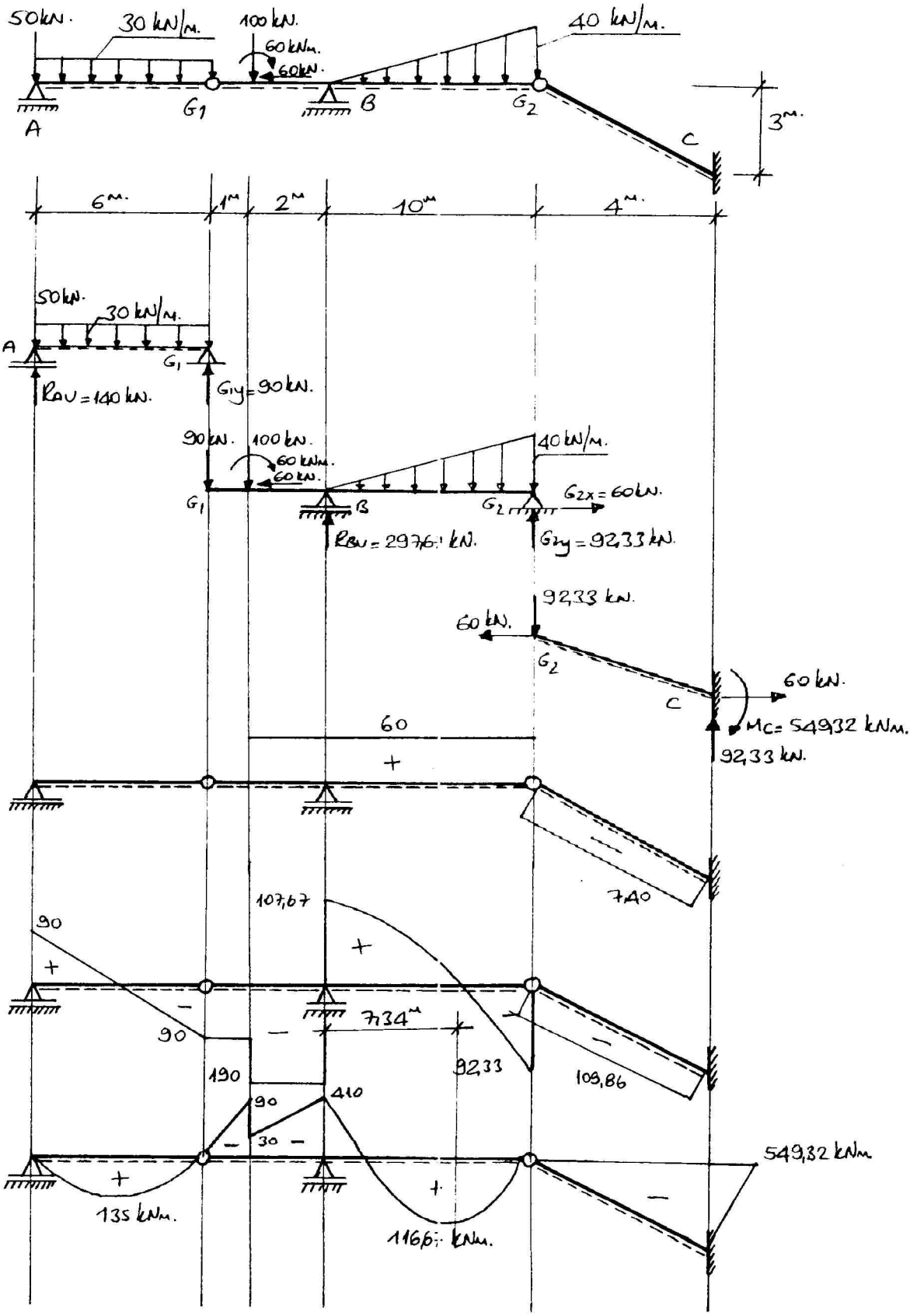
(V)



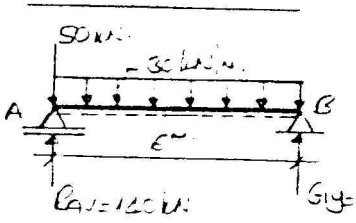
(M)



3)


50kN.
30 kN/m.
100 kN.
60 kNm.
60kN.
40 kN/m.
A
G1
B
G2
C
3m.
6m.
1m
2m
10m
4m.
50kN.
30 kN/m.
A
G1
RAV = 140 kN.
G1y = 90 kN.
90 kN.
100 kN.
60 kNm.
60 kN.
40 kN/m.
G1
B
G2
G2x = 60kN.
RBV = 297,67 kN.
G2y = 92,33 kN.
92,33 kN.
60 kN.
G2
C
60 kN.
MC = 549,32 kNm.
92,33 kN.
(N)
60
+
7,40
(V)
107,67
90
+
+
90
7,34m
92,33
190
108,86
90
410
(M)
30
549,32 kNm
135 kNm.
116,67 kNm.

A - G1 arası

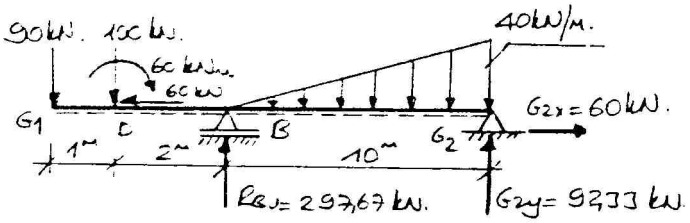


$\Sigma M_{G_1}=0 \rightarrow R_{AV}=\frac{30.6.3}{6}+90=120$ kN. $\Sigma M_A=0 \rightarrow G_{1y}=\frac{30.6.3}{6}=90$ kN.

$V_{AB}=90$ kN. $V_{BA}=-90$ kN.

$x_0=\frac{90}{30}=3^m \rightarrow \max M=\frac{90^2}{2.30}=135$ kNm.

G_1-G_2 arası



$\Sigma M_{G_2}=0 \rightarrow R_{BV}=\frac{90.13+100.12-60+\frac{40.10}{2}.\frac{10}{3}}{10}=297,67$

$\Sigma M_B=0 \rightarrow G_{2y}=\frac{-90.3-100.2+60+\frac{40.10}{2}.\frac{2}{3}.10}{10}=92,33$ kN

$\Sigma Y=0 \rightarrow 90+100+\frac{40.10}{2}=297,67+92,33$

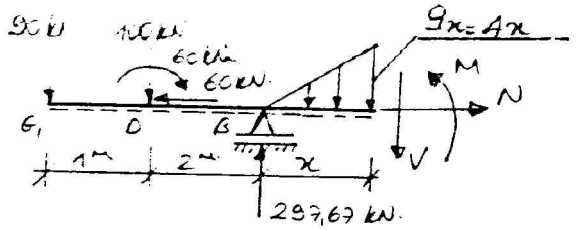
$390=390$

G_1-B arası

$V_{G_1C}=V_{CG_1}=-90$ kN

$V_{DC}=V_{BC}=-190$ kN., $V_{BG_2}=-190+297,67=107,67$ kN.

$N_{G_1C}=N_{CG_1}=0$, $N_{DC}=N_{CB}=+60$ kN.



$N_{BG_2}=N_{G_2B}=+60$ kN.

$V_x=-90-100+297,67-4x.\frac{x}{2}=107,67-2x^2$

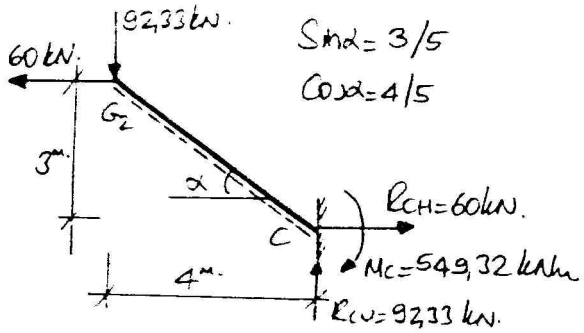
$M_x=-90(3+x)-100(2+x)+60+297,67.x-\frac{4x.x}{2}.\frac{x}{3}$

$\max M \rightarrow V_x=0 \Rightarrow x_0=\sqrt{\frac{107,67}{2}}=7,34^m$

$x=0 \begin{cases} V_{BG_2}=107,67 \text{ kN.} \\ M_B=-410 \text{ kNm.} \end{cases}$ $x=10^m \begin{cases} V_{G_2B}=-92,33 \text{ kN.} \\ M_{G_2}=0 \end{cases}$

$\max M=-90(3+7,34)-100(2+7,34)+60+297,67.7,34-\frac{4}{6}(7,34)^3=$

$\max M=116,67$ kNm.



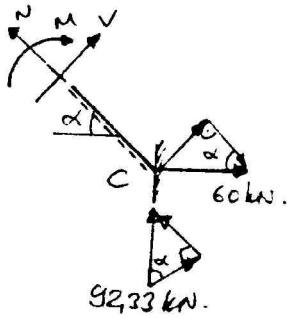
$\sin\alpha=3/5$

$\cos\alpha=4/5$

$\Sigma X=0 \rightarrow R_{CH}=60$ kN.

$\Sigma Y=0 \rightarrow R_{CV}=92,33$ kN.

$\Sigma M_C=0 \rightarrow M_C=92,33.4+60.3=549,32$ kNm.



$N_{CG_2}=N_{G_2C}=60.\cos\alpha-92,33.\sin\alpha$

$=-71,40$ kN.

$V_{CG_2}=V_{G_2C}=-60.\sin\alpha-92,33.\cos\alpha$

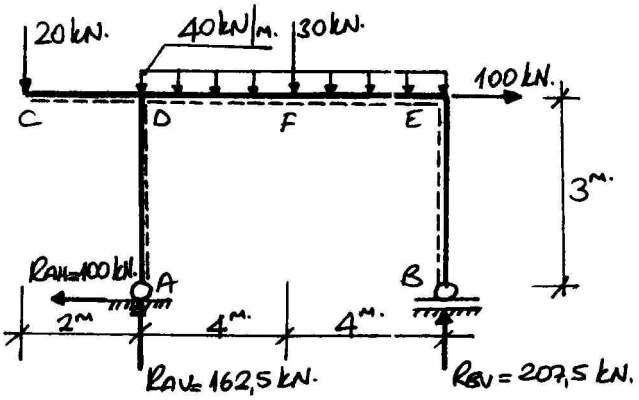
$=-109,86$ kN.

$M_C=-549,32$ kNm.

IV.6) GERGEVELER

IV.6.1) BASIT GERGEVELER

1)



Mesnet Tepkileri

$\Sigma X=0 \rightarrow R_{AH}=100\ kN.$

$\Sigma M_B=0 \rightarrow R_{AV}=\frac{20.10+40.8.4+30.4-100.3}{8}$

$R_{AV}=162,5\ kN.$

$\Sigma M_A=0 \rightarrow R_{BV}=\frac{-20.2+40.8.4+30.4+100.3}{8}$

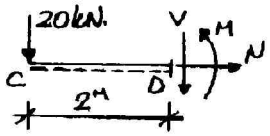
$R_{BV}=207,5\ kN.$

$\Sigma Y=0 \rightarrow 20+40.8+30=162,5+207,5$

$370=370$ ✓

Kesit Tesirleri

C-D arası

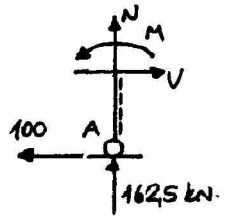


$N_{CD}=N_{DC}=0$

$V_{CD}=V_{DC}=-20\ kN.$

$M_{DC}=-20.2=-40\ kNm.$

A-D arası

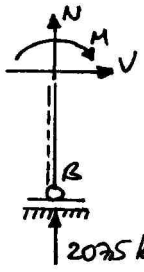


$N_{AD}=N_{DA}=-162,5\ kN.$

$V_{AD}=V_{DA}=100\ kN.$

$M_{DA}=100.3=300\ kNm.$

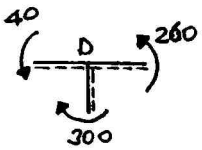
B-E arası



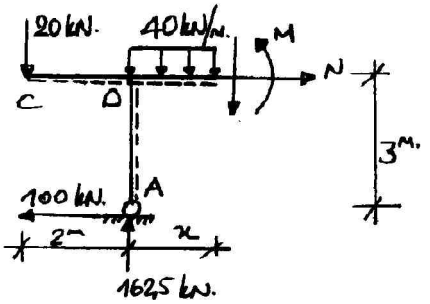
$N_{BE}=N_{EB}=-207,5\ kN.$

$V_{BE}=V_{EB}=0$

$M_{EB}=0$



D-E ARASI



$N_{DE}=N_{ED}=100\ kN.$

$V_{x_1}=-20+162,5-40x=142,5-40x$

$M_{x_1}=-20(2+x)+100.3+162,5(x)-40.x.\frac{x}{2}$

$x=0 \begin{cases} V_{DF}=142,5\ kN. \\ M_{DF}=260\ kNm. \end{cases}$ $x=4^m. \begin{cases} V_{FD}=-17,5\ kN. \\ M_F=510\ kNm. \end{cases}$

$V_{x_2}=142,5-30-40x=112,5-40x$

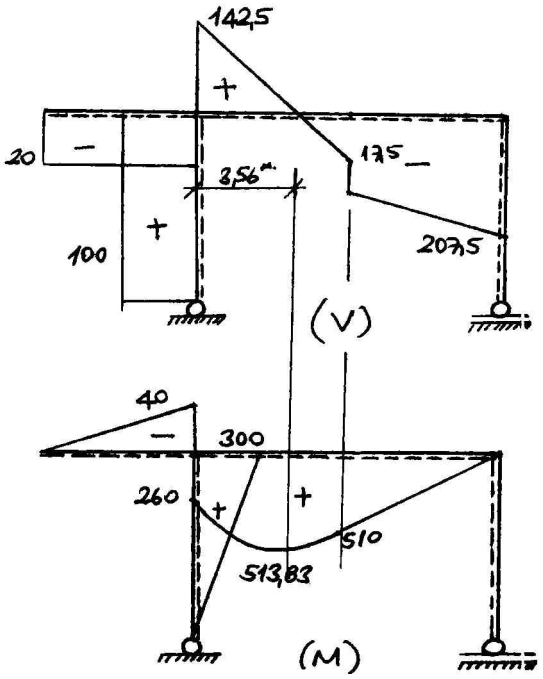
$M_{x_2}=-20(2+x)+100.3+162,5(x)-20x^2-30(x-4)$

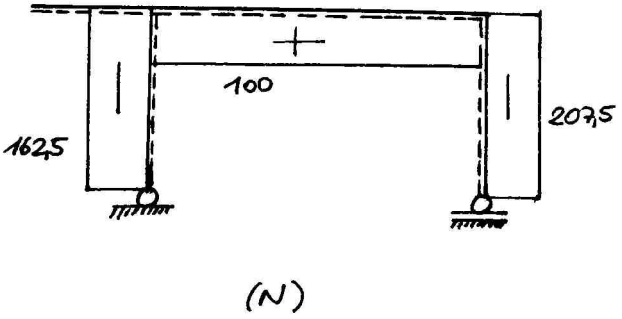
$x=4^m \begin{cases} V_{FE}=-47,5\ kN. \\ M_F=510\ kNm. \end{cases}$ $x=8^m. \begin{cases} V_{EF}=-207,5\ kN. \\ M_{EF}=0 \end{cases}$

max M. C-F arasında

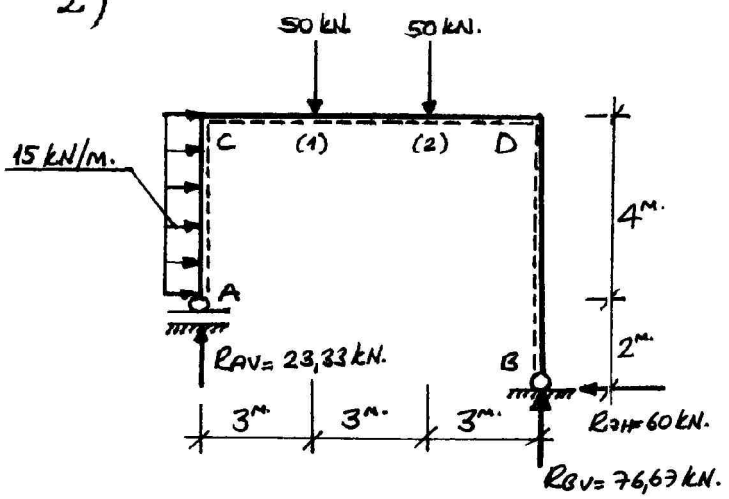
$x_0=\frac{142,5}{40}=3,56^m.$

$\max M=\frac{142,5^2}{2.40}+260=513,83\ kNm.$





2)



Mesnet Tepkileri

$\Sigma X=0 \rightarrow R_{AH}=15.4=60\ kN.$

$\Sigma M_B=0 \rightarrow R_{AV}=\frac{-15.4.4+60.6+50.3}{9}=23,33\ kN$

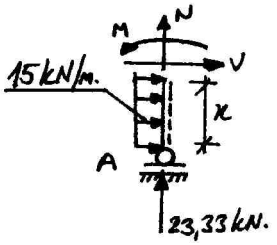
$\Sigma M_A=0 \rightarrow R_{BV}=\frac{60.2+15.4.2+50.3+50.6}{9}=76,67\ kN$

$\Sigma Y=0 \rightarrow 23,33+76,67=50+50$

$100=100$ ✓

KESİT TESİRLERİ

A-C ARASI



$N_{AC}=N_{CA}=-23,33\ kN.$

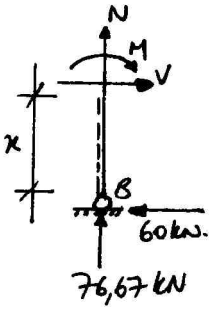
$V_x=15.x$

$M_x=-15\,x.\frac{x}{2}=-7,5x^2$

$x=0 \begin{cases} V_{AC}=0 \\ M_A=0 \end{cases}$

$x=4^m. \begin{cases} V_{CA}=60\ kN \\ M_{CA}=-120\ kNm \end{cases}$

B-D ARASI



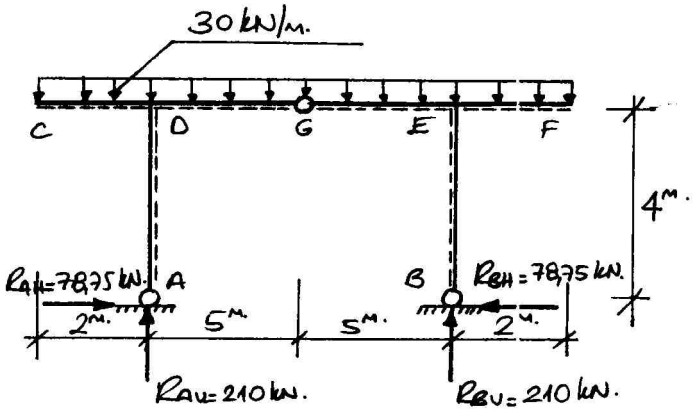
$N_{BD}=N_{DB}=-76,67\ kN.$

$V_{BD}=V_{DB}=60\ kN.$

$M_{DB}=-60.6=-360\ kNm.$

IV. 6.2) ÜÇ MAFSALLI GERGEVELER

3)



MESNET TEPKİLERİ

$\Sigma M_B=0 \Rightarrow R_{AV}=\frac{30.14.5}{10}=210$ kN.

$\Sigma M_A=0 \Rightarrow R_{BV}=\frac{30.14.5}{10}=210$ kN.

$\Sigma Y=0 \Rightarrow 30.14=210+210$

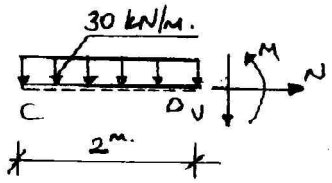
$420=420$ ✓

$\Sigma M_{G_{sol}}=0 \Rightarrow R_{AH}=\frac{210.5-30.7.3,5}{4}=78,75$ kN

$\Sigma M_{G_{sağ}}=0 \Rightarrow R_{BH}=\frac{210.5-30.7.3,5}{4}=78,75$ kN

* Sistem simetrik - yük simetrik olduğundan sistenin yarısına göre hesap yapılacak.

C - D ARASI

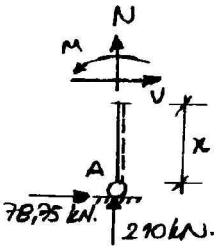


$N_{CD}=N_{DC}=0$

$V_{CD}=0$, $V_{DC}=-30.2=-60$ kN.

$M_{DC}=-30.2.1=-60$ kNm.

A - D ARASI

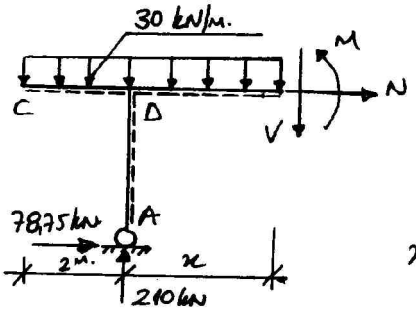


$N_{AD}=N_{DA}=-210$ kN.

$V_{AD}=V_{DA}=-78,75$ kN.

$M_{DA}=-78,75.4=-315$ kNm

D - G ARASI



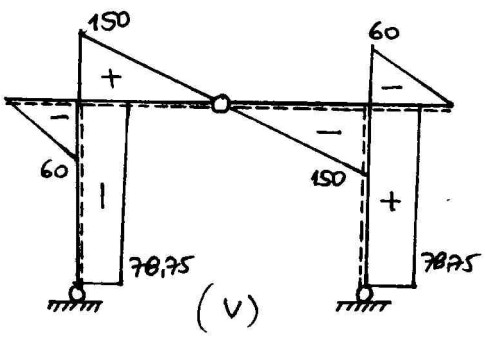
$N_{DE}=N_{ED}=-78,75$ kN.

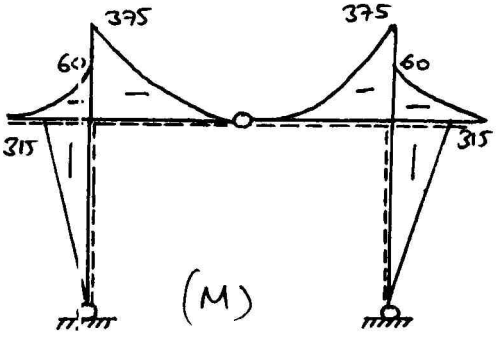
$V_x=210-30(2+x)$

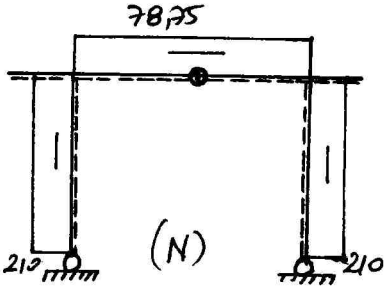
$M_x=210.x-78,75.4-30\frac{(2+x)^2}{2}$

$\max M \rightarrow V_x=0 \Rightarrow x_0=5^m$.

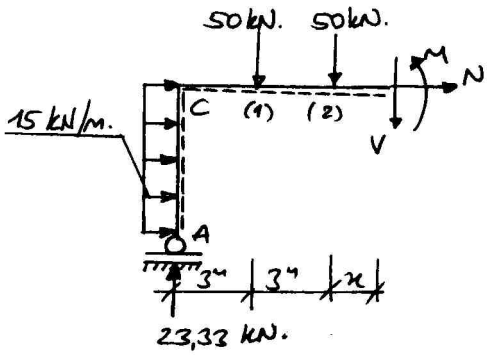
$x=0 \begin{cases} V_{DG}=150 \text{ kN.} \\ M_{DG}=-375 \text{ kNm} \end{cases}$ $x=5^m \begin{cases} V_{GD}=0 \\ M_G=0 \end{cases}$







C-O ARASI



$N_{C1} = N_{OC} = -15 \cdot 4 = -60$ kN.

$V_{C1} = V_{1C} = 23{,}33$ kN, $V_{12} = V_{21} = 23{,}33 - 50 = -26{,}67$ kN.

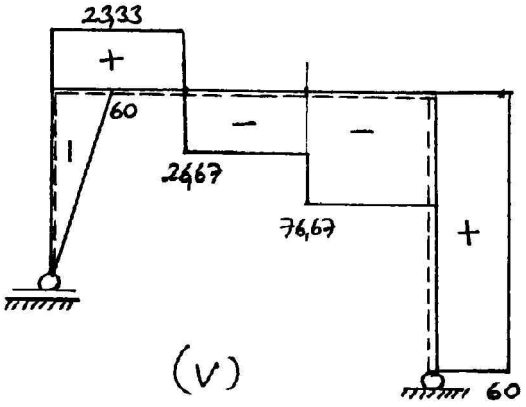
$V_{21} = V_{O2} = 23{,}33 - 100 = -76{,}67$ kN.

$M_{C1} = -15 \cdot 4 \cdot 2 = -120$ kNm.

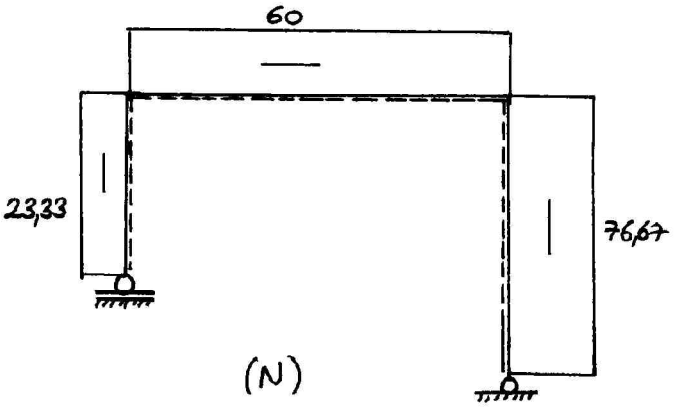
$M_1 = 23{,}33 \cdot 3 - 15 \cdot 4 \cdot 2 = -50$ kNm.

$M_2 = 23{,}33 \cdot 6 - 15 \cdot 4 \cdot 2 - 50 \cdot 3 = -130$ kNm.

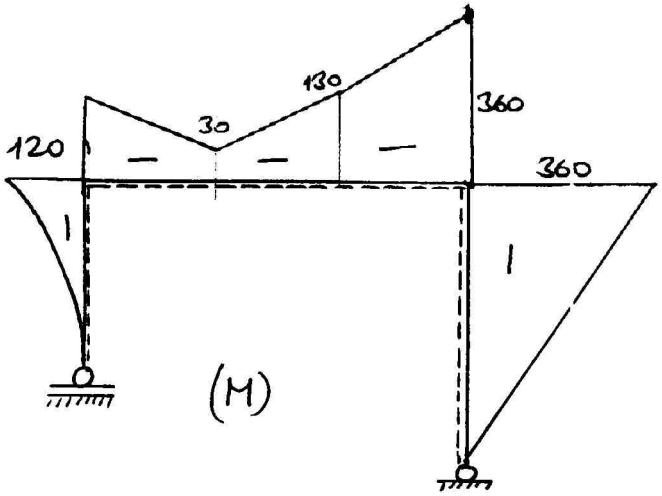
$M_{O3} = 23{,}33 \cdot 9 - 15 \cdot 42 - 50 \cdot 6 - 50 \cdot 3 = -360$ kNm.



(V)

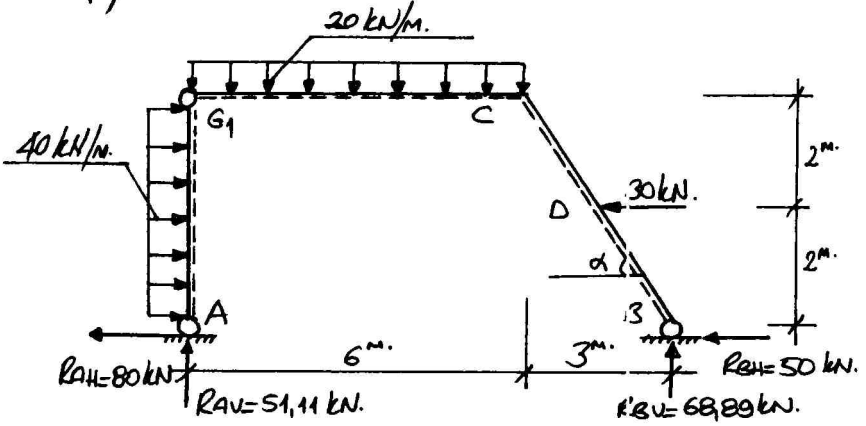


(N)



(M)

4)



Mesnet Tepkileri

$\Sigma M_B=0 \rightarrow R_{AV}=\frac{-40.4.2+20.6.6+30.2}{9}$

$R_{AV}=51,11$ kN.

$\Sigma M_A=0 \rightarrow R_{BV}=\frac{40.4.2+20.6.3-30.2}{9}$

$R_{BV}=68,89$ kN.

$\Sigma Y=0 \rightarrow 20.6=68,89+51,11 \Rightarrow 120=120$ ✓

$\Sigma M_{G,sol}=0 \rightarrow R_{AH}=\frac{40.4.2}{4}=80$kN.

$\Sigma M_{G,sağ}=0 \rightarrow R_{BH}=\frac{68,89.9-30.2-20.6.3}{4}$

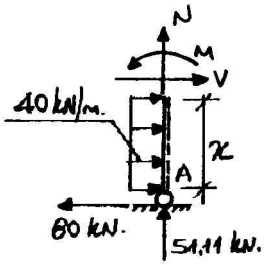
$R_{BH}=50$ kN.

$\sin\alpha=4/5$

$\cos\alpha=3/5$

KESİT TESİRLERİ

A-G1 ARASI

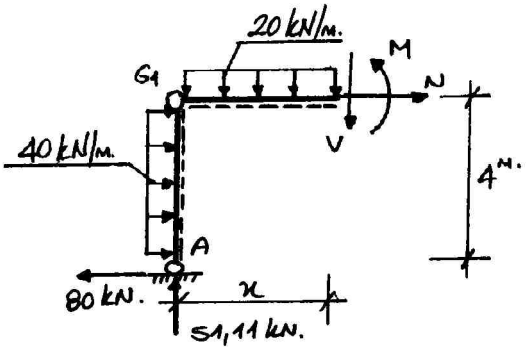


$N_{AG_1}=N_{G_1A}=-51,11$ kN.

$V_{AG_1}=80$ kN, $V_{G_1A}=80-40.4=-80$ kN.

$x_0=\frac{80}{40}=2^m \rightarrow max M=\frac{80^2}{2.40}=80$ kNm.

G1-C ARASI



$N_{G_1C}=N_{CG_1}=80$ kN.

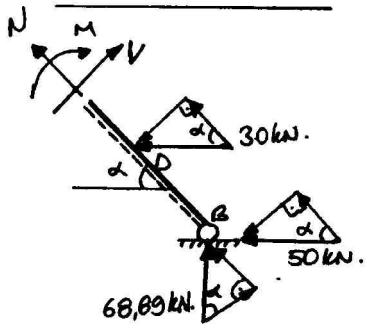
$V_x=51,11-20.x$

$M_x=51,11.x+80.4-40.4.2-20.x.\frac{x}{2}$

$x=0 \begin{cases} V_{G_1C}=51,11 \text{ kN.} \\ M_{G_1}=0 \end{cases}$ $x=6^m \begin{cases} V_{CG_1}=51,11-20.6=-68,89 \text{ kN} \\ M_C=-53,34 \text{ kNm.} \end{cases}$

$max M \rightarrow V_x=0 \Rightarrow x_0=\frac{51,11}{20}=2,56^m \Rightarrow max M=65,31$ kNm.

C-B ARASI



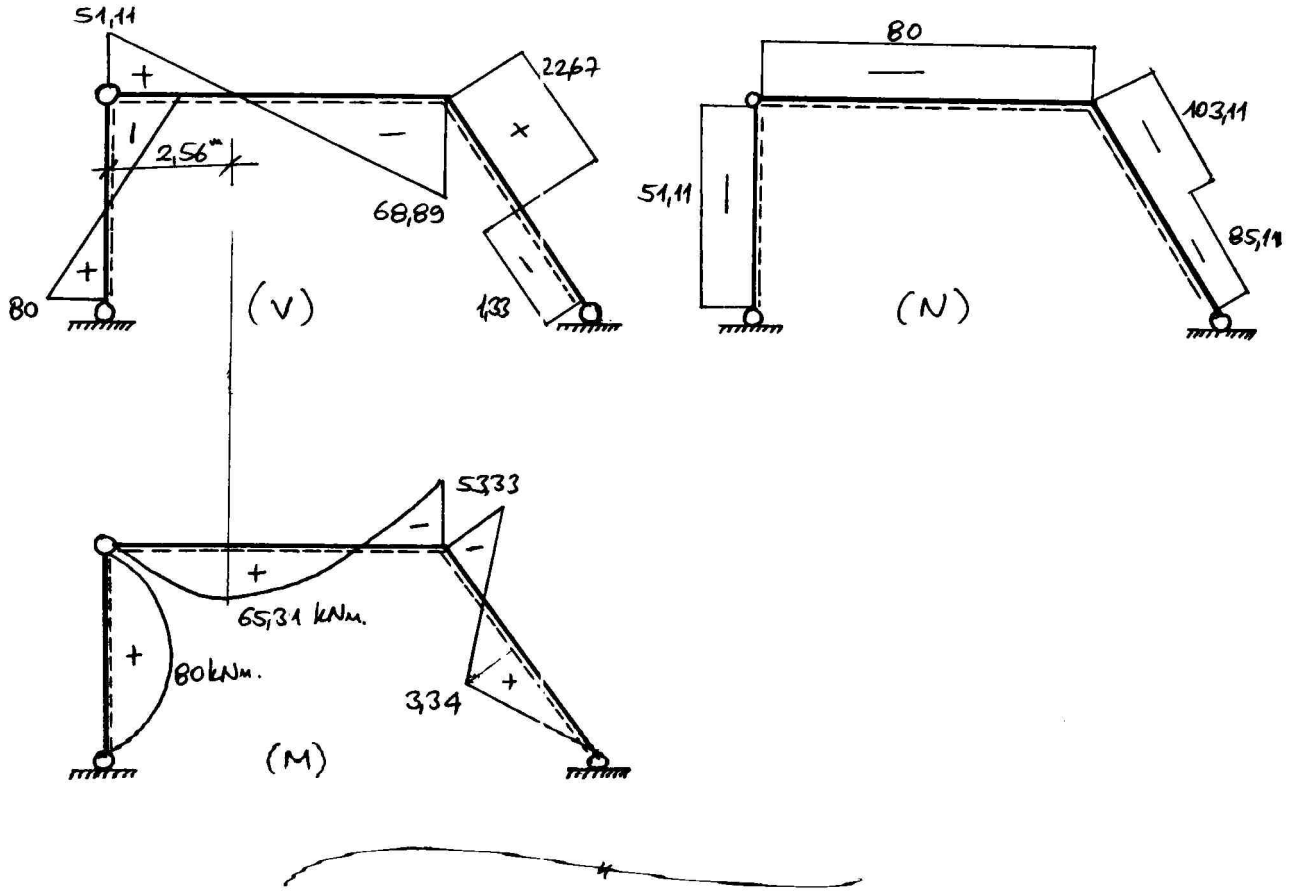
$N_{BO}=N_{OB}=-68,89.\sin\alpha-50.\cos\alpha=-85,11$ kN.

$V_{BO}=V_{OB}=-68,89.\cos\alpha+50.\sin\alpha=-1,33$ kN.

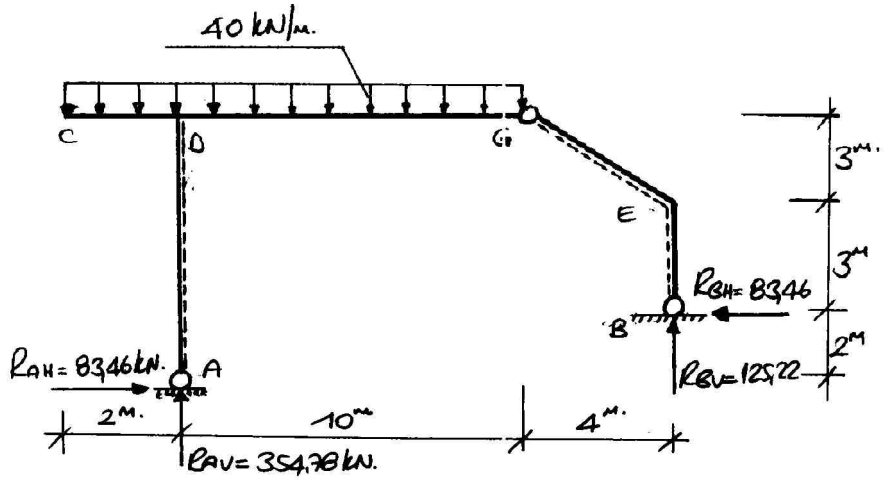
$N_{OC}=N_{CO}=-85,11-30.\cos\alpha=-103,11$ kN.

$V_{OC}=V_{CO}=-1,33+30.\sin\alpha=22,67$ kN.

$max M=68,89.1,5-50.2=3,34$ kNm.



5)



1) $\Sigma M_B = 0 \rightarrow R_{AV}.14 - R_{AH}.2 - 40.12.10 = 0$

$$7R_{AV} - R_{AH} = 2400$$

2) $\Sigma M_A = 0 \rightarrow R_{BV}.14 + R_{BH}.2 + 40.2.1 - 40.10.5 = 0$

$$7R_{BV} + R_{BH} = 960$$

3) $\Sigma M_{G\text{sol}} = 0 \rightarrow R_{AV}.10 - R_{AH}.8 - 40.12.6 = 0$

$$5R_{AV} - 4R_{AH} = 1440$$

4) $\Sigma M_{G\text{sağ}} = 0 \rightarrow R_{BV}.4 - R_{BH}.6 = 0$

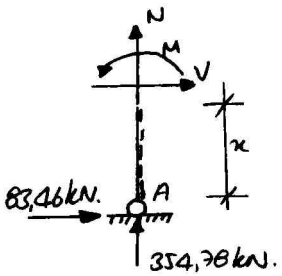
$$2R_{BV} - 3R_{BH} = 0$$

(1), (2), (3), (4) denklemlerinden,

$$\left\{ \begin{array}{l} R_{AV} = 354,78\,kN,\ R_{BV} = 125,22\,kN. \\ R_{AH} = 83,46\,kN,\ R_{BH} = 83,46\,kN. \end{array} \right\}$$

KESİT TESİRLERİ

A-D ARASI

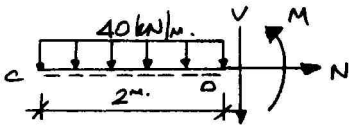


$N_{AD} = N_{DA} = -354,78\ kN.$

$V_{AD} = V_{DA} = -83,46\ kN.$

$M_{DA} = -83,46.8 = -667,68\ kNm.$

C-D ARASI

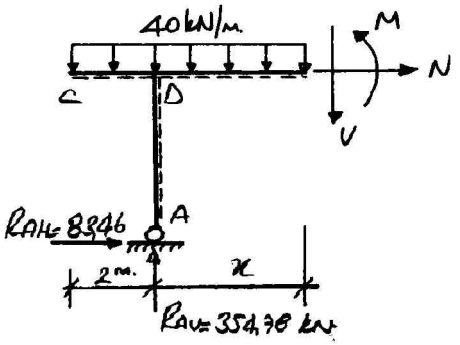


$N_{CD} = N_{DC} = 0$

$V_{DC} = -40.2 = -80\ kN.$

$M_{DC} = -40.2.1 = -80\ kNm.$

D-G ARASI



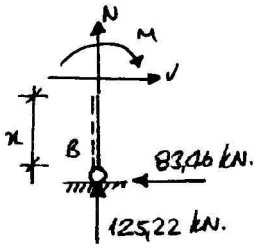
$N'_{DG} = N_{GD} = -83,46\ kN.$, $M_{DG} = -83,46.8 - 40.2.1 = -747,68\ kNm$

$V'_{DG} = 354,78 - 40.2 = 274,78\ kN.$

$V'_{GD} = 274,78 - 40.10 = -125,22\ kN.$

$x_0 = \frac{274,78}{40} = 6,87^{m} \rightarrow M_{max} N = \frac{274,78^2}{2.40} - 747,68 = 196,12\ kNm$

B-E ARASI

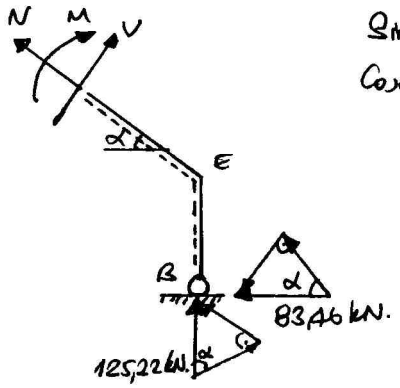


$N_{BE} = N_{EB} = -125,22\ kN.$

$V_{BE} = V_{EB} = 83,46\ kN.$

$M_B = 0$, $M_E = -83,46.3 = -250,38\ kNm.$

E-G ARASI

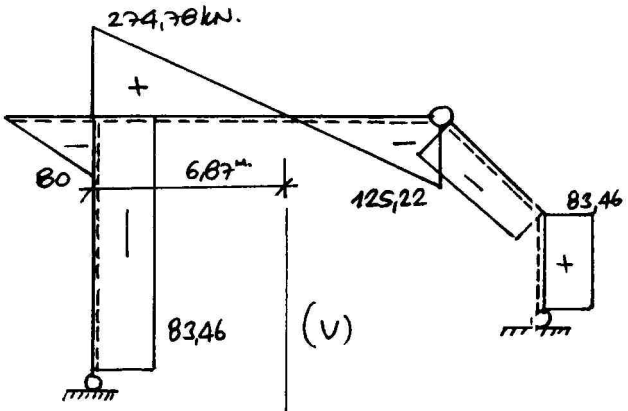


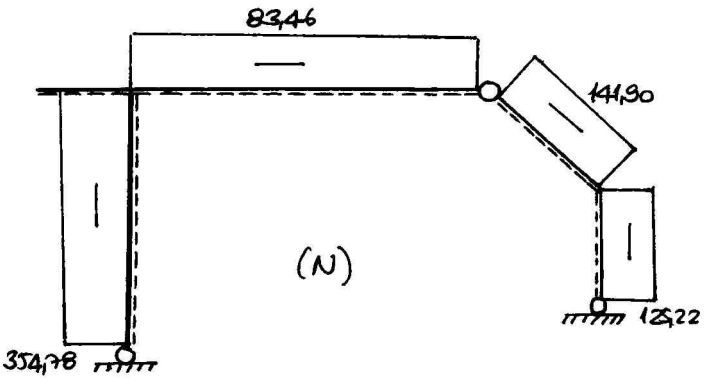
$\sin\alpha = 3/5$

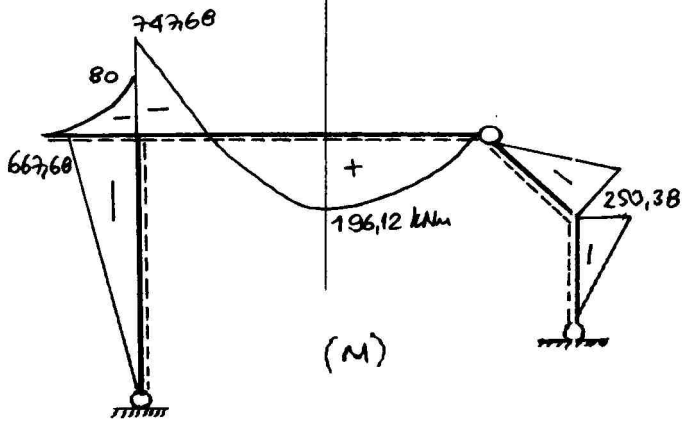
$\cos\alpha = 4/5$

$N_{EG} = N_{GE} = -83,46.\cos\alpha - 125,22.\sin\alpha = -141,90\ kN.$

$V_{EG} = V_{GE} = 83,46.\sin\alpha - 125,22.\cos\alpha = -50,10\ kN.$

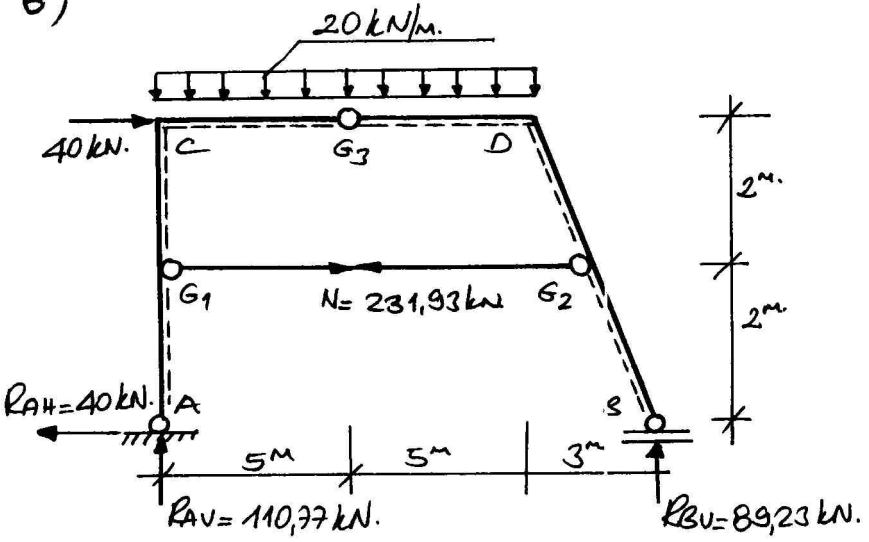


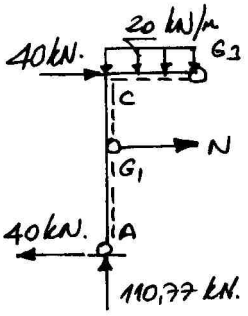




IV.6.3) GERGİLİ GERÇEVELER

6)





MESNET TEPKİLERİ

$\Sigma X=0 \rightarrow R_{AH}=40\ kN.$

$\Sigma M_B=0 \Rightarrow R_{AV}=\frac{20.10.8-40.4}{13}=110,77\ kN.$

$\Sigma M_A=0 \Rightarrow R_{BV}=\frac{20.10.8+40.4}{13}=89,23\ kN.$

$\Sigma Y=0 \Rightarrow 20.10=110,77+89,23$

$200=200$ ✓

$\Sigma M_{G_3 sol}=0 \rightarrow N=\frac{110,775+40.4-20.5.2,5}{2}$

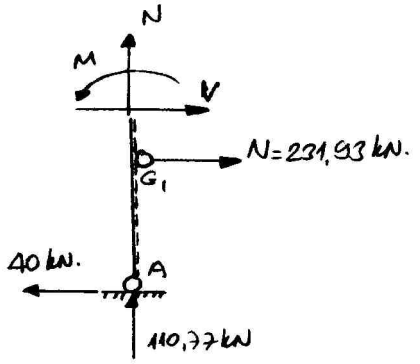
$N=231,93\ kN.$

$\Sigma M_{G_3 sağ}=0 \rightarrow N=\frac{89,23.8-20.5.2,5}{2}$

$N=231,93\ kN.$ ✓

KESİT TESİRLERİ

A-C ARASI



$N_{AC} = N_{CA} = -110,77$ kN.

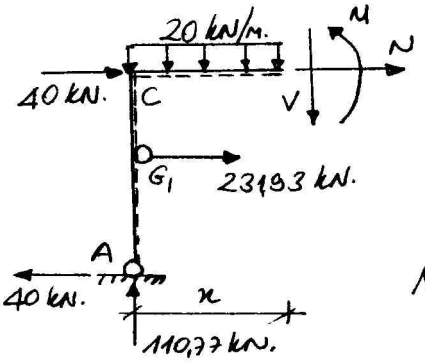
$V_{AG_1} = V_{G_1A} = 40$ kN.

$V_{G_1C} = V_{CG_1} = 40 - 231,93 = -191,93$ kN.

$M_{G_1} = 40.2 = 80$ kNm, $M_{CG_1} = 40.4 - 231,93.2$

$M_{CG_1} = -303,86$ kNm. ✓

C-O ARASI



$N_{CO} = N_{OC} = -231,93$ kN.

$V_x = 110,77 - 20.x$

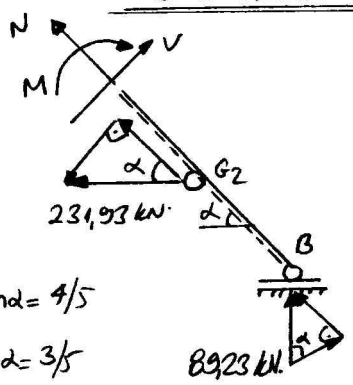
$M_x = 110,77x + 40.4 - 231,93.2 - 20.x.\frac{x}{2}$

$x=0 \begin{cases} V_{CO} = 110,77 \text{ kN.} \\ M_{CO} = -303,86 \text{ kNm.} \end{cases}$ ✓

$x = 5^m \begin{cases} V_{G_2C} = 10,77 \text{ kN.} \\ M_{G_2} = 20 \end{cases}$ $x = 10^m \begin{cases} V_{OG_2} = -89,23 \text{ kN.} \\ M_O = -196,16 \text{ kNm} \end{cases}$

$x_0 = \frac{110,77}{20} = 5,54^m \Rightarrow \underline{\max M = 2,89 \text{ kNm.}}$

B-O ARASI



$\sin\alpha = 4/5$

$\cos\alpha = 3/5$

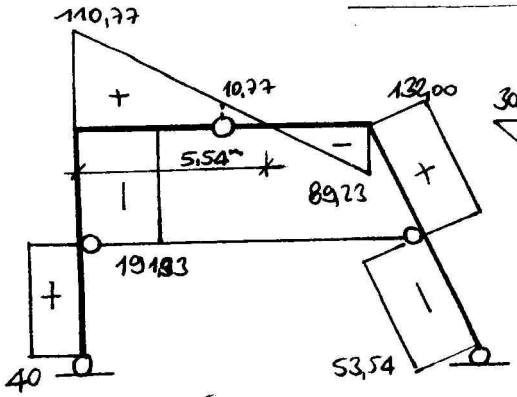
$N_{G_2B} = N_{BG_2} = -89,23.\sin\alpha = -71,38$ kN.

$N_{OG_2} = N_{G_2O} = -89,23.\sin\alpha - 231,93.\cos\alpha = -210,54$ kN.

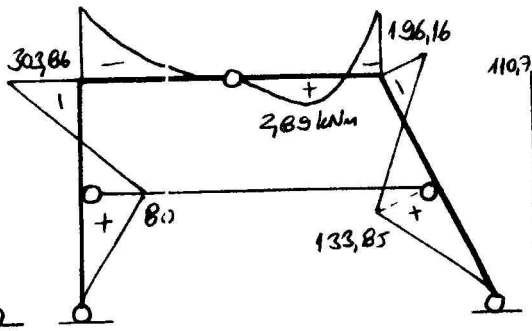
$V_{BG_2} = V_{G_2B} = -89,23.\cos\alpha = -53,54$ kN.

$V_{OG_2} = V_{G_2O} = -89,23.\cos\alpha + 231,93.\sin\alpha = 132,00$ kN.

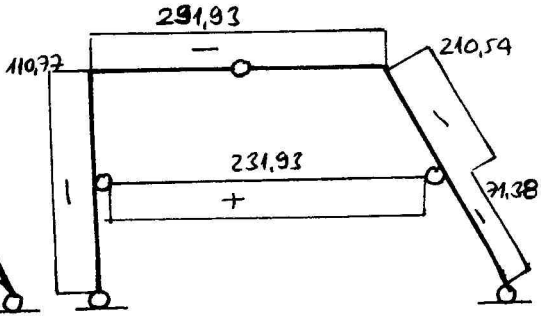
$M_{G_2} = 89,23.1,5 = 133,85$ kNm, $M_O = 89,23.3 - 231,93.2 = -196,17$ kNm.



(V)

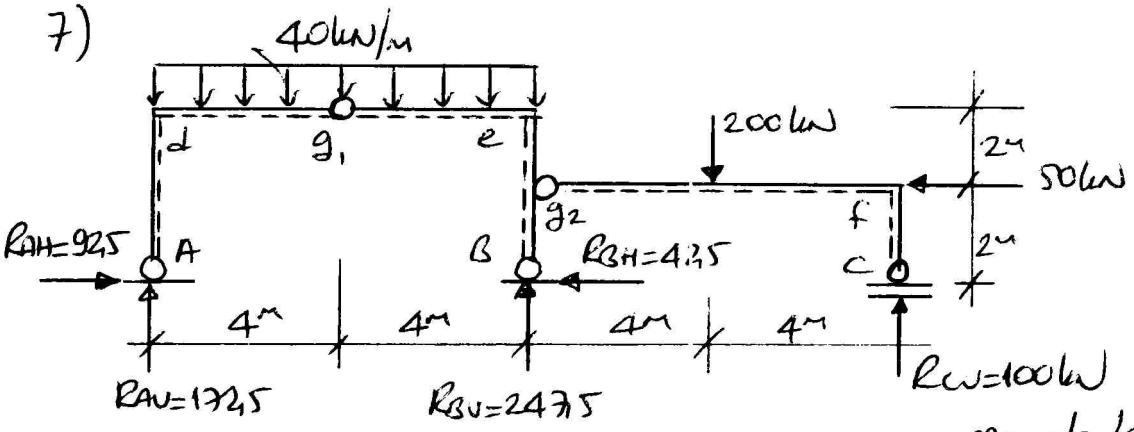


(M)



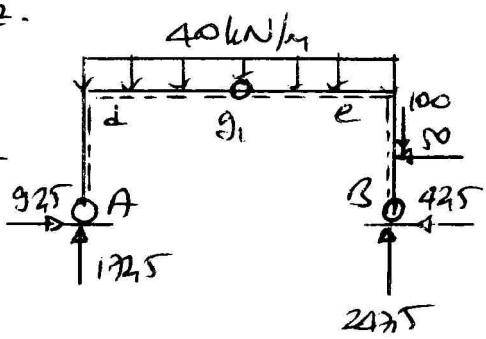
(N)

7)



* Mesnet tepkileri için sayfa 33'e bakınız.

Kesit Tesirleri



A-d arası

$N_{Ad} = N_{dA} = -172,5\ kN$

$V_{Ad} = V_{dA} = -92,5\ kN$

$M_{dA} = -92,5 \times 4 = -370\ kNm$

d-g1 arası

$N_{dg_1} = N_{g_1 d} = -92,5\ kN$

$V_{dg_1} = 172,5\ kN$

$V_{g_1 d} = 172,5 - 40 \times 4 = 12,5\ kN$

$M_{dg_1} = -370\ kNm$

g1e arası

$N_{g_1 e} = N_{eg_1} = -92,5\ kN$

$V_{g_1 e} = 12,5\ kN$

$V_{eg_1} = 12,5 - 40 \times 4 = -147,5\ kN$

max. momentin yeri ve değeri (g1-e arası)

$x_0 = \frac{12,5}{40} = 0,312\ m.$

$max\ M = \frac{12,5^2}{2 \times 40} + 0 = 1,95\ kNm$

B-e arası

$N_{Bg_2} = N_{g_2 B} = -247,5\ kN$

$N_{g_2 e} = N_{eg_2} = -247,5 + 100 = -147,5\ kN$

$V_{Bg_2} = V_{g_2 B} = 42,5\ kN$

$V_{g_2 e} = V_{eg_2} = 42,5 + 50 = 92,5\ kN$

$M_{g_2} = -42,5 \times 2 = -85\ kNm$

$M_e = -42,5 \times 4 - 50 \times 2 = -270\ kNm$

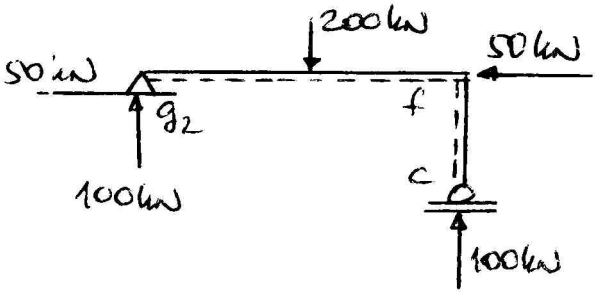
g2-f arası

$N_{g_2 f} = N_{f g_2} = -50\ kN$

$V_{g_2 f} = 100\ kN$

$V_{f g_2} = -100\ kN$

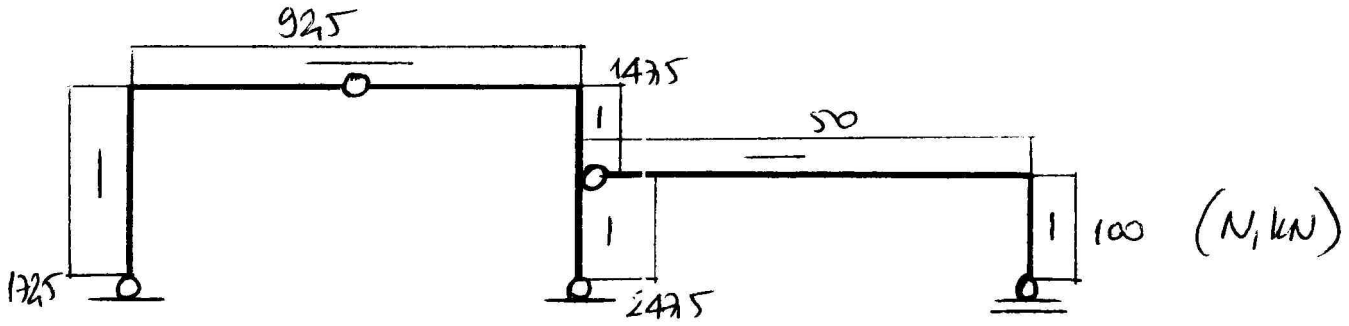
$max\ M = 100 \times 4 = 400\ kNm$

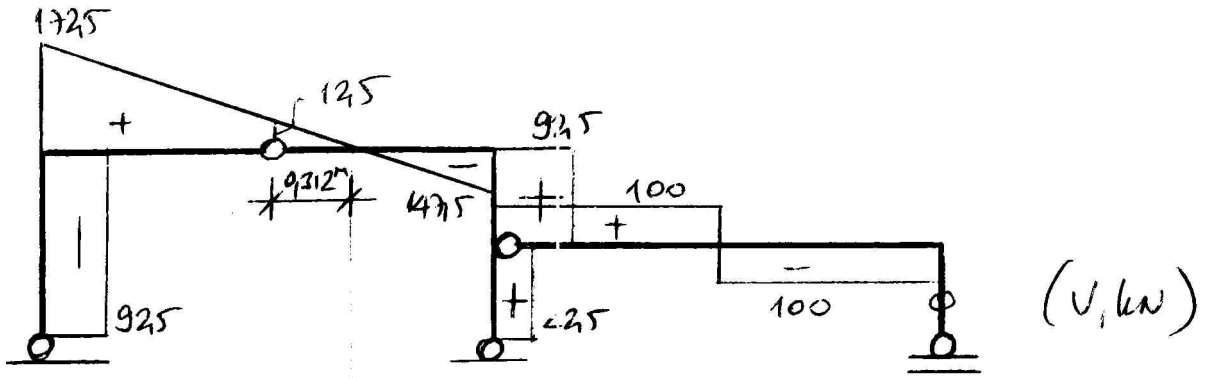


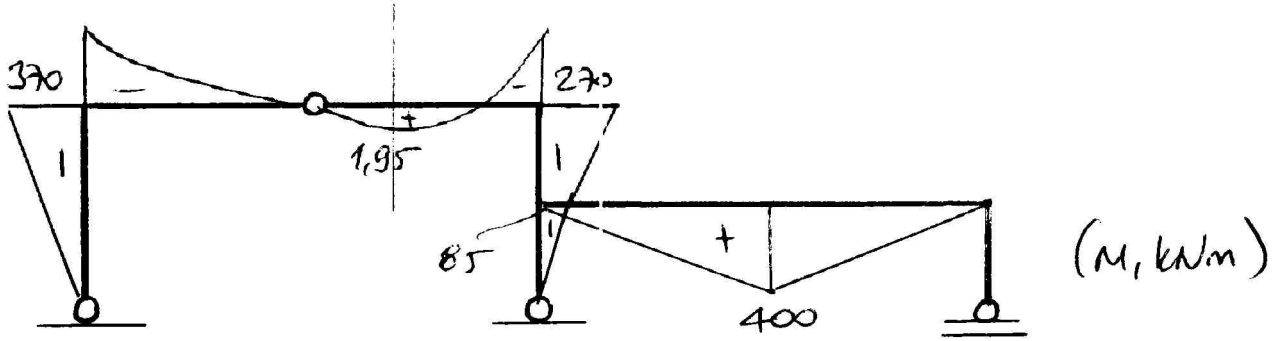
C-f arası

$N_{Cf} = N_{fC} = -100\ kN$

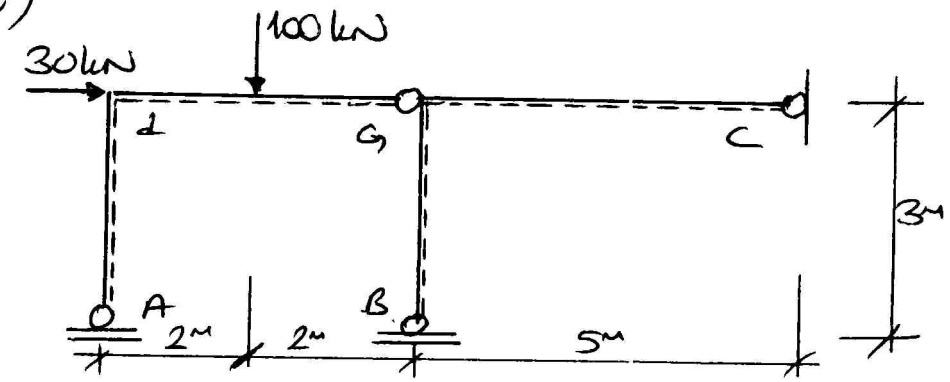
$V_{Cf} = V_{fC} = 0$


925
1475
50
1725
2475
100
(N, kN)

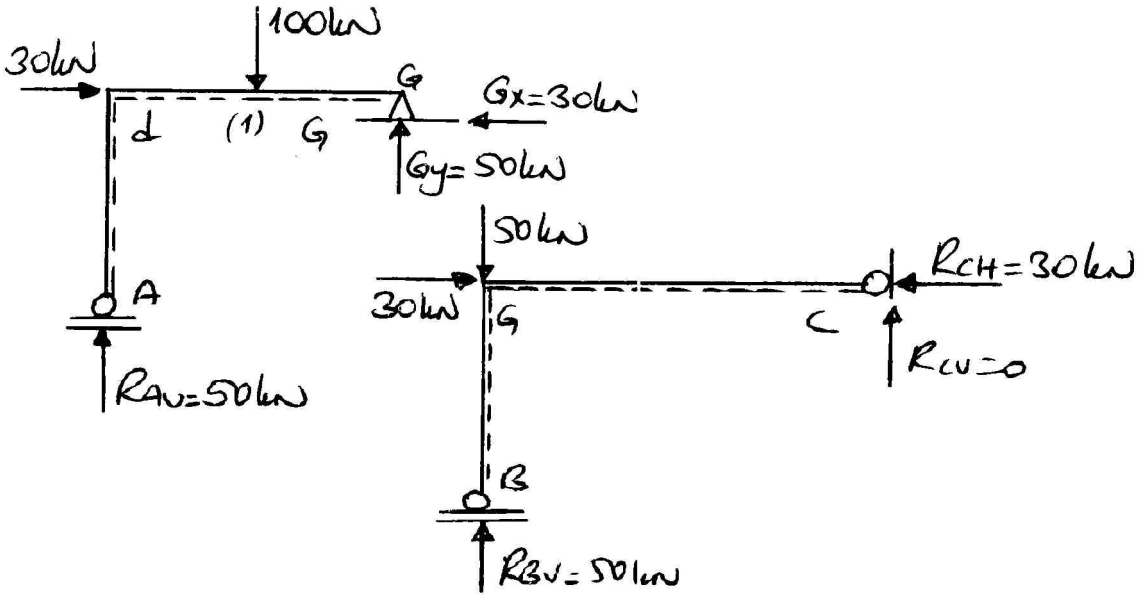

1725
125
925
9,312
475
100
925
225
100
(V, kN)


370
270
1,95
85
400
(M, kNm)

8)



*Mesnet tepkilerinin hesabı için Sayfa 34'e bakınız.



Kesit Tesirleri

A-d Arası

$N_{Ad} = N_{dA} = -50\ kN$

$V_{Ad} = V_{dA} = 0$

$M_{dA} = M_{dG} = 0$

d-G Arası

$N_{dG} = N_{Gd} = -50\ kN$

$V_{d1} = V_{1d} = 50\ kN$

$V_{1G} = V_{G1} = -50\ kN$

$M_1 = 50 \times 2 = 100\ kNm$

B-G arası

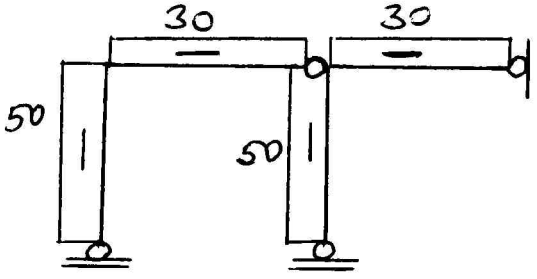
$N_{BG} = N_{GB} = -50\ kN$

$V_{BG} = V_{GB} = 0$

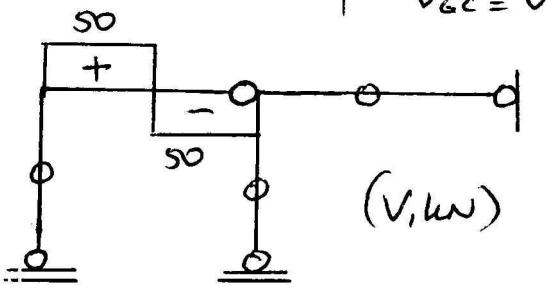
G-C Arası

$N_{GC} = N_{CG} = -50\ kN$

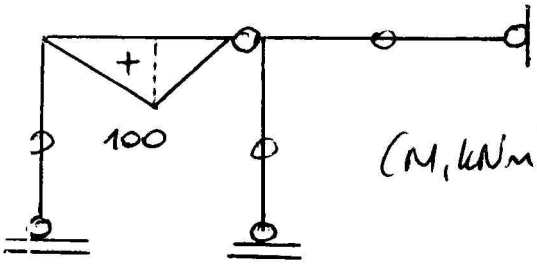
$V_{GC} = V_{CG} = 0$



(N, kN)

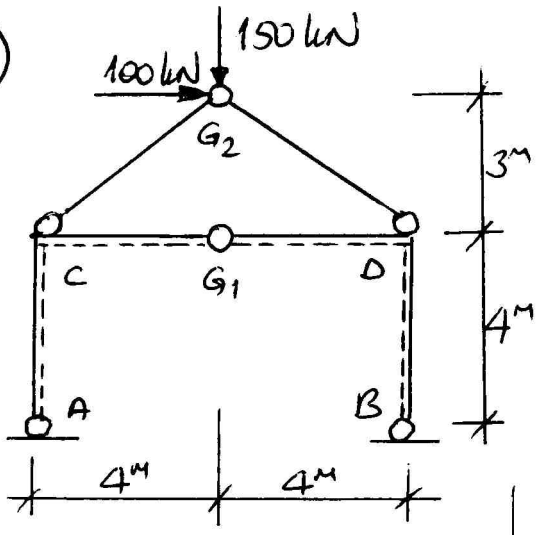


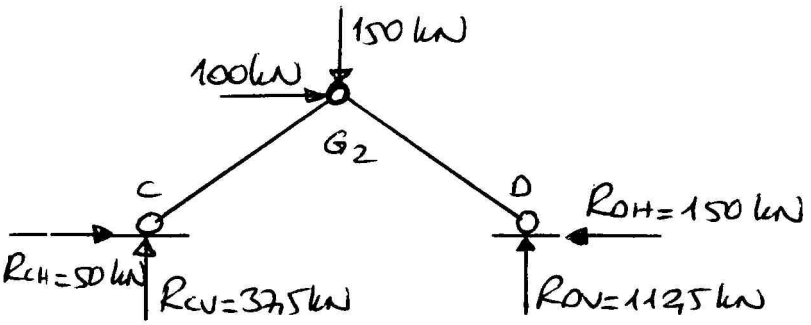
(V, kN)

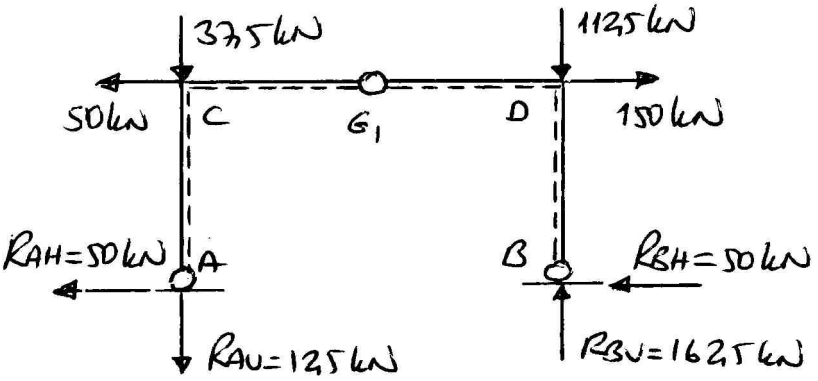


(M, kNm)

9)



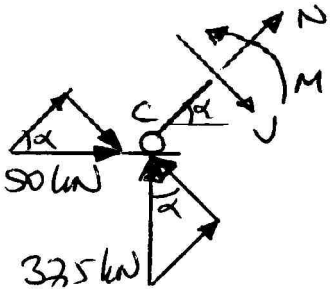




* Mesnet Tepkilerinin hesabı için sayfa 34 ve sayfa 35'e bakınız.

Kesit Tesirlerinin Hesabı

C-G2 arası ≡ G2-D arası



$\sin\alpha = 3/5$

$\cos\alpha = 4/5$

$N_{CG_2} = N_{G_2C} = -50 \cdot \cos\alpha - 375 \cdot \sin\alpha = -625\ kN$

$V_{CG_2} = V_{G_2C} = -50 \cdot \sin\alpha + 375 \cdot \cos\alpha = 0$

A-C Arası

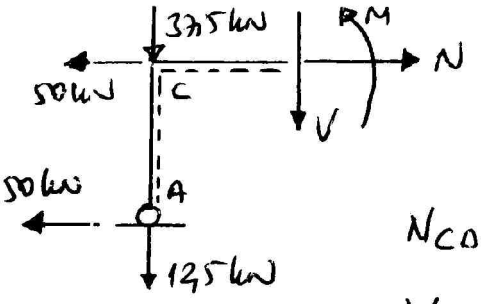
$N_{AC} = N_{CA} = 125\ kN$

$V_{AC} = V_{CA} = 50\ kN$

$M_{CA} = M_{CG_1} = 50 \times 4$

$= 200\ kNm$

C-D arası



$N_{CD} = N_{DC} = 100\ kN$

$V_{CD} = V_{DC} = -375 - 125$

$= -50\ kN$

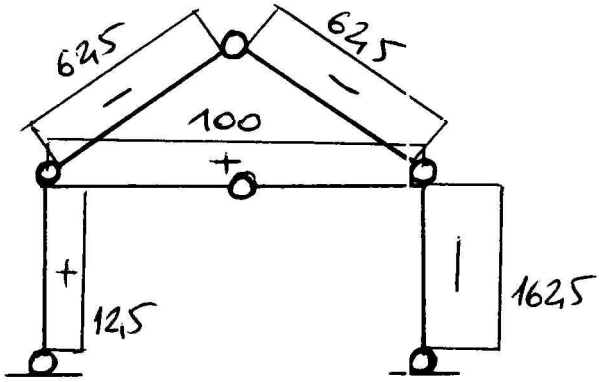
$M_C = 50 \times 4 = 200\ kNm$

$M_D = -50 \times 4 = -200$ "

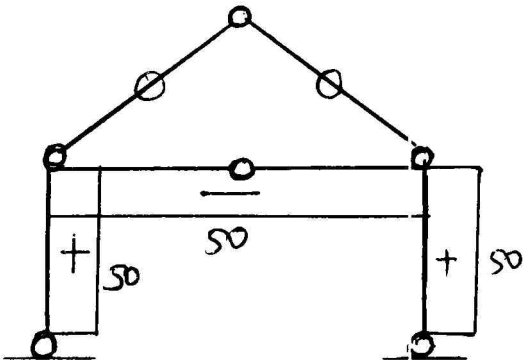
B-D arası

$N_{BD} = N_{DB} = -1625\ kN$

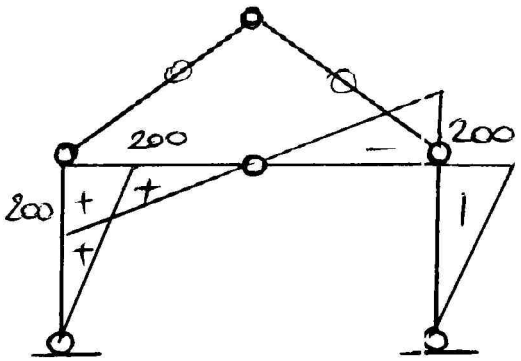
$V_{BD} = V_{DB} = 50\ kN$



(N, kN)

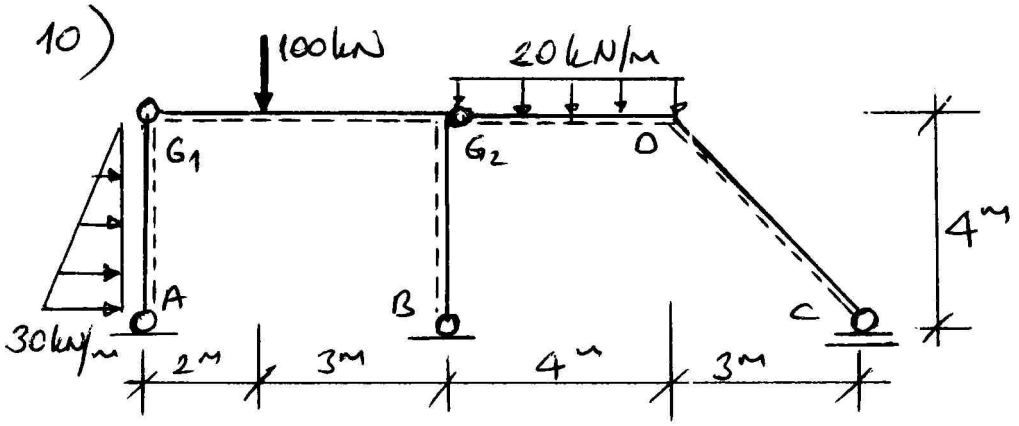


(V, kN)

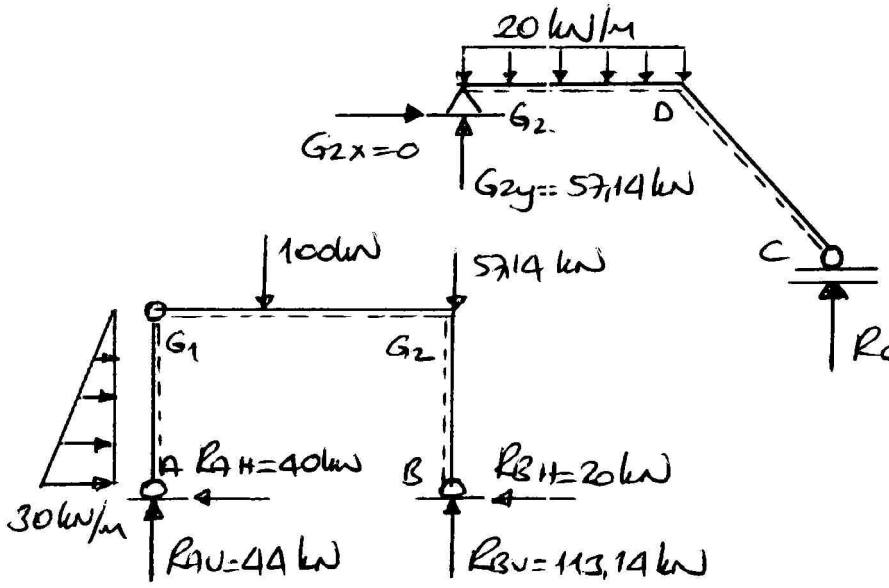


(M, kNm)

10)



Mesnet tepkilerinin hesabı



$\Sigma X=0 \rightarrow G_{2x}=0$

$\Sigma M_C=0 \rightarrow G_{2y}=\frac{20\times4\times5}{7}=57,14\ kN$

$\Sigma M_{G_2}=0 \rightarrow R_{CV}=\frac{20\times4\times2}{7}=22,86\ kN$

$$\Sigma M_B=0 \rightarrow R_{AV}=\frac{-\frac{30\times4}{2}\times\frac{4}{3}+100\times3}{5}=44\ kN$$

$$\Sigma M_A=0 \rightarrow R_{BV}=\frac{\frac{30\times4}{2}\times\frac{4}{3}+100\times2}{5}+57,14=113,14\ kN$$

$$\Sigma Y=0 \rightarrow 100+57,14=44+113,14$$

$$157,14=157,14$$

$$\Sigma X=0 \rightarrow \frac{30\times4}{2}=40+20 \rightsquigarrow 60=60$$

Kesit Tesirlerinin Hesabı

G_2-D arası

$N_{G_2D}=N_{DG_2}=0$

$V_{G_2D}=57,14\ kN$

$V_{DG_2}=57,14-20\times4=-22,86\ kNm$

$M_D=22,86\times3=68,58\ kNm$

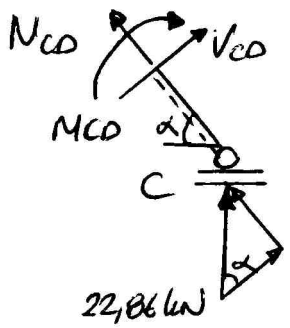
$x_0=\frac{57,14}{20}=2,86\ m$

$max M=\frac{57,14^2}{2\times20}=81,62\ kNm$

veya

$max M=57,14\times2,86-20\times\frac{2,86^2}{2}=81,62\ kNm$

C-D arası

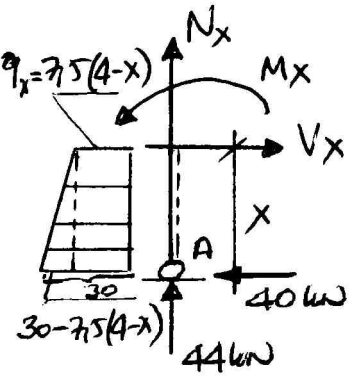


$N_{CD} = N_{DC} = -22,86 \cdot \sin\alpha = -18,29\ kN$

$V_{CD} = V_{DC} = -22,86 \cdot \cos\alpha = -13,72\ kN$

$M_D = 22,86 \times 3 = 68,58\ kNm$

A-G1 Arası



$\frac{30}{4} = \frac{q(x)}{4-x} \rightarrow q_x = \frac{30(4-x)}{4} = 7,5(4-x)$

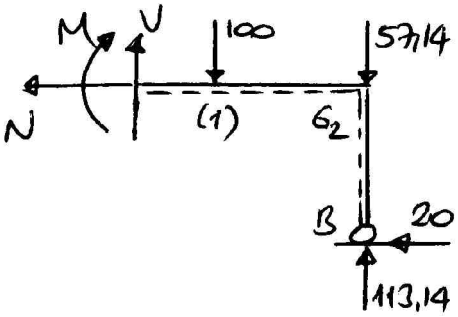
$X/AG_1 = N_{G_1A} = -44\ kN$

$V_x = -\frac{30+7,5(4-x)}{2} \cdot x + 40 = 40 - 3,75x^2$

$M_x = 40x - 7,5(4-x) \cdot x \cdot \frac{x}{2} - \frac{(30-7,5(4-x)) \cdot x}{2} \cdot \frac{2}{3} \cdot x$

$max\,M \longrightarrow /x=0 \longrightarrow x_0 = 3,27\,m \longrightarrow max M = 14,11\ kNm.$

G1 - G2 arası



$N_{G_1G_2} = N_{G_2G_1} = -20\ kN$

$V_{G_1G_2} = 44\ kN \quad (100 + 57,14 - 113,14 = 44\ kN)$

$V_{G_2G_1} = 44 - 100 = -56\ kN$

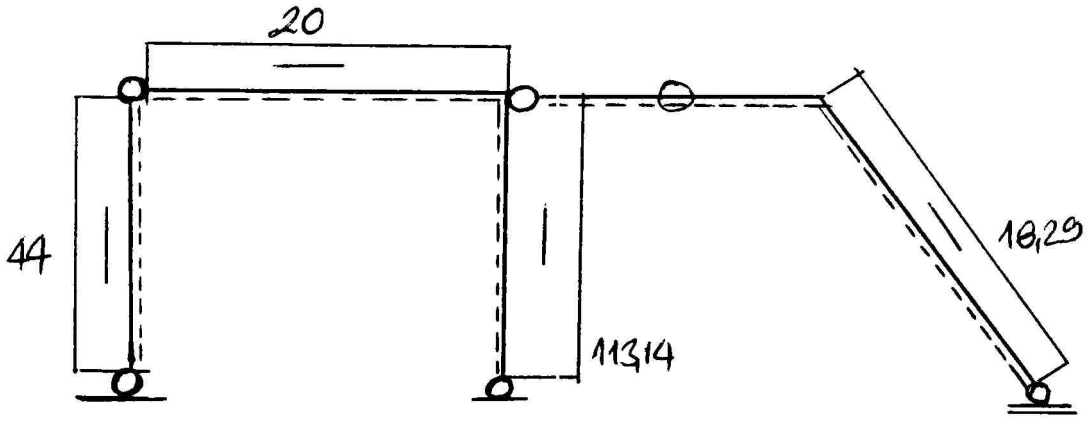
$max M = M_1 = 113,14 \times 3 - 20 \times 4 - 57,14 \times 3 = 88\ kNm$

B - G2 arası

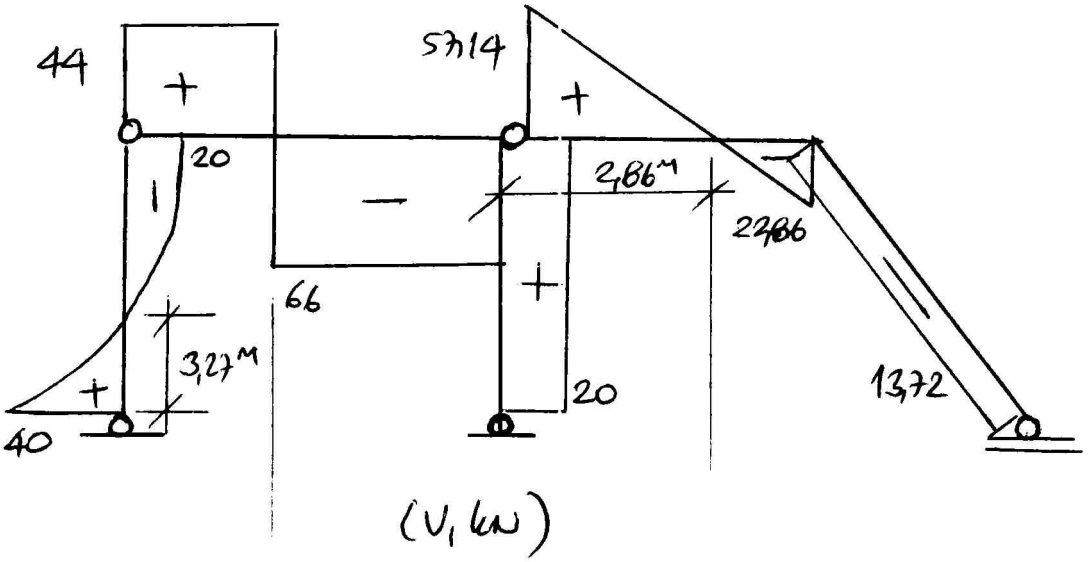
$N_{BG_2} = N_{G_2B} = -113,14\ kN$

$V_{BG_2} = V_{G_2B} = 20\ kN$

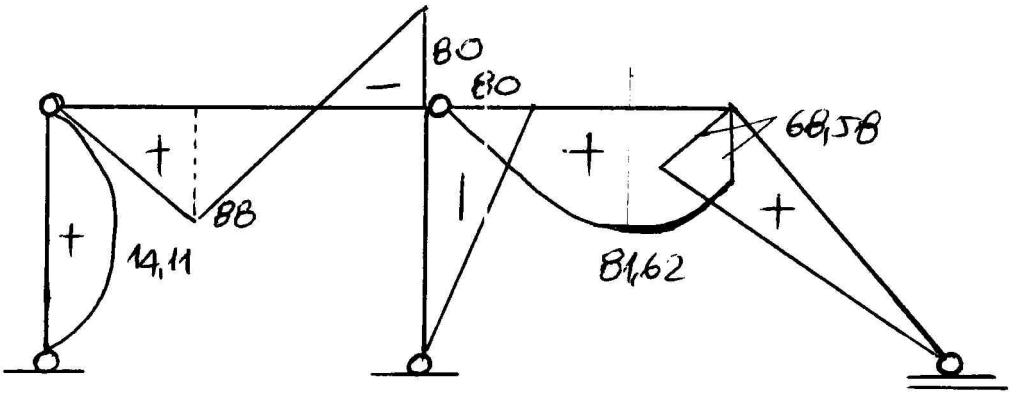
$M_{G_2} = -20 \times 4 = -80\ kNm$


20
44
18,29
113,14

(N, kN)

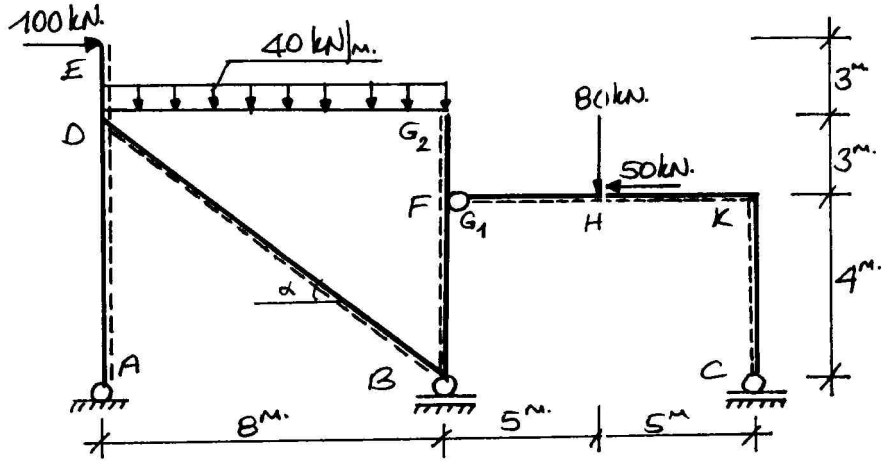

44
57,14
20
2,86 m
22,86
66
3,27 m
40
20
13,72

(V, kN)


80
80
68,58
88
14,11
81,62

(M, kNm)

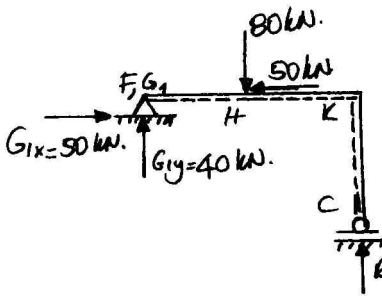
11)



$l_{DB} = \sqrt{64+49} = 10{,}63$ m.

$\cos\alpha = 8/10{,}63$

$\sin\alpha = 7/10{,}63$



$\Sigma X=0 \rightarrow G_{1X}=50$ kN

$\Sigma M_{G_1}=0 \rightarrow R_{CV} = \frac{80.5}{10} = 40$ kN.

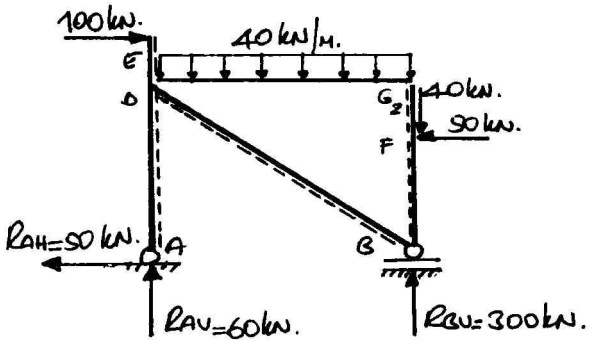
$\Sigma M_C=0 \rightarrow G_{1y} = \frac{80.5}{10} = 40$ kN

$N_{G,H} = N_{HG_1} = -50$ kN. ; $V_{G,H} = V_{HG_1} = 40$ kN.

$N_{HK} = N_{KH} = 0$; $V_{HK} = V_{KH} = -40$ kN.

$N_{CK} = N_{KC} = -40$ kN. ; $V_{CK} = V_{KC} = 0$

$M_H = 40.5 = 200$ kNm, $M_K = 0$



Mesnet Tepkileri

$\Sigma X=0 \rightarrow R_{AH} = 50$ kN

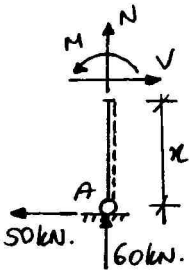
$\Sigma N_B=0 \rightarrow R_{AV} = \frac{-100.10+40.8.4+50.4}{8} = 60$ kN.

$\Sigma M_A=0 \rightarrow R_{BV} = \frac{100.10+40.8.4-50.4}{8} + 40 = 300$ kN.

$\Sigma Y=0 \rightarrow 60+300 = 40.8+40 \Rightarrow 360=360$ ✓

Kesit Tesirleri

A-O arası

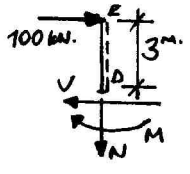


$N_{AO} = N_{OA} = -60$ kN.

$V_{AO} = V_{OA} = 50$ kN.

$M_{OA} = 50.7 = 350$ kNm.

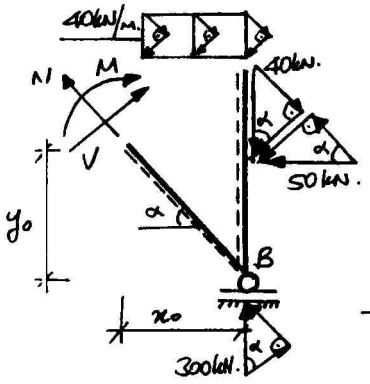
E-O arası



$N_{ED} = N_{DE} = 0$

$V_{ED} = V_{DE} = 100$ kN.

$M_{DE} = -100.3 = -300$ kNm.



$V_x = -300.\cos\alpha + 50.\sin\alpha + 40.\cos\alpha + 40.\cos\alpha.x$

$V_x = -162{,}75 + 30{,}10\,x$

$N_x = -300.\sin\alpha - 50.\cos\alpha + 40.\sin\alpha + 40.\sin\alpha.x$

$N_x = -208{,}84 + 26{,}34\,x$

$x=0 \begin{cases} N_{BO} = -208{,}84 \text{ kN.} \\ V_{BO} = -162{,}75 \text{ kN.} \end{cases}$

$x=8^m \begin{cases} N_{OB} = 1{,}88 \text{ kN.} \\ V_{OB} = 78{,}05 \text{ kN.} \end{cases}$

$\max M \rightarrow V_x = 0 \Rightarrow x_0 = \frac{162{,}75}{30{,}10} = 5{,}41$ m. , $\operatorname{tg}\alpha = \frac{y_0}{x_0}$

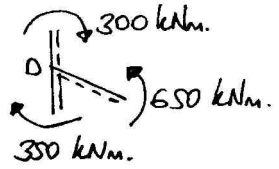
$y_0 = 4{,}73^m$.

$\max M = 300.5{,}41 - 40.5{,}41 - 50.0{,}7\ldots - 40.\frac{5{,}41^2}{2} = 784{,}74\ kNm.$

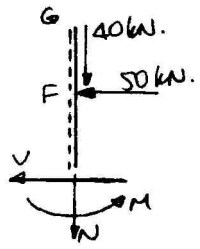
$M_B = -50.4 = -200\ kNm$

$M_{OB} = 300.8 - 40.8 - 50.3 - 40.8.4 = 650\ kNm.$

$M_{OA} = 50.7 = 350\ kNm$, $M_{OE} = -100.3 = -300\ kNm.$



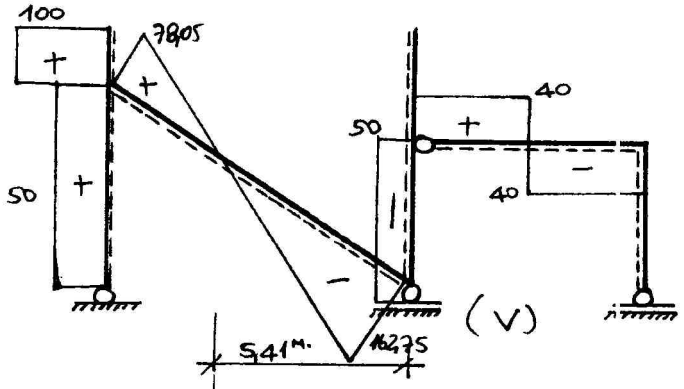
G-B arası



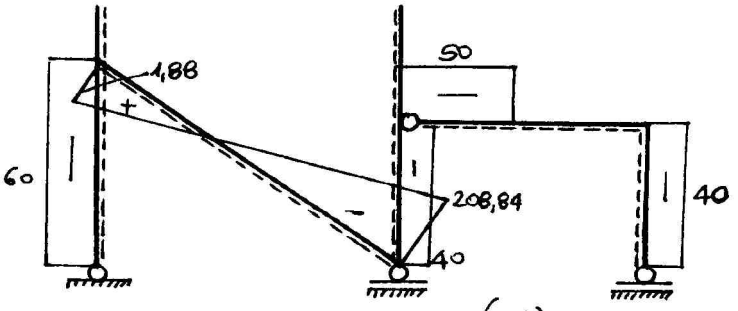
$N_{GF} = N_{FG} = 0$ $\quad N_{FB} = N_{BF} = -40\ kN.$

$V_{GF} = V_{FG} = 0$ $\quad V_{FB} = V_{BF} = -50\ kN.$

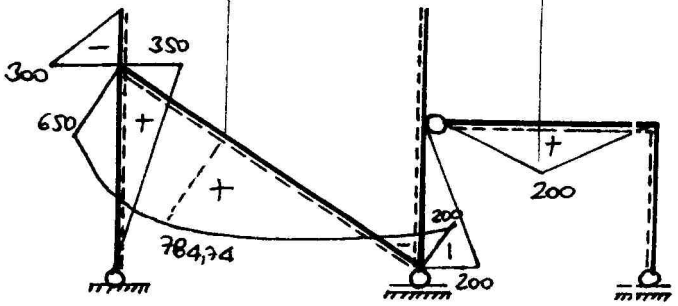
$M_F = 0$ $\quad M_{BF} = -50.4 = -200\ kNm.$



(V)

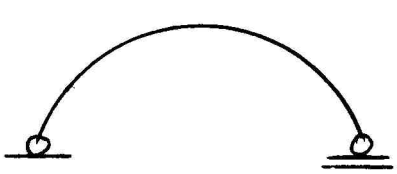


(N)

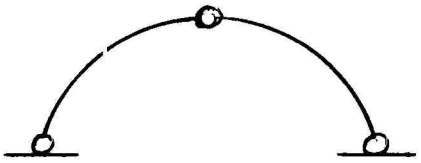


(M)

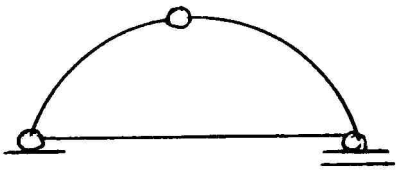
IV_7- KEMERLER



Basit Kemer

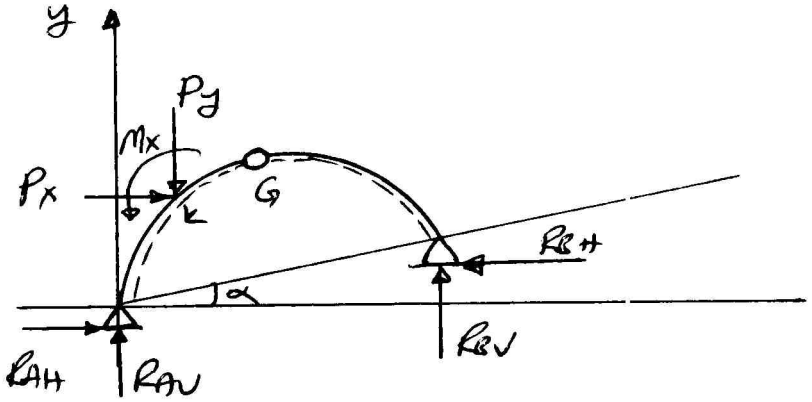


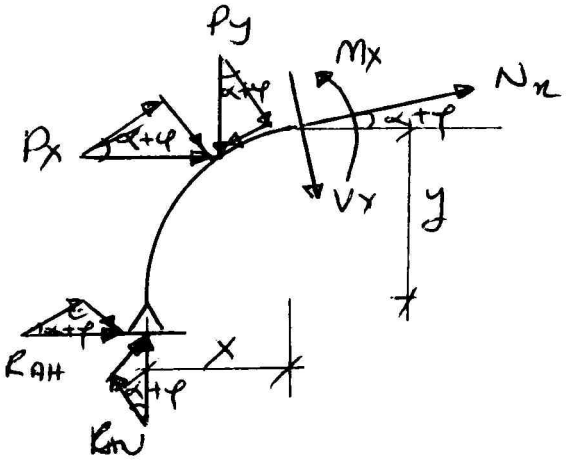
Üç mafsallı Kemer



Gergili Kemer

Genel Çözüm Durumu





Mesnet tepkileri

1) $\Sigma M_A = 0 \rightarrow f(R_{BH}, R_{BV})$

2) $\Sigma M_{G_{sag}} = 0 \rightarrow f(R_{BH}, R_{BV})$

$R_{BV} = !$

$R_{BH} = !$

3) $\Sigma M_B = 0 \rightarrow f(R_{AV}, R_{AH})$

4) $\Sigma M_{G_{sol}} = 0 \rightarrow f(R_{AV}, R_{AH})$

$R_{AV} = !$

$R_{AH} = !$

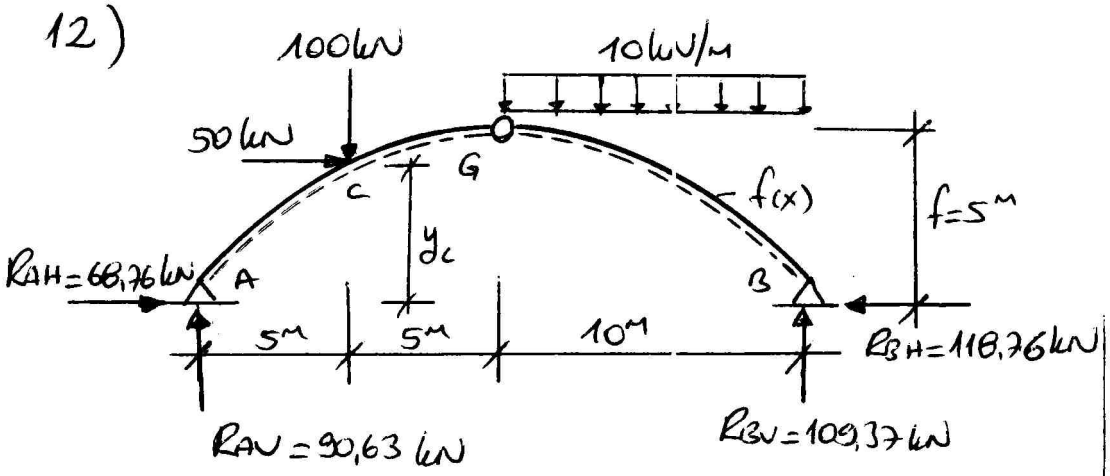
Kontrol : $\Sigma X = 0$, $\Sigma Y = 0$

$$M_x = R_{AV} \cdot x - R_{AH} \cdot y - P_x \cdot a_y - P_y \cdot a_x$$

$$V_x = R_{AV} \cos(\alpha+\varphi) - R_{AH} \cdot \sin(\alpha+\varphi) - P_x \cdot \sin(\alpha+\varphi) - P_y \cos(\alpha+\varphi)$$

$$N_x = -R_{AV} \cdot \sin(\alpha+\varphi) - R_{AH} \cdot \cos(\alpha+\varphi) - P_x \cdot \cos(\alpha+\varphi) + P_y \cdot \sin(\alpha+\varphi)$$

12)



$f(x) = Ax^2 + Bx + C$

$x=0 \rightarrow f(x)=0 \rightarrow C=0$

$x = L/2 \rightarrow f(x)=f \rightarrow f = \frac{AL^2}{4} + \frac{BL}{2}$

$x=L \rightarrow f(x)=0 \rightarrow 0 = AL^2 + BL$

$B = -AL$

$f = \frac{AL^2}{4} - \frac{AL^2}{2} = -\frac{AL^2}{4}$

$B = \frac{4f}{L}$, $A = -\frac{4f}{L^2}$

$f(x) = -\frac{4f}{L^2}x^2 + \frac{4f}{L}x$

$f(x) = \frac{4f}{L^2}x(L-x)$

$y_c = \frac{4\times5}{20^2} \cdot 5(20-5)$

$= 3{,}75\,m$

Mesnet tepkilerinin hesabı

$\Sigma M_B = 0 \rightarrow R_{AV} = \frac{10\times10\times5 + 100\times15 - 50\times3{,}75}{20} = 90{,}63\,kN$

$\Sigma M_{G_{sol}} = 0 \rightarrow R_{AH} = \frac{90{,}63\times10 - 100\times5 - 50\times1{,}25}{5} = 68{,}76\,kN$

$\Sigma M_A = 0 \rightarrow R_{BV} = \frac{10\times10\times15 + 100\times5 + 50\times3{,}75}{20} = 109{,}37\,kN$

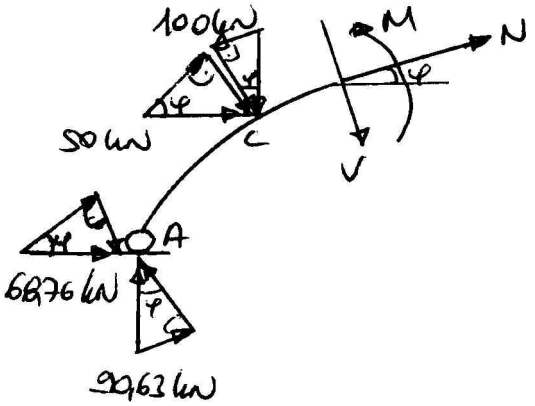
$\Sigma M_{G_{sağ}} = 0 \rightarrow R_{BH} = \frac{109{,}37\times10 - 10\times10\times5}{5} = 118{,}76\,kN$

$\Sigma X = 0 \rightarrow 50 + 68{,}76 = 118{,}76 \rightarrow 118{,}76 = 118{,}76$

$\Sigma Y = 0 \rightarrow 100 + 10\times10 = 90{,}63 + 109{,}37 \rightarrow 200 = 200$

Kesit Tesirlerinin Hesabı

A-G arası



$f'(x) = tg\varphi = -\frac{8f}{L^2}x + \frac{4f}{L}$

$tg\varphi_A = \frac{4f}{L} = \frac{4\times5}{20} = 1$ ($x=0$) → $\sin\varphi = 0{,}707$, $\cos\varphi = 0{,}707$

$\varphi_A = 45°$

$N_{AC} = -68{,}76\cdot\cos\varphi_A - 90{,}63\cdot\sin\varphi_A$

$= -112{,}68\,kN$

$V_{AC} = -68{,}76\cdot\sin\varphi_A + 90{,}63\cdot\cos\varphi_A$

$= 15{,}46\,kN$

$$tg\varphi_c = -\frac{8f}{L^2}x + \frac{4f}{L} = -\frac{8\times5}{20^2}\cdot5 + \frac{4\times5}{20} = 0{,}5$$

$$\varphi_c = 26{,}56 \longrightarrow \sin\varphi_c = 0{,}447,\ \cos\varphi_c = 0{,}894$$

$$N_{CA} = -68{,}76\cdot\cos\varphi_c - 90{,}63\cdot\sin\varphi_c = -101{,}98\ kN$$

$$V_{CA} = -68{,}76\cdot\sin\varphi_c + 90{,}63\cdot\cos\varphi_c = 50{,}29\ kN$$

$$N_{CG} = -101{,}98 - 50\cdot\cos\varphi_c + 100\cdot\sin\varphi_c = -101{,}98\ kN$$

$$V_{CG} = 50{,}29 - 50\cdot\sin\varphi_c - 100\cdot\cos\varphi_c = -61{,}46\ kN$$

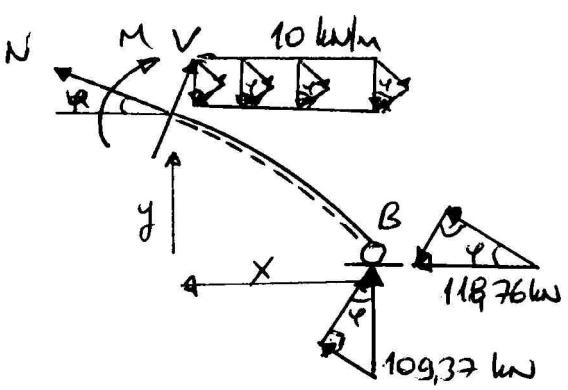
$$M_C = 90{,}63\times5 - 68{,}76\times3{,}75 = 195{,}3\ kNm$$

G noktası

$$N_G = -68{,}76 - 50 = -118{,}76\ kN$$

$$V_G = 90{,}63 - 100 = -9{,}37\ kN$$

B-G arası



$$tg\varphi_B = 1 \longrightarrow \varphi_B = 45° \longrightarrow \sin\varphi_B = \cos\varphi_B = 0{,}707$$

$$N_{BG} = -118{,}76\cdot\cos\varphi_B - 109{,}37\cdot\sin\varphi_B = -161{,}28\ kN$$

$$V_{BG} = 118{,}76\cdot\sin\varphi_B - 109{,}37\cdot\cos\varphi_B = 6{,}64\ kN$$

$$V_x = +118{,}76\cdot\sin\varphi - 109{,}37\cdot\cos\varphi + 10\cdot\cos\varphi\cdot x$$

$$N_x = 109{,}37\cdot x - 118{,}76\cdot y - 10\cdot x\cdot\frac{x}{2}$$

$$tg\varphi = f'(x) = -\frac{8\times5}{20^2}x + \frac{4\times5}{20}$$

$$= 1 - 0{,}1x$$

$$x = 5^m \longrightarrow y = 3{,}75^m$$

$$max M \rightarrow V_x = 0 \rightarrow 118{,}76\cdot\frac{\sin\varphi}{\cos\varphi} - 109{,}37\frac{\cos\varphi}{\cos\varphi} + 10\cdot\frac{\cos\varphi}{\cos\varphi}\cdot x = 0$$

$$118{,}76\ tg\varphi - 109{,}37 + 10x = 0$$

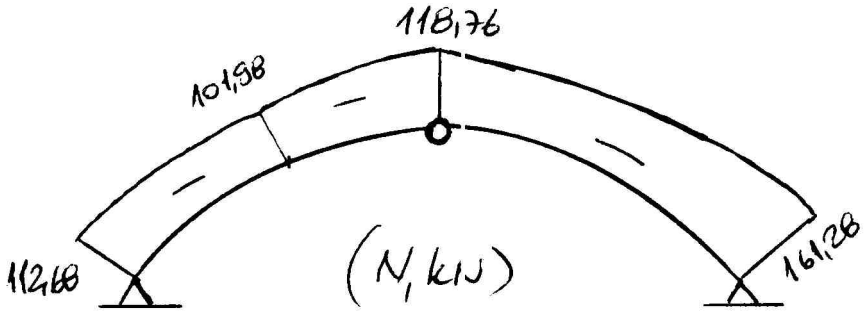
$$118{,}76\,(1 - 0{,}1x) - 109{,}37 + 10x = 0$$

$$118{,}76 - 11{,}876x - 109{,}37 + 10x = 0$$

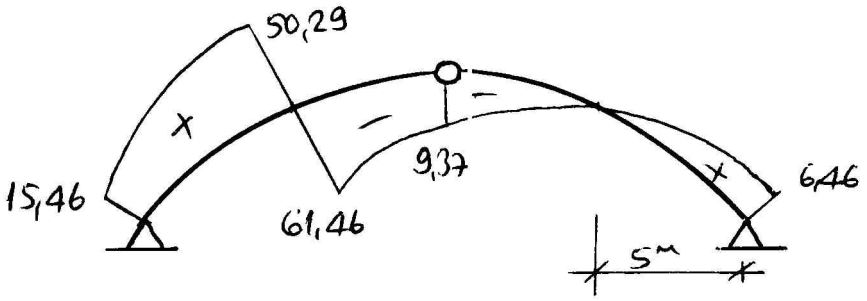
$$9{,}39 - 1{,}876x = 0 \rightarrow x = 5^m$$

$$Max M = 109{,}37\cdot5 - 118{,}76\cdot3{,}75 - 10\frac{5^2}{2}$$

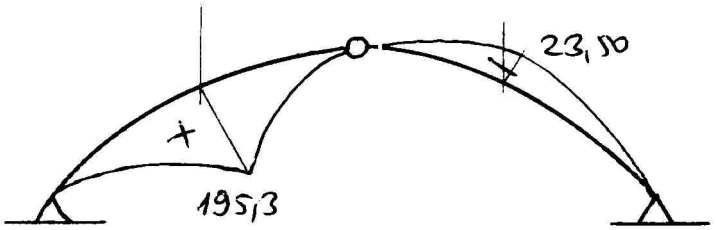
$$= -23{,}50\ kNm$$



(N, kN)

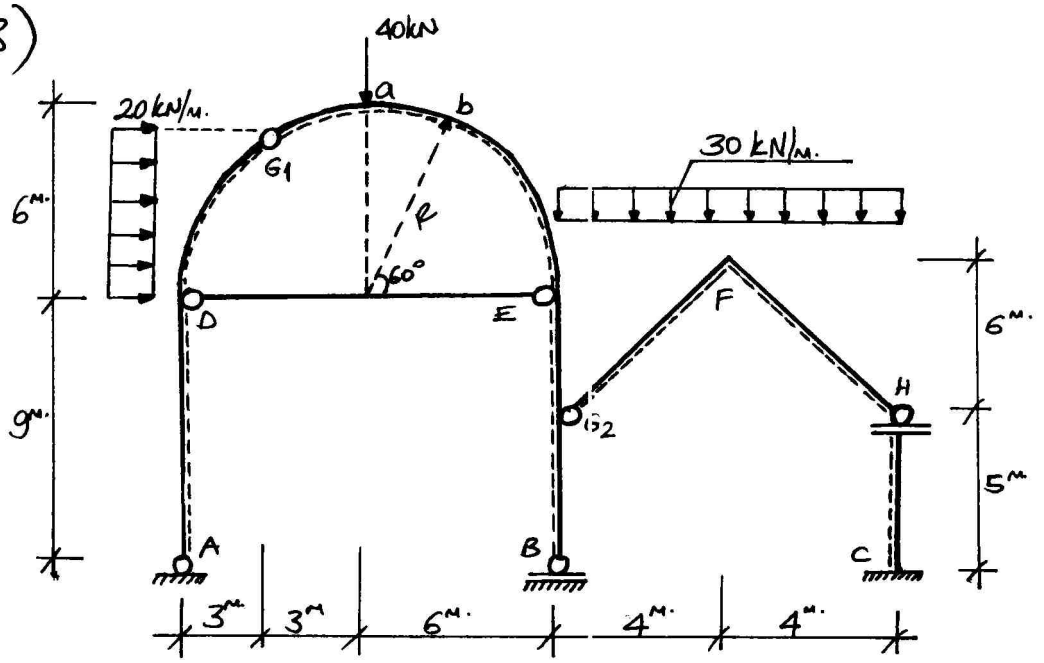


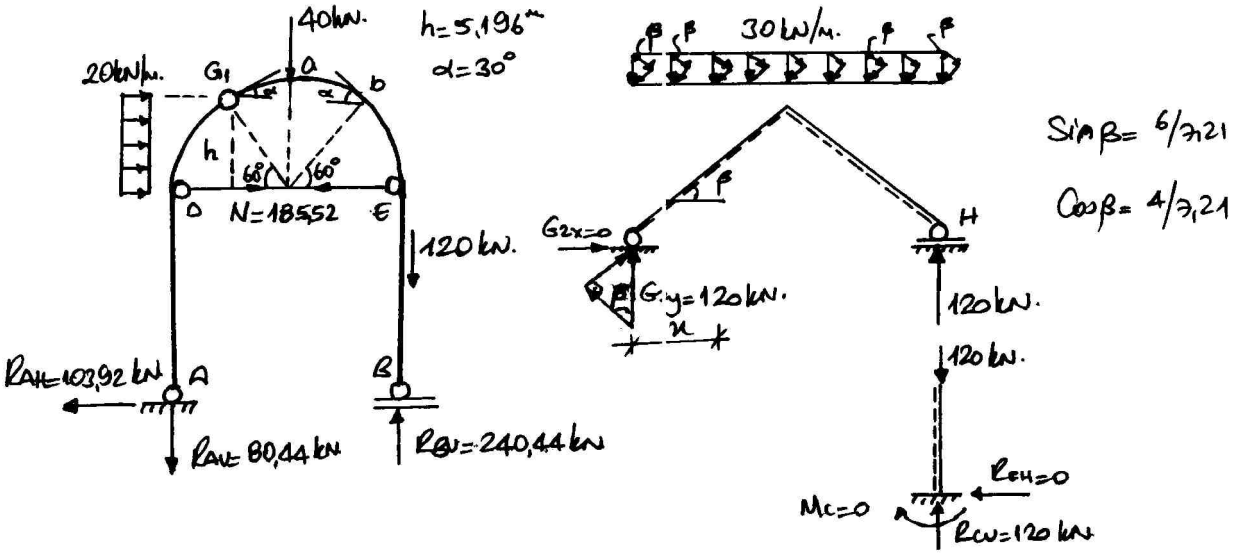
(V, kN)



(M, kNm)

13)





MESNET TEPKİLERİ

$\Sigma M_H = 0 \Rightarrow G_{2y} = \frac{30.8.4}{8} = 120\ kN.$

$\Sigma M_{G_2} = 0 \Rightarrow R_{HV} = 120\ kN.$

$\Sigma M_B = 0 \Rightarrow R_{AV} = \frac{20.5,196.11,598 - 40.6}{12} = 80,44\ kN.$

$\Sigma M_A = 0 \Rightarrow R_{BV} = \frac{20.5,196.11,598 + 40.6}{12} + 120 = 240,44\ kN.$

$\Sigma X = 0 \Rightarrow R_{AH} = 20.5,196 = 103,92\ kN.$

$\Sigma Y = 0 \Rightarrow 240,44 - 80,44 = 40 + 120$

$160 = 160$ ✓

KESİT TESİRLERİ

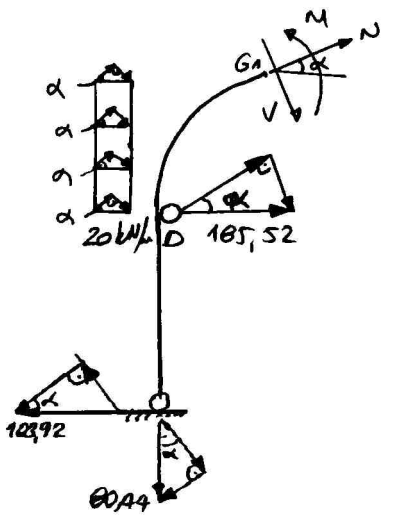
G_2 – H arası

$N_x = 30.\sin\beta.x - 120.\sin\beta$ $\quad x=0 \Rightarrow \begin{cases} N_{G_2F} = -100\ kN. \\ V_{G_2F} = 66,57\ kN. \end{cases}$ $\quad x = 4^m \Rightarrow \begin{cases} N_{FG_2} = 0 \\ V_{FG_2} = 0 \end{cases}$

$V_x = 120.\cos\beta - 30.\cos\beta.x$

$M_F = 120.4 - 30.4.2 = 240\ kNm.$

A – G_1 arası



$N_{AD} = N_{DA} = +80,44\ kN.$

$V_{AD} = V_{DA} = 103,92\ kN.$

$M_D = 103,92.9 = 935,28\ kNm.$

$N_G = 80,44.\sin\alpha + 103,92.\cos\alpha - 20.5,196.\cos\alpha - 185,52.\cos\alpha$

$N_G = -120,14\ kN.$

$V_G = -80,14.\cos\alpha + 103,92.\sin\alpha - 20.5,196.\sin\alpha - 185,52.\sin\alpha$

$V_G = -162,59\ kN.$

$M_G = 0$

$V_{aG} = -80,44\ kN, \quad V_{Ga} = -80,44 - 40 = -120,44\ kN.$

$N_a = 103,92 - 20.5,196 - 185,52 = -185,52\ kN.$

$M_a = -80,44.6 + 103,92.15 - 20.5,196.(5,196/2 + 0,804) - 185,52.6$

$M_a = -3901,50\ kNm.$

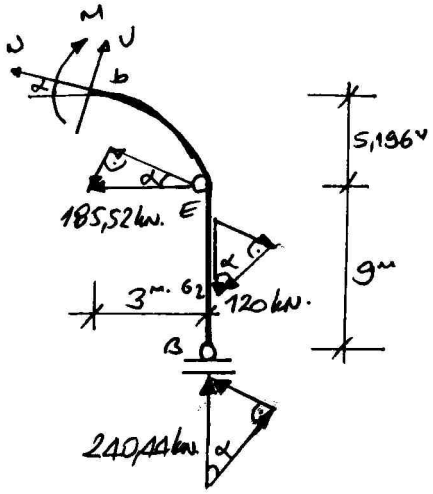
$N_{BG_2} = N_{G_2B} = -240,44\ kN.$

$N_{G_2E} = N_{EG_1} = -240,44 + 120 = -120,44\ kN.$

$V_{BE} = V_{EB} = 0$

$M_B = M_E = 0$

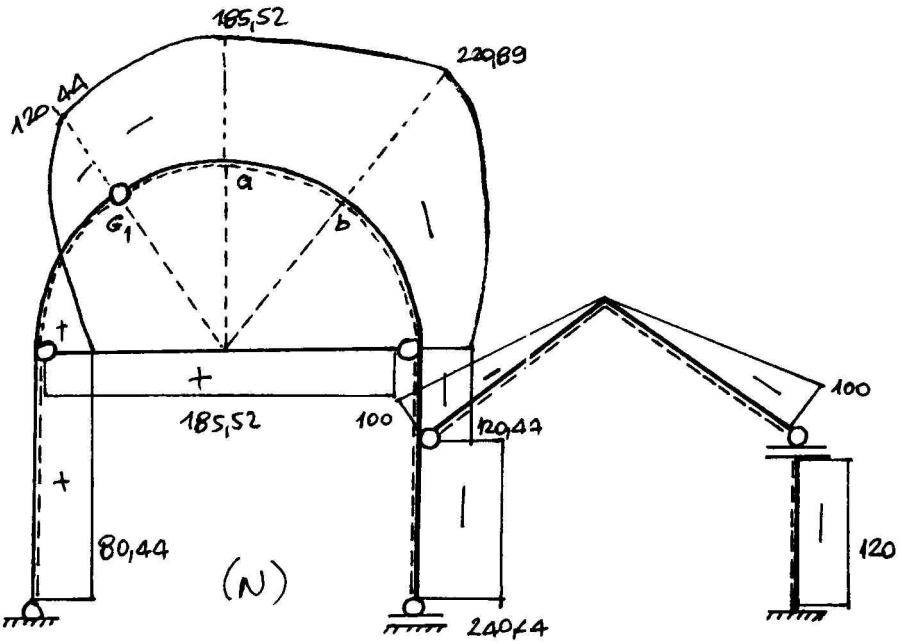
B – b arası

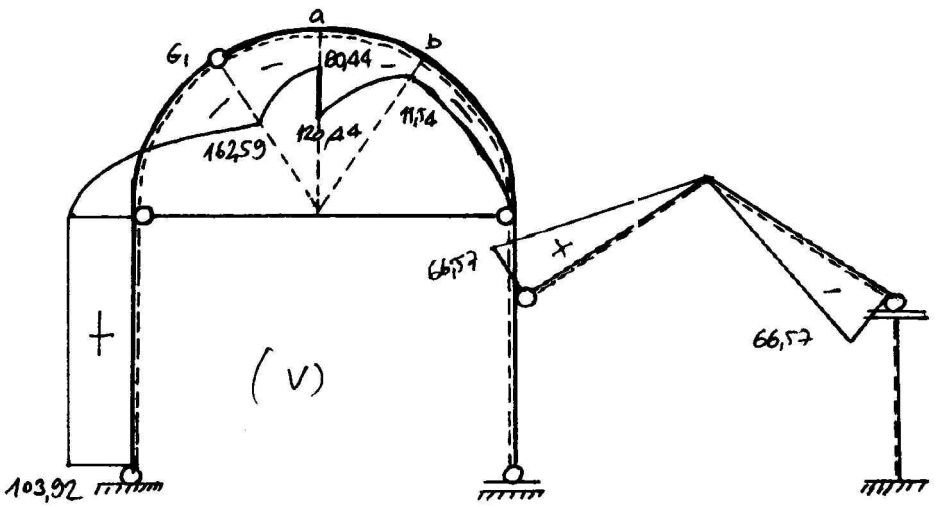


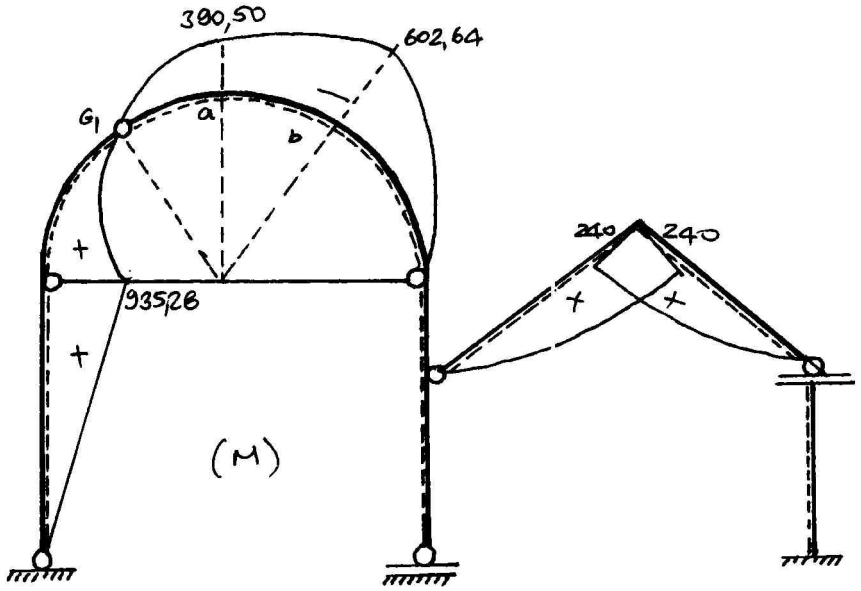
$N_b = -240,44.\sin\alpha + 120.\sin\alpha - 185,52.\cos\alpha = -220,89\ kN.$

$V_b = (-240,44 + 120)\cos\alpha + 185,52.\sin\alpha = -11,54\ kN.$

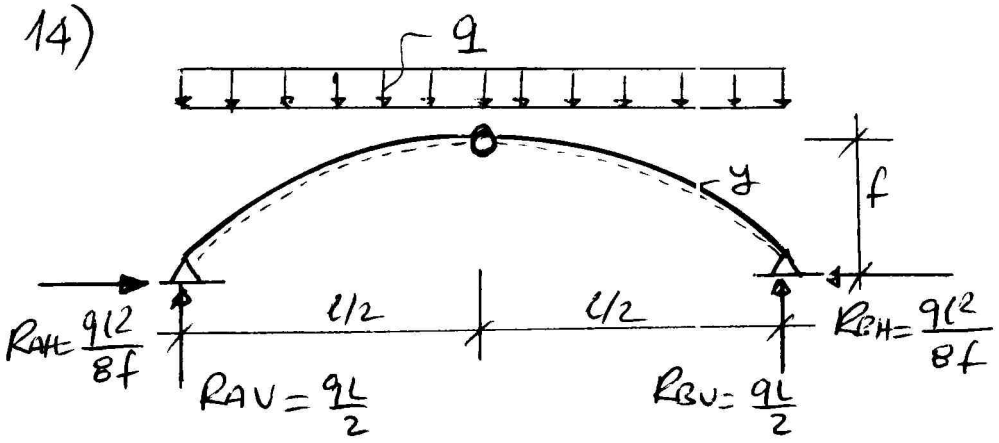
$M_b = 240,41.3 - 120.3 - 185,52.5,196 = -602,64\ kNm.$


185,52
229,89
120,44
a
b
G1
185,52
100
120,44
100
80,44
120
240,44
(N)


a
b
G1
80,44
162,59
120,44
11,54
66,57
66,57
103,92
(V)


380,50
602,64
G1
a
b
240
240
93,28
(M)

14)



Parabolik Kemer

$$y=\frac{4f}{l^2}x(l-x)$$

Mesnet Tepkileri

$$R_{AV}=R_{BV}=\frac{ql}{2}$$

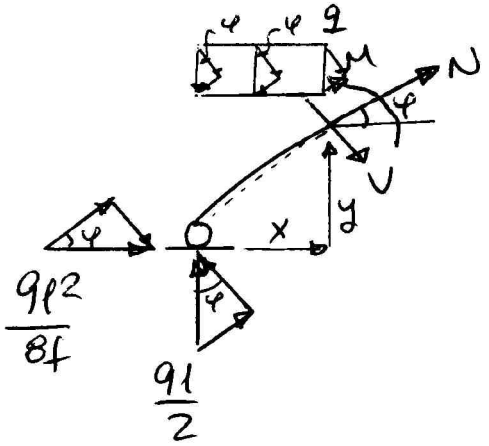
$$R_{AH}=R_{BH}=\frac{ql/2\cdot l/2-q\cdot l/2\cdot l/4}{f}$$

$$=\frac{ql^2}{8f}$$

$$y=-\frac{4f}{l^2}x^2+\frac{4f}{l}x$$

$$tg\varphi=y'=-\frac{8f}{l^2}x+\frac{4f}{l}$$

Kesit Tesirlerinin Hesabı



$$N_x=-\frac{ql^2}{8f}\cos\varphi-\frac{ql}{2}\sin\varphi+q\cdot x\cdot\sin\varphi$$

$$V_x=-\frac{ql^2}{8f}\cdot\sin\varphi+\frac{ql}{2}\cos\varphi-q\cdot x\cdot\cos\varphi$$

$$M_x=-\frac{ql^2}{8f}\cdot y+\frac{ql}{2}\cdot x-\frac{qx^2}{2}$$

$$V_x=-\frac{ql^2}{8f}\frac{\sin\varphi}{\cos\varphi}+\frac{ql}{2}-qx$$

$$=-\frac{ql^2}{8f}\left(-\frac{8f}{l^2}x+\frac{4f}{l}\right)+\frac{ql}{2}-qx$$

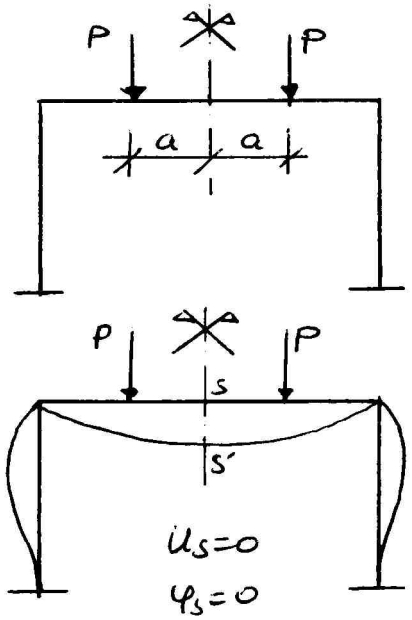
$$=qx-\frac{ql}{2}+\frac{ql}{2}-qx=0$$

$$M_x=-\frac{ql^2}{8f}\left(-\frac{4f}{l^2}x^2+\frac{4f}{l}x\right)+\frac{ql}{2}x-\frac{qx^2}{2}$$

$$=\frac{qx^2}{2}-\frac{ql}{2}x+\frac{ql}{2}x-\frac{qx^2}{2}=0$$

V. SİMETRİK VE ANTİMETRİK SİSTEMLER

Sistem ve yüklemesi Simetrik Sistemler



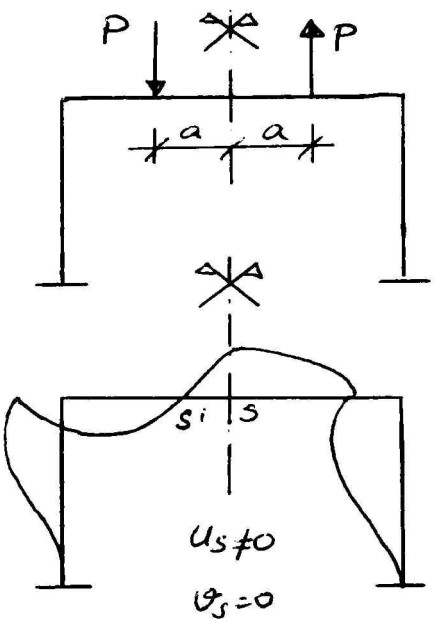
$u_S = 0$

$\varphi_S = 0$

$\vartheta_S \neq 0$

$$\begin{bmatrix} M_M = M_{M'} \\ N_M = N_{M'} \\ V_M = -V_{M'} \end{bmatrix}$$

Sistem simetrik yüklü Antimetrik Sistemler



$u_S \neq 0$

$\vartheta_S = 0$

$\varphi_S \neq 0$

$$\begin{bmatrix} M_M = -M_{M'} \\ N_M = -N_{M'} \\ V_M = V_{M'} \end{bmatrix}$$

Simetrik Sistem

Simetrik yük

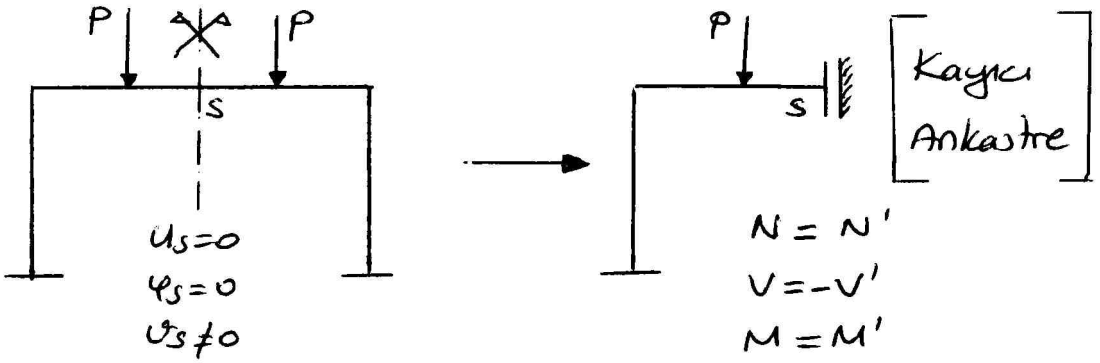
Simetrik	Antimetrik
Tepkiler	V
M	
N	

Antimetrik yük

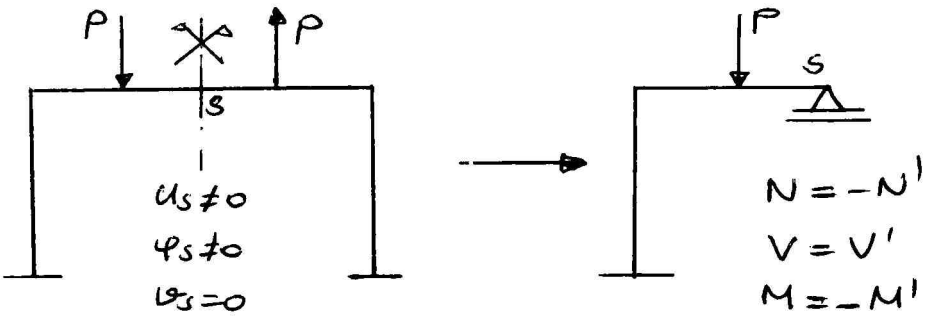
Simetrik	Antimetrik
V	Tepkiler
	M
	N

a) Simetri Ekseni Açıklık Ortasından Geçen Sistemler

Simetrik yük

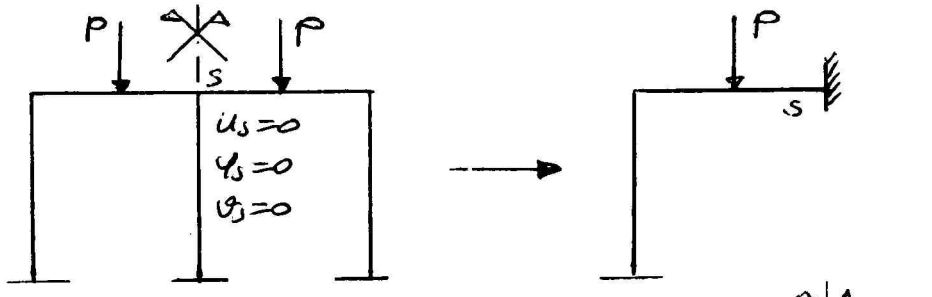


Antimetrik yük

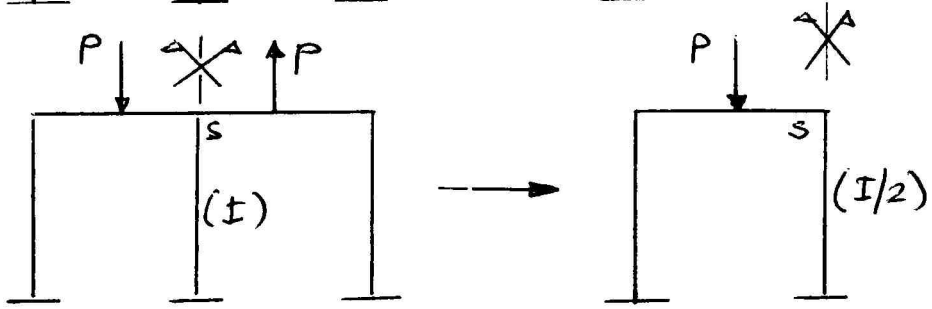


b) Simetri Ekseni Kolondan Geçen Sistemler

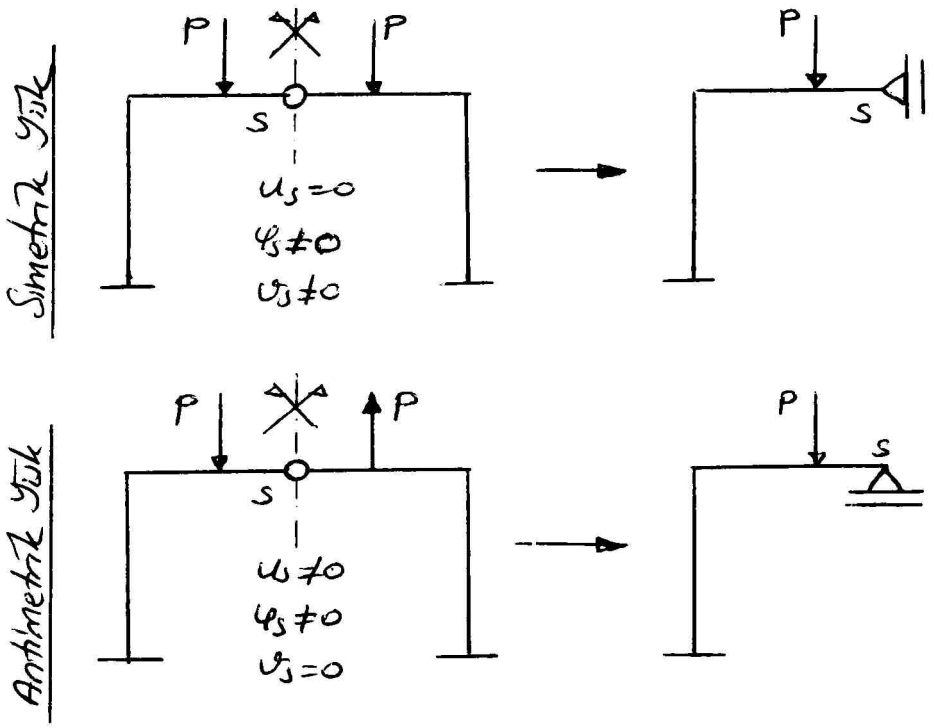
Simetrik yük



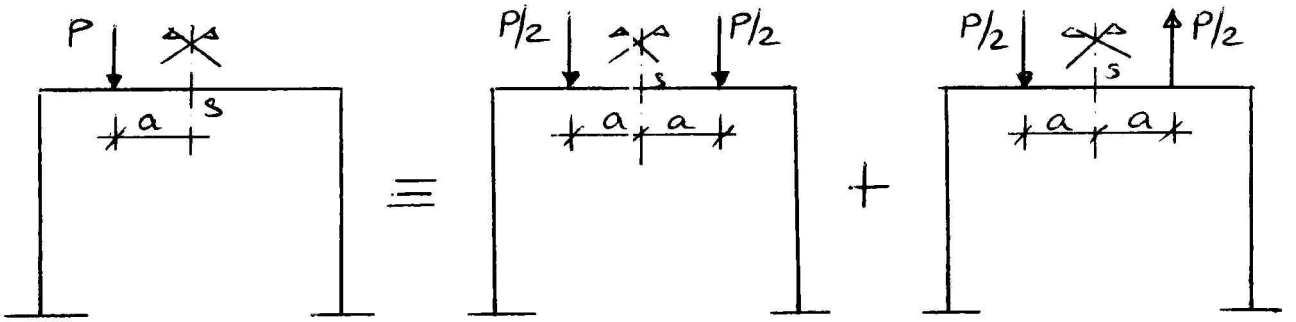
Antimetrik yük



c) Simetri Kesitinde Mafsal Bulunması

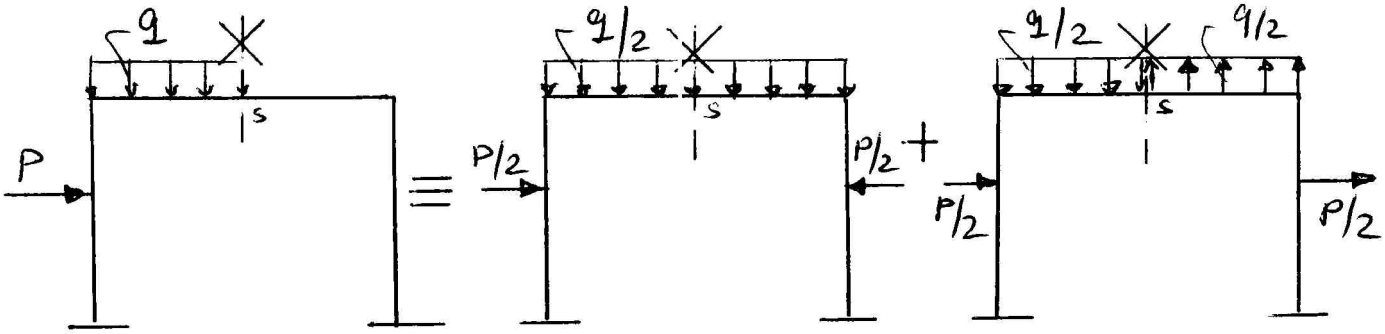


V-1) Simetri ve Antimetriden Faydalanarak Sistem Çözümü



Sistem sim. + yükleme simetrik

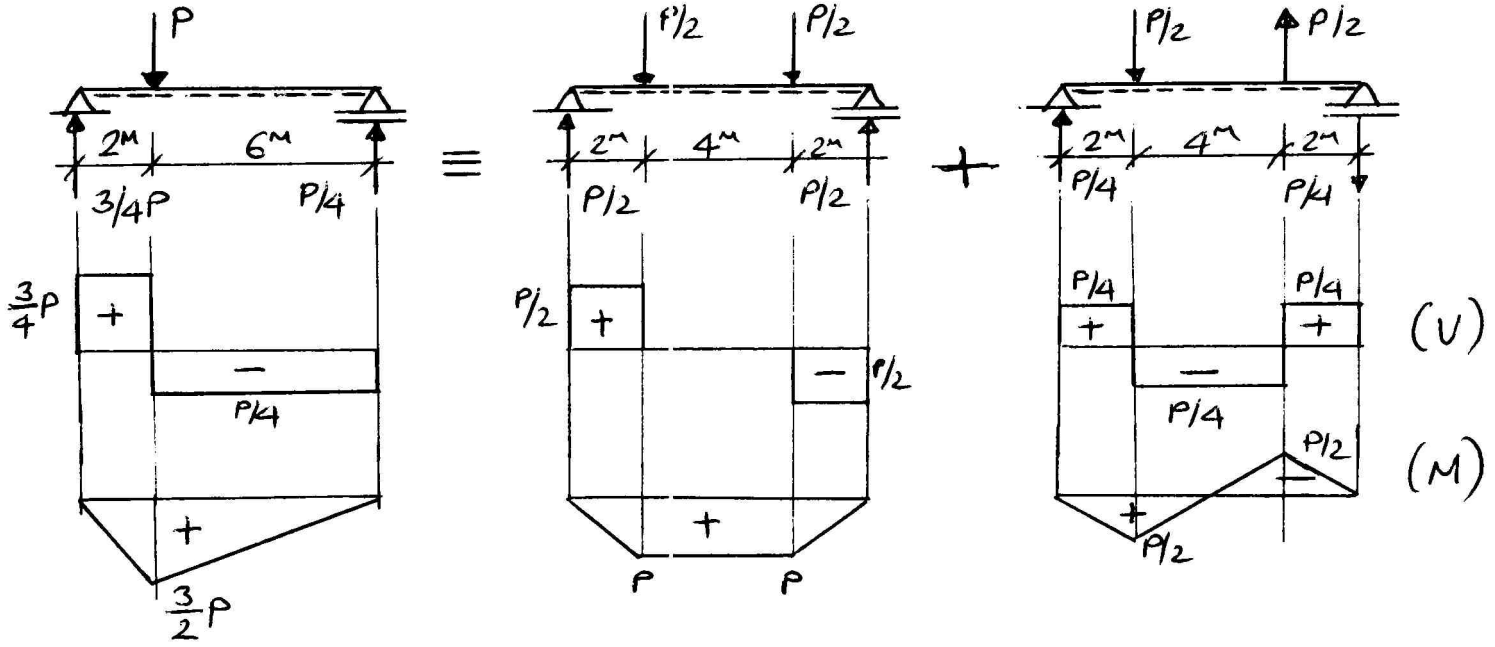
Sistem sim. + yükleme antimetrik



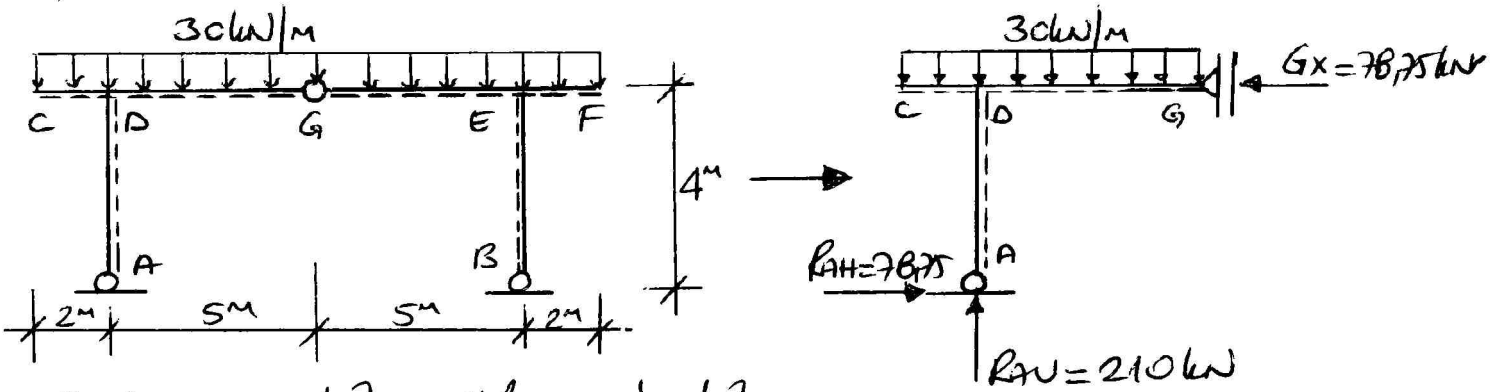
Sistem sim. + yükleme sim.

Sistem sim. + yükleme antimetrik

ÖRN. 1)



ÖRN. 2)



Sistem simetrik + yükleme simetrik

$M = M'$

$N = N'$

$V = -V'$

Kesit Tesirlerinin Hesabı

A-D arası

$N_{AD} = -210\,kN$, $N_{DA} = -210\,kN$

$V_{AD} = V_{DA} = -78{,}75\,kN$

$M_{DA} = -78{,}75 \times 4 = -315\,kN.m$

Mesnet Tepkilerinin Hesabı

$\Sigma Y = 0 \rightarrow R_{AV} = 30 \times 7 = 210\,kN$

$\Sigma M_G = 0 \rightarrow R_{AH} = \dfrac{-30 \times 7 \times 3{,}5 + 210 \times 5}{4}$

$= 78{,}75\,kN$

$\Sigma M_A = 0 \rightarrow G_x = \dfrac{30 \times 7 \times 1{,}5}{4}$

$= 78{,}75\,kN$

C-O arası

$N_{CO} = N_{OC} = 0$

$V_{CO} = 0$, $V_{OC} = -30 \times 2 = -60$ kN

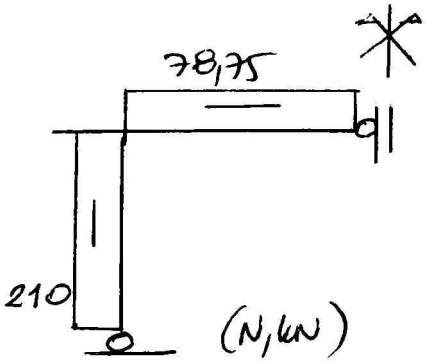
$M_{OC} = -30 \times 2 \times 1 = -60$ kNm

O-G arası

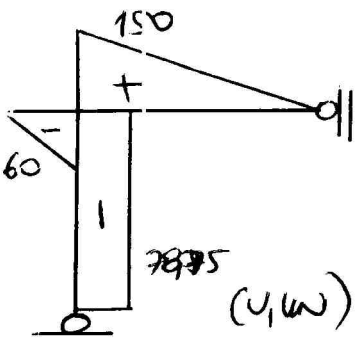
$V_{OG} = 210 - 30 \times 2 = 150$ kN

$N_{OG} = N_{GO} = -78{,}75$ kN

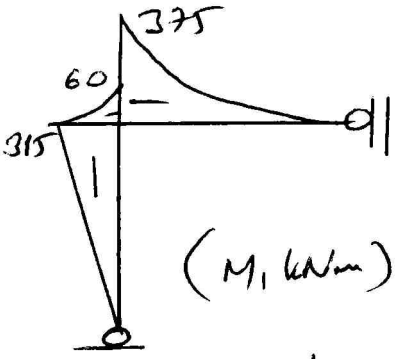
$V_{GO} = 0$, $M_{OG} = -78{,}75 \times 4 - 30 \times 2 \times 1 = -375$ kNm



$N = N'$



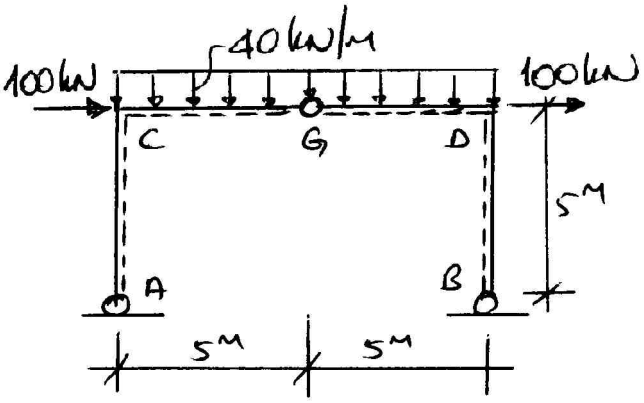
$V = -V'$



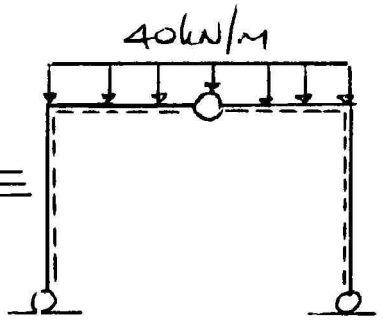
$M = M'$

* Sistemdeki tam kesit tesirlerini görmek için lütfen sayfa 82'ye bakınız.

Örnek.3) Şekilde verilen sistemi simetri ve antimetri özelliğini kullanarak çözünüz.

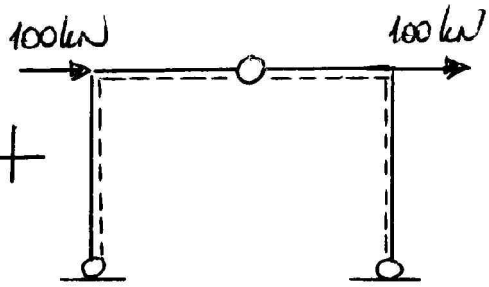


≡

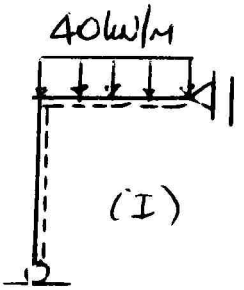


S.S+Y.S

+



S.S+Y.A

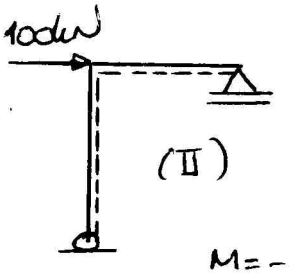


$M = M'$

$N = N'$

$V = -V'$

+

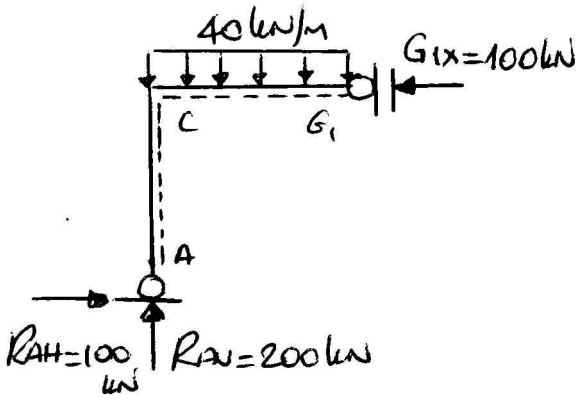


$M = -M'$

$N = -N'$

$V = V'$

(I) Nolu Sistemin Çözümü



Mesnet Tepkilerinin Hesabı

$\Sigma Y=0 \rightarrow R_{AV}=40\times5=200\ kN$

$\Sigma M_A=0 \rightarrow G_{1x}=\frac{40\times5\times2,5}{5}=100\ kN$

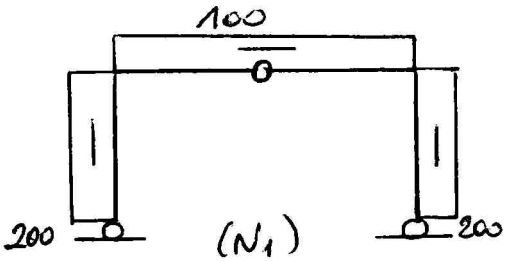
$\Sigma M_{G_1}=0 \rightarrow R_{AH}=\frac{200\times5-40\times5\times2,5}{5}=100\ kN$

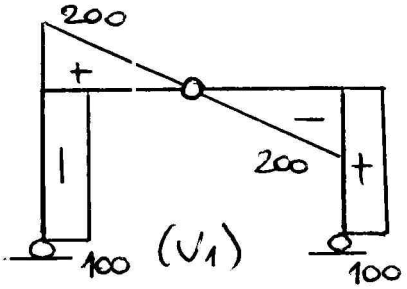
Kesit Tesirlerinin Hesabı

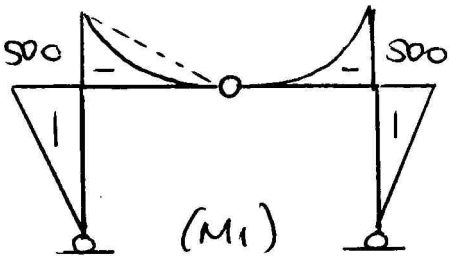
$N_{AC}=N_{CA}=-200\ kN$, $N_{CG}=N_{G_1C}=-100\ kN$

$V_{AC}=V_{CA}=-100\ kN$, $V_{CG_1}=200\ kN$, $V_{G_1C}=0$

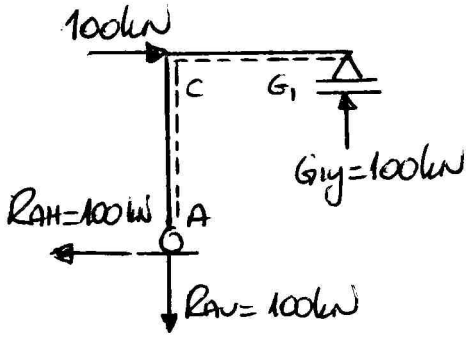
$M_{CA}=M_{CG_1}=-100\times5=-500\ kNm$







(II) Nolu Sistemin Çözümü



Mesnet Tepkilerinin Hesabı

$\Sigma X=0 \rightarrow R_{AH}=100\ kN$

$\Sigma M_{G_1}=0 \rightarrow R_{AV}=\frac{100\times5}{5}=100\ kN$

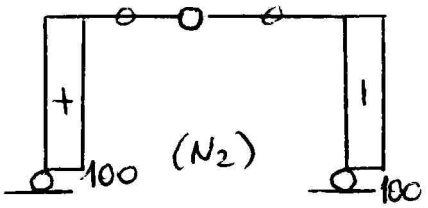
$\Sigma M_A=0 \rightarrow G_{1y}=\frac{100\times5}{5}=100\ kN$

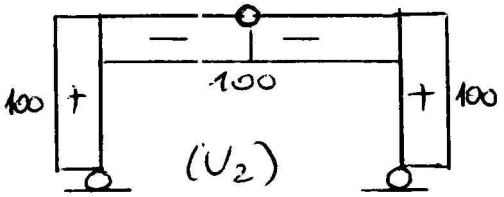
Kesit Tesirlerinin Hesabı

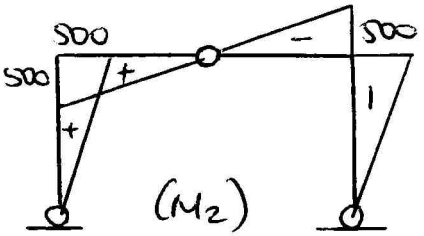
$N_{AC}=N_{CA}=100\ kN$, $N_{CG_1}=N_{G_1C}=0$

$V_{AC}=V_{CA}=100\ kN$, $V_{CG_1}=V_{G_1C}=-100\ kN$

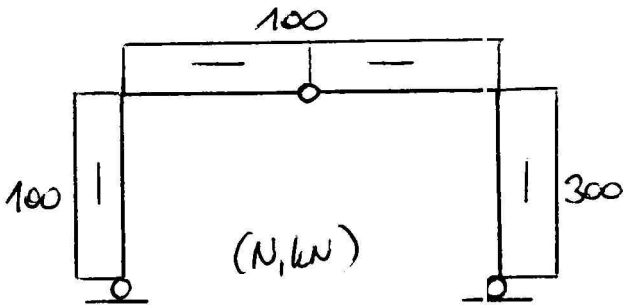
$M_{CA}=M_{CG_1}=100\times5=500\ kNm$

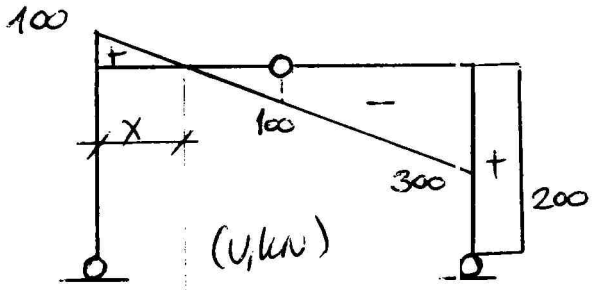






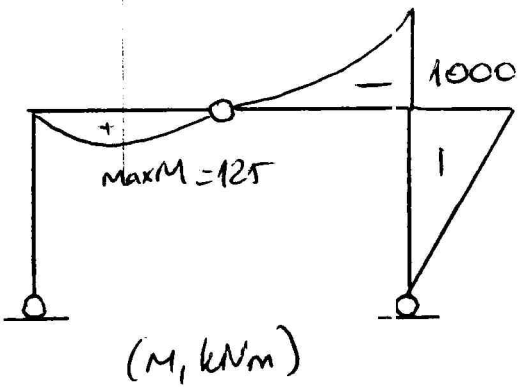
(I) ve (II) Sisteminin Süperpozisyonu



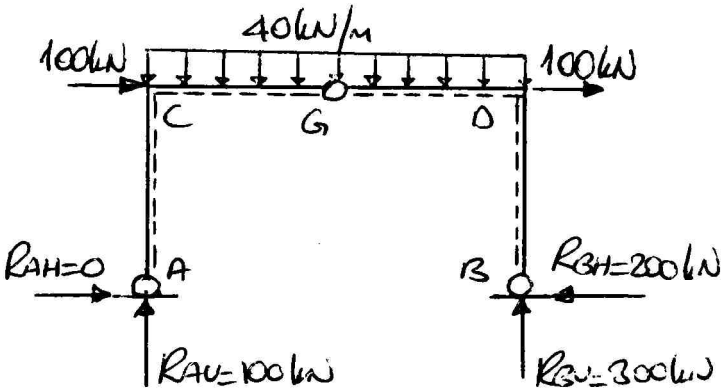


$$X = \frac{100}{40} = 2{,}5\ m.$$

$$Max M = \frac{100^2}{2 \cdot 40} = 125\ kNm$$



Sistemin Çözümü



Mesnet Tepkilerinin Hesabı

$\Sigma M_B = 0 \rightarrow R_{AV} = \frac{40 \times 10 \times 5 - 100 \times 5 \times 2}{10} = 100\,kN$

$\Sigma M_A = 0 \rightarrow R_{BV} = \frac{40 \times 10 \times 5 + 100 \times 5 \times 2}{10} = 300\,kN$

$\Sigma M_{G_{sol}} = 0 \rightarrow R_{AH} = \frac{100 \times 5 - 40 \times 5 \times 2{,}5}{5} = 0$

$\Sigma M_{G_{sağ}} = 0 \rightarrow R_{BH} = \frac{300 \times 5 - 40 \times 5 \times 2{,}5}{5} = 200\,kN$

$\Sigma Y = 0 \rightarrow 40 \times 10 = 100 + 300 \rightarrow 400 = 400$

$\Sigma X = 0 \rightarrow 100 + 100 = 200 \rightarrow 200 = 200$

Kesit Tesirlerinin Hesabı

A-C arası

$N_{AC} = N_{CA} = -100\,kN$

$V_{AC} = V_{CA} = 0$

$M_{CA} = M_{CG} = 0$

C-D arası

$N_{CD} = N_{DC} = -100\,kN$

$V_{CG} = 100\,kN, \quad V_G = 100 - 40 \times 5 = -100\,kN$

$V_{DG} = -100 - 40 \times 5 = -300\,kN$

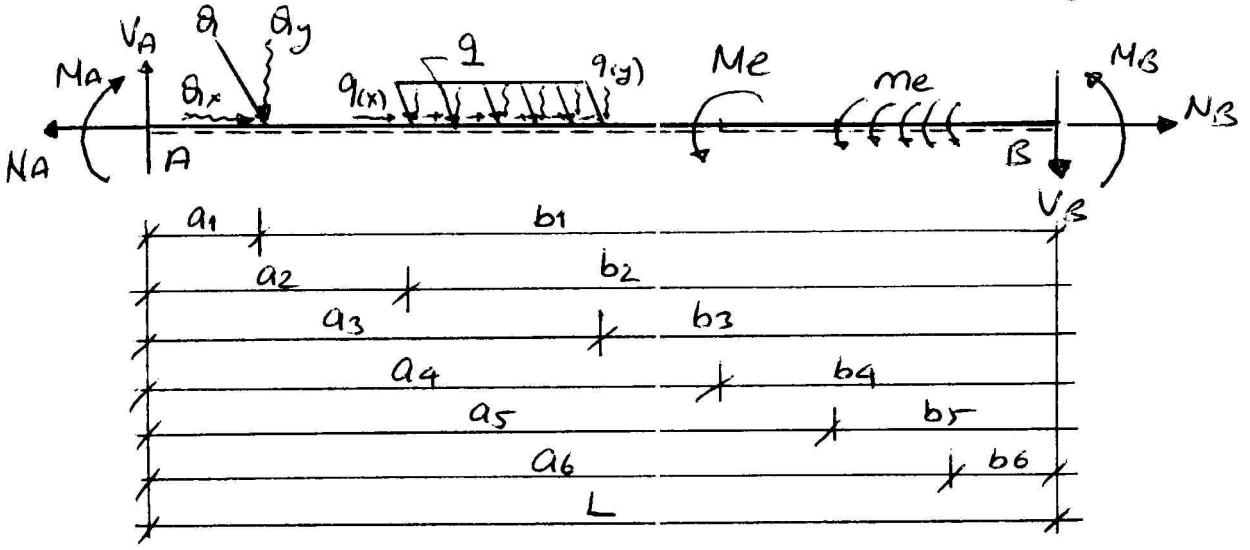
B-D arası

$N_{BD} = N_{DB} = -300\,kN$

$V_{BD} = V_{DB} = 200\,kN$

$M_D = -200 \times 5 = -1000\,kNm$

III- Kesit Tesirlerinin Çözümünde Kolaylık Sağlayan Yöntemler



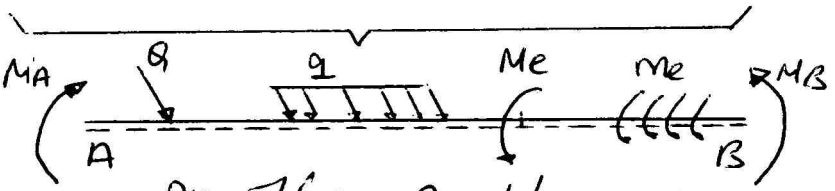
Bir eleman üzerinde herhangi bir kesitteki kesit tesirleri (M, N, V) belli iken başka bir kesitteki kesit tesirleri bulunabilir.

Sistemin B ucundaki kesit tesirleri (V_B, N_B, M_B) belli !

$$\Sigma X=0 \rightarrow N_A = N_B + Q_x + \int_{a_2}^{a_3} q(x)\,dx = N_B + \Sigma Q_{x_i} + \Sigma \int_{a_i}^{a_{ii}} q(x_i)\,dx$$

$$\Sigma Y=0 \rightarrow V_A = V_B + Q_y + \int_{a_2}^{a_3} q(y)\,dx = V_B + \Sigma Q_{y_i} + \Sigma \int_{a_i}^{a_{ii}} q(y)_i\,dx$$

$$\Sigma M_A=0 \rightarrow M_A = M_B - V_B \cdot L - Q_y\, a_1 - \int_{a_2}^{a_3} q(y)\, x\,dx + Me + \int_{a_5}^{a_6} me\,dx$$



Dış yüklerin A noktasına göre momenti

$$\overrightarrow{M}_{oAB} = -Q_y\, a_1 - \int_{a_2}^{a_3} q(y)\, x\,dx + Me + \int_{a_5}^{a_6} me\,dx$$

Bu durumda bağıntı;

$$\boxed{M_A = M_B - V_B \cdot L - \overrightarrow{M}_{oAB}}$$ şeklini alır.

Sistemin A ucundaki kesit tesirleri (V_A, N_A, M_A) belli ise;

$$\Sigma X=0 \rightarrow N_B = N_A - \delta_x - \int_{a_2}^{a_3} q(x)\,dx$$

$$\Sigma Y=0 \rightarrow V_B = V_A - \delta_y - \int_{a_2}^{a_3} q(y)\,dx$$

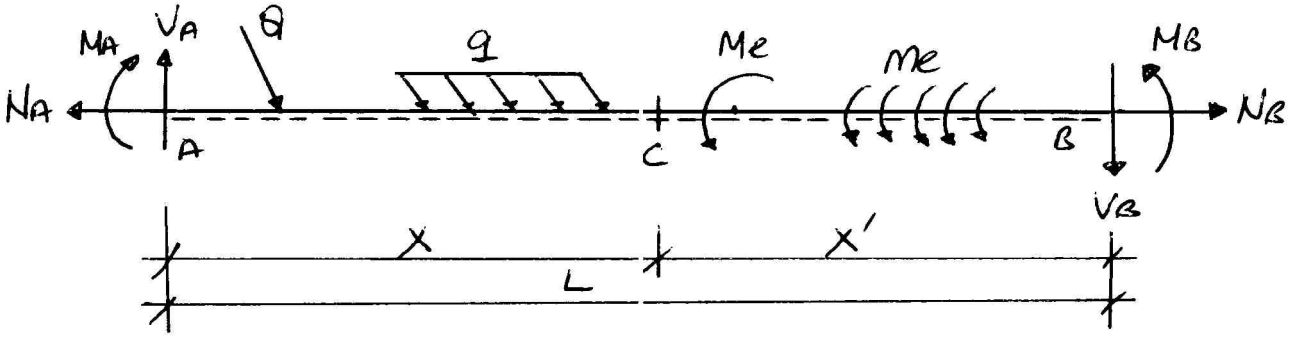
$$\Sigma M_B=0 \rightarrow M_B = M_A + V_A \cdot L - \delta_y \cdot b_1 - \int_{a_2}^{a_1} q(y)\cdot x'\,dx - M_e - \int_{a_5}^{a_6} m_e\,dx$$

Dış yüklerin B noktasına göre momenti

Bu durumda bağıntı; $\overleftarrow{M_{0BA}}$

$$\boxed{M_B = M_A + V_A \cdot L - \overleftarrow{M_{0BA}}}$$ şeklini alır.

Bir çubuk parçasının herhangi bir kesitindeki kesit tesirleri, bu çubuk parçasının iki ucundaki kesit tesirleri yardımıyla bulunabilir.



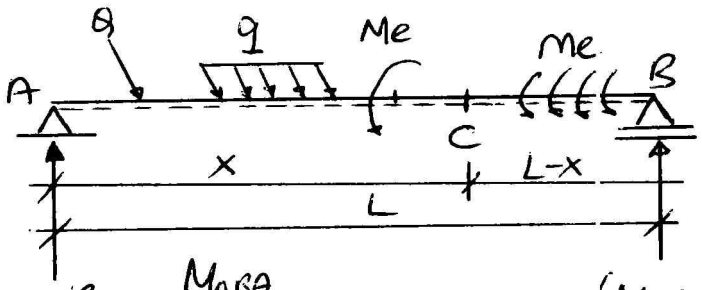
$$M_X = M_A + V_A \cdot x - \overleftarrow{M_{0CA}}$$

$$M_B = M_A + V_A \cdot L - \overleftarrow{M_{0BA}} \;/\; \text{2. denklemi } \left(-\frac{x}{L}\right) \text{ ile çarpıp 2 denklemi toplarsak;}$$

$$M_X - M_B \frac{x}{L} = M_A - M_A \cdot \frac{x}{L} + V_A \cdot x - V_A \cdot L \cdot \frac{x}{L} - M_{0CA} + M_{0BA} \cdot \frac{x}{L}$$

$$M_X = M_A\left(\frac{L-x}{L}\right) + M_B\left(\frac{x}{L}\right) + M_{0BA} \cdot \frac{x}{L} - M_{0CA}$$

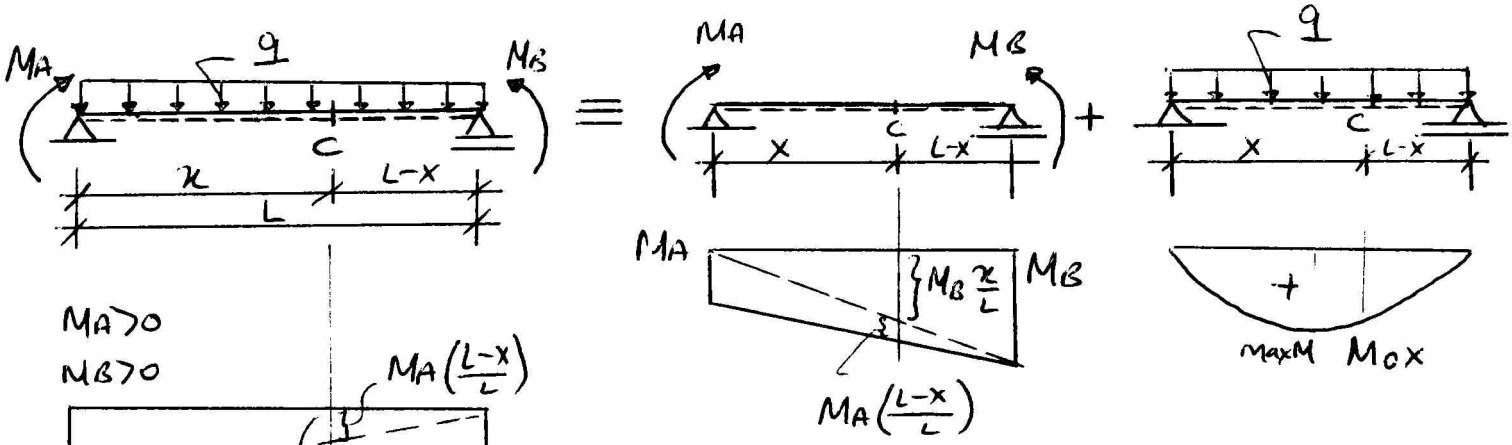
Bağıntıdaki bu üçlü ifade basit kirişin herhangi bir noktasına göre momenti göstermektedir.



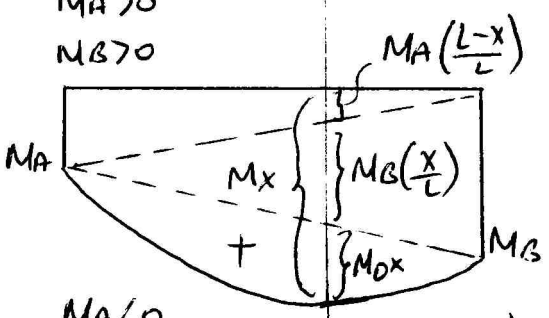
$$R_{AV} = \frac{M_{OBA}}{L} \qquad M_{Ox} = \left(\frac{M_{OBA}}{L}\right)x - M_{OCA}$$

Bu durumda bağıntı;

$$M_x = M_A\left(\frac{L-x}{L}\right) + M_B\left(\frac{x}{L}\right) + M_{Ox}$$ şeklini alır.

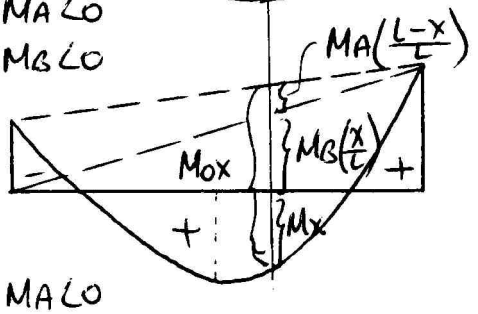


MA>0
MB>0



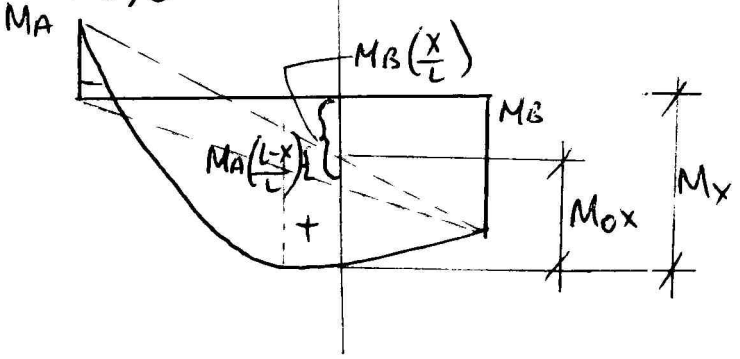
$$M_x = M_A\left(\frac{L-x}{L}\right) + M_B\left(\frac{x}{L}\right) + M_{Ox}$$

MA<0
MB<0



$$M_x = M_{Ox} - M_A\left(\frac{L-x}{L}\right) - M_B\left(\frac{x}{L}\right)$$

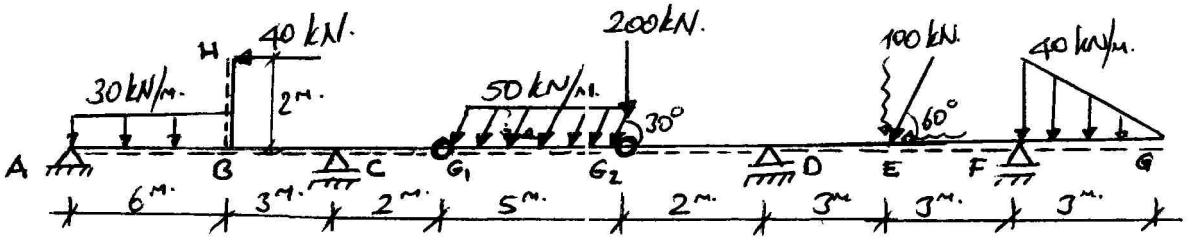
MA<0
MB>0

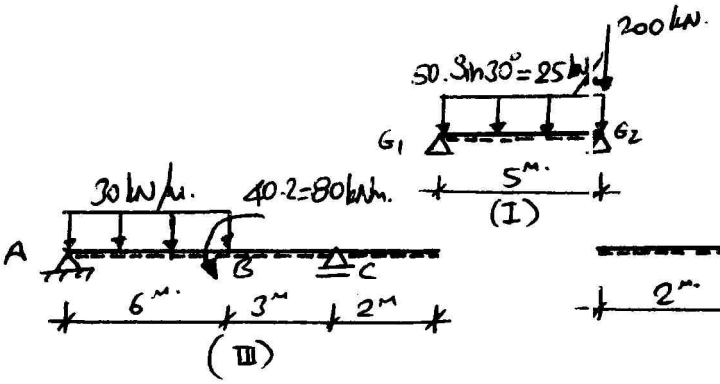


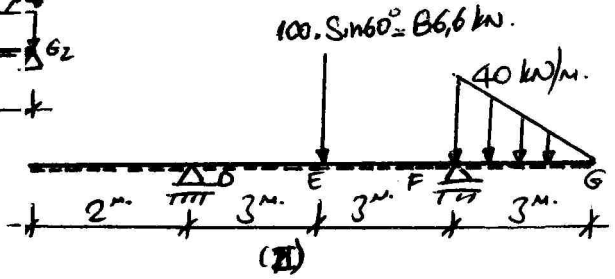
$$M_x = M_{Ox} + M_B\left(\frac{x}{L}\right) - M_A\left(\frac{L-x}{L}\right)$$

KESİT TESİRİ UYGULAMALARI

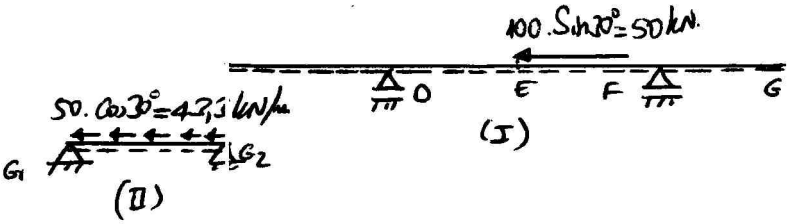
UYGULAMA : 1)



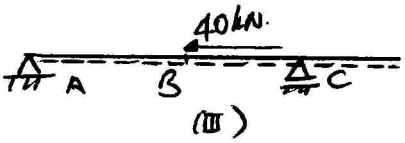




Düşey Yük Taşıma Şeması

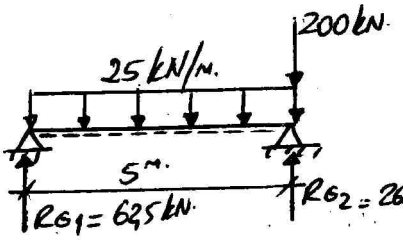


Yatay Yük Taşıma Şeması



a) Düşey Yüklere göre çözüm.

(I) parçası



$\Sigma M_{G_2}=0 \rightarrow R_{G_1}=\frac{25\cdot 5\cdot 2,5}{5}=62,5$ kN.

$\Sigma M_{G_1}=0 \rightarrow R_{G_2}=\frac{25\cdot 5\cdot 2,5+200\cdot 5}{5}=262,5$ kN.

$\Sigma P_y=0$

$200+25\cdot 5=62,5+262,5$

$325=325$ ✓

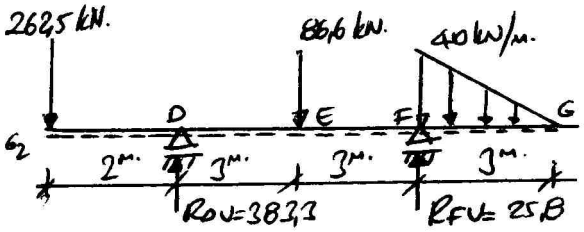
$V_{G_1G_2}=62,5$ kN.

$V_{G_2G_1}=-62,5$ kN.

max M 'in yeri' $\Rightarrow x_0=\frac{62,5}{25}=2,5$ m.

$\max M=\frac{62,5^2}{2\cdot 2,5}=78,125$ kNm.

(II) parçası



$\Sigma M_F=0 \rightarrow R_{OV}=\frac{262,5\cdot 8+86,6\cdot 3-40\cdot\frac{3}{2}\cdot\frac{1}{3}}{6}=383,3$ kN

$\Sigma M_D=0 \rightarrow R_{FV}=\frac{86,6\cdot 3-262,5\cdot 2+\frac{40\cdot 3}{2}\left(\frac{3}{3}+6\right)}{6}=25,8$ kN.

$\Sigma P_y=0 \rightarrow 262,5+86,6+40\cdot 3/2=25,8+383,3$

$409,1=409,1$ ✓

$V_{G_2D}=V_{DG_2}=-262,5$ kN.

$V_{DE}=V_{ED}=120,80$ kN.

$V_{EF}=V_{FE}=120,8-86,6=34,20$ kN.

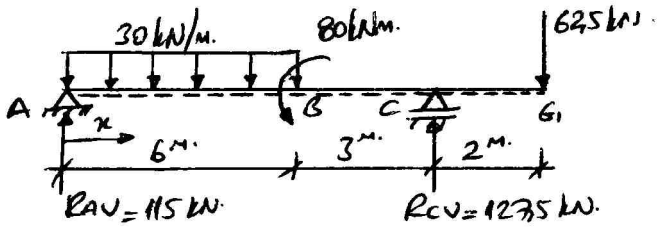
$V_{F\ldots}=34,2+25,8=60$ kN.

$M_D=-262,5\cdot 2=-525$ kNm.

$M_E=-262,5\cdot 5+383,3\cdot 3=-162,6$ kNm.

$M_{F\ldots}=-\frac{40\cdot 3}{2}\cdot 1=-60$ kNm.

(III) parçası



$\Sigma M_C = 0 \rightarrow R_{AV} = \frac{30 \cdot 6 \cdot 6 + 80 - 625 \cdot 2}{9} = 115$ kN.

$\Sigma M_A = 0 \rightarrow R_{CV} = \frac{30 \cdot 6 \cdot 3 - 80 + 625 \cdot 11}{9} = 1273,5$ kN.

$\Sigma P_y = 0 \rightarrow 115 + 1273,5 = 30 \cdot 6 + 625$

$2242,5 = 2242,5$ ✓

$V_x = 115 - 30 \cdot x$

$M_x = 115 \cdot x - \frac{30 \cdot x^2}{2} = 115x - 15x^2$

$x = 0 \Rightarrow \begin{cases} V_{AB} = 115 \text{ kN.} \\ M_{AB} = 0 \end{cases}$

$x = 6^m \Rightarrow \begin{cases} V_{BA} = -65 \text{ kN.} \\ M_{BA} = 150 \text{ kNm.} \end{cases}$

max M için $V_x = 0$ olmalı

$115 - 30x = 0$

$x = 3,83$ m.

$M_{max} = 115 \cdot 3,83 - 15 \cdot (3,83)^2 = 220,42$ kNm.

$\begin{cases} V_{BC} = V_{CB} = -65 \text{ kN.} \\ V_{CG_1} = -65 + 1273,5 = 625 \text{ kN.} \\ V_{G_1C} = 625 \text{ kN.} \end{cases}$

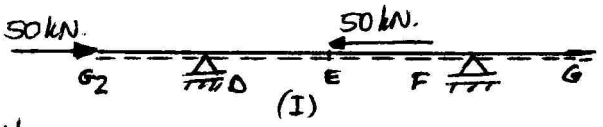
$M_{AB} = 0$

$M_{BA} = 150$ kNm.

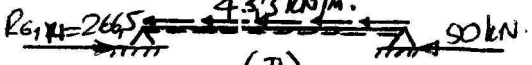
$M_{BC} = 150 - 80 = 70$ kNm.

$M_C = -625 \cdot 2 = -125$ kNm.

b) Yatay yüklere göre çözüm:



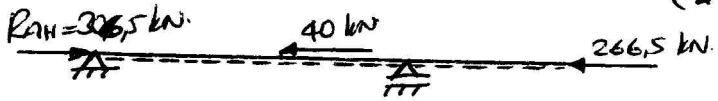
(I)

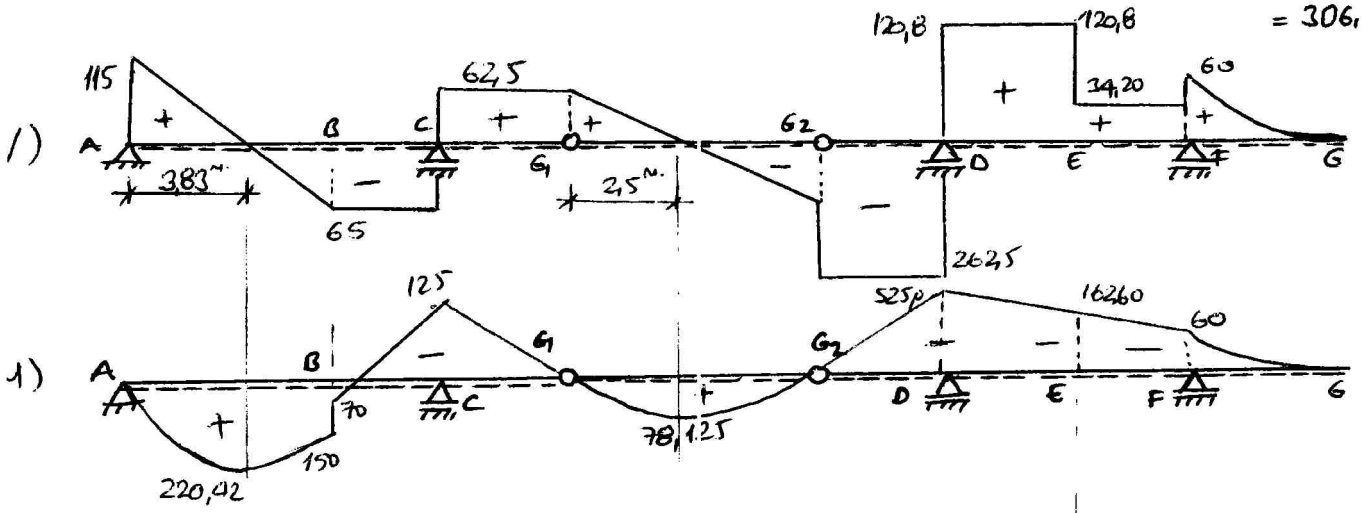


(II)

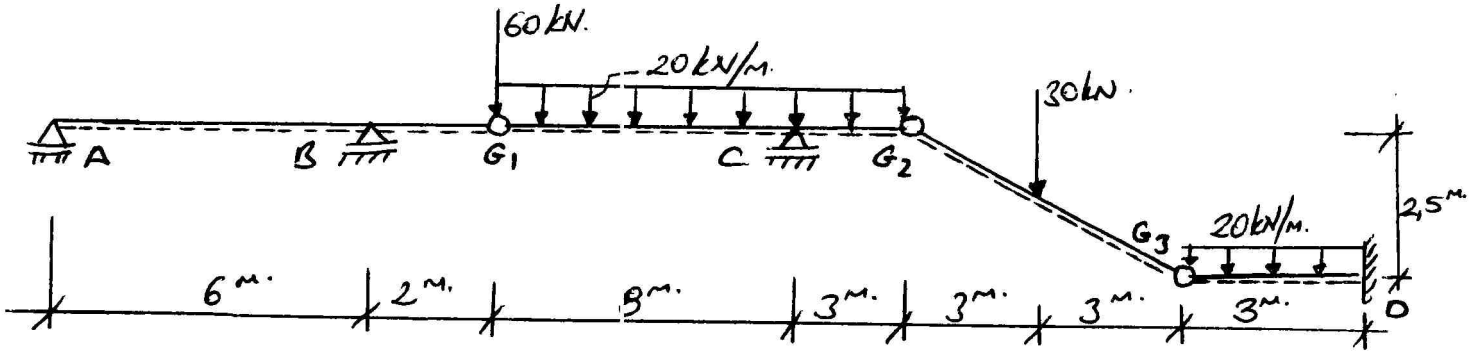
$\Sigma X = 0 \rightarrow R_{G_1H} = 50 + 43,75 = 266,5$ kN.

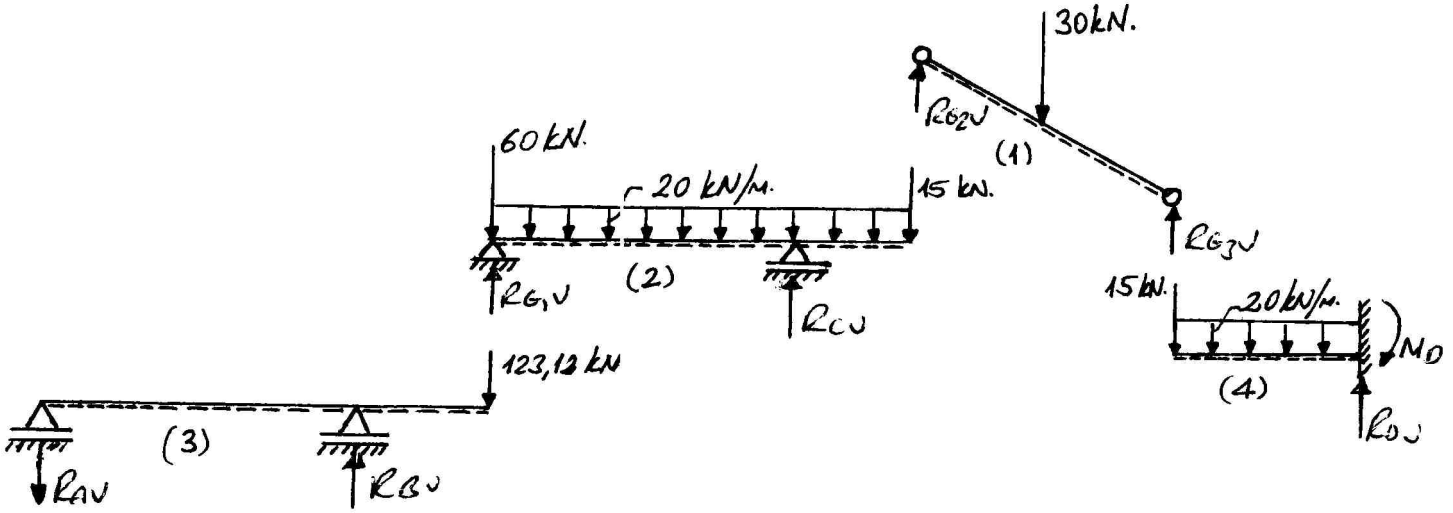
$R_{AH} = 40 + 266,5 = 306,5$ kN.





UYGULAMA: 2)





* Mesnet Tepkileri

(1) parçası: $\Sigma M_3=0 \rightarrow G_{2y}=\frac{30\cdot 3}{6}=15\,kN$

$\Sigma M_2=0 \rightarrow G_{3y}=\frac{30\cdot 3}{6}=15\,kN.$

$\Sigma P_y=0 \quad 15+15=30 \quad 30=30$ ✓

(2) parçası: $\Sigma M_C=0 \rightarrow R_{G_1V}=\frac{60\cdot 8+20\cdot 11\cdot 2,5-15\cdot 3}{8}=123,12\,kN.$

$\Sigma M_{G_1}=0 \rightarrow R_{CV}=\frac{20\cdot 11\cdot 5,5+15\cdot 11}{8}=171,88\,kN.$

$\Sigma P_y=0 \quad 220+60=123,12+171,88$

$295=295$ ✓

(3) parçası: $\Sigma M_B=0 \rightarrow R_{AV}=\frac{123,12\cdot 2}{6}=41,04\,kN.$

$\Sigma M_A=0 \rightarrow R_{BV}=\frac{123,12\cdot 8}{6}=164,16\,kN$

$\Sigma P_y=0 \quad -41,04+164,16=123,12$

$123,12=123,12$ ✓

(4) parçası: $\Sigma P_y=0 \rightarrow R_{DV}=15+20\cdot 3=75\,kN.$

$\Sigma M_D=0 \rightarrow M_D=15\cdot 3+20\cdot 3\cdot 1,5=135\,kNm.$

* Kesit Tesirleri

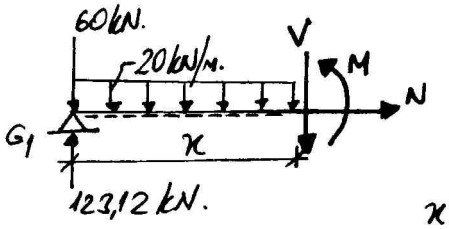
A-G1 GUBUĞU

$V_{AB} = V_{BA} = -41{,}04$ kN.

$M_B = -41{,}04 \cdot 6 = -246{,}24$ kNm.

$V_{BG_1} = V_{G_1B} = 123{,}12$ kN.

G1-G2 GUBUĞU



$V_x = 123{,}12 - 60 - 20.x = 63{,}12 - 20x$

$M_x = 123{,}12.x - 60.x - 20.x.\frac{x}{2} = 63{,}12x - 10x^2$

$x=0 \begin{cases} V_{G_1C} = 63{,}12 \text{ kN.} \\ M_{G_1} = 0 \end{cases}$ $\quad x=8^{m.} \begin{cases} V_{CG_1} = -96{,}88 \text{ kN.} \\ M_C = -135{,}04 \text{ kNm.} \end{cases}$

max M ve max M 'in yeri:

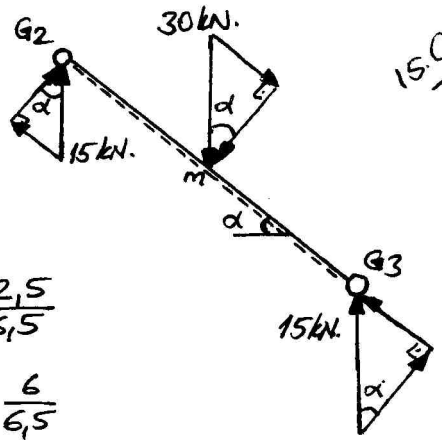
$V_{x=0} \rightarrow 63{,}12 - 20x_0 = 0$

$x_0 = 3{,}156$ m. ; $\max M = 63{,}12(3{,}156) - 10.(3{,}156)^2 = 99{,}60$ kNm.

$V_{CG_2} = 75$ kN.

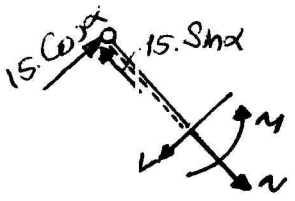
$V_{G_2C} = 15$ kN.

G2-G3 GUBUĞU



$\sin\alpha = \frac{2{,}5}{6{,}5}$

$\cos\alpha = \frac{6}{6{,}5}$



$N_{mG_2} = N_{G_2m} = 15.\sin\alpha = 5{,}77$ kN.

$N_{mG_3} = N_{G_3M} = -15.\sin\alpha = -5{,}77$ kN

$V_{G_2m} = 15.\cos\alpha = 13{,}85$ kN.

$V_{mG_2} = 13{,}85$ kN.

$V_{mG_3} = V_{G_3M} = -15.\cos\alpha = -13{,}85$ kN.

$\max M_m = 15 \cdot 3 = 45$ kNm.

G3-D GUBUĞU

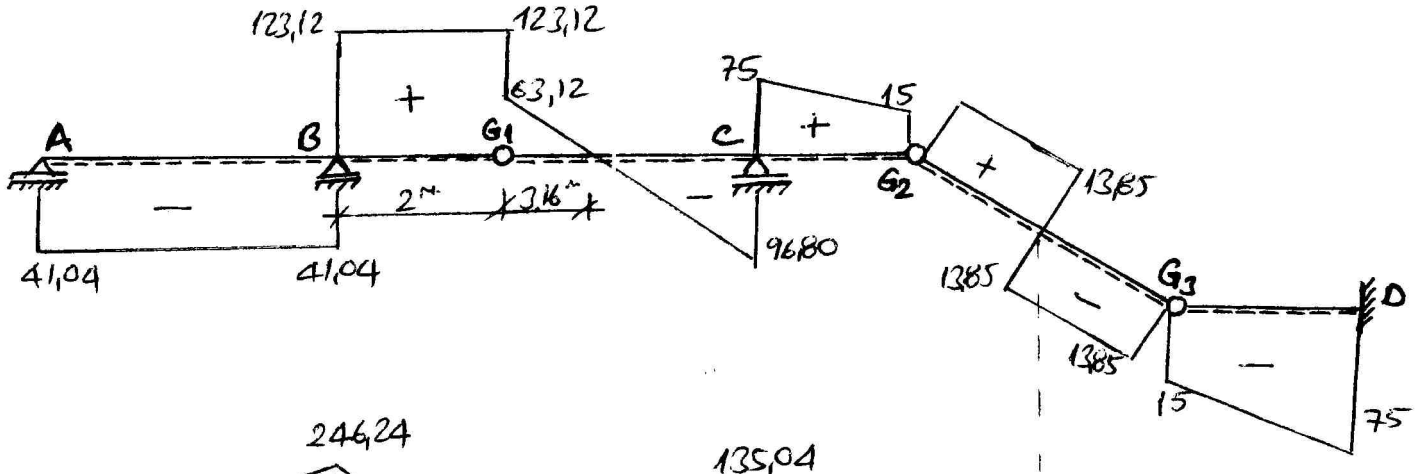
$V_{G_3D} = -15$ kN.

$V_{DG_3} = -15 - 20 \cdot 3 = -75$ kN.

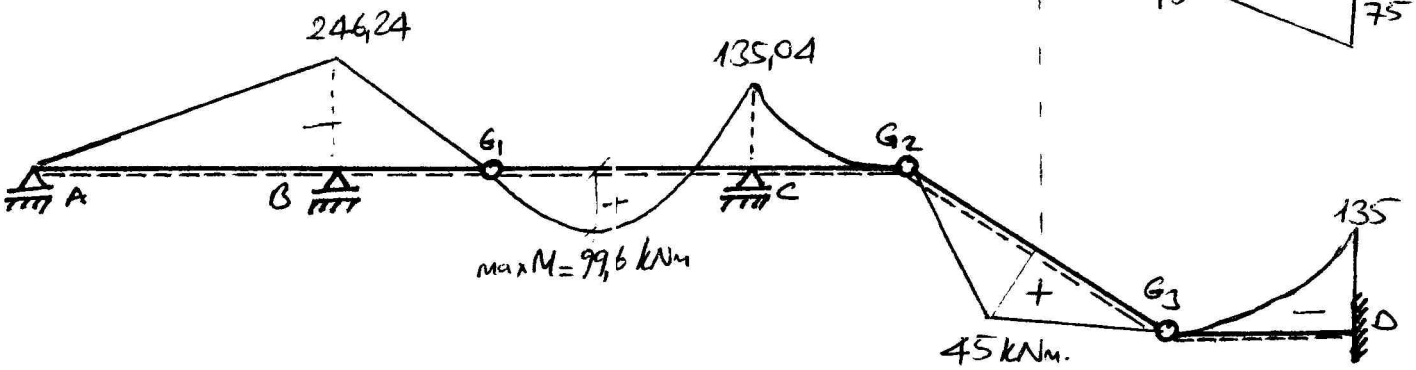
$M_{G_3} = 0$

$M_D = -135$ kNm.

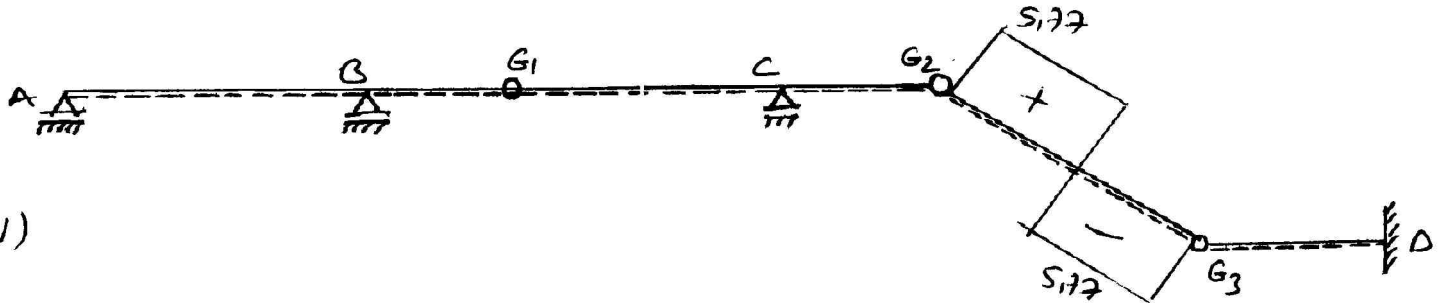
V)


123,12
123,12
63,12
75
15
A
B
G1
C
G2
G3
D
2m
3,16m
41,04
41,04
96,80
13,85
13,85
13,85
15
75

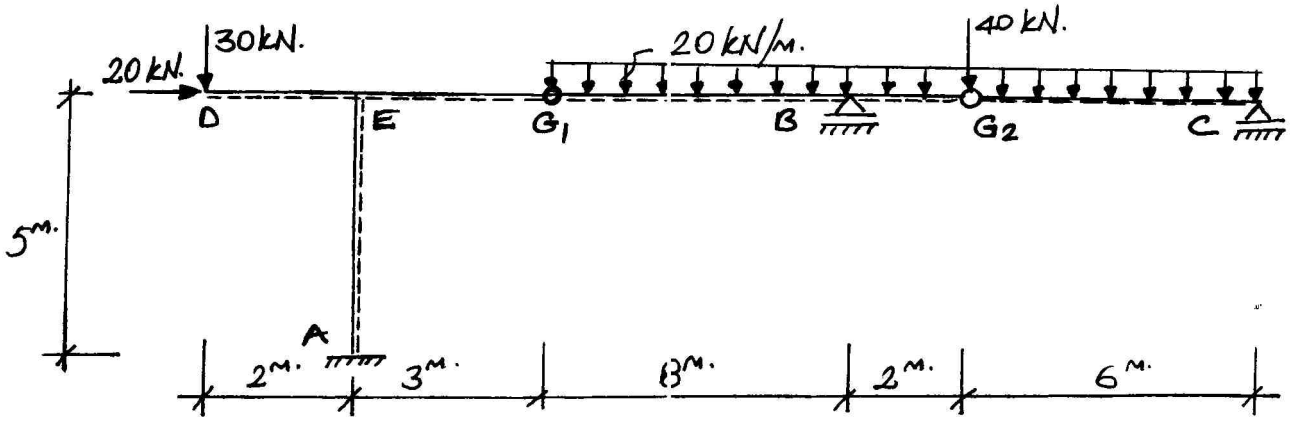
M)

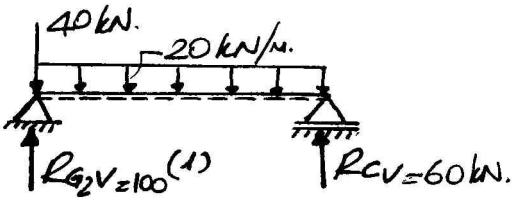

246,24
135,04
A
B
G1
C
G2
G3
D
maxM=99,6 kNm
135
45 kNm.

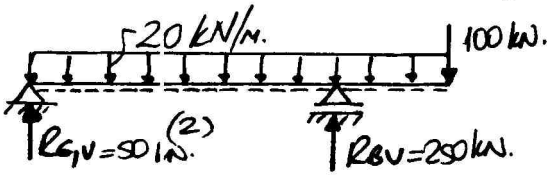
N)

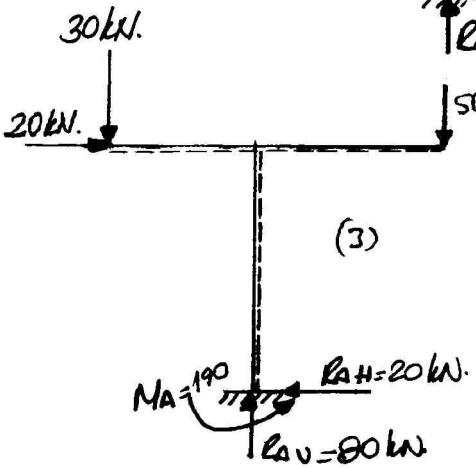

A
B
G1
C
G2
G3
D
5,17
5,17

DYGUCAMA : 3)

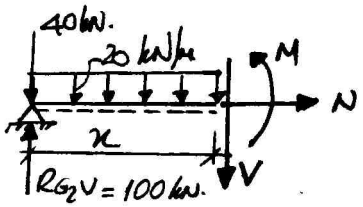








(1) parcial :



$\Sigma M_C = 0 \rightarrow R_{G_2V} = \frac{20 \cdot 6 \cdot 3 + 40 \cdot 6}{6} = 100\,kN.$

$\Sigma M_{G_2} = 0 \rightarrow R_{CV} = \frac{20 \cdot 6 \cdot 3}{6} = 60\,kN.$

$\Sigma F_y = 0 \rightarrow 100 + 60 = 20 \cdot 6 + 40$

$160 = 160$ ✓

$N = 0$

$V_x = 100 - 40 - 20 \cdot x = 60 - 20x$

$M_x = 100 \cdot x - 40 \cdot x - 20 \cdot x \cdot \frac{x}{2} = 60x - 10x^2$

$x = 0 \begin{cases} V_{G_2C} = 60\,kN. \\ M_{G_2} = 0 \end{cases}$

$x = 6^M. \begin{cases} V_{CG_2} = -60\,kN. \\ M_C = 0 \end{cases}$

$x = 0 \Rightarrow \max M$

$V_x = 0 \rightarrow 60 - 20x_0 = 0$

$x_0 = 3M.$

$\max M = 60 \cdot 3 - 10 \cdot 9 = 90\,kNm.$

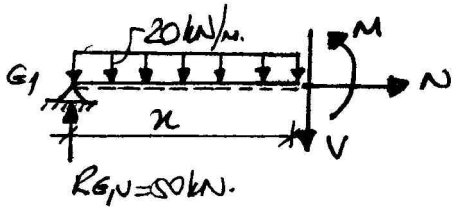
2 parçası:

$\Sigma M_B = 0 \rightarrow R_{G1V} = \frac{10 \cdot 20 \cdot 3 - 100 \cdot 2}{8} = 50\,kN.$

$\Sigma M_{G_1} = 0 \rightarrow R_{BV} = \frac{20 \cdot 10 \cdot 5 + 100 \cdot 10}{8} = 250\,kN.$

$\Sigma P_y = 0 \rightarrow 50 + 250 = 20 \cdot 10 + 100$

$300 = 300$ ✓



$N = 0$

$V_x = 50 - 20.x$

$m_x = 50x - 20.x.\frac{x}{2} = 50x - 10x^2$

$x=0 \begin{cases} V_{G_1B} = 50\,kN. \\ M_{G_1} = 0 \end{cases}$ $\quad x=8^M \begin{cases} V_{BG_1} = 50 - 20.8 = -110\,kN. \\ M_B = 50.8 - 10.64 = -240\,kNm. \end{cases}$

$V_{BG_2} = -110 + 250 = 140\,kN.$

$V_{G_2B} = 100\,kN.$

$x_0 = \frac{V}{q} = \frac{50}{20} = 2{,}5^M.$

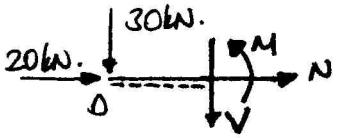
$m_{max} M = \frac{50^2}{2 \cdot 20} + 0 = 625\,kNm.$

3 parçası:

$\Sigma M_A = 0 \rightarrow H_A = 20 \cdot 5 - 30 \cdot 2 + 50 \cdot 3 = 190\,kNm.$

$\Sigma P_y = 0 \rightarrow R_{AV} = 30 + 50 = 80\,kN$

$\Sigma P_x = 0 \rightarrow R_{AH} = 20\,kN.$

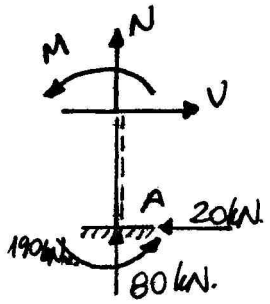


$N_{DE} = X_{ED} = -20\,kN.$

$V_{DE} = V_{ED} = -30\,kN.$

$N_{G_1E} = N_{EG_1} = 0$

$V_{G_1E} = V_{EG_1} = 50\,kN.$

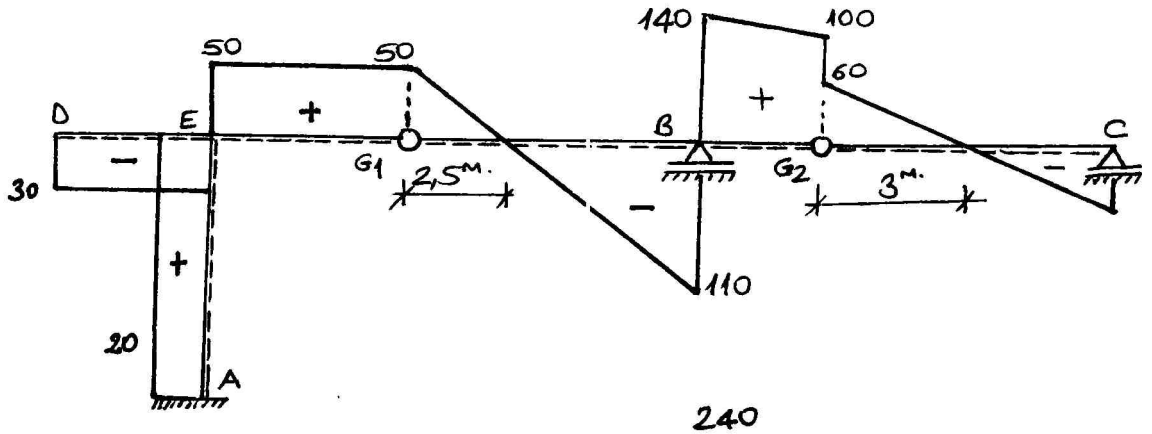


$N_{AE} = N_{EA} = -80\,kN.$

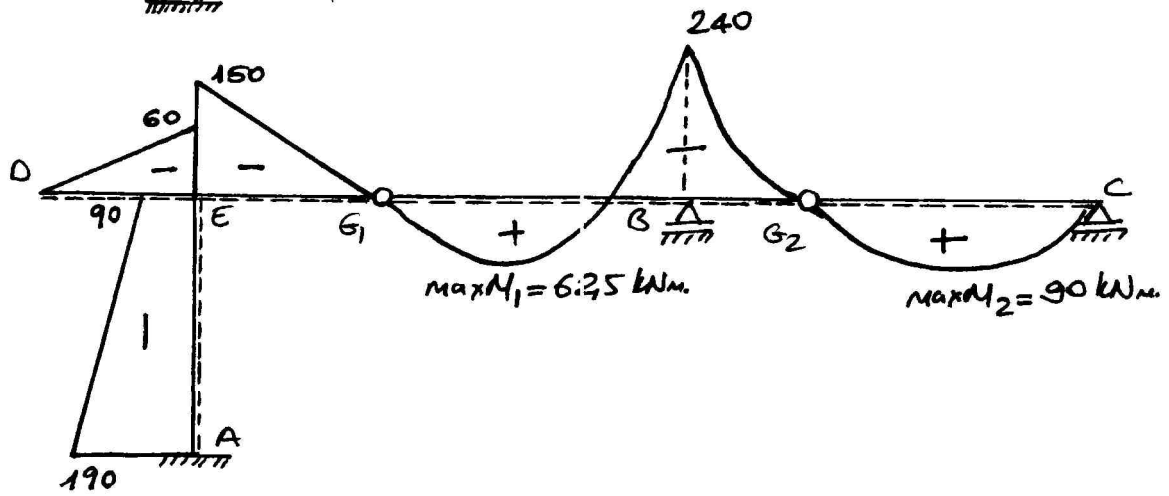
$V_{AE} = V_{EA} = 20\,kN.$

$M_A = -80\,kNm.$

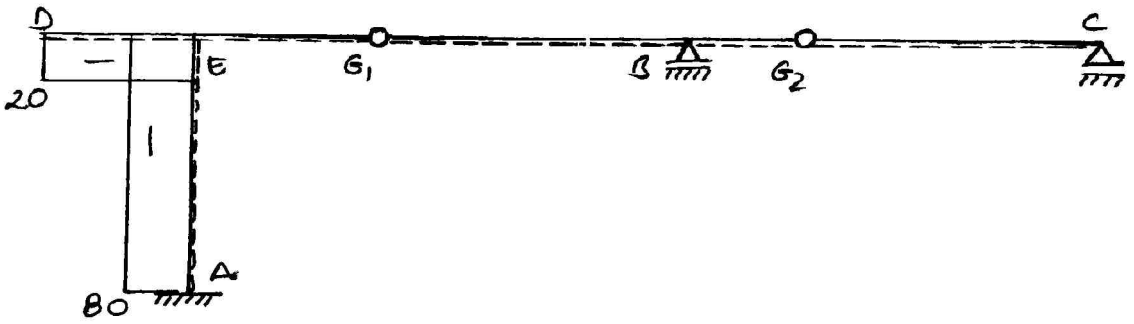
$M_E = -190 + 20 \cdot 5 = -90\,kNm.$


140
100
50
50
60
D
E
B
C
G1
2,5m.
G2
3m.
30
110
20
A

(V)

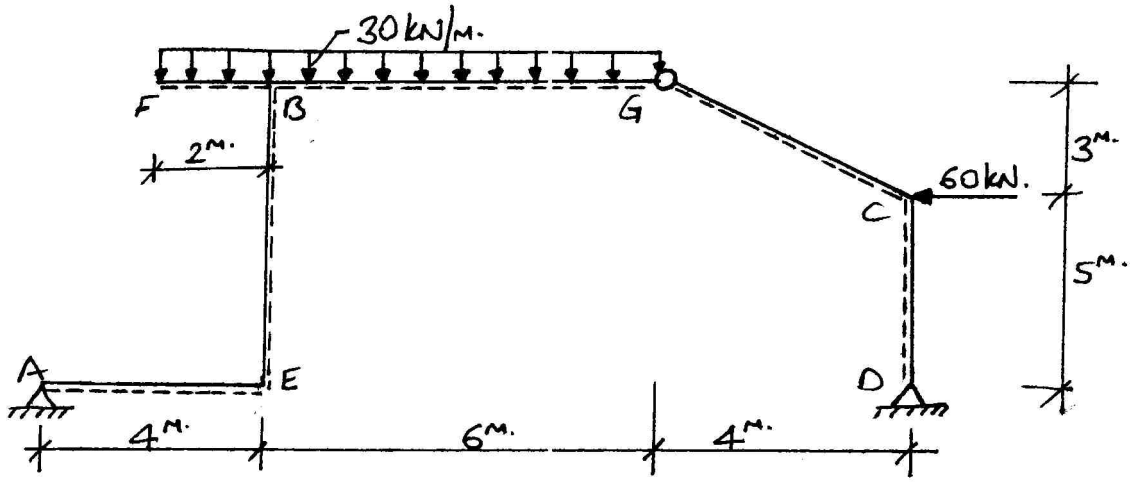

240
160
60
D
90
E
G1
B
G2
C
A
190
maxM1 = 6,25 kNm.
maxM2 = 90 kNm.

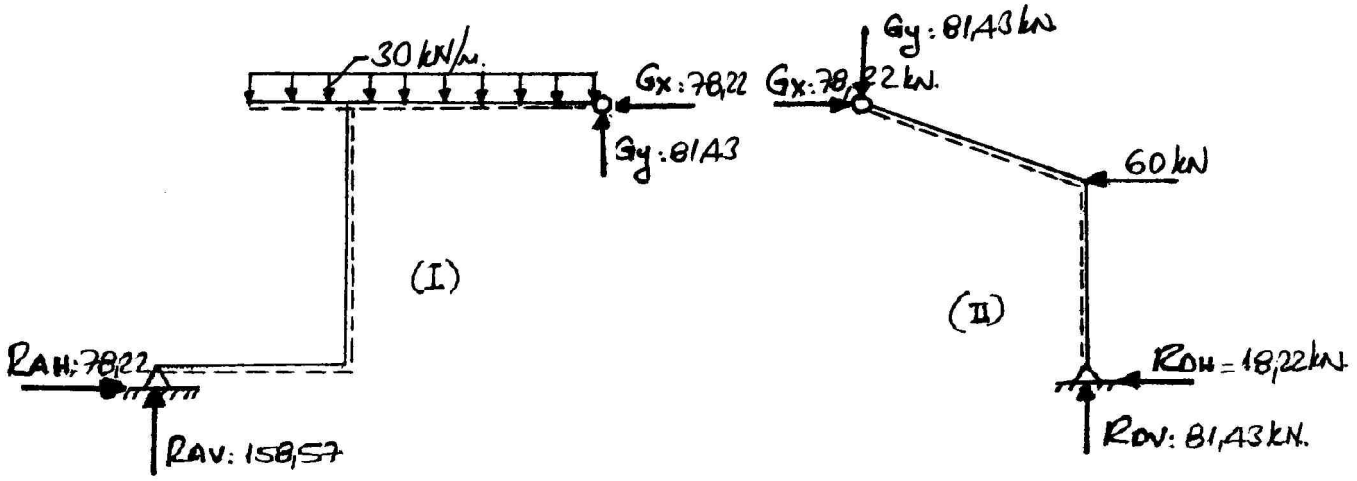
(M)


D
E
G1
B
G2
C
20
A
80

(N)

UYGULAMA : 4)





Tüm sistemde; $\Sigma M_D = 0 \rightarrow R_{AV} = \frac{30 \cdot 8 \cdot 8 + 60 \cdot 5}{14} = 158,57\ kN.$

$\Sigma M_A = 0 \rightarrow R_{DV} = \frac{30 \cdot 8 \cdot 6 - 60 \cdot 5}{14} = 81,43\ kN.$

Kontrol: $\Sigma P_y = 0 \rightarrow 30 \cdot 8 = 158,57 + 81,43$

$240 = 240$ ✓

(I) parçasında;

$\Sigma M_G = 0 \rightarrow R_{AH} = \frac{158,57 \cdot 10 - 30 \cdot 8 \cdot 4}{8} = 78,22\ kN.$

$\Sigma X = 0 \rightarrow G_x = 78,22\ kN.$

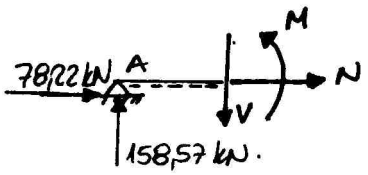
$\Sigma Y = 0 \rightarrow G_y = 81,43\ kN.$

(II) parçasında;

$\Sigma M_G = 0 \rightarrow R_{DH} = \frac{81,43 \cdot 4 - 60 \cdot 3}{8} = 18,22\ kN.$

Kesit tesirleri:

A-E arası

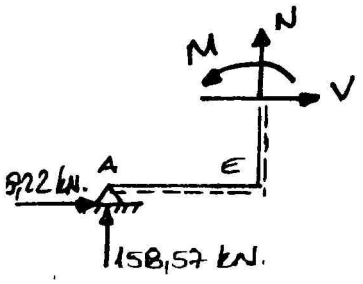


$N_{AE} = N_{EA} = -78,22$ kN.

$V_{AE} = V_{EA} = 158,57$ kN.

$M_A = 0$; $M_E = 158,57 * 4 = 634,28$ kNm.

E-B arası

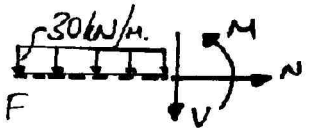


$N_{EB} = N_{BE} = -158,57$ kN.

$V_{EB} = V_{BE} = -78,22$ kN.

$M_E = 634,28$ kN. ; $M_B = 158,57 * 4 - 78,22 * 8 = -8,52$ kNm.

F-B arası

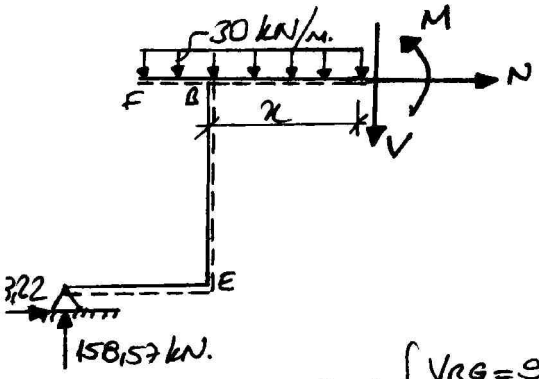


$N_{FB} = N_{BF} = 0$

$V_{FB} = 0$; $V_{BF} = -30 * 2 = -60$ kN.

$M_F = 0$; $M_{BF} = -30 * 2 * 1 = -60$ kNm.

B-G arası



$N_{BG} = N_{GB} = -78,22$ kN.

$V_x = 158,57 - 30.2 - 30.x = 98,57 - 30x$

$M_x = 158,57.(4+x) - 78,22 * 8 - 30(2+x)^2 . 1/2$

$M_x = 634,28 + 158,57x - 625,76 - 60 - 60x - 15x^2$

$M_x = -51,48 + 98,57x - 15x^2$

$x = 0 \begin{cases} V_{BG} = 98,57 \text{ kN.} \\ M_{BG} = -51,48 \text{ kNm.} \end{cases}$

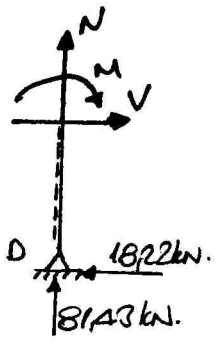
$V_x = 0 \rightarrow x_0 = \frac{98,57}{30} = 3,29$ m.

$x = 6^{m.} \begin{cases} V_{GB} = -81,43 \text{ kN.} \\ M_{GB} = 0 \end{cases}$

$max M = -51,48 + 98,57 * 3,29 - 15(3,29)^2$

$max M = 110,45$ kNm.

D-C arası

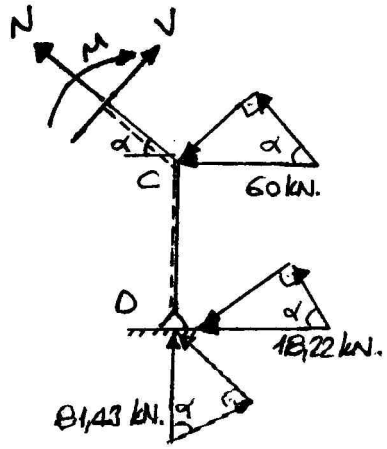


$N_{DC}=N_{CD}=-81,43$ kN.

$V_{DC}=V_{CD}=18,22$ kN.

$M_D=0$; $M_C=-18,22\cdot 5=-91,10$ kNm.

C-G arası

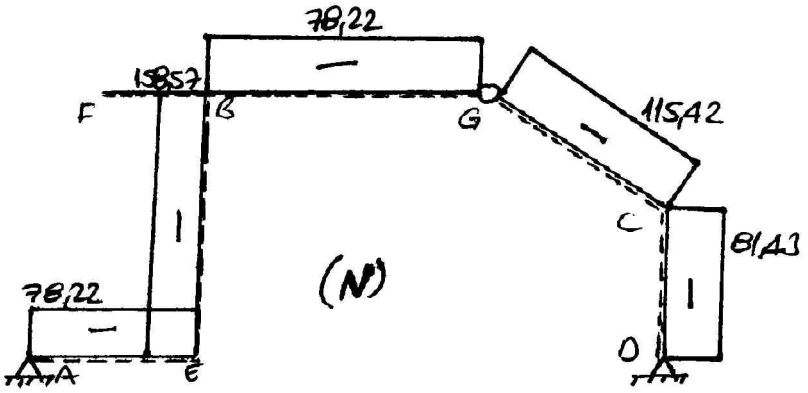


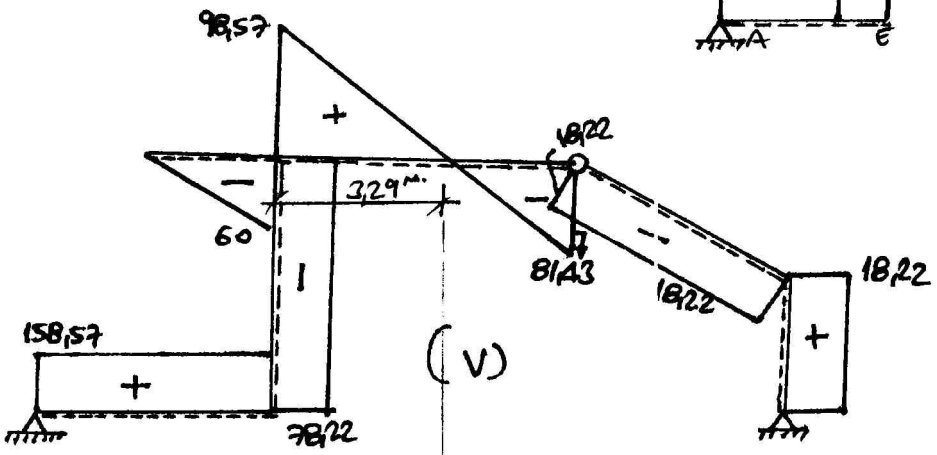
$N_{CG}=N_{GC}=-60\cdot\cos\alpha-18,22\cdot\cos\alpha-81,43\cdot\sin\alpha=-111,43$ kN.

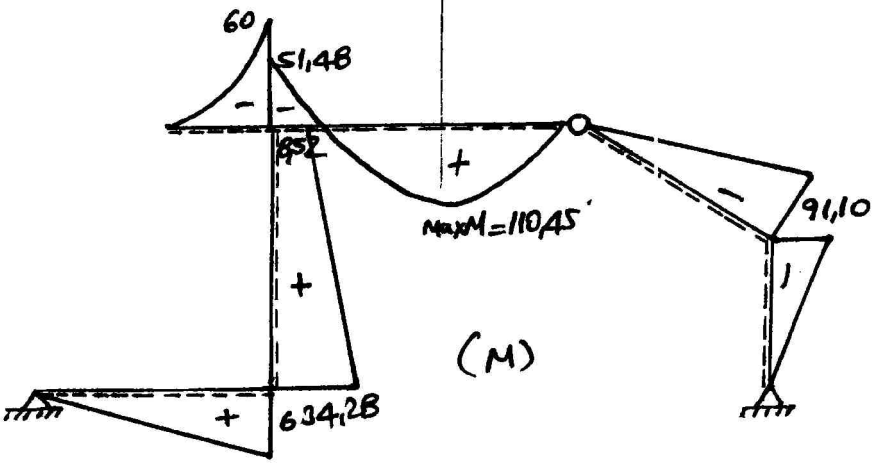
$V_{CG}=V_{GC}=-81,43\cdot\cos\alpha+18,22\cdot\sin\alpha+60\cdot\sin\alpha=-18,22$ kN.

$M_C=-18,22\cdot 5=-91,10$ kNm.

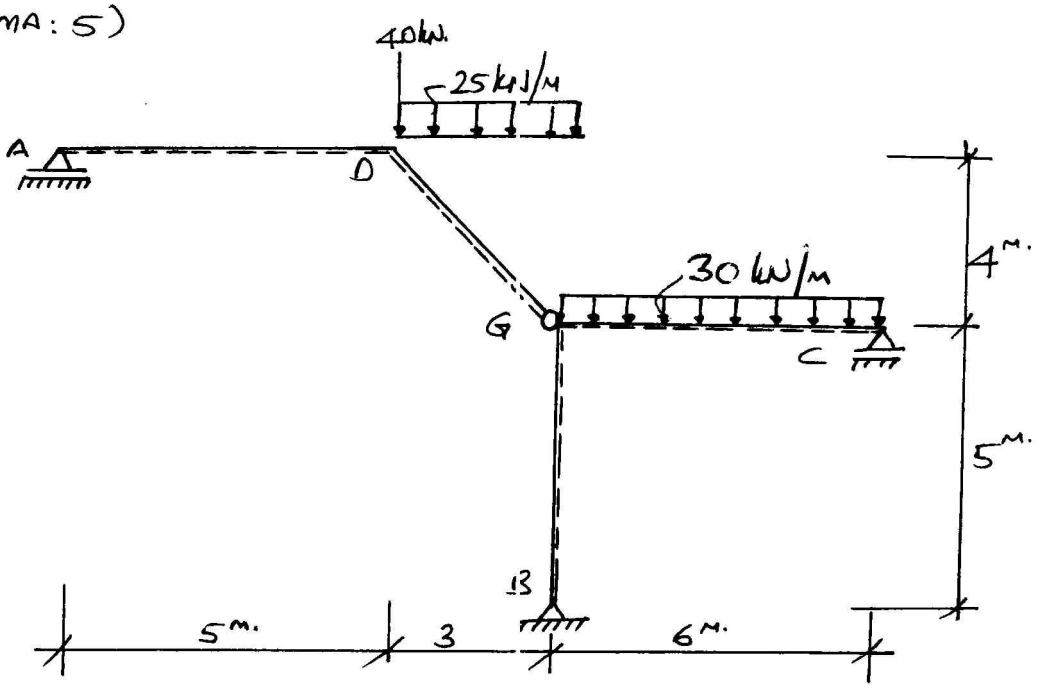
$\cos\alpha=\frac{4}{5}$; $\sin\alpha=\frac{3}{5}$

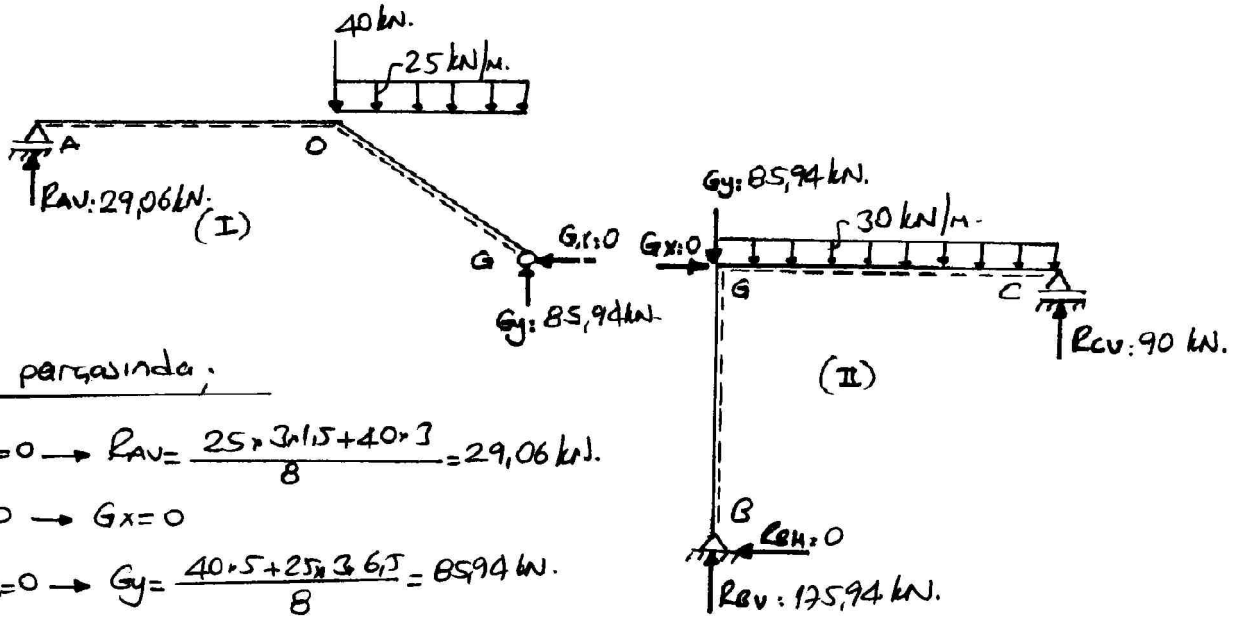






UYGULAMA: 5)





(I) parçasında;

$\Sigma M_G = 0 \rightarrow R_{AV} = \frac{25 \cdot 3 \cdot 1{,}5 + 40 \cdot 3}{8} = 29{,}06$ kN.

$\Sigma X = 0 \rightarrow G_x = 0$

$\Sigma M_A = 0 \rightarrow G_y = \frac{40 \cdot 5 + 25 \cdot 3 \cdot 6{,}5}{8} = 85{,}94$ kN.

Tüm sistemde

$\Sigma M_B = 0 \rightarrow R_{CV} = \frac{29{,}06 \cdot 8 - 40 \cdot 3 - 25 \cdot 3 \cdot 1{,}5 + 30 \cdot 6 \cdot 3}{6} = 90$ kN.

(II) parçasında

$\Sigma M_{G_1} = 0 \rightarrow R_{BH} = \frac{-29{,}06 \cdot 8 + 40 \cdot 3 + 25 \cdot 3 \cdot 1{,}5 - 30 \cdot 6 \cdot 3 + 90 \cdot 6}{5} = 0$

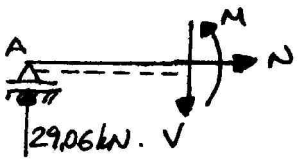
$\Sigma M_A = 0 \rightarrow R_{BV} = \frac{40 \cdot 5 + 25 \cdot 3 \cdot 6{,}5 + 30 \cdot 6 \cdot 11 - 90 \cdot 14}{8} = 175{,}94$ kN.

Kontrol: $\Sigma P_y = 0$

$29{,}06 + 175{,}94 + 90 = 40 + 25 \cdot 3 + 30 \cdot 6$

$295 = 295$ ✓

A-D arası

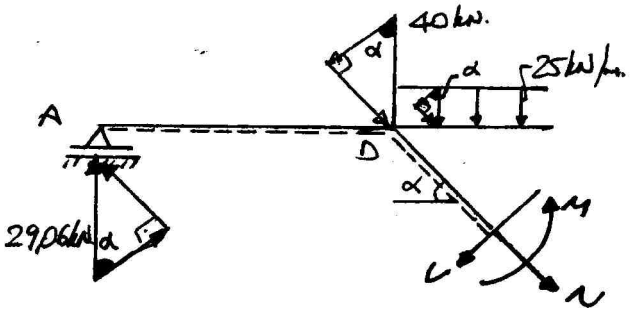


$N_{AD}=N_{DA}=0$

$V_{AD}=V_{DA}=29,06\ kN.$

$N_A=0$; $M_D=29,06\cdot 5=145,3\ kNm.$

D-G arası



$N_x=29,06\cdot\sin\alpha-40\cdot\sin\alpha-25\cdot x\cdot\sin\alpha$

$N_x=-8,752-20x$

$V_x=29,06\cdot\cos\alpha-40\cdot\cos\alpha-25\cdot\cos\alpha\cdot x$

$V_x=-6564-15x$

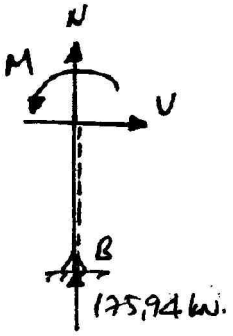
$x=0\begin{cases}N_{DG}=-8,75\ kN.\\ V_{DG}=-6,56\ kN.\end{cases}$

$x=3^m\begin{cases}N_{GD}=-8,752-20\cdot 3=-68,75\ kN.\\ V_{GD}=-6,564-15\cdot 3=-51,56\ kN.\end{cases}$

$\sin\alpha=\frac{4}{5}$; $\cos\alpha=\frac{3}{5}$

B-G arası

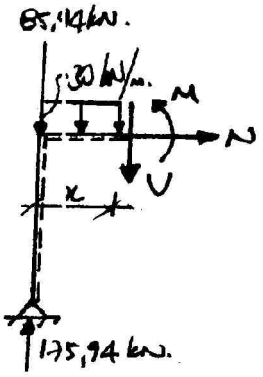
$N_{BG}=N_{GB}=175,94\ kN.$



$V_{GB}=V_{BG}=0$

$M_B=0$; $M_G=0$

G-C arası



$N_{GC}=N_{CG}=0$

$V_x=175,94-85,94-30\cdot x$

$V_x=90-30x$

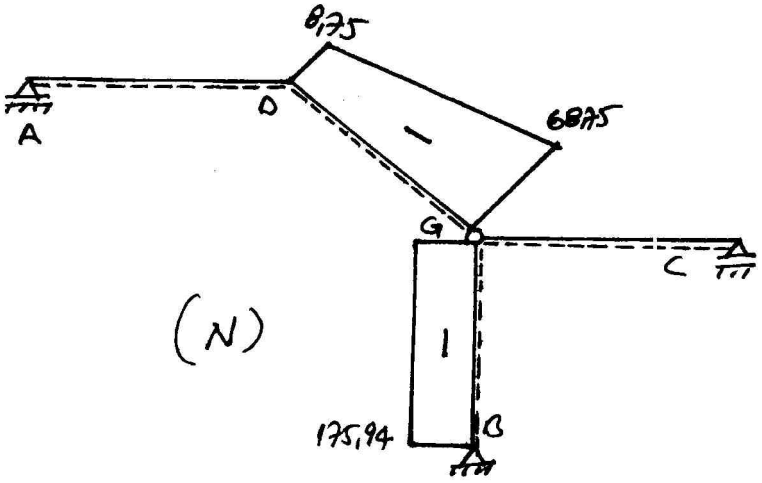
$M_x=175,94\cdot x-85,94\cdot x-30\cdot x\cdot\frac{x}{2}$

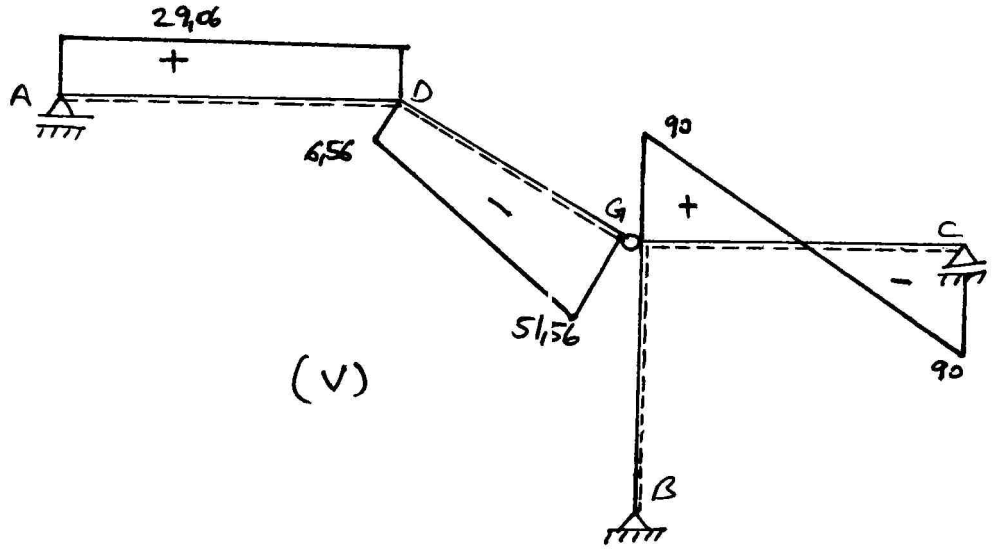
$M_x=90x-15x^2$

$x=0\begin{cases}V_{GC}=90\ kN.\\ M_{GC}=0\end{cases}$

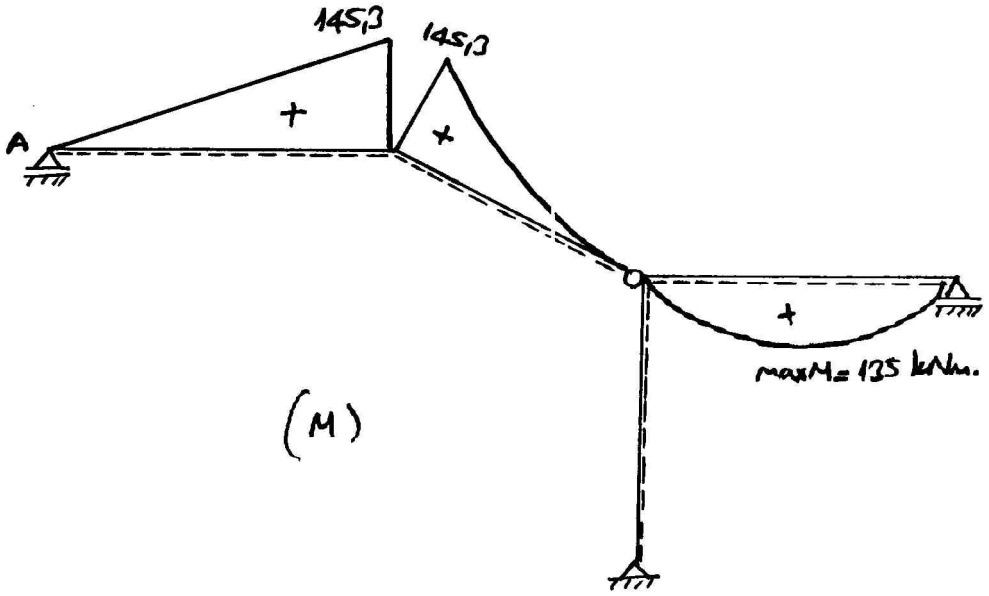
$x=6^m\begin{cases}V_{CG}=90-180=-90\ kN.\\ M_{GG}=0\end{cases}$

$M_{max}=\frac{90^2}{2\cdot 30}=135\ kNm.$



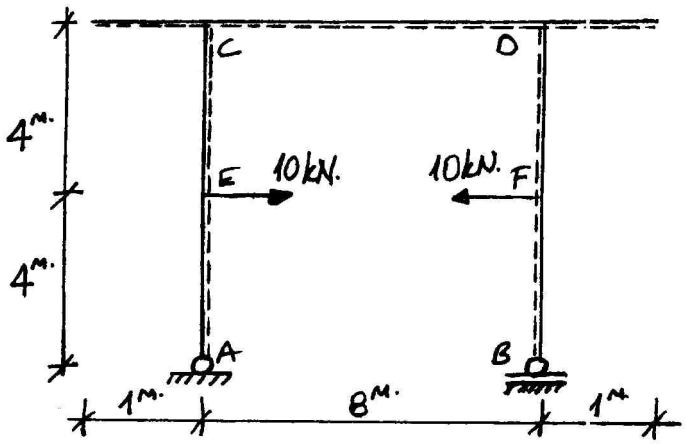


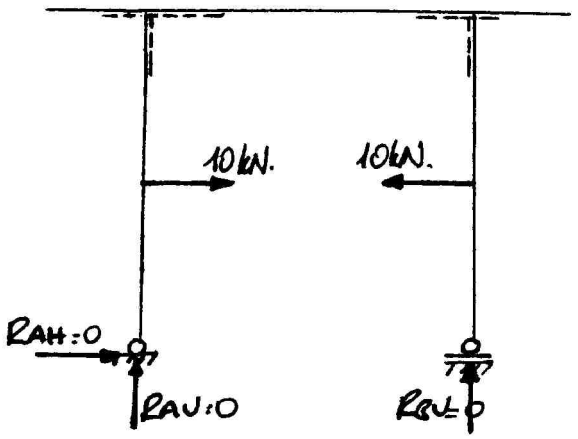
(V)



(M)

UYGULAMA : 6)



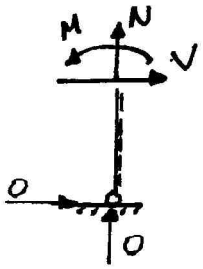


$\Sigma M_A = 0 \rightarrow R_{BV} = 0$

$\Sigma M_B = 0 \rightarrow R_{AV} = 0$

$\Sigma X = 0 \rightarrow R_{AH} = 0$

A-E arası * B-F arası



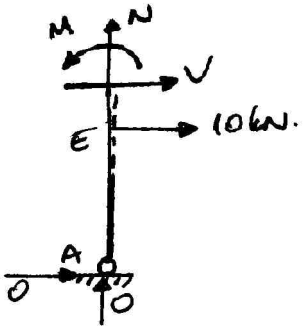
$N_{AE} = N_{EA} = 0$; $V_A = 0$

$N_{BF} = N_{FB} = 0$; $V_E = 0$

$V_{AE} = V_{EA} = 0$; $V_B = 0$

$V_{BF} = V_{FB} = 0$; $V_F = 0$

E-C arası * F-D arası



$N_{EC} = N_{CE} = 0$

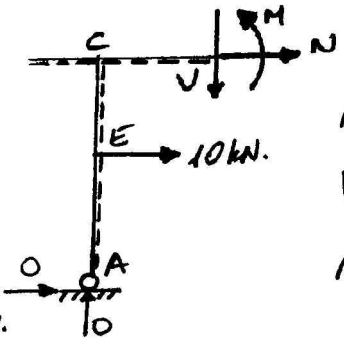
$N_{FD} = N_{DF} = 0$

$V_{EC} = V_{CE} = -10\ kN.$

$V_{FD} = V_{DF} = 10\ kN.$

$M_C = M_D = -10 \cdot 4 = -40\ kNm.$

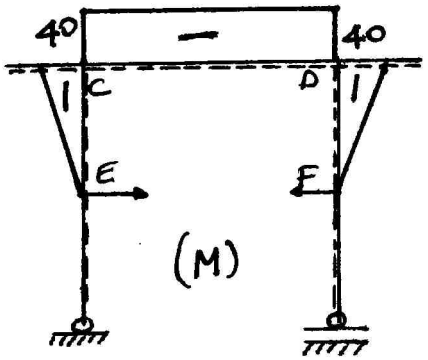
C-D arası

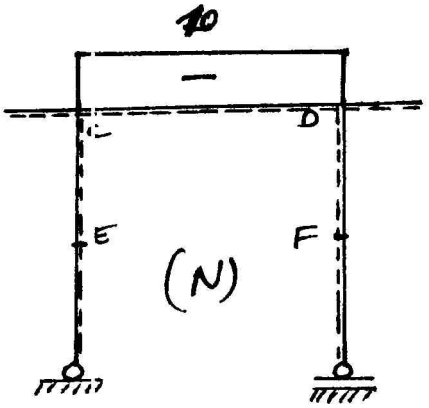


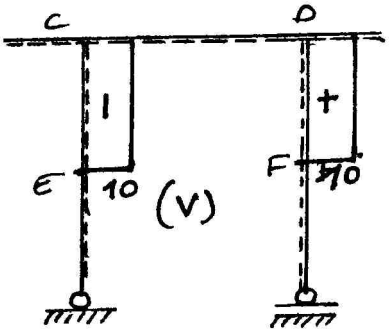
$N_{CD} = N_{DC} = -10\ kN.$

$V_{CD} = V_{DC} = 0$

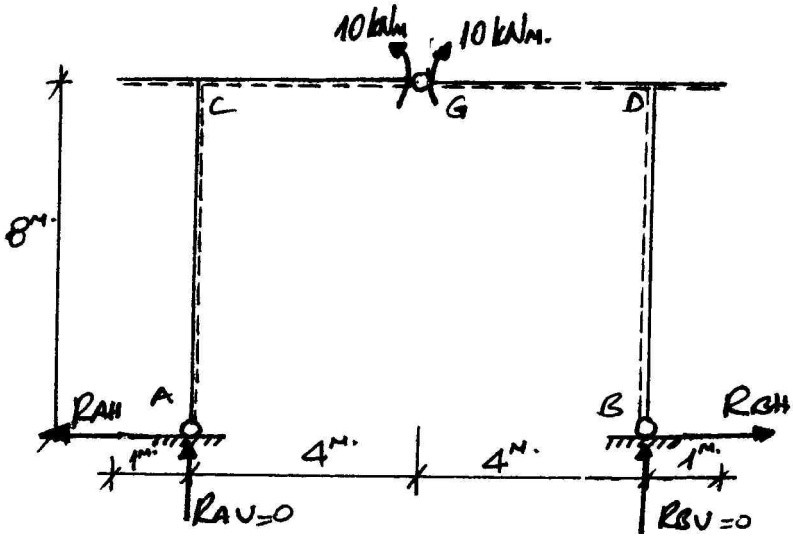
$M_{CD} = M_{DC} = -40\ kNm$







UYGULAMA : 7)



Tüm sistemde;

$\Sigma M_A = 0 \rightarrow R_{BV} = 0$

$\Sigma M_B = 0 \rightarrow R_{AV} = 0$

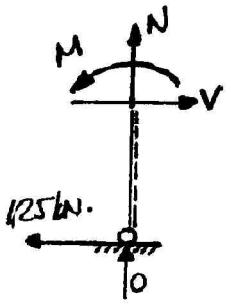
Sol sistemde;

$\Sigma M_G = 0 \rightarrow R_{AH} = \frac{10}{8} = 1{,}25\,kN.$

Sağ sistemde

$\Sigma M_G = 0 \rightarrow R_{BH} = \frac{10}{8} = 1{,}25\,kN$

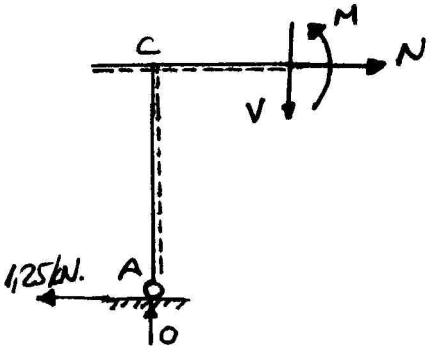
A-C arası * B-D arası



$N_{AC} = N_{CA} = 0$; $V_{AC} = V_{CA} = 1{,}25\,kN.$; $M_C = M_D = 1{,}25 \times 8 = 10\,kNm.$

$N_{BD} = N_{DB} = 0$; $V_{BD} = V_{DB} = -1{,}25\,kN.$; $M_A = M_B = 0$

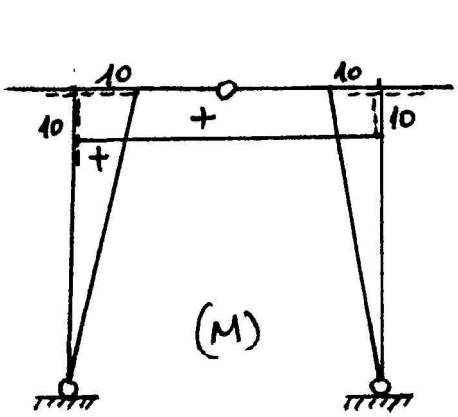
C-G arası * G-D arası

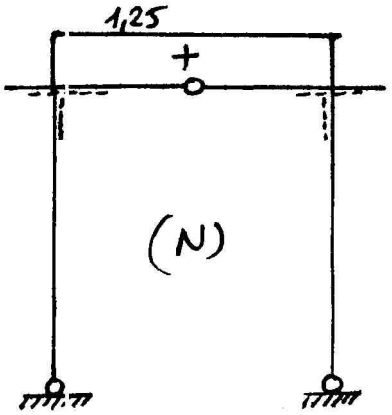


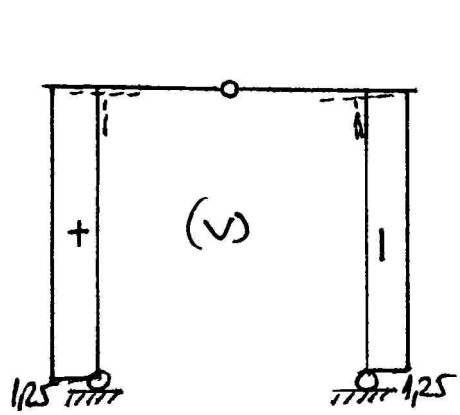
$N_{CD} = N_{DC} = 1{,}25\,kN.$

$V_{CD} = V_{DC} = 0$

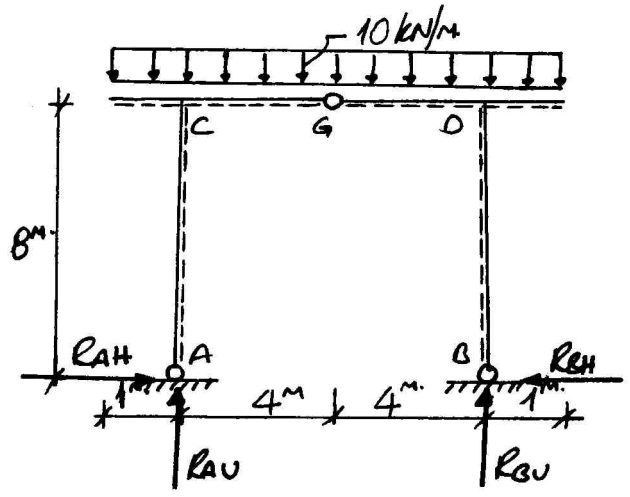
$M_{CD} = M_{DC} = 10\,kNm.$







UYGULAMA : 8)



Tüm sistemde;

$\Sigma M_B = 0 \rightarrow R_{AV} = \frac{10 \times 10 \times 4}{8} = 50\ kN.$

$\Sigma M_A = 0 \rightarrow R_{BV} = \frac{10 \times 10 \times 4}{8} = 50\ kN.$

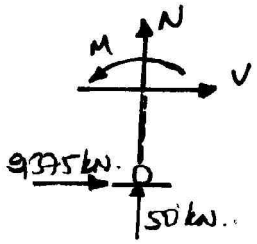
Sol parçada;

$\Sigma M_G = 0 \rightarrow R_{AH} = \frac{50 \cdot 4 - 10 \cdot 5 \cdot 2{,}5}{8} = 9{,}375\ kN.$

Sağ parçada;

$\Sigma M_G = 0 \rightarrow R_{BH} = \frac{50 \cdot 4 - 10 \cdot 5 \cdot 2{,}5}{8} = 9{,}375\ kN.$

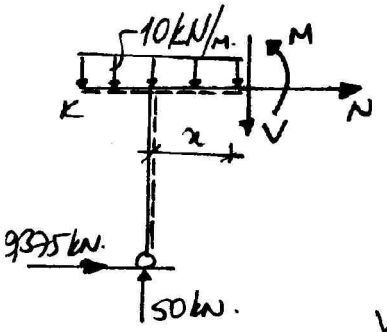
A-C arası * B-D arası



$N_{AC} = N_{CA} = -50 kN.$; $V_{AC} = V_{CA} = -9{,}38\ kN.$; $M_{CA} = M_{DB} = 9375 \cdot 8 = 75\ kNm.$

$N_{BD} = N_{DB} = -50 kN.$, $V_{BD} = V_{DB} = 9{,}38\ kN.$;

C-G arası * G-D arası



$N_{CD} = N_{DC} = -9{,}38\ kN.$

$V_x = 50 - 10.1 - 10.x = 40 - 10x$

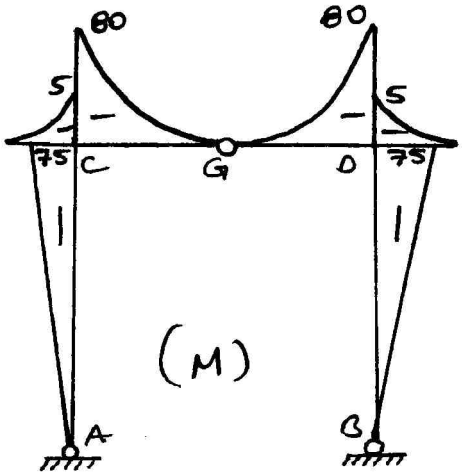
$M_x = 9{,}375 \cdot 8 + 50 \cdot x - 10(1+x) \cdot 1/2$

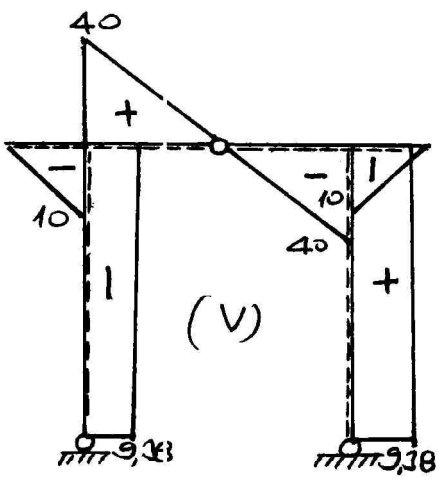
$M_x = -80 + 40x - 5x^2$

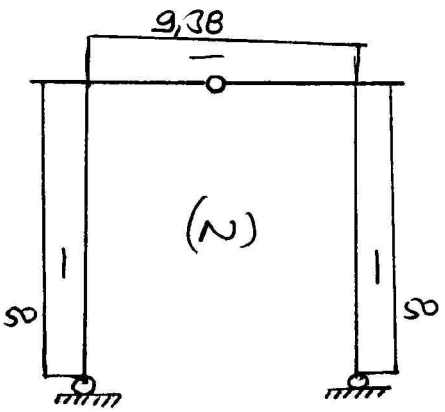
$x = 0 \begin{cases} V_{CG} = 40\ kN. \\ M_C = -80\ kNm. \end{cases}$

$x = 4^m \Rightarrow \begin{cases} V_{GC} = 40 - 10.4 = 0 \\ M_G = -80 + 160 - 165 \\ M_G = 0 \end{cases}$

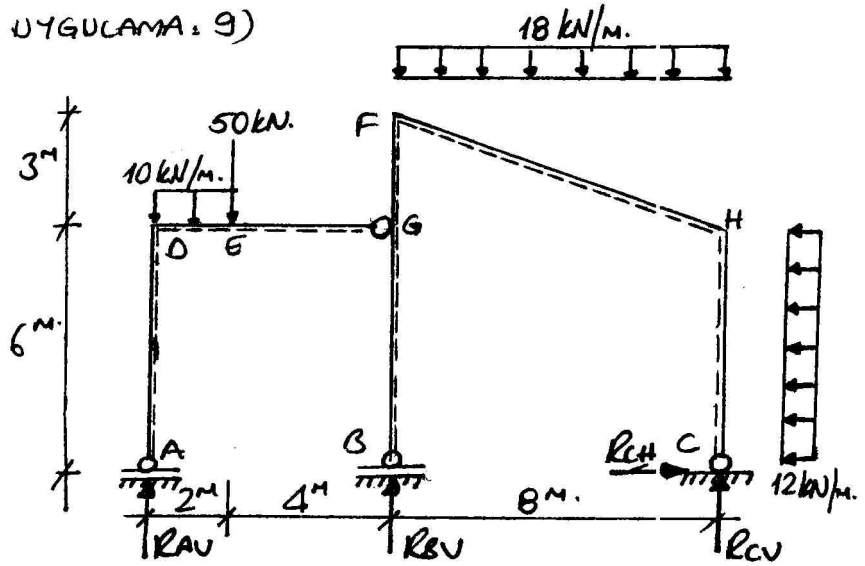
$V_{CK} = -10.1 = -10\ kN$, $M_{CK} = -10.1.0{,}5 = -5\ kNm.$







UYGULAMA: 9)



A-D-G gövdesinde,

$\Sigma M_G = 0$

$R_{AV} = \frac{10 \cdot 2 \cdot 5 + 50 \cdot 4}{6} = 50\ kN.$

Tüm sistemde;

$\Sigma M_B = 0$

$R_{CV} = \frac{50 \cdot 6 - 10 \cdot 2 \cdot 5 - 50 \cdot 4 - 12 \cdot 6 \cdot 3 + 18 \cdot 8}{8}$

$R_{CV} = 45\ kN.$

$\Sigma M_C = 0$

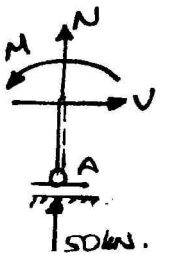
$R_{BV} = \frac{10 \cdot 2 \cdot 13 + 50 \cdot 12 - 50 \cdot 14 + 18 \cdot 8 \cdot 4 + 12 \cdot 6 \cdot 3}{8}$

$R_{BV} = 119\ kN.$

$\Sigma M_G = 0$

$R_{CH} = \frac{-45 \cdot 8 + 12 \cdot 6 \cdot 3 + 18 \cdot 8 \cdot 4}{6} = 72\ kN.$

A-D arası

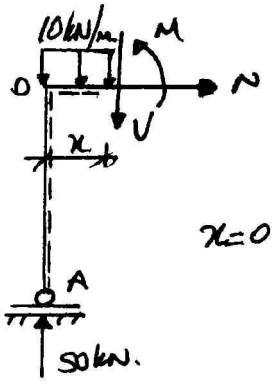


$N_{AD} = N_{DA} = -50\ kN.$

$V_{AD} = V_{DA} = 0$

$M_A = 0\ ;\ M_D = 0$

D-G arası



$N_{DG} = N_{GD} = 0$

$V_x = 50 - 10.x$

$M_x = 50x - \frac{10x^2}{2} = 50x - 5x^2$

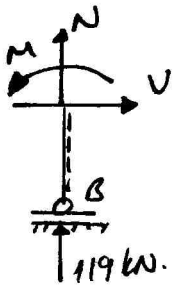
$x=0 \begin{cases} V_{DE} = 50\ kN. \\ M_D = 0 \end{cases}$ $\quad x=2^M \begin{cases} V_{ED} = 30\ kN. \\ M_E = 100 - 20 = 80\ kNm. \end{cases}$

E-G arası

$V_{GE} = V_{EG} = 30 - 50 = -20\ kN$

$\max M_E = 80\ kNm.$

B-G arası

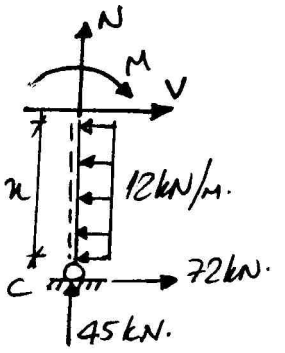


$N_{BG} = N_{GB} = -119\ kN.$

$V_{BG} = V_{GB} = 0$

$M_B = M_G = M_F = 0$

C-H arası



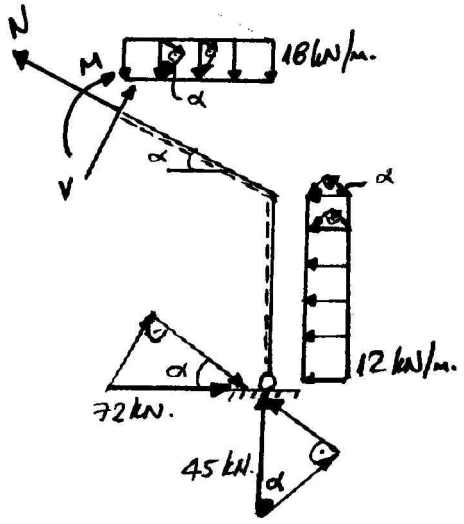
$N_{CH} = N_{HC} = -45\,kN.$

$V_x = -72 + 12.x$

$m_x = 72x - 12.\frac{x^2}{2} = +72x + 6x^2$

$x=0^M \begin{cases} V_{CH} = -72\ kN. \\ M_C = 0 \end{cases}$ $x=6^M \begin{cases} V_{HC} = -72+12.6 = 0 \\ M_H = 72.6 - 6.36 = 216\ kNm. \end{cases}$

H-F arası



$V_x = -72.\sin\alpha - 45.\cos\alpha + 12.6.\sin\alpha + 18.\cos\alpha.x$

$N_x = 72\cos\alpha - 45\sin\alpha - 12.6.\cos\alpha + 18.\sin\alpha.x$

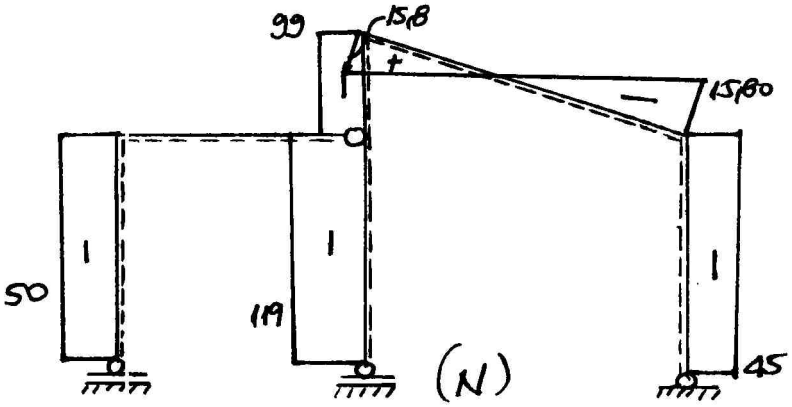
$x=0 \begin{cases} V_{FE} = -42{,}13\ kN. \\ N_{FE} = -15{,}80\ kN. \end{cases}$

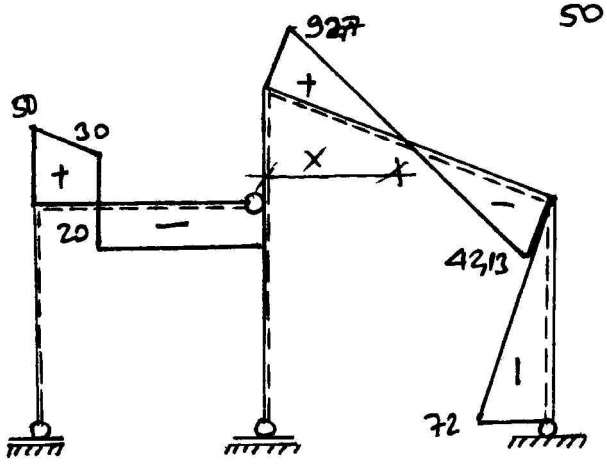
$x=6^M. \begin{cases} V_{EF} = 92{,}70\ kN. \\ N_{EF} = 15{,}80\ kN. \end{cases}$

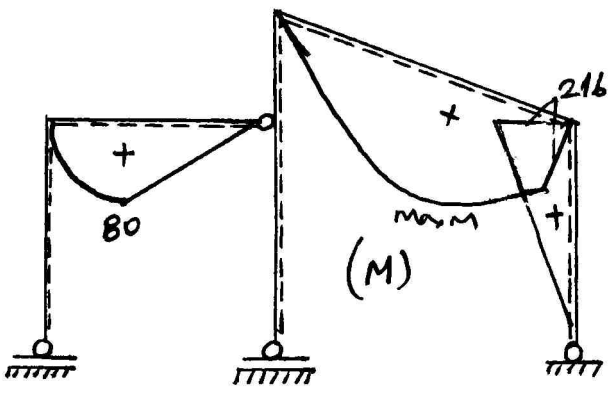
G-F arası

$V_{GF} = V_{FG} = 0$

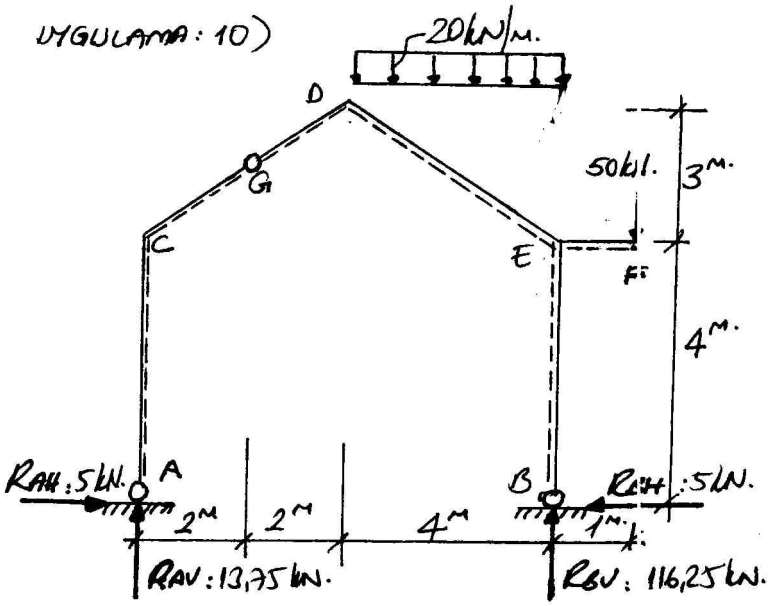
$N_{GF} = N_{FG} = -99\ kN.$







UYGULAMA: 10)



Tüm sistemde,

$\Sigma M_B = 0 \rightarrow R_{AV} = \frac{20 \cdot 4 \cdot 2 - 50 \cdot 1}{8}$

$R_{AV} = 13{,}75\ kN.$

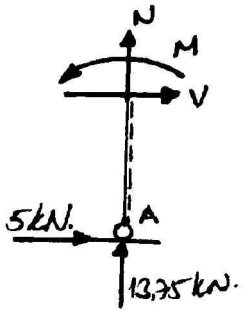
$\Sigma N_{A=0} \rightarrow R_{BV} = \frac{20 \cdot 4 \cdot 6 + 50 \cdot 9}{8} = 116{,}25\ kN.$

$\Sigma M_{G=0}^{sol} \rightarrow R_{AH} = \frac{13{,}75 \cdot 2}{5{,}5} = 5\ kN.$

$\Sigma N_{G\,sağ=0} \rightarrow R_{BH} = \frac{116{,}25 \cdot 6 - 20 \cdot 4 \cdot 4 - 50 \cdot 7}{5{,}5} =$

$R_{BH} = 5\ kN.$

A-C arası

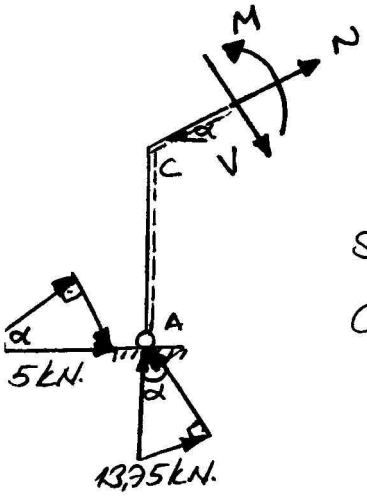


$N_{AC} = N_{CA} = -13{,}75\ kN.$

$V_{AC} = V_{CA} = -5\ kN.$

$M_A = 0,\ M_C = -5 \cdot 4 = -20\ kNm.$

C-D arası



$N_{CD} = N_{DC} = -5 \cdot \cos\alpha - 13{,}75 \cdot \sin\alpha = -12{,}25\ kN.$

$V_{CD} = V_{DC} = -5 \cdot \sin\alpha + 13{,}75 \cdot \cos\alpha = 8\ kN.$

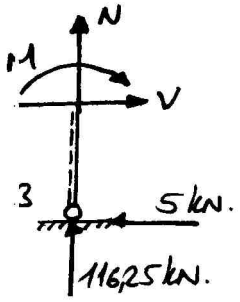
$M_C = -5 \cdot 4 = -20\ kNm.$

$M_D = 13{,}75 \cdot 4 - 5 \cdot 7 = 20\ kNm.$

$\sin\alpha = \frac{3}{5}$

$\cos\alpha = \frac{4}{5}$

B-E arası

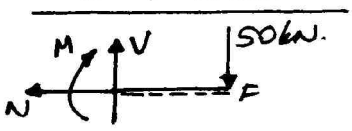


$N_{BE} = N_{EB} = -116{,}25\ kN.$

$V_{BE} = V_{EB} = 5\ kN.$

$M_B = 0;\ M_{EB} = -5 \cdot 4 = -20\ kNm.$

E-F arası

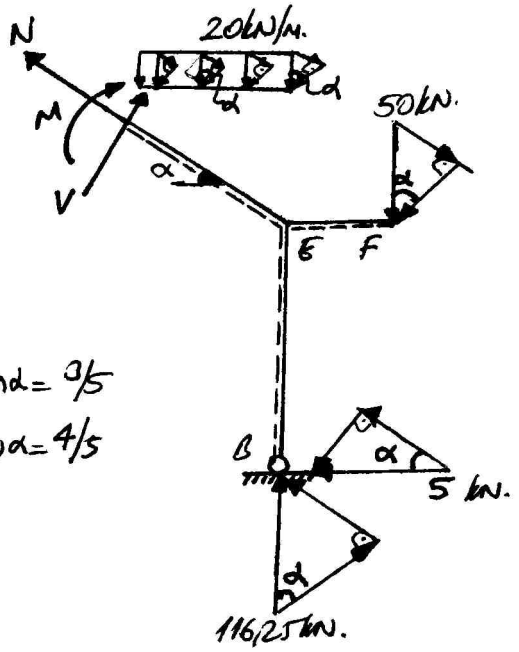


$N_{EF} = N_{FE} = 0$

$V_{EF} = V_{FE} = 50\ kN.$

$M_E = -50 \cdot 1 = -50\ kNm.$

E-D arco I



$\sin\alpha = 3/5$

$\cos\alpha = 4/5$

$Nx = -5.\cos\alpha - 116{,}25.\sin\alpha + 50.\sin\alpha + 20.x.\sin\alpha$

$Nx = -43{,}75 + 12x$

$Vx = -116{,}25.\cos\alpha + 5.\sin\alpha + 50.\cos\alpha + 20.x\cos\alpha$

$Vx = -50 + 16x$

$x=0 \begin{cases} N_{ED} = -43{,}75 \\ V_{ED} = -50\ kN. \end{cases}$ $x=4^m \begin{cases} N_{DE} = 4{,}25\ kN. \\ V_{DE} = 14\ kN. \end{cases}$

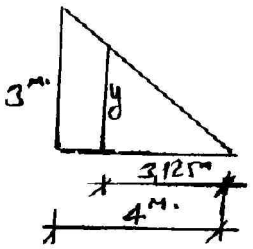
$-50 + 16.x_0 = 0$

$x_0 = 3{,}125\ m.$

$\max M = 116{,}25 \times 3{,}125 - 5 \times 6{,}34 - 50 \times 4{,}125 - 20.\overline{3{,}125}^2.1/2$

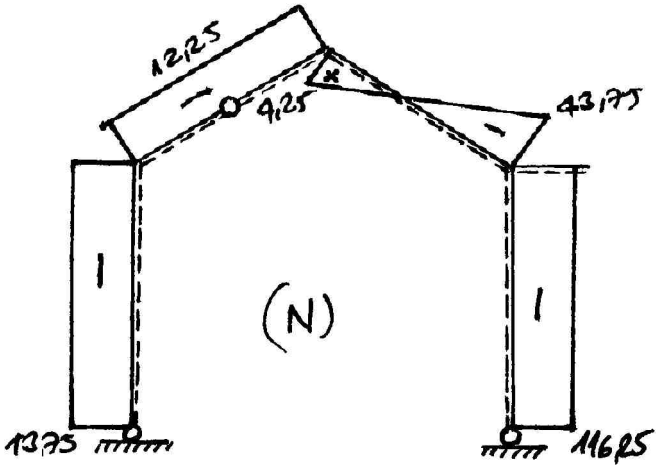
$\max M = 27{,}675\ kNm.$

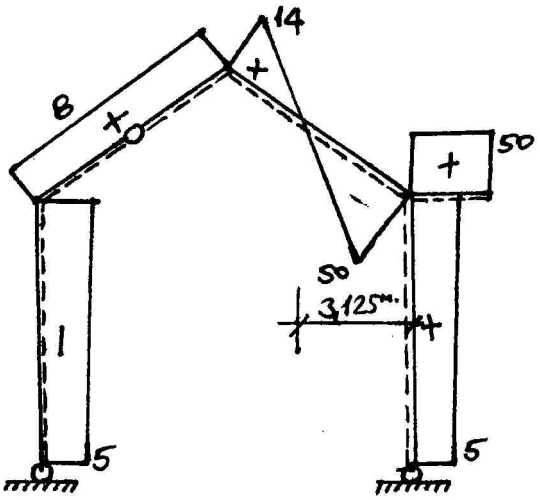
$M_{ED} = -5 \times 4 - 50.1 = -70\ kNm.$

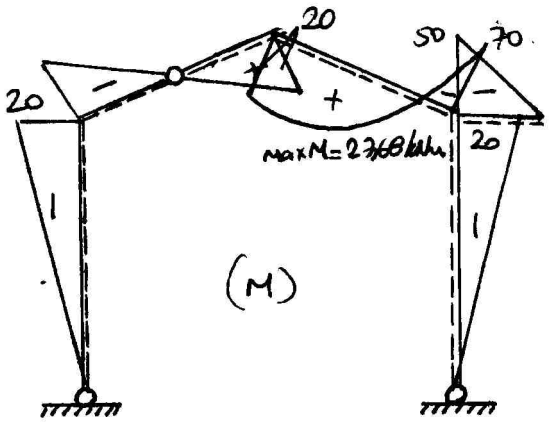


$\frac{3}{y} = \frac{4}{3{,}125}$

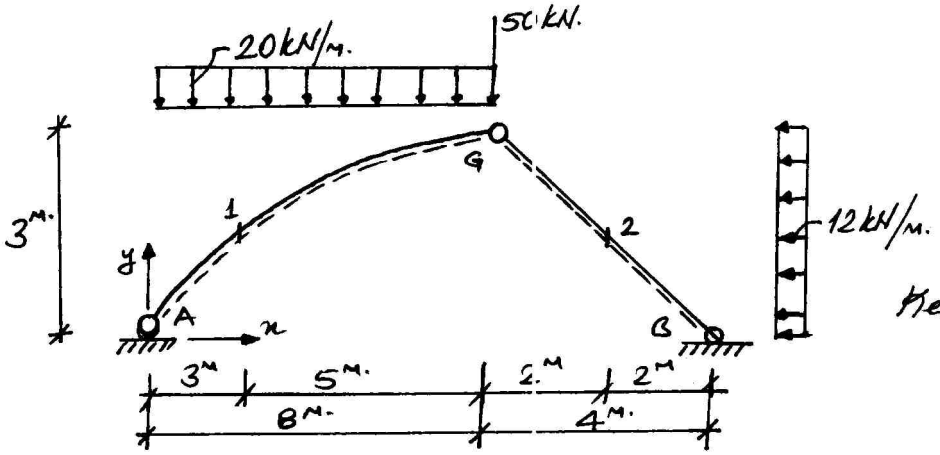
$y = 2{,}34\ m.$







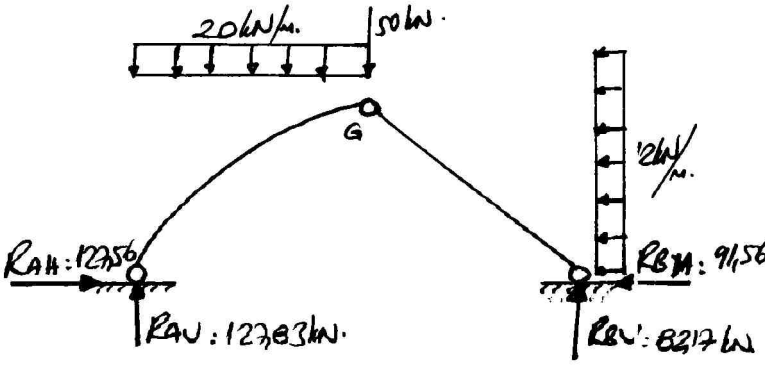
UYGULAMA: 11)



Kemer eksen eğrisi' formülüyle:

$$y = \frac{3}{64} x (16 - x)$$

Şekildeki sistemin 1 ve 2 kesitindeki kesit tesirleri değerlerini bulunuz.



Tüm sistemde,

$$\Sigma M_B = 0 \rightarrow R_{AV} = \frac{20 \cdot 8 \cdot 8 + 50 \cdot 4 + 12 \cdot 3 \cdot 1,5}{12}$$

$$R_{AV} = 127,83 \text{ kN.}$$

$$\Sigma M_A = 0 \rightarrow R_{BV} = \frac{20 \cdot 8 \cdot 4 + 50 \cdot 8 - 12 \cdot 3 \cdot 1,5}{12}$$

$$R_{AV} = 82,17 \text{ kN.}$$

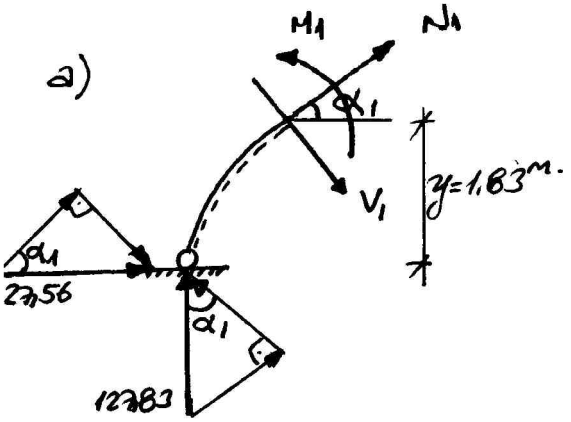
$$\Sigma M_{G_{sol}} = 0 \rightarrow R_{AH} = \frac{127,83 \cdot 8 - 20 \cdot 8 \cdot 4}{3}$$

$$R_{AH} = 127,56 \text{ kN.}$$

$$\Sigma M_{G_{sağ}} = 0 \rightarrow R_{BH} = \frac{-12 \cdot 3 \cdot 1,5 + 82,17 \cdot 4}{3}$$

$$R_{BH} = 91,56 \text{ kN.}$$

a)



$$y = \frac{3}{64} \cdot 3 \cdot (16 - 3) = 1,83 \text{ m.}$$

$$y = 0,75x - \frac{3}{64} x^2$$

$$y' = 0,75 - \frac{6}{64} x$$

$$y' = 0,75 - \frac{6}{64} \cdot 3 = \operatorname{tg}\alpha$$

$$\alpha = 25,11^\circ$$

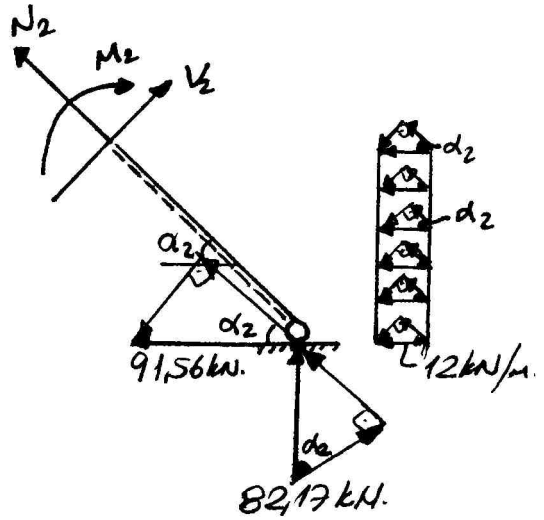
$$N_1 = -127,83 \cdot \operatorname{Sin}\alpha - 127,56 \cdot \operatorname{Cos}\alpha + 20 \cdot 3 \cdot \operatorname{Sin}\alpha$$

$$V_1 = 127,83 \cdot \operatorname{Cos}\alpha - 127,56 \cdot \operatorname{Sin}\alpha - 20 \cdot 3 \cdot \operatorname{Cos}\alpha$$

$$M_1 = 127,83 \cdot 3 - 127,56 \cdot 1,83 - 20 \cdot 3 \cdot 1,5$$

$$\left.\begin{array}{l} N_1 = -144,29 \text{ kN.} \\ V_1 = 7,29 \text{ kN.} \\ +M_1 = 60,06 \text{ kNm.} \end{array}\right\}$$

b)



$\sin\alpha_2 = \frac{3}{5}$; $\cos\alpha_2 = \frac{4}{5}$

$N_2 = -82{,}17 \cdot \sin\alpha_2 - 91{,}56 \cdot \cos\alpha_2 - 12 \cdot 1{,}5 \cdot \cos\alpha_2$

$V_2 = -82{,}17 \cdot \cos\alpha_2 + 91{,}56 \cdot \sin\alpha_2 + 12 \cdot 1{,}5 \cdot \sin\alpha_2$

$M_2 = 82{,}17 \cdot 2 - 91{,}56 \cdot 1{,}5 - 12 \cdot 1{,}5^2 \cdot 1/2$

$$\begin{cases} N_2 = -136{,}95 \text{ kN} \\ V_2 = 0 \\ M_2 = 13{,}5 \text{ kNm.} \end{cases}$$